

Lecture Note for MAT8030: Advanced Probability *

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1 Measure theory preliminaries

In this section, we review fundamental concepts in measure theory and examine their integration into modern probability theory. Most of the material remains within the scope of the seminal work, **Foundations of the Theory of Probability**, by Kolmogorov in 1933 ([Kol33]).

1.1 Random variables, σ -fields and measures

Kolmogorov introduced the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ as a unified framework for all probabilistic models. To motivate, we first recall some examples of random variables (r.v.s) that the reader should be familiar with from elementary probability. Two types of r.v.s are considered in elementary probability: *discrete* and *continuous*.

Example 1.1 (discrete r.v.s.) • **Bernoulli:** $X \sim \text{Ber}(p)$, with $\mathbb{P}(X = 1) = p$, $\mathbb{P}(X = 0) = 1 - p$.

- **binomial:** $X \sim \text{Binom}(n, p)$ with $\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, 1, \dots, n$.
- **geometry:** $X \sim \text{Geo}(p)$, with $\mathbb{P}(X = k) = (1 - p)^{k-1} p$, $k = 1, 2, \dots$.
- **Poisson:** $X \sim \text{Poi}(\lambda)$, with $\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$, $k = 0, 1, \dots$.

Example 1.2 (continuous r.v.s) Continuous r.v.s are described by the density function $\mathbb{P}(X \leq a) = \int_{-\infty}^a p(x) dx$.

- **exponential:** $X \sim \text{Exp}(\lambda)$, with $p(x) = \mathbb{1}_{[0, \infty)}(x) \cdot \lambda e^{-\lambda x}$.
- **uniform:** $X \sim \text{Unif}[a, b]$, with $p(x) = \mathbb{1}_{[a, b]}(x) \cdot \frac{1}{b-a}$.
- **normal/Gaussian:** $X \sim \mathcal{N}(\mu, \sigma^2)$, with $p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$.

Since density functions may not exist for discrete r.v.s, a unified way to study these two kinds of random variables is to use the cumulative distribution functions (c.d.f.). The c.d.f. of a r.v. X is $F_X(a) = \mathbb{P}(X \leq a)$. The sets of the form $\{X \leq a\}$ are examples of *events*, of which we can evaluate the probability. An event is simply a subset of Ω , the *sample space*. A key insight from Kolmogorov is that, not all subsets of Ω should be events. An event A will have $\mathbb{P}(A)$ as its probability. There are certain axioms that all events and $\mathbb{P}$ must satisfy, and the axioms give restriction on the events as well.

To be more precise, the collection of all the subsets that we call events is $\mathcal{F}$, and the probability is a *set function*: $\mathcal{F} \rightarrow [0, 1]$, which satisfies the following properties:

1. **Non-negativity:** $\mathbb{P}(A) \geq 0$ for every A .

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2. $P(\emptyset) = 0$.

3. **Countable additivity**: for any *disjoint* $A_1, A_2, \dots$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n). \quad (1.1)$$

Of these three properties, the last one is the most important. We call it “ σ -additivity”, where the prefix “ σ -” often indicates connection to some countable condition in measure theory. It is only with σ -additivity, not finite additivity, that one can establish non-trivial limit theorems for integration/expectation, and eventually prove various limit theorems in modern probability. More generally, we call P a *measure* if the range is $[0, \infty]$ instead of $[0, 1]$. This will be related to the general measure theory set up.

We can derived other important properties of measures from **Items 1** and **3**.

4. **Finite additivity** from **Items 2** and **3**: let $A_{n+1} = A_{n+2} = \dots = \emptyset$ in (1.1); then

$$P\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n P(A_k).$$

5. **Monotonicity** from **Items 1** and **4**: if $A \subset B$, then $A \cap (B \setminus A) = \emptyset$, and hence

$$P(B) = P(A) + P(B \setminus A) \geq P(A).$$

6. **Sub-additivity** from **Items 3** and **5**: let $\tilde{A}_n = A_n \setminus (\bigcup_{k=1}^{n-1} A_k) \subset A_n$; then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(\tilde{A}_n) \leq \sum_{n=1}^{\infty} P(A_n).$$

7. **Continuity from above** from **Items 2** and **3**: if $A_n \downarrow A$ and $P(A_1) < \infty$, then $P(A) = \lim_{n \rightarrow \infty} P(A_n)$. (We call it “continuity” since $A = \bigcap_{n=1}^{\infty} A_n$ is the limit of A_n .)

First, we have that A_1 is the disjoint union of

$$A_1 = A \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup \dots.$$

Indeed, for $\omega \in A_1$, either $\omega \in A$, or ω appears only in finitely many A_k , then $\omega \in A_k \setminus A_{k+1}$ where k is the index of the set of its last appearance. Thus, by σ -additivity, we have

$$P(A_1) = P(A) + \sum_{k=1}^{\infty} P(A_k \setminus A_{k+1}).$$

All the terms are positive, and the LHS is finite, so the tail of the infinite sum must converges to 0. Similarly, we have also

$$P(A_N) = P(A) + \sum_{k=N}^{\infty} P(A_k \setminus A_{k+1}).$$

Hence,

$$P(A) = \lim_{N \rightarrow \infty} \left(P(A_N) - \sum_{k=N}^{\infty} P(A_k \setminus A_{k+1}) \right) = \lim_{n \rightarrow \infty} P(A_n).$$

8. **Continuity from below** from **Item 3**: if $A_n \uparrow A$, then $P(A) = \lim_{n \rightarrow \infty} P(A_n)$. By assumption, we have the disjoint union

$$A = A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus A_2) \cup \dots$$

based on the first occurrence of any element $\omega \in A$. Hence, by σ -additivity,

$$\begin{aligned} P(A) &= P(A_1) + \sum_{k=1}^{\infty} P(A_{k+1} \setminus A_k) \\ &= P(A_1) + \lim_{n \rightarrow \infty} \sum_{k=1}^n P(A_{k+1} \setminus A_k) \\ &= P(A_1) + \lim_{n \rightarrow \infty} P(A_{n+1} \setminus A_1) \\ &= \lim_{n \rightarrow \infty} P(A_{n+1}), \end{aligned}$$

as desired.

These σ -additivity property of measures also impose a restriction on $\mathcal{F}$, the domain of the set function P . At the very least, the domain $\mathcal{F}$ should behave well under countable union/intersection. This leads to the definition of σ -algebras.

Definition 1.1 Let Ω be any non-empty set. A collection of its subsets $\mathcal{F}$ is a σ -algebra (or σ -field), if

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$,
3. (closure under countable union) $A_n \in \mathcal{F}$ implies $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Example 1.3 1. The smallest σ -algebra: $\mathcal{F} = \{\emptyset, \Omega\}$.

2. The largest σ -algebra: $\mathcal{F} = \{\text{all subsets of } \Omega\}$.

We call a set Ω equipped with a σ -algebra $\mathcal{F}$ on it a *measurable space*, written in a pair $(\Omega, \mathcal{F})$. The definition of the σ -algebra leads to the following properties.

Proposition 1.1 Let $\mathcal{F}$ be a σ -algebra. Then

- $\emptyset \in \mathcal{F}$,
- $A \subset B$ and $A, B \in \mathcal{F}$ imply $B \setminus A \in \mathcal{F}$,
- (closure under countable intersection) $A_n \in \mathcal{F}$ implies $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.

Definition 1.2 A probability space $(\Omega, \mathcal{F}, P)$ is such that $(\Omega, \mathcal{F})$ is a measurable space and $P : \mathcal{F} \rightarrow [0, 1]$ is a measure with $P(\Omega) = 1$.

A measure μ is finite if $\mu(\Omega) < \infty$, and σ -finite if there exists $A_n \uparrow \Omega$ such that $\mu(A_n) < \infty$. For example, the Lebesgue measure on $\mathbb{R}$ is not finite but σ -finite, since $A_n = [-n, n] \uparrow \mathbb{R}$ and $\text{Leb}(A_n) < \infty$.

Definition 1.3 A random variable (r.v.) $X = X(\omega) : \Omega \rightarrow \mathbb{R}$ is a map from a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ to $\mathbb{R}$, such that

$$\{\omega : X(\omega) \leq a\} \in \mathcal{F}, \quad \forall a \in \mathbb{R},$$

or written more compactly, $X^{-1}(-\infty, a] \in \mathcal{F}$ for all $a \in \mathbb{R}$.

The map φ^{-1} denotes the *pre-image map* of φ : for a map $\varphi : U \rightarrow V$, we define φ^{-1} to be

$$\varphi^{-1}(W) := \{u \in U : \varphi(u) \in W\}.$$

The map φ^{-1} commutes with common set operations.

Proposition 1.2 • $\varphi^{-1}(W_1 \cap W_2) = \varphi^{-1}(W_1) \cap \varphi^{-1}(W_2)$.

- $\varphi^{-1}(W_1 \cup W_2) = \varphi^{-1}(W_1) \cup \varphi^{-1}(W_2)$.
- $\varphi^{-1}(W^c) = (\varphi^{-1}(W))^c$.

For a r.v. X on $(\Omega, \mathcal{F}, \mathbf{P})$, we can evaluate the probability of *events*, which are sets of the form

$$\{\omega : X(\omega) \in A\} =: X^{-1}(A). \quad (1.2)$$

To have a probability, the sets in (1.2) must be in the domain of $\mathbf{P}$, which is $\mathcal{F}$. This imposes a restriction on A . Let us put all such sets A into a collection

$$\mathcal{B} = \{A \text{ s.t. } X^{-1}(A) \in \mathcal{F}\}.$$

Definition 1.3 implies that $\mathcal{B}$ contains all half-infinite interval $(-\infty, a]$, and by **Proposition 1.2** it contains all intervals $(a, b]$ since

$$X^{-1}(a, b] = X^{-1}(-\infty, b] \setminus X^{-1}(-\infty, a] \in \mathcal{F}.$$

In fact, the collection $\mathcal{B}$ is a σ -algebra, since we can verify the closure under countable union using **Proposition 1.2**:

$$X^{-1}(I_n) \in \mathcal{F} \quad \Rightarrow \quad X^{-1}\left(\bigcup_{n=1}^{\infty} I_n\right) = \bigcup_{n=1}^{\infty} X^{-1}(I_n) \in \mathcal{F}.$$

In the next section, we will see that $\mathcal{B}$ will contain a common sub- σ -algebra, called the *Borel σ -algebra*, no matter what the r.v. X is. The Borel σ -algebra is the “smallest” σ -algebra containing all the intervals $(a, b]$; it is the most important class of σ -algebras in probability theory.

Counterexample 1.4 (Finite additivity vs. σ -additive.) Let $\Omega = [0, 1] \cap \mathbb{Q}$ and $\mathcal{F}_1$ be the class of subsets of Ω of the form (a, b) , $[a, b)$, $(a, b]$ or $[a, b]$ where $a, b \in \mathbb{Q}$. Let $\mathcal{F}_2$ be the class of all **finite** sum of disjoint sets of $\mathcal{F}_1$. Let $\mathbf{P}$ be define by

$$\mathbf{P}(A) = \sum_{k=1}^n (b_k - a_k), \quad A = \bigcup_{k=1}^n A_k, \quad A_k \text{ has endpoints } a_k, b_k.$$

Then $\mathbf{P}(\cdot)$ is finitely additive.

But $\mathbf{P}(\cdot)$ is not σ -additive. To see this, we can prove by contradiction. Notice that $\mathbf{P}(\{r\}) = 0$ for $r \in \mathbb{Q}$. If $\mathbf{P}(\cdot)$ is σ -additive, then

$$\mathbf{P}(\Omega) = \sum_{r \in [0, 1] \cap \mathbb{Q}} \mathbf{P}(\{r\}) = 0.$$

But $\mathbf{P}(\Omega) = 1 - 0 = 1$ by definition. This is a contradiction.

1.2 Construction of σ -algebra and (probability) measures

The Borel σ -algebra is the *smallest* σ -algebra containing by open sets. The meaning of “smallest” will be clear after the following proposition.

Lemma 1.3 1. If $\mathcal{F}_1$ and $\mathcal{F}_2$ are two σ -algebras on Ω , then $\mathcal{F}_1 \cap \mathcal{F}_2$ is also a σ -algebra.

2. If $\mathcal{F}_\gamma, \gamma \in \Gamma$ are σ -algebras on Ω , where Γ is an arbitrary index set (countable or uncountable), then $\bigcap_{\gamma \in \Gamma} \mathcal{F}_\gamma$ is also a σ -algebra.

Proposition 1.4 Let $\mathcal{A}$ be a collection of subsets in Ω . Then there exists a smallest σ -algebra containing $\mathcal{A}$, called the σ -algebra generated by $\mathcal{A}$ and written $\sigma(\mathcal{A})$, in the sense that if $\mathcal{G} \supset \mathcal{A}$ is a σ -algebra, then $\sigma(\mathcal{A}) \subset \mathcal{G}$.

Proof: Take $\sigma(\mathcal{A}) = \bigcap_{\mathcal{F} \text{ } \sigma\text{-algebra: } \mathcal{F} \supset \mathcal{A}} \mathcal{F}$. □

Definition 1.4 (Borel σ -algebra) Let M be a metric space (or more generally, a topological space). Its Borel σ -algebra $\mathcal{B}(M)$ is the σ -algebra generated by all the open sets in M .

Example 1.5 • $\mathcal{B}(\mathbb{R}) = \sigma((-\infty, a], a \in \mathbb{R})$.

• $\mathcal{B}(\mathbb{R}^d) = \sigma((-\infty, a_1] \times \cdots \times (-\infty, a_d], a_i \in \mathbb{R})$.

Remark 1.6 Here, one need to first show that any open sets in $\mathbb{R}^d$ can be obtained from countable union of sets of the form $(-\infty, a_1] \times \cdots \times (-\infty, a_d]$. The construction requires some ideas from point-set topology, but it is elementary, and thus omitted here.

Proposition 1.5 A map $X(\omega)$ on $(\Omega, \mathcal{F}, \mathbf{P})$ is a r.v. if and only if $X^{-1}(A) \in \mathcal{F}$ for any $A \in \mathcal{B}(\mathbb{R})$.

Remark 1.7 In fact, this is usually taken as the definition for r.v.s.

Now let us take about the distribution of a r.v. X . One can check that $\mu = \mathbf{P} \circ X^{-1}$ defined by

$$\mu(A) = \mathbf{P}(\{\omega : X(\omega) \in A\}), \quad A \in \mathcal{B}(\mathbb{R}),$$

is a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We call μ the *distribution/law* of X . Clearly, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ is a probability space. For most practical application, like computing expectation, variance and so on, it is enough to understand the distribution of a r.v., not the original probability measure $\mathbf{P}$ on some abstract space that can be potentially be complicate. Another obvious advantage is that the distributions of all r.v.s are probability measures live on the *same* measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Note that the *cumulative distribution function (c.d.f.)* of a r.v. can be read from its distribution:

$$F_X(a) = \mathbf{P}(X \leq a) = \mu((-\infty, a]), \quad a \in \mathbb{R}.$$

The central topic for this section is to understand how the c.d.f. determines μ . Along the way we will learn how to construct σ -algebras and (probability) measures. Some of the presentation here is from [Shi96, Chap. 2.3]. The next theorem is a fundamental and important result.

Theorem 1.6 Every increasing, right continuous function $F : \mathbb{R} \rightarrow [0, 1]$ with $F(-\infty) = 0$ and $F(\infty) = 1$ uniquely determines a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

We start by introducing some notions on collections of sets.

Definition 1.5 A collection of sets $\mathcal{S}$ is a semi-algebra if first, it is closed under intersection, that is, $A \cap B \in \mathcal{S}$ whenever $A, B \in \mathcal{S}$ and second, for every $A \in \mathcal{S}$, its complement A^c is disjoint union of some $A_1, A_2, \dots, A_n$ in $\mathcal{S}$.

A collection of sets $\mathcal{S}$ is an algebra, or field, if $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$ and $A^c \in \mathcal{S}$.

We can relate these two notions by the following proposition.

Proposition 1.7 Let $\mathcal{S}$ be a semi-algebra. Then

$$\bar{\mathcal{S}} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$$

is an algebra.

Example 1.8 All the d -dimensional half-open, half-closed rectangles forms a semi-algebra:

$$\mathcal{S}_d = \{\emptyset, (a_1, b_1] \times \dots \times (a_d, b_d], -\infty \leq a_i < b_i \leq \infty\}.$$

Definition 1.6 A collection of sets $\mathcal{S}$ is a monotone class, if $\lim_{n \rightarrow \infty} A_n \in \mathcal{S}$ for every monotone sequence of sets $A_n \in \mathcal{S}$.

Here, for an increasing sequence $A_n \subset A_{n+1} \subset \dots$, we define its limit to be $A := \bigcup_{n=1}^{\infty} A_n$, and for an decreasing sequence $A_n \supset A_{n+1} \supset \dots$, its limit $A := \bigcap_{n=1}^{\infty} A_n$.

It is easy to see that any intersection of monotone classes is still an m -class. Therefore, it makes sense to talk about the *smallest* monotone classes containing any collection of sets $\mathcal{A}$ (c.f. **Proposition 1.4**). We denote this smallest monotone class by $m(\mathcal{A})$.

The monotone class condition basically bridges the difference between σ -algebras and algebras.

Proposition 1.8 Let $\mathcal{A}$ be a collection of subsets of Ω . Then $\mathcal{A}$ is a σ -algebra if and only if $\mathcal{A}$ is both an algebra and a monotone class.

Theorem 1.9 (Monotone Class Theorem) Let $\mathcal{A}$ be an algebra. Then $\sigma(\mathcal{A}) = m(\mathcal{A})$.

Proof: By **Proposition 1.8**, $\sigma(\mathcal{A})$ is necessarily a monotone class, and by the minimum property we have the inclusion $m(\mathcal{A}) \subset \sigma(\mathcal{A})$.

To show the other direction $\sigma(\mathcal{A}) \subset m(\mathcal{A})$, it suffices to show that $m(\mathcal{A})$ is an algebra, and hence a σ -algebra (using **Proposition 1.8** again). To establish that $m(\mathcal{A})$ is an algebra, we will use the *principle of appropriate sets*.

First, $m(\mathcal{A})$ is closed under complement. Let

$$\mathcal{S} = \{A : A, A^c \in m(\mathcal{A})\} \subset m(\mathcal{A}).$$

Our goal is to show that $m(\mathcal{A}) = \mathcal{S}$. Clearly, by definition we have $\mathcal{A} \in \mathcal{S}$. Moreover, $\mathcal{S}$ is a monotone class: if $A_n \uparrow A$ and $A_n \in \mathcal{S}$, then (A_n) and (A_n^c) are both monotone sequences in $m(\mathcal{A})$, and hence their respective limits A and A^c are in $m(\mathcal{A})$; if $A_n \downarrow A$ it is similar. Therefore, $\mathcal{S}$ must contain the smallest monotone class that contains $\mathcal{A}$, which is $m(\mathcal{A})$. This shows $\mathcal{S} = m(\mathcal{A})$, and hence by the definition of $\mathcal{S}$, the collection of set $m(\mathcal{A})$ is closed under complement.

Second, $m(\mathcal{A})$ is closed under intersection. Since intersection involves two sets, the proof is slightly more complicated and we will do it in two steps. In the first step, for a fixed $A \in \mathcal{A}$, let

$$\mathcal{S}_A = \{B : B \in m(\mathcal{A}), A \cap B \in m(\mathcal{A})\} \subset m(\mathcal{A}).$$

It is clear that $\mathcal{A} \subset \mathcal{S}_A$ since A is an algebra and $m(\mathcal{A})$ contains $\mathcal{A}$. Also, one can check that $\mathcal{S}_A$ is a monotone class since $B_n \downarrow B$ or $B_n \uparrow B$ implies $A \cap B_n \downarrow A \cap B$ or $A \cap B_n \uparrow A \cap B$. Therefore, we have $m(\mathcal{A}) \subset \mathcal{S}_A$, and this means that $A \cap B \in m(\mathcal{A})$ whenever $A \in \mathcal{A}$ and $B \in m(\mathcal{A})$.

In the second step, let

$$\mathcal{S} = \{A \in m(\mathcal{A}) : A \cap B \in m(\mathcal{A}), \forall B \in m(\mathcal{A})\}.$$

The first step implies that $\mathcal{A} \subset \mathcal{S}$. Again, it is not hard to check that $\mathcal{A}$ is a monotone class. Hence $m(\mathcal{A}) = \mathcal{S}$ and this proves that $m(\mathcal{A})$ is closed under intersection.

In conclusion, $m(\mathcal{A})$ is an algebra and hence a σ -algebra, this completes the proof. $\square$

A related concept is the Dynkin system (d-system, λ -class).

Definition 1.7 Let $\mathcal{D}$ be a collection of subsets of Ω . We say that $\mathcal{D}$ is a Dynkin system if

1. $\Omega \in \mathcal{D}$,
2. $A, B \in \mathcal{D}, A \subset B \Rightarrow B \setminus A \in \mathcal{D}$,
3. $A_n \uparrow A, A_n \in \mathcal{D} \Rightarrow A \in \mathcal{D}$.

We say that $\mathcal{A}$ is a π -system if it is closed under intersection. One can check that $\mathcal{A}$ is a σ -algebra if and only if it is both a π -system and Dynkin system. Moreover, analogous to [Theorem 1.9](#), the following is true.

Theorem 1.10 (π - λ Theorem; Dynkin Theorem) If $\mathcal{A}$ is a π -system, then $\sigma(\mathcal{A})$ is the smallest Dynkin system containing $\mathcal{A}$.

Proof: The proof can be done via the principle of appropriate sets. $\square$

Given a distribution function F as in [Theorem 1.6](#), we can introduce a (probability) measure μ_0 on the algebra

$$\bar{\mathcal{S}} = \left\{ \bigcup_{k=1}^n (a_k, b_k], \text{ disjoint union} \right\},$$

given by

$$\mu_0(A) = \sum_{k=1}^n [F(b_k) - F(a_k)].$$

It is easy to check that μ_0 is finitely additive. An important step is the following.

Proposition 1.11 The finitely additive measure μ_0 is σ -additive on $\bar{\mathcal{S}}$, that is, if $A_n \in \bar{\mathcal{S}}$ are disjoint and $\bigcup_{n=1}^{\infty} A_n \in \bar{\mathcal{S}}$, then

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

Proof: We will use the fact that σ -additivity is equivalent to continuity at $\emptyset$, that is, μ_0 is σ -additive if and only if $\lim_{n \rightarrow \infty} \mu_0(A_n) = \mu_0(\emptyset) = 0$ whenever $A_n \downarrow \emptyset$.

Suppose that there is some $L > 0$ such that $A_n \in [-L, L]$. Let $\varepsilon > 0$. We claim that there exists $B_n \in \bar{\mathcal{S}}$ such that $\overline{B_n} \subset A_n$ and

$$\mu_0(A_n) - \mu_0(B_n) \leq \varepsilon \cdot 2^{-n}.$$

The existence of B_n is a consequence of the right continuity of F . In fact, writing $A_n = \bigcup_{i=1}^m (a_i^{(n)}, b_i^{(n)}]$, and $B_n = \bigcup_{i=1}^m (a_i^{(n)} + \delta, b_i^{(n)}]$, we have

$$\mu_0(A_n) - \mu_0(B_n) = \sum_{i=1}^m (F(b_i^{(n)} + \delta) - F(b_i^{(n)})) \rightarrow 0, \quad \delta \downarrow 0.$$

By choosing δ small enough we can make the sum less than $\varepsilon \cdot 2^{-n}$.

We claim that

$$[-L, L] \subset \bigcup_{n=1}^{\infty} [-L, L] \setminus \overline{B_n}. \quad (1.3)$$

Indeed, for $x \in [-L, L]$, since $A_n \downarrow \emptyset$, there exists n_0 such that $x \notin A_{n_0}$. Since $\overline{B_{n_0}} \subset A_{n_0}$, and hence $x \in [-L, L] \setminus \overline{B_{n_0}}$. This proves the claim. Note that (1.18) is an open cover for $[-L, L]$. By the Finite Open Cover Theorem, there exists a finite sub-cover, that is, there exists n_0 such that

$$[-L, L] \subset \bigcup_{n=1}^{n_0} [-L, L] \setminus \overline{B_n},$$

and hence $\bigcap_{n=1}^{n_0} \overline{B_n} = \emptyset$. Therefore,

$$\mu_0(A_{n_0}) = \mu_0\left(A_{n_0} \setminus \bigcap_{n=1}^{n_0} \overline{B_n}\right) = \mu\left(\bigcap_{n=1}^{n_0} A_n \setminus \bigcap_{n=1}^{n_0} \overline{B_n}\right) \leq \mu_0\left(\bigcup_{n=1}^{n_0} (A_n \setminus \overline{B_n})\right) \leq \sum_{n=1}^{n_0} \mu_0(A_n \setminus B_n) \leq \varepsilon \sum_{n=1}^{\infty} 2^{-n} \leq \varepsilon.$$

Since $\mu_0(A_n)$ is decreasing and ε is arbitrary, we obtain $\lim_{n \rightarrow \infty} \mu_0(A_n) = 0$.

When A_n are unbounded, since $F(-\infty) = 0$ and $F(\infty) = 1$, for every $\varepsilon > 0$, we can choose L large enough so that $\mu_0(-L, L] \geq 1 - \varepsilon$. Let $\tilde{A}_n = A_n \cap (-L, L]$. Then $\tilde{A}_n \downarrow \emptyset$ and $\tilde{A}_n$ are bounded. Then $\lim_{n \rightarrow \infty} \mu_0(\tilde{A}_n) = 0$ as previously proved, and hence

$$\limsup_{n \rightarrow \infty} \mu_0(A_n) \leq \limsup_{n \rightarrow \infty} \mu_0(\tilde{A}_n) + \limsup_{n \rightarrow \infty} \mu_0(A_n \setminus (-L, L]) \leq 0 + \varepsilon = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we obtain $\lim_{n \rightarrow \infty} \mu_0(A_n) = 0$ as desired. $\square$

After establishing σ -additivity of μ_0 on $\tilde{\mathcal{S}}$ using [Proposition 1.11](#), we can extend μ_0 to a probability measure on $\sigma(\tilde{\mathcal{S}}) = \mathcal{B}(\mathbb{R})$ with the help of the next theorem.

Theorem 1.12 (Carathéodory's Extension Theorem) *Let μ_0 be a σ -additive measure on an algebra $\mathcal{A}$. Then μ_0 has a unique extension to $\sigma(\mathcal{A})$.*

Here, an extension of μ_0 to $\sigma(\mathcal{A})$ is a measure μ on $\sigma(\mathcal{A})$ such that $\mu_0(A) = \mu(A)$ for every $A \in \mathcal{A}$.

Remark 1.9 We will use [Theorem 1.12](#) in the case where μ_0 (and hence the resulting extension μ) is a *probability* measure. But the theorem also holds when μ_0 is σ -finite, which means that there exist $A_n \uparrow \Omega$ such that $\mu_0(A_n) < \infty$.

Proof of Uniqueness: Let $\mu, \tilde{\mu}$ be two extensions and $\mathcal{S} = \{A : \mu(A) = \tilde{\mu}(A)\}$. We will show (i) $\mathcal{A} \subset \mathcal{S}$; (ii) $\mathcal{A}$ is a monotone class. Then, by [Theorem 1.9](#), $\mathcal{S}$ contains $\sigma(\mathcal{A})$, so $\mu = \tilde{\mu}$ on $\sigma(\mathcal{A})$, which is the uniqueness.

The first statement $\mathcal{A} \subset \mathcal{S}$ follows from definition of the extension.

To prove the second statement, let $A_n \uparrow A$ and $A_n \in \mathcal{S}$. Since μ and $\tilde{\mu}$ are measures, and measures are continuous from below, we have $\mu(A_n) \rightarrow \mu(A)$ and $\tilde{\mu}(A_n) \rightarrow \tilde{\mu}(A)$, and thus $\mu(A) = \tilde{\mu}(A)$. Similarly, if $A_n \downarrow A$ and $A_n \in \mathcal{S}$, since μ is continuous from above, we have $\mu(A_n) \rightarrow \mu(A)$ and $\tilde{\mu}(A_n) \rightarrow \tilde{\mu}(A)$, and thus $\mu(A) = \tilde{\mu}(A)$. This completes the proof of uniqueness. $\square$

To prove the existence we need to use the outer measure, which is also a standard procedure in constructing the Lebesgue measure. We will only sketch the most important steps in this note.

Given a σ -additive measure μ_0 on an algebra $\mathcal{A}$, the *outer measure*, defined for *any* sets, is

$$\mu_*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu_0(A_n) : A \subset \bigcup_{n=1}^{\infty} A_n, A_n \in \mathcal{A} \right\}.$$

For the Lebesgue measure, $\mathcal{A}$ consists of nice sets like intervals, rectangles, etc, and the outer measure is the generalization of length, area, volume and so on. But the outer measure cannot be measure, since the latter is not defined for arbitrary sets. A key point is to defined what is “measurable” w.r.t. the outer measure μ_* . We say a set A is measurable, if it satisfies the *Carathéodory's condition*:

$$\mu_*(D) = \mu_*(D \cap A) + \mu_*(D \cap A^c), \quad \forall D. \quad (1.4)$$

With some more efforts, one can show:

1. every set $A \in \mathcal{A}$ satisfies (1.4) and $\mu_*(A) = \mu_0(A)$;
2. the collection of sets that satisfy (1.4), denoted by $\mathcal{F}$, forms a σ -algebra, and moreover, μ_* is a measure on $\mathcal{F}$.

The desired extension is then defined by $\mu := \mu_*|_{\sigma(\mathcal{A})}$.

Remark 1.10 Typically, $\sigma(\mathcal{A})$ is a proper subset of $\mathcal{F}$. For example, in the case of constructing *Lebesgue measure*, we have $F(x) = x$ and

$$\sigma(\mathcal{A}) = \{\text{Borel sets}\}, \quad \mathcal{F} = \{\text{Lebesgue measurable sets}\}.$$

In **Proposition 1.16** we will see that there exist Lebesgue measurable sets which are not Borel.

However, if we complete $(\Omega, \sigma(\mathcal{A}), \mu)$, then the result is $(\Omega, \mathcal{F}, \mu|_{\mathcal{F}})$. Here, a *complete* measure space $(\Omega, \mathcal{F}, \mathbb{P})$ means that if $B \subset A \in \mathcal{F}$ such that $\mathbb{P}(A) = 0$, then $B \in \mathcal{F}$.

Counterexample 1.11 (Fields and σ -fields) Let $\Omega = [0, \infty)$. Let $\mathcal{F}_1$ be the class of intervals of the type $[a, b]$ where $0 \leq a < b \leq \infty$. Let $\mathcal{F}_2$ be the class of all finite (disjoint) sums of intervals of $\mathcal{F}_1$.

It is easy to check that $\mathcal{F}_1$ is not a field: let $0 < a < b < \infty$, then $A = [a, b] \in \mathcal{F}_1$, but $A^c = [0, a) \cup [b, \infty) \notin \mathcal{F}_1$ since all elements in $\mathcal{F}_1$ are connected sets.

One can verify easily that $\mathcal{F}_2$ is a field. But $\mathcal{F}_2$ is not a σ -field: if so, then $\{0\} = \bigcap_{n=1}^{\infty} [0, 1/n] \in \mathcal{F}_2$, but $\{0\} \notin \mathcal{F}_2$, because if $0 \in A \in \mathcal{F}_1$, then A contains $[0, \delta)$ for some $\delta > 0$.

Counterexample 1.12 (A Dynkin system that is not a field.) Let $\Omega = \{\omega_1, \dots, \omega_{2n}\}$, and $\mathcal{D}_e$ be the subsets $D \subset \Omega$ that contains only even number of elements. The $\mathcal{D}_e$ is a Dynkin system. But it is not a field, not to mention a σ -field. To see this, take $A = \{\omega_1, \omega_2\}$ and $B = \{\omega_2, \omega_3\}$. Then $A \cap B = \{\omega_2\} \notin \mathcal{D}_e$.

Of course, if a Dynkin system is closed under intersection, then it is a σ -field.

Counterexample 1.13 (Carathéodory's extension theorem fails on a general class which is not an algebra.) Let $\Omega = \{a, b, c, d\}$, and $\mathbb{P}_1, \mathbb{P}_2$ be defined by

$$\mathbb{P}_1(a) = \mathbb{P}_1(d) = \mathbb{P}_2(b) = \mathbb{P}_2(c) = \frac{1}{6}, \quad \mathbb{P}_1(b) = \mathbb{P}_1(c) = \mathbb{P}_2(a) = \mathbb{P}_2(d) = \frac{1}{3}.$$

Let $\mathcal{F} = \mathbb{P}(\Omega)$, and

$$\mathcal{C} = \{\{a, b\}, \{c, d\}, \{a, c\}, \{b, d\}\}.$$

Then $\mathcal{F} = \sigma(\mathcal{C})$ and $\mathbb{P}_1 = \mathbb{P}_2$ on $\mathcal{C}$. But $\mathbb{P}_1 \neq \mathbb{P}_2$ on $\mathcal{F}$.

1.3 Decomposition of distribution functions and singular measures

Let $F(x)$ be an increasing, right continuous function, e.g., the c.d.f. of some r.v. In this section, we first demonstrate how to decompose every such function into the jumping (or discontinuous) part, the absolutely continuous part and the singularly continuous part, written as

$$F = F_d + F_{ac} + F_{sc}.$$

The singularly continuous part also gives rise the so-called *singular random variables*. Then, we will look at a similar decomposition from the point of view of measures.

First, let us look at the discontinuous part. Since F is right continuous and increasing, F only has discontinuity points of the first kind. This leads to the following definition.

Definition 1.8 A point x is a point of jump/discontinuity of F if $F(x) - F(x-) > 0$.

Proposition 1.13 The points of jump for an increasing, right continuous function are countable.

Proof: On any compact set $[-L, L]$,

$$\{x \in [-L, L] \text{ is a jump}\} = \bigcup_{n=1}^{\infty} \left\{x \in [-L, L] : F(x) - F(x-) > \frac{1}{n}\right\}.$$

All sets in the union are finite, since each contains at most $n(F(L) - F(L-))$ points. The conclusion then follows. $\square$

Let a_i , $i = 1, 2, \dots$, be the points of jump for the function $F(x)$ and let $b_i = F(a_i) - F(a_i-)$ be the “size of jumps”. Define

$$F_d(x) = \sum_{i=1}^{\infty} b_i \mathbb{1}_{[a_i, \infty)}(x).$$

After subtracting the jump part F_d , the remaining part $F_c(x) = F(x) - F_d(x)$ is increasing and continuous.

Next we need to classify increasing and continuous functions.

Definition 1.9 (Absolute Continuity) An increasing, continuous function $F(x)$ is absolutely continuous if there exist $f \in L^1_{\text{loc}}(\mathbb{R})$ such that

$$F(b) - F(a) = \int_a^b f(x) dx. \quad (1.5)$$

Remark 1.14 This is the generalized Newton–Leibniz formula. By Lebesgue Differentiability Theorem, if (1.5) holds, then F' exists almost everywhere and $F' = f$.

On the other hand, using the Vitali covering theorem from real analysis, we know that an increasing functions is differentiable almost everywhere.

Proposition 1.14 If F is increasing, then F' exists almost everywhere.

Note that non-differentiable points in **Proposition 1.14** could be points of jumps. But if we are looking at continuous, increasing functions, we have the following.

Proposition 1.15 An increasing and continuous function F can be uniquely decomposed as

$$F = F_{ac} + F_{sc},$$

where F_{ac} is absolutely continuous and $F_{ac} = \int_{-\infty}^x F'(x) dx$, and F_{sc} is increasing and continuous but $F'_{sc} \stackrel{\text{a.e.}}{=} 0$.

Remark 1.15 The function F_{sc} appearing in **Proposition 1.15** is singularly continuous. One may ask if there exists non-trivial singularly continuous function. A famous example is the Cantor function, or the “Devil’s staircase”.

Recall that the *Cantor set*, denoted by $\mathcal{C}$, is constructed by starting with the interval $[0, 1] \subset \mathbb{R}$, then dividing it into three intervals of equal length and removing the middle interval, and repeating this process of division and removal. In the end, we obtain

$$\mathcal{C} = [0, 1] \setminus \bigcup_{n,k} I_n^{(k)},$$

where $I_n^{(k)}$, $1 \leq k \leq 2^{n-1}$, $n \geq 1$, are the intervals that we remove in the n -th steps, that is,

$$I_1^{(1)} = \left(\frac{1}{3}, \frac{2}{3}\right), \quad I_2^{(1)} = \left(\frac{1}{9}, \frac{2}{9}\right), \quad I_2^{(2)} = \left(\frac{7}{9}, \frac{8}{9}\right), \dots$$

As the complement of an open set the set $\mathcal{C}$ is a closed set, and from a direct calculation of the total length of the removed intervals, one can show that $\mathcal{C}$ has Lebesgue measure 0.

The *Cantor function*, denoted by $\varphi(x)$, is an increasing function constructed as follows. Set $\varphi(x) = 0$ for $x \leq 0$ and $\varphi(x) = 1$ for $x \geq 1$. When $x \in (0, 1)$, set $\varphi(x) = \frac{1}{2}$ for $x \in (\frac{1}{3}, \frac{2}{3}) = I_1^{(1)}$, $\varphi(x) = \frac{1}{4}$ for $x \in (\frac{1}{9}, \frac{2}{9}) = I_2^{(1)}$, and $\varphi(x) = \frac{3}{4}$ for $x \in (\frac{7}{9}, \frac{8}{9}) = I_2^{(2)}$ and so on. Then define φ on $\mathcal{C}$ by monotonicity. It follows from the construction that φ is also continuous. See also [Dur19, Fig. 1.5].

We can use the Cantor set and the Cantor function to show the following.

Proposition 1.16 *There exists a Lebesgue measurable set which is not Borel measurable.*

We first prove a lemma using the Vitali-set argument.

Lemma 1.17 *Every Lebesgue measurable set $A \subset [0, 1]$ with $m(A) > 0$ contains a non-Lebesgue measurable subset.*

Proof: We define an equivalence relation on A : $x \sim y$ if and only if $x - y \in \mathbb{Q}$. Then, by the axiom of selection, there is a subset $E \subset A$ where it contains one element from each equivalence class. We will argue that E is non-measurable.

First, we notice that $E + q$, $q \in \mathbb{Q}$ are disjoint, by the construction of E . We have

$$A \subset \bigcup_{q \in [-1, 1] \cap \mathbb{Q}} (E + q) \subset [-1, 2]$$

If $m(E) = 0$, then $m(E + q) = 0$ for all q . But then

$$m(A) \leq \sum_{q \in \mathbb{Q}} m(E + q) = 0,$$

which contradicts with $m(A) > 0$. If $m(E) > 0$, then

$$\infty = \sum_{q \in [-1, 1] \cap \mathbb{Q}} m(E + q) \leq m([-1, 2]) = 3,$$

which is also a contradiction. Hence E is non-measurable. $\square$

Proof of Proposition 1.16 : We will prove the statement by contradiction.

Let $\psi(x) = \frac{1}{2}(x + \varphi(x))$. Then $\psi(x)$ is a continuous, strictly increasing function from $[0, 1]$ onto itself. Let $H = \psi^{-1}$. Then H is also continuous and strictly increasing.

It is easy to check that for any $E \subset [0, 1]$,

$$\mathbb{1}_{H(E)}(H(x)) = \mathbb{1}_E(x).$$

Note that the Lebesgue measure of $\psi(\mathcal{C})$ is $1/2$. Hence, there exists a set $E \subset \psi(\mathcal{C})$ which is NOT Lebesgue measurable by Lemma 1.17. On the other hand, $H(E) = \psi^{-1}(E) \subset \mathcal{C}$ is a subset of Lebesgue measure 0 set, and hence by completeness of the Lebesgue measure space (as a consequence of using outer measure in Theorem 1.12), it is also Lebesgue measurable.

Now, if all Lebesgue measurable sets are Borel, then $\mathbb{1}_{H(E)}$ will be Borel measurable as the indicator function of a Borel set. Therefore, $\mathbb{1}_E = \mathbb{1}_{H(E)} \circ H$ is the composition of two Borel measurable functions,

and is also Borel measurable. But this contradicts with the fact that E is chosen to be non-measurable. $\square$

In the second part of this section, we will derive the same decomposition from the perspective of measures. This can be generalized to measures on $\mathbb{R}^d$, $d > 1$.

Let μ be a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 1.10 A point x is a point of mass if $\mu(\{x\}) > 0$.

Let $I = \{x : \mu(\{x\}) > 0\}$ be the set of points of mass. We can define $\mu_d(A) = \sum_{x \in I} \delta_x(A) \cdot \mu(\{x\})$.

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

is the *Dirac measure* on x . We call μ_d the discrete part of the measure μ , and this corresponds to the jumping part of the distribution functions.

The remaining part $\mu_c = \mu - \mu_d$ will not have points of mass. To further decompose it, we need to introduce the notion of absolute continuity and singularity for measures. Let P, Q are two probability measures on $(\Omega, \mathcal{F})$. For the simplest example, one can take $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 1.11 A measure P is absolutely continuous w.r.t. Q , written $P \ll Q$, if $Q(A) = 0$ implies $P(A) = 0$.

We recall the Radon–Nikodym derivative.

Theorem 1.18 (Radon–Nikodym Theorem) Let ν and μ be two σ -finite measures on a measurable space $(\Omega, \mathcal{F})$ such that $\nu \ll \mu$. Then there exists a function f , measurable w.r.t. $\mathcal{F}$, such that

$$\int_A f d\mu = \nu(A).$$

We call $f = \frac{d\nu}{d\mu}$ the Radon–Nikodym derivative, and μ the reference measure.

Theorem 1.18 holds on general measure spaces. For r.v.s, the reference measure is the Lebesgue measure.

Definition 1.12 A r.v. X is continuous if its distribution μ is absolutely continuous with respect to the Lebesgue measure. In this case, the density of X is $\frac{d\mu}{d\text{Leb}}$.

The last definition is mutual singularity.

Definition 1.13 Two measures P, Q are mutually singular, denoted by $P \perp Q$, if there exists A such that $P(A) = 0$ and $Q(A^c) = 0$.

Example 1.16 Cantor set induce a distribution $\mu_C = d\varphi$. Since

$$\mu_C(\mathcal{C}^c) = 0, \quad \text{Leb}(\mathcal{C}) = 0,$$

we have $\mu_C \perp \text{Leb}$. In fact, an increasing function F is singularly continuous if and only if $dF \perp \text{Leb}$.

Definition 1.14 A r.v. X is singular if $\mu_X \perp \text{Leb}$.

How common are singular measures and Cantor-like sets? Surprisingly, they are ubiquitous in probability theory. They usually arise from self-similarities or fractal structures, or from infinite dimensional spaces.

Example 1.17 The example is about Brownian motion, which is a important object to study in stochastic analysis. Without getting into too many details, a Brownian motion $B_t(\omega)$ is a random continuous function.

For each $a \in \mathbb{R}$,

$$\mathcal{Z}_a(\omega) := \{t : B_t(\omega) = a\}.$$

be the level set of the Brownian motion; note the level set is also a random set. For almost every ω and every a , the level $\mathcal{Z}_a(\omega)$ has a similar structure as a Cantor set, in the sense that it is the complement of the union of nested open intervals, but the interval length is random.

To get singular measures, consider the maximal process $B_t^* = \sup_{0 \leq s \leq t} B_s$. Since $t \mapsto B_t$ is continuous, the maximal process B_t^* is increasing and continuous. One can show that $dB_t^* \perp \text{Leb}$.

Example 1.18 Let us consider i.i.d. Bernoulli r.v.s $\text{Ber}(1/3)$ and $\text{Ber}(2/3)$. More precisely, let $(\Omega, \mathcal{F})$ be

$$\Omega = \{\omega = (\omega_1, \omega_2, \dots), \omega_i \in \{0, 1\}\}, \quad \mathcal{F} = \mathcal{P}(\Omega).$$

We can define two probability measures on $(\Omega, \mathcal{F})$:

1. one corresponding to i.i.d. $\text{Ber}(1/3)$: $\mathbb{P}_1(\omega_i = 1) = \frac{1}{3}$ and $\mathbb{P}_1(\omega_i = 0) = 2/3$;
2. the other one corresponding to i.i.d. $\text{Ber}(2/3)$: $\mathbb{P}_2(\omega_i = 1) = \frac{2}{3}$ and $\mathbb{P}_2(\omega_i = 0) = 1/3$.

Let

$$A_1 = \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \omega_k = \frac{1}{3} \right\}, \quad A_2 = \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \omega_k = \frac{2}{3} \right\}.$$

Then by the Strong Law of Large Numbers, we have $\mathbb{P}_1(A_1) = 1$ and $\mathbb{P}_2(A_2) = 1$. On the other hand, we have $A_1 \cap A_2 = \emptyset$. It follows that $\mathbb{P}_1(A_2) = 0$ and $\mathbb{P}_2(A_1^c) = 0$, so $\mathbb{P}_1 \perp \mathbb{P}_2$.

Counterexample 1.19 (Nowhere differentiable functions.) Weierstrass example:

$$W(x) = \sum_{n=1}^{\infty} a^n \cos(b^n x), \quad a \in (0, 1), \quad b = 2n + 1, \quad ab > 1 + \frac{3\pi}{2}.$$

Another famous example from probability: Brownian motion with probability one is nowhere differentiable. One can represent Brownian motion on $[0, 1]$ as

$$B_t = \sum_{n=1}^{\infty} \xi_n \int_0^t e_n(x) dx,$$

where ξ_n are i.i.d. $\mathcal{N}(0, 1)$ and $\{e_n\}$ an orthonormal basis of $L^2(0, 1)$. For example, if we take $\{e_n\}$ to be the Haar basis

$$h_{n,k}(t) = 2^{n/2} \begin{cases} 1, & \frac{k}{2^n} \leq t < \frac{k+1/2}{2^n}, \\ -1, & \frac{k+1/2}{2^n} \leq t < \frac{k+1}{2^n}, \\ 0, & \text{otherwise.} \end{cases}$$

This representation is very similar to the Weierstrass function, but a randomized version. Brownian motion is also nowhere monotone.

1.4 Random variables and measurable maps

Let $(S, \mathcal{S})$ be a measurable space. We say that a map $\varphi : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ is *measurable* if $\varphi^{-1}(A) \in \mathcal{F}$, $\forall A \in \mathcal{S}$. Random variables and vectors are measurable maps between appropriate spaces.

Definition 1.15 A r.v. X is a measurable map from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. A random vector $X = (X_1, \dots, X_d)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

Since the Borel σ -algebra is generated by open sets, we have a simple criterion to check whether a map defines a r.v.

Proposition 1.19 A map X is a random variable if and only if $X^{-1}(O) \in \mathcal{F}$ for every open set O .

Definition 1.16 A function f is a Borel measurable if f is measurable map from $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ onto itself.

Similar to [Proposition 1.19](#), we have the following.

Proposition 1.20 A function f is Borel measurable if and only if $f^{-1}(O) \subset \mathcal{B}(\mathbb{R})$ for every open set O .

To compare with the Lebesgue measurability: f is Lebesgue measurable if and only if $f^{-1}(O)$ is Lebesgue measurable set for every open set O .

Proposition 1.21 If f is Borel measurable and X is a random variable, then $f(X)$ is a r.v.

Proof: Let O be a open set. Then $f^{-1}(O) \in \mathcal{B}(\mathbb{R})$ since f is Borel measurable. Hence,

$$\{\omega : f(X(\omega)) \in O\} = X^{-1}(f^{-1}(O)) \in \mathcal{F}.$$

This shows that $f(X)$ is a r.v. □

Remark 1.20 In this example, if “ f is Borel measurable” is replaced by “ f is Lebesgue measurable”, then the conclusion is false, as seen from the proof of [Proposition 1.16](#).

We often drop the word “measurable” and simply say “Borel sets” or “Borel functions”.

Proposition 1.22 If $f : \mathbb{R} \rightarrow \mathbb{R}^d$ is a Borel map and $X = (X_1, \dots, X_d)$ is a random vector, then $f(X) = f(X_1, \dots, X_d)$ is a random variable.

Example 1.21 We can use [Proposition 1.22](#) to create new r.v.s. For example, if X_1, X_2 are r.v.s, then $X_1 + X_2$, $\min\{X_1, X_2\}$ are also r.v.s.

Next, we need to understand the limits of r.v.s.

Proposition 1.23 Let X_n , $n = 1, 2, \dots$ be r.v.s. Then

$$\sup_{n \geq 1} X_n, \quad \inf_{n \geq 1} X_n, \quad \limsup_{n \rightarrow \infty} X_n, \quad \liminf_{n \rightarrow \infty} X_n$$

are r.v.s.

Proof:

(i) Let $Y_1(\omega) = \sup_n X_n(\omega)$. We need to show that $Y_1^{-1}(-\infty, a] \in \mathcal{F}$ for every $a \in \mathbb{R}$. Indeed,

$$Y_1^{-1}(-\infty, a] = \{\omega : \sup_n X_n(\omega) \leq a\} = \bigcap_{n=1}^{\infty} \{\omega : X_n(\omega) \leq a\} \in \mathcal{F}.$$

Therefore, $Y_1(\omega) = \sup_n X_n(\omega)$ is a r.v.

(ii) Let $Y_2(\omega) = \inf_n X_n(\omega)$. We need to show that $Y_2^{-1}([a, \infty)) \in \mathcal{F}$ for every $a \in \mathbb{R}$. Indeed,

$$Y_2^{-1}[a, \infty) = \{\omega : \inf_n X_n(\omega) \geq a\} = \bigcap_{n=1}^{\infty} \{\omega : X_n(\omega) \geq a\} \in \mathcal{F}.$$

Therefore, $Y_2(\omega) = \inf_n X_n(\omega)$ is a r.v.

(iii) By definition of $\limsup$, for every ω , we have

$$\limsup_{n \rightarrow \infty} X_n(\omega) = \inf_{n \geq 1} \sup_{m \geq n} X_m(\omega).$$

By part (i), for every $n \geq 1$, the map $\omega \mapsto \sup_{m \geq n} X_m(\omega)$ is measurable. Hence, for every $a \in \mathbb{R}$,

$$\{\omega : \limsup_{n \rightarrow \infty} X_n(\omega) \geq a\} = \{\omega : \inf_{n \geq 1} \sup_{m \geq n} X_m(\omega) \geq a\} = \bigcap_{n=1}^{\infty} \{\omega : \sup_{m \geq n} X_m(\omega) \geq a\} \in \mathcal{F}.$$

(iv) By definition of $\liminf$, for every ω , we have

$$\liminf_{n \rightarrow \infty} X_n(\omega) = \sup_{n \geq 1} \inf_{m \geq n} X_m(\omega).$$

By part (ii), for every $n \geq 1$, the map $\omega \mapsto \inf_{m \geq n} X_m(\omega)$ is measurable. Hence, for every $a \in \mathbb{R}$,

$$\{\omega : \liminf_{n \rightarrow \infty} X_n(\omega) \leq a\} = \{\omega : \sup_{n \geq 1} \inf_{m \geq n} X_m(\omega) \leq a\} = \bigcap_{n=1}^{\infty} \{\omega : \inf_{m \geq n} X_m(\omega) \leq a\} \in \mathcal{F}.$$

□

Corollary 1.24 *Let X_n , $n = 1, 2, \dots$, be r.v.s. The set $\Omega_0 = \{\omega : \lim_{n \rightarrow \infty} X_n(\omega) \text{ exists}\}$ belongs to $\mathcal{F}$.*

Proof: Note that

$$\Omega_0 = \{\omega : \lim_{n \rightarrow \infty} X_n(\omega)\} = \{\omega : \limsup_{n \rightarrow \infty} X_n(\omega) - \liminf_{n \rightarrow \infty} X_n(\omega) = 0\}.$$

By **Proposition 1.23**, $Y_1 = \limsup_{n \rightarrow \infty} X_n(\omega)$ and $Y_2 = \liminf_{n \rightarrow \infty} X_n(\omega)$ are r.v.s, and hence $Y_1 - Y_2$ is a r.v. Therefore, $\Omega_0 = \{Y_1 - Y_2 = 0\} \in \mathcal{F}$. □

Does a random variable with a given distribution function F exists? By **Theorem 1.6** and the usage of Carathéodory's Extension Theorem there gives the construction of a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, where $\mu(-\infty, a] = F(a)$. To construct a r.v. X with distribution μ , we take $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ and $X(\omega) = \omega$.

Another way to construct a r.v. with given a c.d.f. $F(x)$ is to use the *generalized inverse* F^{-1} :

$$F^{-1}(x) = \sup\{y : F(y) \leq x\}.$$

One can check that F^{-1} is increasing and left continuous. In addition, if F is strictly increasing and continuous, then F^{-1} is the normal inverse function of F .

Proposition 1.25 *Let $U \sim \text{Unif}[0, 1]$ be a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$. Then $F^{-1}(U)$ is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$ with c.d.f. F .*

Proof: Since F^{-1} is left continuous and increasing, it is Borel measurable. Hence, $\omega \mapsto F^{-1}(U(\omega))$ is measurable and $F^{-1}(U)$ is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$.

To check that the c.d.f. of $F^{-1}(U)$ is F , we will use without proof that

$$\{y : F^{-1}(y) \leq x\} = \{y : y \leq F(x)\}. \quad (1.6)$$

Indeed, assuming (1.6), we have

$$\mathbf{P}(F^{-1}(U) \leq a) = \mathbf{P}(U \leq F(a)) = F(a).$$

as desired. $\square$

Proposition 1.25 plays an important role in computer science when it comes to stochastic simulation. On computers, one can use pseudo random number generators to produce i.i.d. uniform integers X in the set $\{1, 2, \dots, N\}$ where N is sufficiently large. Then, X/N will approximate the uniform distribution on $[0, 1]$, and thus $F^{-1}(X/N)$ is closed to a r.v. with c.d.f. F . Of course, it is often the case where F^{-1} is costly to compute, and some other sampling methods will be efficient. But this algorithm is good enough to generate the exponential and Gaussian random variables, which are building blocks of the jump process and diffusions.

In the last part, we briefly discuss the distribution function of a random vector. Recall that $F : \mathbb{R} \rightarrow \mathbb{R}$ is a c.d.f., that is, $F(a) = \mathbf{P}(X \leq a)$ for some r.v. X on $(\Omega, \mathcal{F}, \mathbf{P})$ if and only if it has the following properties:

1. F is non-decreasing.
2. F is right-continuous with left limits.
3. $F(-\infty) = 0$, $F(\infty) = 1$.

For dimension $d \geq 2$, we say that $G(x_1, \dots, x_d)$ is a d -dimensional d.f. if the following holds.

1. $G(x_1, \dots, x_d)$ is non-decreasing in each of the arguments.
2. $G(x_1, \dots, x_n)$ is right-continuous and has left limits in each of its arguments.
3. $G(x_1, \dots, x_d) \rightarrow 0$ as $x_j \rightarrow -\infty$ for at least one j , and $G(x_1, \dots, x_d) \rightarrow 1$ as $x_j \rightarrow \infty$ for at least one j .
4. For $a_j \leq b_j$, $j = 1, 2, \dots, d$, define

$$\Delta_{a_j, b_j} G(x_1, \dots, x_d) = G(x_1, \dots, b_j, \dots, x_d) - G(x_1, \dots, a_j, \dots, x_d).$$

Then for all $a_j \leq b_j$.

$$\Delta_{a_1, b_1} \Delta_{a_2, b_2} \cdots \Delta_{a_d, b_d} G(x_1, \dots, x_n) \geq 0.$$

These are also sufficient and necessary conditions for existence of a random vector $(X_1, \dots, X_d)$ on a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ such that $\mathbf{P}(X_1 \leq a_1, \dots, X_d \leq a_d) = G(a_1, \dots, a_d)$. The last condition **Item 4** is saying

$$\Delta_{a_1, b_1} \Delta_{a_2, b_2} \cdots \Delta_{a_d, b_d} G(x_1, \dots, x_n) = \mathbf{P}(X_j \in (a_j, b_j), j = 1, \dots, d) \geq 0,$$

which gives the probability of the half-open, half-closed rectangles in the standard choice of π -system/semi-algebra needed for Carathéodory's Extension Theorem. The proof will be omitted here. We note that **Items 1 to 3** are direct generalization of **Items 1 to 3** in high dimensions, but **Item 4** does not follow from **Items 1 to 3** unless in dimension, as we will see in the counterexamples below.

Practically, proving a functions satisfying **Item 4** is quite hard. Construction of random vectors usually starts from independent random variables (see **Section 2.2**) or some conditional structures (see **Section 8.1**), then use a measurable map $\Phi : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^{d_2}$ to construct new variables.

Counterexample 1.22 We present two examples where $F(x_1, x_2)$ satisfies **Items 1 to 3** but not **Item 4**.
For the first example, consider

$$F(x, y) = \begin{cases} 0, & x \leq 0 \text{ or } x + y \leq 1 \text{ or } y \leq 0, \\ 1, & \text{else.} \end{cases}$$

Then

$$F(1, 1) - F(1, 1/3) - F(1/3, 1) + F(1/3, 1/3) = -1.$$

For the second example, consider

$$F(x, y) = \begin{cases} 0, & x \leq 0 \text{ or } y \leq 0, \\ \min\{1, \max\{x, y\}\}, & \text{else.} \end{cases}$$

Then for any $0 \leq x_1 < y_1 < x_2 \leq y_2 \leq 1$, we have

$$F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) = y_2 - y_1 - x_2 + y_1 < 0.$$

1.5 Integration and expectation

In this section, we will briefly present the theory of integration of measurable functions, or in the context of probability theory, the mathematical expectation. The main difference is that in probability theory, the probability measure has total mass 1 and is a finite measure.

Let X be a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$. We will denote its expectation X by $\mathbb{E}(X)$, or using a more measure theory oriented notation, sometimes we also write

$$\mathbb{E}X = \int_{\Omega} X(\omega) \mathbb{P}(d\omega). \quad (1.7)$$

The definition of (1.7) is through approximation via simple r.v.s (simple functions in measure theory). To start, we say that a r.v. $X(\omega)$ is *simple*, if there exists finitely many $A_1, \dots, A_n \in \mathcal{F}$ and $c_1, \dots, c_n \in \mathbb{R}$ such that

$$X(\omega) = \sum_{k=1}^n c_k \mathbb{1}_{A_k}(\omega). \quad (1.8)$$

In the case of (1.8), unquestionably we should define

$$\mathbb{E}(X) = \sum_{k=1}^n c_k \mathbb{P}(A_k).$$

It is routine to verify common integral properties for expectation of simple r.v.s, e.g., linearity, monotonicity, order preserving, etc, so we omit it in this note.

For a non-negative r.v. $X(\omega)$, we define

$$\mathbb{E}X = \int_{\Omega} X(\omega) \mathbb{P}(d\omega) := \sup \left\{ \int Y(\omega) \mathbb{P}(d\omega) : Y \text{ simple, } 0 \leq Y(\omega) \leq X(\omega) \right\} \in [0, \infty]. \quad (1.9)$$

For the general case, we write $X(\omega) = X_+(\omega) - X_-(\omega)$, where

$$X_+(\omega) = X(\omega) \mathbb{1}_{\{X > 0\}}, \quad X_-(\omega) = -X(\omega) \mathbb{1}_{\{X \leq 0\}}$$

are the positive and negative parts of X . If $\mathbb{E}(X_+) < \infty$ or $\mathbb{E}(X_-) < \infty$, then we define

$$\mathbb{E}(X) = \mathbb{E}(X_+) - \mathbb{E}(X_-).$$

Otherwise, EX is undefined since $\infty - \infty$ cannot be defined.

Next, we will discuss conditions that justifies exchanging order of limit and integration, that is,

$$\mathbb{E} \lim_{n \rightarrow \infty} X_n = \lim_{n \rightarrow \infty} \mathbb{E} X_n. \quad (1.10)$$

Lemma 1.26 *Let $X_n \uparrow X$ such that $X_n \geq 0$ and X_n are simple. Then (1.10) holds.*

Remark 1.23 If “ $X_n \uparrow X$ ” is replaced by “ $X_n \leq X$ and $X_n \rightarrow X$ ”, we can still get an increasing sequence by considering $Y_n = \max_{1 \leq k \leq n} X_k$. It is easy to see that Y_n are also simple and $Y_n \uparrow X$.

Proof: From the definition (1.9), we have $\mathbb{E}(X) \geq \mathbb{E}(X_n)$. It remains to establish the inequality in the other direction:

$$\mathbb{E} X \leq \lim_{n \rightarrow \infty} \mathbb{E} X_n. \quad (1.11)$$

Note that the limit on the right hand side always exists, since X_n , and hence $\mathbb{E} X_n$, are increasing in n .

If $\mathbb{E} X < \infty$, then for every $\varepsilon > 0$, by the definition of supremum, there exists a non-negative simple r.v. Y_ε such that $Y_\varepsilon \leq X$ and $\mathbb{E}(Y_\varepsilon) \geq \mathbb{E}(X) - \varepsilon$. For every $\delta > 0$, let $A_n = \{\omega : X_n(\omega) \geq Y_\varepsilon(\omega) - \delta\}$. Since $X_n(\omega) \uparrow X(\omega) \geq Y_\varepsilon(\omega)$, we have $A_n \uparrow \Omega$ and hence $A_n^c \downarrow \emptyset$. We have

$$\begin{aligned} \mathbb{E} X_n &= \mathbb{E} X_n \mathbb{1}_{A_n} + \mathbb{E} X_n \mathbb{1}_{A_n^c} \geq \mathbb{E}(Y_\varepsilon - \delta) \mathbb{1}_{A_n} \\ &= \mathbb{E} Y_\varepsilon \mathbb{1}_{A_n} - \delta \mathbb{P}(A_n) \\ &= \mathbb{E} Y_\varepsilon - \mathbb{E} Y_\varepsilon \mathbb{1}_{A_n^c} - \delta \mathbb{P}(A_n) \\ &\geq \mathbb{E} X - \varepsilon - \sup_{\omega} Y_\varepsilon(\omega) \cdot \mathbb{P}(A_n^c) - \delta \end{aligned}$$

Since Y_ε is simple, it is always bounded, so $\sup_{\omega} Y_\varepsilon(\omega) < \infty$. Letting $n \rightarrow \infty$, we obtain

$$\lim_{n \rightarrow \infty} \mathbb{E} X_n \geq \mathbb{E} X - \varepsilon - \delta.$$

Since $\varepsilon, \delta > 0$ are arbitrary, this implies (1.11).

If $\mathbb{E} X = \infty$, then by (1.9), for every $M > 0$, there exists a simple r.v. Y_M such that $Y_M \leq X$ and $\mathbb{E} Y_M \geq M$. For every $\xi > 0$, let $B_n = \{\omega : X_n(\omega) \geq Y_M(\omega) - \xi\}$. Since $X_n(\omega) \uparrow X(\omega) \geq Y_M(\omega)$, we have $B_n \uparrow \Omega$ and hence $B_n^c \downarrow \emptyset$. Therefore,

$$\begin{aligned} \mathbb{E} X_n &= \mathbb{E} X_n \mathbb{1}_{B_n} + \mathbb{E} X_n \mathbb{1}_{B_n^c} \geq \mathbb{E}(Y_M - \xi) \mathbb{1}_{B_n} \\ &= \mathbb{E} Y_M \mathbb{1}_{B_n} - \xi \mathbb{P}(B_n) \\ &= \mathbb{E} Y_M - \mathbb{E} Y_M \mathbb{1}_{B_n^c} - \xi \mathbb{P}(B_n) \\ &\geq M - \sup_{\omega} Y_M(\omega) \cdot \mathbb{P}(B_n^c) - \xi \end{aligned}$$

Letting $n \rightarrow \infty$, we obtain $\lim_{n \rightarrow \infty} \mathbb{E} X_n \geq M - \xi$. Since $M, \xi > 0$ are arbitrary, this implies (1.11). $\square$

Note that for any non-negative r.v. X , we can explicitly construct simple r.v.s $X_n \uparrow X$ as follows, so that Lemma 1.26 applies:

$$X_n(\omega) = \frac{[2^n X(\omega)]}{2^n} \wedge n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\{X(\omega) \in [\frac{k}{2^n}, \frac{k+1}{2^n})\}} + n \mathbb{1}_{\{X(\omega) \geq n\}},$$

where $a \wedge b := \min(a, b)$ and $[x]$ denotes the integer part of x . To see that $X_n \rightarrow X$, notice that

$$|X(\omega) - X_n(\omega)| \leq \frac{1}{2^n}, \quad \text{uniformly on } \{\omega : X(\omega) \leq n\}.$$

Theorem 1.27 (Monotone Convergence Theorem, MCT) *If $X_n \geq 0$ and $X_n \uparrow X$, then (1.10) holds.*

Proof: Again, it suffices to establish (1.11).

Let $Y_n^{(m)}$ be simple r.v.s that increase to X_n , and $Z^{(m)} = \max(Y_1^{(m)}, \dots, Y_m^{(m)})$. Then $Z^{(m)}$ are simple, and also increasing in m since

$$Z^{(m)} = \max_{1 \leq n \leq m} Y_n^{(m)} \leq \max_{1 \leq n \leq m} Y_n^{(m+1)} \leq \max_{1 \leq n \leq m+1} Y_n^{(m+1)} = Z^{(m+1)}.$$

In addition, we have

$$Y_n^{(m)} \leq Z^{(m)} \leq X_m, \quad \forall m \geq n \geq 1.$$

Taking $m \rightarrow \infty$, we see that

$$X_n \leq \lim_{m \rightarrow \infty} Z^{(m)} \leq X, \quad \forall n \geq 1.$$

Taking $n \rightarrow \infty$, and using that $X_n \uparrow X$, we see that $Z^{(m)} \uparrow X$. Then by Lemma 1.26, we have

$$\mathbb{E}X = \lim_{m \rightarrow \infty} \mathbb{E}Z^{(m)}. \quad (1.12)$$

On the other hand, since $Y_m^{(m)} \leq Z^{(m)} \leq X_m$, we have

$$\lim_{m \rightarrow \infty} \mathbb{E}Z^{(m)} \leq \lim_{m \rightarrow \infty} \mathbb{E}X_m. \quad (1.13)$$

Then (1.11) follows from (1.12) and (1.13), and this completes the proof. $\square$

Remark 1.24 In Theorem 1.27, the condition “ $X_n \geq 0$ ” can be replaced by

$$“X_n \geq -Y, \text{ for some } Y \geq 0 \text{ with } \mathbb{E}Y < \infty”. \quad (1.14)$$

Indeed, if (1.14) holds, then $\tilde{X}_n = X_n + Y \geq 0$. Since $\tilde{X}_n \uparrow \tilde{X} = X + Y$, we have

$$\lim_{n \rightarrow \infty} (\mathbb{E}X_n + \mathbb{E}Y) = \lim_{n \rightarrow \infty} \mathbb{E}\tilde{X}_n = \mathbb{E}\tilde{X} = \mathbb{E}(X + Y).$$

Since $0 \leq \mathbb{E}Y < \infty$, we can subtract $\mathbb{E}Y$ from both sides to obtain $\lim_{n \rightarrow \infty} \mathbb{E}X_n = \lim_{n \rightarrow \infty} \mathbb{E}X$.

Theorem 1.28 (Fatou’s Lemma) *If $X_n \geq 0$ (or (1.14) holds), then*

$$\liminf_{n \rightarrow \infty} \mathbb{E}X_n \geq \mathbb{E} \liminf_{n \rightarrow \infty} X_n.$$

Proof: Let

$$Y_n = \inf_{m \geq n} X_m \uparrow \liminf_{n \rightarrow \infty} X_n.$$

Since Y_n is an infimum we have $Y_n \leq X_n$. By MCT (Theorem 1.27), we have

$$\mathbb{E} \liminf_{n \rightarrow \infty} X_n = \lim_{n \rightarrow \infty} \mathbb{E}Y_n \leq \liminf_{n \rightarrow \infty} \mathbb{E}X_n.$$

$\square$

Theorem 1.29 (Dominated Convergence Theorem, DCT) *If $X_n \rightarrow X$ a.s. and $|X_n| \leq Y$ for some r.v. Y with $\mathbb{E}Y < \infty$, then $\lim_{n \rightarrow \infty} \mathbb{E}X_n = \mathbb{E}X$.*

Proof: By the assumption we have $|X| \leq Y$. Since $2Y - |X_n - X| \geq 0$, by Fatou's Lemma (Theorem 1.28), we have

$$\liminf_{n \rightarrow \infty} \mathbb{E}(2Y - |X_n - X|) \geq \mathbb{E}(2Y).$$

Since $\mathbb{E}(2Y) < \infty$, we can subtract it from both side and obtain

$$0 \geq \limsup_{n \rightarrow \infty} \mathbb{E}|X_n - X| = 0.$$

□

Corollary 1.30 (Bounded Convergence Theorem, BCT) *If $X_n \rightarrow X$ and $|X_n| \leq M$, $n \geq 1$ for some constant M , then $\lim_{n \rightarrow \infty} \mathbb{E}X_n = \mathbb{E}X$.*

Proof: Take $Y(\omega) \equiv M$. □

Next, we will present some useful inequalities for expectation. We try to give proofs that is general enough so that we can adapt them to prove statements about other measurable maps.

Proposition 1.31 (Jensen inequality) *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. If $\mathbb{E}|x| < \infty$, then $\mathbb{E}\varphi(x) \geq \varphi(\mathbb{E}X)$.*

Proof: Let $\mathbb{E}X = a \in (-\infty, \infty)$. By convexity, there exists $k \in \mathbb{R}$ (taking $k \in [\varphi'_-(a), \varphi'_+(a)]$) s.t.

$$\varphi(t) \geq \varphi(a) + k(t - a), \quad \forall t.$$

Plugging in $t = X$ and taking expectation, we have

$$\mathbb{E}\varphi(X) \geq \mathbb{E}\varphi(a) + k\mathbb{E}(X - a) = \varphi(a) - ka + k\mathbb{E}X = \varphi(\mathbb{E}X).$$

□

Example 1.25 Let $\varphi(t) = |t|^p$, $p \geq 1$. Then for every $|X|$, we have

$$\mathbb{E}|X|^p \geq (\mathbb{E}|X|)^p.$$

Proposition 1.32 (Hölder's inequality) *If $p, q \in [1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$ then*

$$\mathbb{E}|XY| \leq (\mathbb{E}|X|^p)^{1/p} \cdot (\mathbb{E}|Y|^q)^{1/q}. \quad (1.15)$$

When $p = q = 2$, this is the Cauchy-Schwartz inequality.

Proof: We recall the *Young's inequality*: if $\frac{1}{p} + \frac{1}{q} = 1$, then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}, \quad x, y \geq 0. \quad (1.16)$$

If X and Y are bounded, then we have $\mathbb{E}|X|^p, \mathbb{E}|Y|^q < \infty$. Let

$$\tilde{X} = \frac{|X|}{(\mathbb{E}|X|^p)^{1/p}}, \quad \tilde{Y} = \frac{|Y|}{(\mathbb{E}|Y|^q)^{1/q}}.$$

By (1.16), we have

$$\mathbb{E}\tilde{X}\tilde{Y} \leq \frac{\mathbb{E}\tilde{X}^p}{p} + \frac{\mathbb{E}\tilde{Y}^q}{q} = \frac{1}{p} + \frac{1}{q} = 1$$

This is (1.15).

If X and Y are not bounded, consider the truncation $X_M = |X| \wedge M$ and $Y_M = |Y| \wedge M$ where $M > 0$. For every fixed M we have

$$\mathbb{E}X_M Y_M \leq (\mathbb{E}X_M^p)^{1/p} \cdot (\mathbb{E}Y_M^q)^{1/q}.$$

Taking $M \uparrow \infty$, since $X_M \uparrow X$ and $Y_M \uparrow |Y|$, (1.15) follows from the MCT. $\square$

The final result in this section is about change of variables when we switch measures when performing integration. We will use a technique called “functional Monotone Class Theorem”, which will be useful later for other problems as well.

Theorem 1.33 (Change of variables) *Let X be a r.v. and f is a Borel function. Assume either $f \geq 0$ or $\mathbb{E}|f(X)| < \infty$. Then*

$$\mathbb{E}f(X) = \int_{\Omega} f(X(\omega)) \mathbb{P}(d\omega) = \int_{\mathbb{R}} f(y) \mu_X(dy), \quad (1.17)$$

where $\mu_X = \mathbb{P} \circ X^{-1}$ is the distribution of X .

Proof: Let

$$\mathcal{H} = \{f : f \text{ is Borel measurable s.t. (1.17) holds}\}.$$

We want to show that $f \in \mathcal{H}$ whenever $f \geq 0$ or $\mathbb{E}|f(X)| < \infty$. We will do this in four steps.

1. $\mathbb{1}_A \in \mathcal{H}$ for every $A \in \mathcal{B}(\mathbb{R})$.

Indeed, by definition of the expectation and μ_X , we have

$$\mathbb{E}\mathbb{1}_A = \int_{\Omega} \mathbb{1}_A(X(\omega)) \mathbb{P}(d\omega) = \mathbb{P}(X \in A) = \mu_X(A) = \int_{\mathbb{R}} \mathbb{1}_A(y) \mu_X(dy)$$

2. Let $f_1, \dots, f_n$ be functions in $\mathcal{H}$. For any $a_1, \dots, a_n \in \mathbb{R}$, we have

$$a_1 f_1 + \dots + a_n f_n \in \mathcal{H},$$

This follows from the linearity of integrals. Combining with Item 1, $\mathcal{H}$ contains all simple functions.

3. $\mathcal{H}$ contains all non-negative functions.

Indeed, for every nonnegative function f , there exists a sequence of simple functions f_n such that $f_n \geq 0$ and $f_n \uparrow f$. By Item 2, we have

$$\int_{\Omega} f_n(X(\omega)) \mathbb{P}(d\omega) = \int_{\mathbb{R}} f_n(y) \mu_X(dy)$$

By MCT, (1.17) follows from

$$\int_{\Omega} f_n(X(\omega)) \mathbb{P}(d\omega) \rightarrow \int_{\Omega} f(X(\omega)) \mathbb{P}(d\omega), \quad \int_{\mathbb{R}} f_n(y) \mu_X(dy) \rightarrow \int_{\mathbb{R}} f(y) \mu_X(dy).$$

4. If $\mathbb{E}|f(X)| < \infty$, then the positive and negative parts $f_+, f_- \in \mathcal{H}$, and hence $f = f_+ - f_- \in \mathcal{H}$.

$\square$

Counterexample 1.26 (DCT is only a sufficient condition.)

For $(\Omega, \mathcal{F}, \mathbf{P}) = ((0, 1), \mathcal{B}(0, 1), \text{Leb})$, consider

$$X_n(\omega) = \frac{n}{\log n} \mathbb{1}_{(0, 1/n)}(\omega), \quad n \geq 2. \quad (1.18)$$

Then $X_n(\omega) \rightarrow 0 =: X_\infty(\omega)$, and

$$\int_{\Omega} X_n(\omega) \mathbf{P}(d\omega) = \frac{1}{\log n} \rightarrow 0 = \int_{\Omega} X_\infty(\omega) \mathbf{P}(d\omega).$$

However, there is no $Y \in L^1(d\mathbf{P})$ such that $Y \geq |X_n|$. Indeed, the minimum of such Y is given by

$$Y(\omega) = \sup_n X_n(\omega) = \mathbb{1}_{[0, 1/2]}(\omega) \cdot \frac{[1/\omega]}{\log[1/\omega]},$$

where $[\cdot]$ denotes the integer part. However,

$$\int_{\Omega} Y(\omega) \mathbf{P}(d\omega) = \sum_{n=2}^{\infty} \frac{n}{\log n} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=2}^{\infty} \frac{1}{(n+1) \log n} = \infty,$$

and hence $Y \notin L^1(d\mathbf{P})$.

2 Independence and product measures

2.1 Definitions of independence

Recall from elementary probability that two events A and B are *independent* if and only if

$$\mathbf{P}(AB) = \mathbf{P}(A)\mathbf{P}(B).$$

We can use this to defined independence of r.v.s.

Definition 2.1 *Two r.v.s X and Y are independent if*

$$\mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B), \quad \forall A, B \in \mathcal{B}(\mathbb{R}), \quad (2.1)$$

Using the definition of independence of evnets, **Definition 2.1** is the most basic definition for independence of r.v.s. But in practice there are other more useful definitions.

Let X be a r.v. The σ -algebra generated by X , denoted by $\sigma(X)$, is the smallest σ -algebra on Ω which makes $X : \Omega \rightarrow \mathbb{R}$ measurable. It is easy to check that $\sigma(X)$ has the explicit form

$$\sigma(X) = \{X^{-1}(A), A \in \mathcal{B}(\mathbb{R})\}.$$

We may also introduce independence of σ -algebras.

Definition 2.2 *Two σ -algebras $\mathcal{F}$ and $\mathcal{G}$ are independent, if*

$$\mathbf{P}(AB) = \mathbf{P}(A) \cdot \mathbf{P}(B), \quad \forall A \in \mathcal{F}, B \in \mathcal{G},$$

Using the independence of σ -algebras, we can reformulate **Definition 2.1** as follows.

Proposition 2.1 *Two r.v.s X and Y are independent if and only if $\sigma(X)$ and $\sigma(Y)$ are independent.*

In practice, it also useful to characterize independence via expectation.

Proposition 2.2 *Two r.v.s X and Y are independent if and only if either*

$$\mathbb{E}f(X)g(Y) = \mathbb{E}f(X)\mathbb{E}g(Y), \quad \forall f, g \text{ bounded and Borel}, \quad (2.2)$$

or,

$$\mathbb{E}f(X)g(Y) = \mathbb{E}f(X)\mathbb{E}g(Y), \quad \forall f, g \text{ bounded and continuous}. \quad (2.3)$$

Proof: (2.2) implies (2.1) since we can take $f = \mathbb{1}_A$ and $g = \mathbb{1}_B$ for any Borel sets A and B . To show the other direction, we will use the idea of “functional Monotone Class Theorem”.

First, for fixed $A \in \mathcal{B}(\mathbb{R})$, let

$$\mathcal{H}_A = \{g : g \text{ bounded and Borel, s.t. } \mathbb{P}\{X \in A\}\mathbb{E}g(Y) = \mathbb{E}\mathbb{1}_A(X)g(Y)\}.$$

We claim that $\mathcal{H}_A$ contains all bounded Borel functions. We will prove this by considering more and more general functions.

1. $\mathcal{H}_A$ contains all indicator functions $\mathbb{1}_B$, $B \in \mathcal{B}(\mathbb{R})$. This follows directly from (2.1).
2. If $g_1, g_2 \in \mathcal{H}_A$, then $\alpha_1 g_1 + \alpha_2 g_2 \in \mathcal{H}_A$. That is, $\mathcal{H}_A$ is closed under linear combination. This implies that $\mathcal{H}_A$ contains all simple functions.
3. If $g_n \geq 0$, $g_n \in \mathcal{H}_A$ and $g_n \uparrow g$, then $g_n(Y) \uparrow g(Y)$ and $\mathbb{1}_A(X)g_n(Y) \uparrow \mathbb{1}_A(X)g(Y)$. By MCT, we have

$$\mathbb{P}(X \in A)\mathbb{E}g(Y) = \lim_{n \rightarrow \infty} \mathbb{P}(X \in A)\mathbb{E}g_n(Y) = \lim_{n \rightarrow \infty} \mathbb{E}\mathbb{1}_A(X)g_n(Y) = \mathbb{E}\mathbb{1}_A(X)g(Y).$$

Therefore, $\mathcal{H}_A$ contains all non-negative Borel functions, and hence all bounded Borel functions by linearity.

Second, let

$$\mathcal{H} = \{f : \text{bounded and Borel s.t. } \mathbb{E}f(X) \cdot \mathbb{E}g(Y) = \mathbb{E}f(X)g(Y)\}.$$

Then $\mathbb{1}_A \in \mathcal{H}$ for every $A \in \mathcal{B}(\mathbb{R})$. Repeating the above argument again, we can show that $\mathcal{H}$ contains all bounded Borel functions. This establishes equivalence between (2.2) and (2.1).

Next, we show that (2.3) and (2.1) are equivalent. First, (2.2) implies (2.3) since continuous functions are Borel. Second, assuming (2.3), for any open intervals A and B , by choosing bounded, non-negative continuous functions f_n and g_n such that $f_n \uparrow \mathbb{1}_A$ and $g_n \uparrow \mathbb{1}_B$, MCT implies that (2.1) holds for such A and B . From open intervals to arbitrary Borel sets we need to use the monotone class theorem. Details are omitted here. $\square$

We can also introduce the notion of a collection of r.v.s being independent.

Definition 2.3 *Let I be a countable index set. A collection of r.v.s $(X_n)_{n \in I}$ are independent, if the σ -algebras $(\sigma(X_n))_{n \in I}$ are independent, that is,*

$$\mathbb{P}\left(\bigcap_{n \in I} A_n\right) = \prod_{n \in I} \mathbb{P}(A_n), \quad \forall A_n \in \sigma(X_n).$$

In probability theory, a fundamental model is a sequence of *independent and identically distributed* (i.i.d.) r.v.s $(X_n)_{n \geq 1}$, which, in addition to X_n being independent, requires that the distribution of X_n is the same. A natural question that we must answer first before delving into nice theories built upon i.i.d. r.v.s like the law of large numbers, central limit theorem and so on, is the existence of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which there live independent r.v.s X_n with given common distribution μ .

The answer is affirmative, and we will discuss its solution in the rest of this section. We break this into three cases.

1. The one-dimensional case: given a c.d.f. $F(x)$, how to construct a r.v. X such that $P(X \leq a) = F(a)$? This is already discussed in [Section 1.4](#).
2. The two/finite-dimensional case: given two probability measures μ_1 and μ_2 on $(\mathbb{R}, \mathcal{B}(R))$, how can we construct two r.v.s X, Y such that $\mathcal{L}(X) = \mu_1$, $\mathcal{L}(Y) = \mu_2$ and X and Y are independent? We will do this in [Section 2.2](#) with the help of product measures.
3. The infinite-dimensional case: given probability measures $(\mu_n)_{n \in I}$ on $(\mathbb{R}, \mathcal{B}(R))$, how can we construct r.v.s $(X_n)_{n \in I}$ such that $\mathcal{L}(X_n) = \mu_n$, $n \in I$, and X_n are independent. In particular, for a sequence of i.i.d. r.v.s, we need $I = \mathbb{N}$. We will do this in [Section 2.3](#) with the help of the celebrated Kolmogorov's Extension Theorem [Theorem 2.7](#).

On the other hand, when the common distribution μ is as simple as the Bernoulli distribution, we have explicit construction of the probability space and r.v.s.

Example 2.1 Let $(\Omega, \mathcal{F}, P) = ((0, 1), \mathcal{B}(0, 1), \text{Leb})$. Every $\omega \in \Omega = (0, 1)$ admits a dyadic expansion:

$$\omega = \sum_{n=1}^{\infty} \xi_n(\omega) \frac{1}{2^n}, \quad \xi_n(\omega) \in \{0, 1\}. \quad (2.4)$$

When $\omega = \frac{k}{2^n}$ is a dyadic rational, the expansion (2.4) is non-unique; in that case, we will choose the expansion with an infinite number of 1's to fix the choice. For example, we choose

$$\frac{1}{2} = 0 \cdot \frac{1}{2^1} + 1 \cdot \frac{1}{2^2} + 1 \cdot \frac{1}{2^3} + 1 \cdot \frac{1}{2^4} + \cdots = \sum_{n=2}^{\infty} \frac{1}{2^n}, \quad \text{rather than} \quad \frac{1}{2} = \frac{1}{2} + \sum_{n=2}^{\infty} 0 \cdot \frac{1}{2^n}.$$

One can verify directly that $(\xi_n)_{n \geq 1}$ are i.i.d. Bernoulli r.v.s with parameter $1/2$.

Counterexample 2.2 (Pairwise independence does not imply mutual independence.) We first give a classical discrete random variable example. Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), \text{Leb})$ and

$$X_1(\omega) = \begin{cases} 1, & \omega \in [\frac{1}{2}, 1], \\ -1, & \omega \in [0, \frac{1}{2}), \end{cases} \quad X_2(\omega) = \begin{cases} 1, & \omega \in [\frac{1}{4}, \frac{1}{2}) \cup [\frac{3}{4}, 1], \\ -1, & \omega \in [0, \frac{1}{4}) \cup [\frac{1}{2}, \frac{3}{4}), \end{cases} \quad X_3(\omega) = X_1(\omega) \cdot X_2(\omega).$$

It is easy to check that X_1, X_2 are r.v.s since they are simple functions, and thus X_3 is a r.v. since it is a product of two r.v.s. It is also easy to check that X_1, X_2, X_3 are pairwise independent. However, they are not independent, since

$$P(X_1 = X_2 = X_3 = -1) = 0 \neq \frac{1}{8} = P(X_1 = -1)P(X_2 = -1)P(X_3 = -1).$$

A more general example is as follows: let X_1, X_2 be independent $\text{Unif}[0, 1)$ r.v.s and let

$$X_3 = -(X_1 + X_2) \mod 1.$$

Then X_3 is also a $\text{Unif}[0, 1)$ r.v. and is independent of either X_1 or X_2 . But X_1, X_2, X_3 are not independent since $X_1 + X_2 + X_3 = 0 \mod 1$. From this example we can build more general r.v.s by taking $Y_j = G_j(X_j)$ for arbitrary increasing functions $G_j : [0, 1] \rightarrow \mathbb{R}$; see also [Proposition 1.25](#).

2.2 Product Measures and Fubini's Theorem

Let $(\Omega_i, \mathcal{F}_i, P_i)$, $i = 1, 2$, be two probability spaces. Let

$$\begin{aligned} \Omega &= \Omega_1 \times \Omega_2 = \{(\omega_1, \omega_2) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\}, \\ \mathcal{F} &= \mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma(A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2). \end{aligned}$$

Then $(\Omega, \mathcal{F})$ is a measurable space. A special case is $(\Omega_i, \mathcal{F}_i) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ where $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$, where the equality is due to the fact that open rectangles

$$(a, b) \times (c, d), \quad -\infty < a < b < \infty, \quad -\infty < c < d < \infty,$$

form a topological basis for open sets in $\mathbb{R}^2$.

Our goal is to construct the product measure $P_1 \times P_2$ on $(\Omega, \mathcal{F})$. We will need to introduce an appropriate algebra generating $\mathcal{F}$ and use Carathéodory's Extension Theorem ([Theorem 1.12](#)). Consider the collection of “rectangles”

$$\mathcal{S} = \{A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}.$$

It is not hard to check that $\mathcal{S}$ forms a semi-algebra:

1. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$,
2. $(A \times B)^c = (A^c \times B) \cup (A \times B^c) \cup (A^c \times B^c)$.

The semi-algebra $\mathcal{S}$ naturally generates an algebra

$$\bar{\mathcal{S}} = \left\{ \bigcup_{i=1}^k I_i, I_i \in \mathcal{S}, I_i \text{ disjoint} \right\}.$$

We note that unless one of $\mathcal{F}_i$ is trivial, $\mathcal{S} \subsetneq \sigma(\mathcal{S}) = \mathcal{F}$ (actually, $\mathcal{S} \subsetneq \bar{\mathcal{S}}$ for nontrivial $\mathcal{F}_i$).

Remark 2.3 Using standard notion of Cartesian products, one may write “ $\mathcal{S} = \mathcal{F}_1 \times \mathcal{F}_2$ ”, but be careful that some authors also use “ $\mathcal{F}_1 \times \mathcal{F}_2$ ” for the product σ -algebra. In this note, we always use the tensor product notation “ $\otimes$ ” to for the product σ -algebra, which is usually larger than the Cartesian product of σ -algebras.

The unique measure μ defined in the next theorem is the desired product measure $P_1 \times P_2$.

Theorem 2.3 *There exists a unique probability measure μ on $(\Omega, \mathcal{F})$ such that*

$$\mu(A \times B) = P_1(A) \cdot P_2(B).$$

Proof: We can define a finitely additive probability measure μ_0 on $\bar{\mathcal{S}}$ by

$$\mu_0(D) = \sum_{i=1}^k P_1(A_i) \cdot P_2(B_i), \quad D = \text{disjoint union of } A_1 \times B_1, \dots, A_k \times B_k.$$

The conclusion follows from [Theorem 1.12](#), if we can show that μ_0 is a σ -additive on $\bar{\mathcal{S}}$. For this, it suffices to check that if $A_n \times B_n$, $n = 1, \dots$, are disjoint and $A \times B = \bigcup_{n=1}^{\infty} (A_n \times B_n)$, then

$$\mu_0(A \times B) = \sum_{n=1}^{\infty} \mu_0(A_n \times B_n). \tag{2.5}$$

(This is σ -additivity on $\mathcal{S}$, not on $\bar{\mathcal{S}}$, but here they are equivalent.)

For $x \in A$, let $I(x) = \{n : x \in A_n\}$. Then

$$B = \bigcup_{n \in I(x)} B_n, \quad \forall x \in A, \tag{2.6}$$

since $\{x\} \times B \subset \bigcup_{n \in I(x)} (A_n \times B_n)$. For $x \in A$, we have

$$\mathbb{1}_A(x) \cdot P_2(B) = \mathbb{1}_A(x) \cdot \sum_{n \in I(x)} P_2(B_n) = \sum_{n \in I(x)} \mathbb{1}_{A_n}(x) P_2(B_n) = \sum_{n \geq 1} \mathbb{1}_{A_n}(x) P_2(B_n). \quad (2.7)$$

The first equality holds since we have (2.6) and B_n are disjoint, the second holds since $\mathbb{1}_A(x) = \mathbb{1}_{A_n}(x) = 1$ for $n \in I(x)$, and the third holds since we are adding more zero terms.

Note that (2.7) also holds for $x \notin A$, since

$$\mathbb{1}_A(x) \cdot P_2(B) = 0 = \sum_{n \geq 1} \mathbb{1}_{A_n}(x) P_2(B_n), \quad x \notin A.$$

Integrating (2.7) over $x \in \Omega$, the left hand side becomes

$$\left[\int_{\Omega} \mathbb{1}_A(x) P_1(dx) \right] \cdot P_2(B) = P_1(A) \cdot P_2(B) = \mu_0(A \times B),$$

and the right hand side becomes

$$\begin{aligned} \int_{\Omega} \left[\sum_{n \geq 1} \mathbb{1}_{A_n}(x) P_2(B_n) \right] P_1(dx) &= \int_{\Omega} \left[\lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{1}_{A_n}(x) P_2(B_n) \right] P_1(dx) \\ &= \lim_{N \rightarrow \infty} \int_{\Omega} \left[\sum_{n=1}^N \mathbb{1}_{A_n}(x) P_2(B_n) \right] P_1(dx) \\ &= \sum_{n=1}^{\infty} P_1(A_n) P_2(B_n) = \sum_{n=1}^{\infty} \mu_0(A_n \times B_n), \end{aligned}$$

where we use MCT in the second line. This proves (2.5) and concludes the proof. $\square$

We can construct two independent r.v.s with given distribution using **Theorem 2.3**. Let X be a r.v. on $(\Omega_1, \mathcal{F}_1, P_1)$ and Y a r.v. on $(\Omega_2, \mathcal{F}_2, P_2)$. On $(\Omega, \mathcal{F}, \mu) = (\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, P_1 \times P_2)$, we define

$$\tilde{X}(\omega_1, \omega_2) = X(\omega_1), \quad \tilde{Y}(\omega_1, \omega_2) = Y(\omega_2).$$

Then

$$\begin{aligned} P(\tilde{X} \in A, \tilde{Y} \in B) &= \mu(X^{-1}(A) \times Y^{-1}(B)) = P_1(X^{-1}(A)) \cdot P_2(Y^{-1}(B)) \\ &= P_1(X \in A) \cdot P_2(Y \in B) = P(\tilde{X} \in A) \cdot P(\tilde{Y} \in B), \end{aligned}$$

that is, $\tilde{X}$ (respectively, $\tilde{Y}$) has the same distribution as X (resp. Y), and $\tilde{X}, \tilde{Y}$ are independent.

Integration on the product measure space can be computed using Fubini's Theorem below. Fubini's Theorem also includes some measurability statements on jointly measurable maps.

Theorem 2.4 (Fubini's Theorem) *Let $(\Omega_i, \mathcal{F}_i, P_i)$, $i = 1, 2$, be two measure spaces, where P_i are probability (or σ -finite) measures. Let $f : \Omega \rightarrow \mathbb{R}$ be $\mathcal{F}_1 \otimes \mathcal{F}_2$ -measurable where $\Omega = \Omega_1 \times \Omega_2$. Assume either*

$$f \geq 0, \quad (2.8a)$$

$$\text{or} \quad \int |f(\omega_1, \omega_2)| (P_1 \times P_2)(d\omega_1 d\omega_2) < \infty. \quad (2.8b)$$

Then the following holds.

1. For every $\omega_1 \in \Omega$, the function $f(\omega_1, \cdot)$ is $\mathcal{F}_2$ -measurable. And if (2.8b) holds,

$$\int_{\Omega_2} |f(\omega_1, \omega_2)| P_2(d\omega_2) < \infty, \quad \text{for almost every } \omega_1 \in \Omega. \quad (2.9)$$

2. The function $g(\omega_1) = \int_{\Omega_2} f(\omega_1, \omega_2) P_2(d\omega_2)$ is $\mathcal{F}_1$ -measurable. And if (2.8b) holds,

$$\int_{\Omega_1} |g(\omega_1)| P_1(d\omega_1) < \infty. \quad (2.10)$$

3. The double integral is equal to either iterated integral, that is,

$$\begin{aligned} \iint_{\Omega_1 \times \Omega_2} f(\omega_1, \omega_2) (P_1 \times P_2)(d\omega_1 d\omega_2) &= \int_{\Omega_1} P_1(d\omega_1) \int_{\Omega_2} f(\omega_1, \omega_2) P_2(d\omega_2) \\ &= \int_{\Omega_2} P_2(d\omega_2) \int_{\Omega_1} f(\omega_1, \omega_2) P_1(d\omega_1). \end{aligned} \quad (2.11)$$

Proof: Let $\mathcal{H}$ be the collection of all $\mathcal{F}_1 \otimes \mathcal{F}_2$ -measurable functions f such that Items 1 to 3 hold. As usual, we will show that $\mathcal{H}$ contains more and more general functions, in particular, all $\mathcal{F}_1 \otimes \mathcal{F}_2$ -measurable functions f such that either (2.8a) or (2.8b) holds.

1. Indicator functions of rectangles are in $\mathcal{H}$.

Let $f(\omega_1, \omega_2) = \mathbb{1}_A(\omega_1) \mathbb{1}_B(\omega_2)$ where $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$. We have

$$f(\omega_1, \cdot) = \begin{cases} 0, & \omega_1 \notin A, \\ \mathbb{1}_B(\cdot), & \omega_1 \in A, \end{cases}$$

so $f(\omega_1, \cdot)$ is $\mathcal{F}_2$ -measurable for every ω_1 . In addition, direct computation gives

$$g(\omega_1) = \begin{cases} 0, & \omega_1 \notin A \\ P_2(B), & \omega_1 \in A \end{cases} = \mathbb{1}_A(\omega_1) \cdot P_2(B),$$

and hence g is $\mathcal{F}_1$ -measurable. It is easy to verify (2.9) to (2.11).

2. The indicator function $\mathbb{1}_D(\omega_1, \omega_2) \in \mathcal{F}$ for every $D \in \mathcal{F}$.

We will use the method of appropriate sets. Let

$$\mathcal{G} = \{D \in \mathcal{F}_1 \otimes \mathcal{F}_2, \mathbb{1}_D \in \mathcal{H}\}.$$

We note that $\mathcal{G}$ contains the algebra $\bar{\mathcal{S}}$ as a consequence of the first part, and that $\mathcal{G}$ is a monotone class, since the measurability conditions are preserved by taking limits, and the integral conditions are preserved by the MCT. Hence, by the monotone class theorem $\mathcal{G} = \mathcal{F}_1 \otimes \mathcal{F}_2$.

3. Simple functions of the form $\varphi(\omega) = \sum_{i=1}^n c_i \mathbb{1}_{D_i}(\omega)$ are in $\mathcal{H}$, since Items 1 to 3 are preserved by taking finite linear combination.

4. All nonnegative, $\mathcal{F}_1 \otimes \mathcal{F}_2$ -measurable functions f are in $\mathcal{H}$.

Recall that there exist simple functions $\{f_n\}$ such that $f_n(\omega) \uparrow f(\omega)$ for every ω . We have already shown that $f_n \in \mathcal{H}$.

Since for every ω_1 , the function $f_n(\omega_1, \cdot)$ is $\mathcal{F}_2$ -measurable, the limit $f(\omega_1, \cdot) = \lim_{n \rightarrow \infty} f_n(\omega_1, \cdot)$ is also $\mathcal{F}_2$ -measurable. By MCT,

$$g(\omega_1) = \int_{\Omega_2} f(\omega_1, \omega_2) P_2(d\omega_2) = \lim_{n \rightarrow \infty} \int_{\Omega_2} f_n(\omega_1, \omega_2) P_2(d\omega_2) = \lim_{n \rightarrow \infty} g_n(\omega_1).$$

Since $g_n(\omega_1)$ are $\mathcal{F}_1$ -measurable, their increasing limit $g(\omega_1)$ is also $\mathcal{F}_1$ -measurable. Finally, by MCT applied to both (g_n) and (f_n) ,

$$\begin{aligned} \int_{\Omega_1} g(\omega_1) P_1(d\omega_1) &= \lim_{n \rightarrow \infty} \int_{\Omega_1} g_n(\omega_1) P_1(d\omega_1) = \lim_{n \rightarrow \infty} \int_{\Omega} f_n(\omega_1, \omega_2) (P_1 \times P_2)(d\omega_1 d\omega_2) \\ &= \int_{\Omega} f(\omega_1, \omega_2) (P_1 \times P_2)(d\omega_1 d\omega_2), \end{aligned}$$

and then by symmetry in ω_1 and ω_2 ,

$$\int_{\Omega} f(\omega_1, \omega_2) (P_1 \times P_2)(d\omega_1 d\omega_2) = \int_{\Omega_2} P_2(d\omega_2) \int_{\Omega_1} f(\omega_1, \omega_2) P_1(d\omega_1).$$

This verifies (2.11) and thus $f \in \mathcal{H}$.

5. For general function f , we consider $f = f_+ - f_-$. To show that $f \in \mathcal{H}$, everything is straightforward except (2.9).

Applying Fubini's Theorem to $|f| \geq 0$, we have

$$\int_{\Omega_1} P(d\omega_1) \left[\int_{\Omega_2} |f(\omega_1, \omega_2)| P_2(d\omega_2) \right] = \int_{\Omega} |f(\omega_1, \omega_2)| (P_1 \times P_2)(d\omega_1 d\omega_2) < \infty.$$

This implies (2.9). □

Let $D \subset \Omega$. The cross section of D at x is defined by

$$D_x = \{y : (x, y) \in D\}.$$

As a corollary of Theorem 2.4, we obtain measurability of the cross section.

Proposition 2.5 *Let $D \in \mathcal{F}_1 \otimes \mathcal{F}_2$. Then $D_x \in \mathcal{F}_2$ for every $x \in \Omega_1$.*

Proof: Note that $y \in D_x$ if and only if $\mathbb{1}_D(x, y) > 0$. For every $x \in \Omega_1$, by Theorem 2.4, the function $\mathbb{1}_D(x, \cdot)$ is $\mathcal{F}_2$ -measurable, and thus

$$D_x = \{y : \mathbb{1}_D(x, y) > 0\} \in \mathcal{F}_2. \quad \square$$

Counterexample 2.4 (Product of complete σ -algebras may not be complete.) Recall that the completion of a probability space $(\Omega, \mathcal{F}, P)$ is a complete probability space $(\Omega, \bar{\mathcal{F}}, \bar{P})$ such that

$$\bar{\mathcal{F}} = \{A : \exists B_1 \subset A \subset B_2, B_1, B_2 \in \mathcal{F} \text{ s.t. } P(B_1) = P(B_2), P(B_1 \setminus B_2) = 0\},$$

and for $A \in \bar{\mathcal{F}}$, we define $\bar{P}(A) = P(B_1)$ where B_1 is given above. Note that $\overline{\mathcal{B}(\mathbb{R})} = \{\text{Lebesgue sets}\}$.

We claim that

$$\overline{\mathcal{B}(\mathbb{R})} \otimes \overline{\mathcal{B}(\mathbb{R})} \neq \overline{\mathcal{B}(\mathbb{R}^2)},$$

and in general,

$$\overline{\mathcal{F}_1 \otimes \mathcal{F}_2} \neq \overline{\mathcal{F}_1} \otimes \overline{\mathcal{F}_2}.$$

Indeed, Let $A \subset [0, 1]$ be a non-Lebesgue set and $D = A \times \{0\}$. We have $D \subset [0, 1] \times \{0\}$ and

$$\text{Leb}([0, 1] \times \{0\}) = \lim_{n \rightarrow \infty} \text{Leb}([0, 1] \times [0, 1/n]) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Hence $D \in \overline{\mathcal{B}(\mathbb{R}^2)}$ by the definition of completion. But $D \notin \overline{\mathcal{B}(\mathbb{R})} \otimes \overline{\mathcal{B}(\mathbb{R})}$, otherwise by [Proposition 2.5](#),

$$A = \{x \in \mathbb{R} : (x, 0) \in D\} = D_0 \in \overline{\mathcal{B}(\mathbb{R})},$$

which is absurd.

In general, completion of probability spaces has to be done in the final step, after the construction of product spaces.

There is a version of Fubini's Theorem stated for the completion of the σ -algebra $\overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$. The proof is technical, but this version will be useful in certain cases. This is also the version of the Fubini's Theorem that is stated in a real analysis course, in which Lebesgue sets rather than Borel sets are the primary interest. We include it here and sketch the main technicalities in the proof, from which the reader can also learn how to deal with zero measure sets.

Theorem 2.6 (Fubini's Theorem for complete measure spaces) *Let $f : \Omega \rightarrow \mathbb{R}$ be $\overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$ -measurable. Assume either [\(2.8a\)](#) or [\(2.8b\)](#). Then*

1. *There exists a set $N \in \mathcal{F}_1$ with $\mathbb{P}(N) = 0$, such that for every $\omega_1 \in N^c$, the function $f(\omega_1, \cdot)$ is $\overline{\mathcal{F}_2}$ -measurable. When [\(2.8b\)](#) holds, the set N can be chosen such that for $\omega_1 \in N^c$,*

$$\int |f(\omega_1, \omega_2)| \mathbb{P}_2(d\omega_2) < \infty.$$

2. *Let*

$$g(\omega_1) = \begin{cases} \int_{\Omega_2} f(\omega_1, \omega_2) \mathbb{P}_2(d\omega_2), & f(\omega_1, \cdot) \text{ is } \overline{\mathcal{F}_2}\text{-measurable,} \\ \text{undefined,} & \text{otherwise.} \end{cases}$$

Then $g(\omega_1)$ is $\overline{\mathcal{F}_1}$ -measurable. If [\(2.8b\)](#) holds, then [\(2.10\)](#) is true.

3. [\(2.11\)](#) holds.

Proof: Let $\mathcal{H}$ be the collection of $\overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$ -measurable functions such that the Fubini's Theorem holds.

There are two key steps. First, we need to show that $\mathbb{1}_D \in \mathcal{H}$ for any $D \in \overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$. Second, we need to show that $\mathcal{H}$ is closed under taking limit, that is, if $f_n \in \mathcal{H}$, $f_n \geq 0$, $f_n \uparrow f$, then $f \in \mathcal{H}$.

To prove the first step, let $D \in \overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$. By the definition of completion, there exists $D^\pm \in \mathcal{F}_1 \otimes \mathcal{F}_2$ such that

$$D^- \subset D \subset D^+, \quad (\mathbb{P}_1 \times \mathbb{P}_2)(D^+ \setminus D^-) = 0.$$

By definition of the cross section, for every $\omega_1 \in \Omega_1$, we have $D_{\omega_1}^- \subset D_{\omega_1} \subset D_{\omega_1}^+$. In addition, by [Proposition 2.5](#) and [Theorem 2.4j](#), for every $\omega_1 \in \Omega_1$, we have $D_{\omega_1}^\pm \in \mathcal{F}_2$ and that $q(\omega_1) = \mathbb{P}_2(D_{\omega_1}^+) - \mathbb{P}_2(D_{\omega_1}^-)$ is $\mathcal{F}_1$ -measurable, and

$$\begin{aligned} \int q(\omega_1) \mathbb{P}_1(d\omega_1) &= \int [\mathbb{P}_2(D_{\omega_1}^+) - \mathbb{P}_2(D_{\omega_1}^-)] \mathbb{P}_1(d\omega_1) \\ &= \int (\mathbb{1}_{D^+}(\omega) - \mathbb{1}_{D^-}(\omega)) (\mathbb{P}_1 \times \mathbb{P}_2)(d\omega_1 d\omega_2) = (\mathbb{P}_1 \times \mathbb{P}_2)(D^+ \setminus D^-) = 0. \end{aligned} \tag{2.12}$$

Since $q(\omega_1) \geq 0$, (2.12) implies that there exists $N \in \mathcal{F}_1$ with $P_1(N) = 0$ such that

$$q(\omega_1) = P_2(D_{\omega_1}^+) - P_2(D_{\omega_1}^-) = 0, \quad \forall \omega_1 \notin N.$$

Hence, for $\omega_1 \notin N$, the set D_{ω_1} is $\overline{\mathcal{F}_2}$ -measurable since

$$P_2(D_{\omega_1}^+) = P_2(D_{\omega_2}^-), \quad D_{\omega_1}^- \subset D_{\omega_1} \subset D_{\omega_1}^+.$$

Note that $g(\omega_1)$ is defined on N^c , so

$$\{\omega_1 : g(\omega_1) \text{ not defined}\} \subset N,$$

and it is an element of $\overline{\mathcal{F}_1}$ by definition. It is easy to verify (2.11).

For the second step, let $\mathcal{H} \in f_n \uparrow f$ and let $N_n \in \mathcal{F}_1$ be the corresponding zero measure sets corresponding to f_n . Let $N = \bigcup_{n=1}^{\infty} N_n$. Then $N \in \mathcal{F}_1$ and $P_1(N) = 0$. If $\omega_1 \notin N$, then $\omega_1 \notin N_n$ for every n , and hence $f_n(\omega_1, \cdot)$ is $\overline{\mathcal{F}_2}$ -measurable, the $f(\omega_1, \cdot)$ as the limit of $f_n(\omega_1, \cdot)$ is $\overline{\mathcal{F}_2}$ -measurable, for $\omega_1 \notin N$. It is easy to check the other conditions. $\square$

2.3 Measures on $\mathbb{R}^\infty$ and Kolmogorov's Extension Theorem

We can generalize the notion of product measures to a finite number of probability spaces. Hence, we can construct a finite number of independent r.v.s with given distribution. More precisely, given probability spaces $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_i)$, $1 \leq i \leq n$, let

$$(\Omega, \mathcal{F}, P) = (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \bigotimes_{i=1}^n \mu_i).$$

We write an element of Ω as $\omega = (\omega_1, \dots, \omega_n)$. Let X_i be r.v.s defined by $X_i(\omega) = \omega_i$, $1 \leq i \leq n$. Then $\{X_i\}_{1 \leq i \leq n}$ are independent and $\mathcal{L}(X_i) = \mu_i$.

In this section, we illustrate how to construct an infinite sequence of independent r.v.s. It is important to understand the structure of the measure space $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$.

The space $\mathbb{R}^\infty$ forms a metric space with the metric

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} (1 \wedge |x_n - y_n|) \leq 1, \quad x = (x_1, x_2, \dots) \in \mathbb{R}^\infty.$$

We say that $O \subset \mathbb{R}^\infty$ is an open set, if for every $x \in O$, there exists $\delta > 0$ such that

$$\{y : d(x, y) < \delta\} \subset O.$$

It is also useful to introduce the projection: $\pi_n : \mathbb{R}^m \rightarrow \mathbb{R}^n$, $n \leq m \leq \infty$, where $\pi_n x$ is the first n coordinates of x . The convergence in $\mathbb{R}^\infty$ can be characterized by convergence in finite dimensional spaces:

$$d(x^{(m)}, x^{(0)}) \rightarrow 0, \quad m \rightarrow \infty \quad \Leftrightarrow \quad \pi_n x^{(m)} \rightarrow \pi_n x^{(0)}, \quad \forall n \geq 1. \quad (2.13)$$

With the definition of open sets, we can define the Borel σ -algebra $\mathcal{B}(\mathbb{R}^\infty)$. It is not hard to check that, instead of open balls, $\mathcal{B}(\mathbb{R}^\infty)$ can also be generated by

$$\mathcal{B}(\mathbb{R}^\infty) = \sigma\left(O_n \times \mathbb{R}^\infty, O_n \text{ open set in } \mathbb{R}^n\right). \quad (2.14)$$

In general, set of the form

$$\pi_n^{-1} A = A \times \mathbb{R}^\infty, \quad A \in \mathcal{B}(\mathbb{R}^n)$$

are called *cylinder sets*.

For $n \geq 1$, let μ_n be probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. We say that μ_n satisfies the *consistency condition*, if

$$\mu_{n+1} \circ \pi_n^{-1} = \mu_n, \quad (2.15)$$

The condition (2.15) is the same as

$$\mu_{n+1}(A \times \mathbb{R}) = \mu_n(A), \quad \forall A \in \mathcal{B}(\mathbb{R}^n),$$

or,

$$\mu_{n+m} \circ \pi_n^{-1} = \mu_n, \quad \forall m, n \geq 1. \quad (2.16)$$

Theorem 2.7 (Kolmogorov's Extension Theorem) Assume (2.16). There exists a unique measure μ on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $\mu \circ \pi_n^{-1} = \mu_n$ for every $n \geq 1$, that is,

$$\mu(A \times \mathbb{R}^\infty) = \mu_n(A), \quad \forall A \in \mathcal{B}(\mathbb{R}^n). \quad (2.17)$$

To construct an infinite sequence of independent r.v.s, we will use Theorem 2.7 in the following way. Given $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda_i)$, $i \geq 1$, let

$$\mu_n = \bigotimes_{i=1}^n \lambda_i$$

be probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Then μ_n satisfies the consistency condition (2.16) by properties of the product measures. Then by Theorem 2.7, there exists a unique probability measure μ on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ so that (2.17) holds. Let

$$X_n(\omega) = \omega_n, \quad n \geq 1.$$

Then $(X_n)_{n \geq 1}$ are independent r.v.s on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty), \mu)$.

Next, we will prove Theorem 2.7. Before that, we need to understand compact sets in $\mathbb{R}^\infty$.

Proposition 2.8 Let F_m , $m \geq 1$, be nonempty compact sets in $\mathbb{R}^m$ such that

$$D_m = \pi_m^{-1}(F_m) = F_m \times \mathbb{R}^\infty$$

are decreasing in m . Then $\bigcap_{m=1}^\infty D_m \neq \emptyset$.

Proof: For every $m \geq 1$, pick $x^{(m)} \in D_m$. Since D_m are decreasing cylinder sets, for every $n \geq 1$, we have $(\pi_n x^{(m)})_{m \geq n} \subset \pi_n(D_n) = F_n$ is a bounded sequence in $\mathbb{R}^n$.

Bounded sequences in $\mathbb{R}^n$ have convergence subsequence. Therefore, there exists $(m_k^1)_{k \geq 1}$ so that $\pi_1 x^{(m_k^1)}$ converges in $\mathbb{R}^1$, and $(m_k^2)_{k \geq 2} \subset (m_k^1)_{k \geq 1}$ so that $\pi_2 x^{(m_k^2)}$ converges in $\mathbb{R}^2$ and so on. Let $y^{(k)} = x^{(m_k^k)}$ be the diagonal sequence. For every $n \geq 1$, the sequence $(\pi_n y^{(k)})_{k \geq 1}$ converges in $\mathbb{R}^n$ by construction. By (2.13), there exists $y^* \in \mathbb{R}^\infty$ such that $y^{(k)} \rightarrow y^*$ in $\mathbb{R}^\infty$. Noting that $\pi_n y^{(k)} \in F_n$ for $k \geq n$, we have $y^* \in D_n$ for every n , and thus $y^* \in \bigcap_{n=1}^\infty D_n$. This proves the conclusion. $\square$

Remark 2.5 A similar argument shows that the metric we put on $\mathbb{R}^\infty$ is such that for any $L_n \in (0, \infty)$, the product set

$$\bigotimes_{n=1}^\infty [-L_n, L_n]$$

is sequentially compact in $\mathbb{R}^\infty$.

We also need a small lemma about the regularity of Borel sets in $\mathbb{R}^d$.

Proposition 2.9 Let λ be a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Let $A \in \mathcal{B}(\mathbb{R}^d)$. For every $\varepsilon > 0$, there exists a closed set F_ε and an open set G_ε such that

$$F_\varepsilon \subset A \subset G_\varepsilon, \quad \lambda(G_\varepsilon) - \lambda(F_\varepsilon) < \varepsilon.$$

In addition, F_ε can be chosen to be compact since

$$\lim_{L \rightarrow \infty} \lambda(F_\varepsilon \cap [-L, L]^d) = \lambda(F_\varepsilon).$$

Proof: Let $\mathcal{S}$ be the collection of sets A that satisfy the condition. Then $\mathcal{S}$ contains all open sets, and thus all rectangles $(a_1, b_1) \times \cdots \times (a_d, b_d)$. It is not hard to show that $\mathcal{S}$ forms a σ -algebra. Therefore, $\mathcal{S} \supset \mathcal{B}(\mathbb{R}^d)$. $\square$

Proof of Theorem 2.7: Let $\mathcal{C} = \{\text{cylinder sets}\}$. We have the following.

1. $\mathcal{C}$ is an algebra.
2. The condition (2.17) specifies the measure μ on $\mathcal{C}$.
3. (2.14) implies that $\mathcal{B}(\mathbb{R}^\infty) = \sigma(\mathcal{C})$.
4. The consistency condition (2.16) implies that (2.17) defines a finitely additive measure μ on $\mathcal{C}$.

Putting all these together, we can use the Carathéodory's Extension Theorem to construct the desired measure μ , provided that we verify that μ is σ -additive on $\mathcal{C}$.

To show σ -additivity, it suffices to show continuity at $\emptyset$, that is, $\mu(D_n) \rightarrow 0$ for every $\mathcal{C} \ni D_n \downarrow \emptyset$.

Without loss of generality, we can assume that $D_n = \pi_n^{-1}(B_n)$ where $B_n \in \mathcal{B}(\mathbb{R}^n)$. We will prove by contradiction.

Assume the contrary that there exists $\delta > 0$ such that $\mu(D_n) = \mu_n(B_n) \geq \delta$ for every n . By Proposition 2.9, there exist compact sets $F_n \subset B_n$ such that $\mu_n(B_n \setminus F_n) \leq \delta 2^{-n-1}$, $n \geq 1$.

Let $\hat{E}_n = \pi_n^{-1}(F_n) \in \mathcal{C}$. Then $\mu(D_n \setminus \hat{E}_n) = \mu_n(B_n \setminus F_n) \leq \delta 2^{-n-1}$. The sets $\hat{E}_n$ may not be decreasing, but if we set

$$E_n = \bigcap_{m=1}^n \hat{E}_m, \quad n \geq 1,$$

then E_n are decreasing. In addition,

$$\mu(D_n \setminus E_n) \leq \mu\left(\bigcup_{m=1}^n (D_n \setminus \hat{E}_m)\right) \leq \sum_{m=1}^n \frac{\delta}{2^{m+1}} \leq \frac{\delta}{2}.$$

Hence, $\mu(E_n) \geq \mu(D_n) - \delta/2 \geq \delta/2$ for all $n \geq 1$. In particular, $E_n \neq \emptyset$ for all n , and hence we can apply Proposition 2.8 to conclude that $\bigcap_{n=1}^\infty E_n \neq \emptyset$. But

$$\bigcap_{n=1}^\infty E_n \subset \bigcap_{n=1}^\infty D_n = \emptyset,$$

and we arrive at a contradiction. $\square$

Remark 2.6 Instead of $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$, Kolmogorov's Extension Theorem can also be stated for general measurable spaces $(\prod_{n=1}^\infty S_n, \bigotimes_{n=1}^\infty \mathcal{S}_n)$. To verify the σ -additivity needed for Carathéodory's Extension Theorem, some topological information is needed for the spaces $(S_n, \mathcal{S}_n)$. A sufficient condition is that all $(S_n, \mathcal{S}_n)$ are Borel spaces: a measurable space $(S, \mathcal{S})$ is called Borel if there is a one-to-one map $\varphi : (S, \mathcal{S}) \rightarrow ([0, 1], \mathcal{B}[0, 1])$ so that φ and φ^{-1} are both measurable. In particular, all complete and separable metric spaces equipped with Borel σ -algebras are Borel.

As a final note, we can also apply Kolmogorov's Extension Theorem on $(\mathbb{R}^T, \mathcal{B}(\mathbb{R}^T))$, where T is any index set, and the Borel σ -algebra $\mathcal{B}(\mathbb{R}^T)$ is generated by all “(finite-dimensional) cylinder sets”

$$\pi_{t_1, t_2, \dots, t_n}^{-1}(A_n), \quad A_n \text{ open set in } \mathbb{R}^n, \quad t_1, \dots, t_n \in T.$$

All cylinder sets form an algebra, and a probability measure μ on this space exists, provided that its “finite-dimension distributions” $\mu \circ \pi_{t_1, \dots, t_n}^{-1}$ satisfy the consistency condition. In fact, we can characterize the resulting σ -algebra: if $A \in \mathcal{B}(\mathbb{R}^T)$, then there are a countable set of points $t_1, t_2, \dots$ of T and a Borel set $B \in \mathcal{B}(\mathbb{R}^\infty)$ such that

$$A = \{x : (x_{t_1}, x_{t_2}, \dots) \in B\}. \quad (2.18)$$

For some discussion of this, see also [Shi96, Chap. II. 2.2, 2.5]. We call $(X_t)_{t \in T}$ a *stochastic process on T* on $(\Omega, \mathcal{F}, \mathbb{P})$ if for every $t \in T$, the map $\omega \mapsto X_t(\omega)$ is measurable (or X_t is a random variable). Every probability measure μ on $(\mathbb{R}^T, \mathcal{B}(\mathbb{R}^T))$ gives rise to a stochastic process given by $X_t(\omega) := \omega_t$.

Counterexample 2.7 (The set where a continuous-time stochastic process is continuous is not measurable.)

Let $(X_t)_{t \in [0,1]}$ be a stochastic process on $[0, 1]$. For every $t_0 \in [0, 1]$, the set

$$\{\omega : \lim_{t \rightarrow t_0} X_t(\omega) = X_{t_0}(\omega)\} \quad (2.19)$$

is not in $\mathcal{B}(\mathbb{R}^{[0,1]})$, since the condition in (2.19) cannot be expressed in countably many coordinates, and hence is not of the form (2.18).

This means that the σ -algebra $\mathcal{B}(\mathbb{R}^T)$ is not suitable to study continuous-time stochastic process. In fact, this is the first technical issue in the construction of Brownian motion in stochastic analysis, which is required to be an almost surely continuous process. For more details, see [KS, Chap. 2.2].

3 Mode of convergence for random variables

3.1 Definitions

There are four basic modes of convergence for r.v.s. We list their definitions below.

1. Almost sure convergence.

We say that $X_n \rightarrow X$ almost surely (a.s.), if

$$\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$$

2. Convergence in probability.

We say that $X_n \rightarrow X$ in probability (in pr.), if

$$\lim_{n \rightarrow \infty} \mathbb{P}\{|X_n - X| > \varepsilon\} = 0, \quad \forall \varepsilon > 0. \quad (3.1)$$

3. Weak convergence or convergence in distribution.

We say that $X_n \rightarrow X$ in distribution, or in law, or weakly, or weakly-*, if for every continuous and bounded function f , have

$$\lim_{n \rightarrow \infty} \mathbb{E}f(X_n) = \mathbb{E}f(X).$$

We also write this as $X_n \Rightarrow X$ or $X_n \Rightarrow_d X$. We will explain the origins of all these different terms in Section 3.4.

4. Convergence in L^p .

We say that $X_n \rightarrow X$ in L^p if

$$\lim_{n \rightarrow \infty} \mathbb{E}|X_n - X|^p = 0.$$

Next, we will explore the relations between these different concepts of convergence.

3.2 Almost sure convergence and convergence in probability

Proposition 3.1 *If $X_n \rightarrow X$ a.s., then $X_n \rightarrow X$ in pr.*

Proof: If $X_n \rightarrow X$ a.s., then for every $\varepsilon > 0$, we have

$$\mathbb{P}\left\{\lim_{n \rightarrow \infty} |X_n - X| > \varepsilon\right\} = 0.$$

On the other hand, since

$$\{\omega : \limsup_{n \rightarrow \infty} |X_n(\omega) - X(\omega)| > \varepsilon\} = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} \{\omega : |X_m(\omega) - X(\omega)| > \varepsilon\},$$

we have

$$\begin{aligned} \mathbb{P}\{\limsup_{n \rightarrow \infty} |X_n - X| > \varepsilon\} &= \mathbb{P}\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} \{\omega : |X_m(\omega) - X(\omega)| > \varepsilon\}\right) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{m=n}^{\infty} \{\omega : |X_m(\omega) - X(\omega)| > \varepsilon\}\right) \\ &\geq \limsup_{n \rightarrow \infty} \mathbb{P}(|X_n(\omega) - X(\omega)| > \varepsilon). \end{aligned}$$

Hence, $X_n \rightarrow X$ in pr. □

Convergence in pr. does NOT imply a.s. convergence. For example, let

$$(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), \text{Leb}), \quad X_{n,k}(\omega) = \mathbb{1}_{\left[\frac{k}{n}, \frac{k+1}{n}\right)}(\omega), \quad 0 \leq k \leq n-1. \quad (3.2)$$

Then $X_{n,k} \rightarrow 0$ in pr. but not a.s.

However, the other direction holds on a subsequence.

Proposition 3.2 *If $X_n \rightarrow X$ in pr., then there exists a subsequence $\{X_{n_k}\}$ such that $X_{n_k} \rightarrow X$ a.s.*

To prove this result we need some preparation. Let $A_1, A_2, \dots \in \mathcal{F}$ be a sequence of events. We define the event where A_n happens *infinitely often* by

$$\{A_n, \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m = \limsup_{n \rightarrow \infty} A_n. \quad (3.3)$$

Lemma 3.3 (First Borel–Cantelli Lemma) *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\{A_n, \text{ i.o.}\}) = 0$.*

Proof: By (3.3), we have

$$\mathbb{P}(\{A_n, \text{ i.o.}\}) \leq \mathbb{P}\left(\bigcup_{m=n}^{\infty} A_m\right) \leq \sum_{m=n}^{\infty} \mathbb{P}(A_m)$$

. Since $\sum_{n=1}^{\infty} P(A_n) < \infty$, we have

$$\lim_{n \rightarrow \infty} \sum_{m=n}^{\infty} P(A_n) = 0$$

and the conclusion follows. $\square$

We also have Cauchy's criterion for convergence in pr.

Proposition 3.4 *There exists a r.v. X such that $X_n \rightarrow X$ in pr. if and only if for every $\varepsilon > 0$,*

$$\lim_{N \rightarrow \infty} \sup_{n, m \geq N} P\{|X_n - X_m| > \varepsilon\} = 0$$

The “only if” part follows immediately from (3.1); we will use this in the proof of Proposition 3.2. The “if” part in Proposition 3.4 will use Proposition 3.2 and is left as an exercise.

Proof of Proposition 3.2: Since $X_n \rightarrow X$ in pr., by Proposition 3.4 with $\varepsilon = 2^{-k}$, there exist $N_k \uparrow \infty$ such that

$$P\{|X_{N_k} - X_{N_{k+1}}| \geq \frac{1}{2^k}\} \leq \frac{1}{2^k}, \quad k \geq 1.$$

Since $\sum_{k=1}^{\infty} 2^{-k} < \infty$, by Borel–Cantelli (Lemma 3.3), we have

$$P(\{|X_{N_k} - X_{N_{k+1}}| > \frac{1}{2^k}, \text{ i.o.}\}) = 0,$$

that is, for almost every ω , there exists $k_0 = k_0(\omega)$ such that

$$|X_{N_k}(\omega) - X_{N_{k+1}}(\omega)| \leq \frac{1}{2^k}, \quad \forall k \geq k_0(\omega).$$

For such ω , the infinite series

$$X_*(\omega) = X_{N_1}(\omega) + \sum_{k=1}^{\infty} (X_{N_{k+1}}(\omega) - X_{N_k}(\omega))$$

converges absolutely. Hence, $X_{N_k}(\omega) \rightarrow X_*(\omega)$ a.s. as $k \rightarrow \infty$.

It remains to show that $X_* = X$ almost surely. Indeed, since $X_{N_k} \rightarrow X_*$ almost surely, we have $X_{N_k} \rightarrow X_*$ in pr. The claim then follows from Proposition 3.5 below, which asserts that the limit in pr. is unique up to a set of measure zero. $\square$

Proposition 3.5 *If $X_n \rightarrow X$ in pr. and $X_n \rightarrow Y$ in pr., then $X = Y$ almost surely.*

Proof: Since $|X - Y| \leq |X_n - X| + |X_n - Y|$, for every $\varepsilon > 0$,

$$P(|X - Y| \geq 2\varepsilon) \leq P(|X_n - X| \geq \varepsilon) + P(|X_n - Y| \geq \varepsilon).$$

Taking $n \rightarrow \infty$, since $X_n \rightarrow X, Y$ in pr., the left-hand side must be 0. Therefore,

$$P(|X - Y| \neq 0) = \lim_{n \rightarrow \infty} P(|X - Y| \geq 1/n) = 0,$$

and this completes the proof. $\square$

As a corollary of Proposition 3.2, we have the following.

Proposition 3.6 *Almost sure convergence is not expressible via a metric.*

Proof: Assume the contrary that there exists a distance $d(\cdot, \cdot)$ such that $X_n \rightarrow X$ a.s. if and only if $d(X_n, X) \rightarrow 0$. Let $X_n \rightarrow X$ in pr. but not a.s. (such example exists by (3.2)). Then, there exists $\varepsilon_0 > 0$ and a sequence (n') such that

$$d(X_{n'}, X) \geq \varepsilon_0 > 0. \quad (3.4)$$

But $X_{n'}$ still converges to X in pr. since it is a subsequence. By Proposition 3.2, there is a further subsequence $(n'') \subset (n')$ such that $X_{n''} \rightarrow X$ a.s. This implies that $d(X_{n''}, X) \rightarrow 0$, which contradicts with (3.4). $\square$

Note that convergence in pr. is expressible via a metric. For example, $X_n \rightarrow 0$ in pr. if and only if $\mathbb{E} \frac{|X_n|}{1+|X_n|} \rightarrow 0$. Therefore, a possible metric for convergence in pr. is

$$d(X, Y) = \mathbb{E} \left[\frac{|X - Y|}{1 + |X - Y|} \right]. \quad (3.5)$$

Of course, one need to verify that (3.5) satisfies the triangle inequality and indeed defines a metric on the space of r.v.s.

We can also relax the condition of a.s. convergence in DCT to convergence in pr.

Proposition 3.7 *If $X_n \rightarrow X$ in pr. and $|X_n| \leq Y$ for some Y with $\mathbb{E}Y < \infty$, then (1.10) holds.*

Proof: For every subsequence $(X_{n_k}) \subset (X_n)$, by Proposition 3.2, there exists a further subsequence $(X_{n_{k_m}}) \subset (X_{n_k})$ such that $X_{n_{k_m}} \rightarrow X$ almost surely, and hence $\mathbb{E}X_{n_{k_m}} \rightarrow \mathbb{E}X$ by DCT. This implies $\mathbb{E}X$ is the unique possible limit point for the sequence $(\mathbb{E}X_n)_{n \geq 1}$, and hence (1.10) holds. $\square$

3.3 Convergence in L^p and uniform integrability

Proposition 3.8 *If $X_n \rightarrow X$ in L^p , then $X_n \rightarrow X$ in pr.*

This proposition follows immediately from the result below.

Lemma 3.9 (Chebyshev's inequality) *For every $\varepsilon > 0$,*

$$\mathbb{P}(|X| \geq \varepsilon) \leq \frac{\mathbb{E}|X|}{\varepsilon}$$

Proof: Since

$$|X| = |X| \mathbb{1}_{\{|X| \geq \varepsilon\}} + |X| \mathbb{1}_{\{|X| < \varepsilon\}} \geq |X| \mathbb{1}_{\{|X| \geq \varepsilon\}} \geq \varepsilon \mathbb{1}_{\{|X| \geq \varepsilon\}},$$

taking expectation on both sides, we have $\mathbb{E}|X| \geq \varepsilon \mathbb{P}\{|X| \geq \varepsilon\}$, and the conclusion follows. $\square$

Proof of Proposition 3.8: Let $X_n \rightarrow X$ in L^p . For every $\varepsilon > 0$, by Lemma 3.9, we have

$$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(|X_n - X|^p \geq \varepsilon^p) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} \rightarrow 0.$$

Therefore, $X_n \rightarrow X$ in pr. $\square$

Limits in L^p are also unique.

Proposition 3.10 *If $X_n \rightarrow X$ in L^p and $X_n \rightarrow Y$ in L^p , then $X = Y$ a.s.*

Proof: By Proposition 3.8, $X_n \rightarrow X, Y$ in pr., and hence by Proposition 3.5, $X = Y$ a.s. $\square$

Other than Proposition 3.1 and Proposition 3.8, there are not more implications between the three modes of convergence. We have seen one counterexample (3.2), and we can obtain other counterexamples by modifying (3.2).

1. $X_n \rightarrow X$ in pr. does not implies $X_n \rightarrow X$ in L^p . For example, let

$$X_{n,k}(\omega) = n^c \mathbb{1}_{[\frac{k}{n}, \frac{k+1}{n}]}(\omega),$$

where $c \geq 1/p$. We have $\mathbb{E}|X_{n,k}|^p \geq 1$ but $X_{n,k} \rightarrow 0$ in pr.

2. $X_n \rightarrow X$ a.s. does not implies $X_n \rightarrow X$ in L^p . For example, let

$$X_n(\omega) = n^c \mathbb{1}_{[0, \frac{1}{n}]}(\omega),$$

where $c \geq 1/p$. We have $X_n(\omega) \rightarrow 0$ but $\mathbb{E}|X_n|^p \geq 1$.

3. $X_n \rightarrow X$ in L^p does not implies $X_n \rightarrow X$ a.s. For example, let

$$X_{n,k}(\omega) = n^c \mathbb{1}_{[\frac{k}{n}, \frac{k+1}{n}]}(\omega),$$

where $c < 1/p$. We have $\mathbb{E}|X_{n,k}|^p \rightarrow 0$ but $X_n \not\rightarrow 0$ a.s.

Convergence in L^p and a.s. convergence are equivalent under some integrability condition. Without loss of generality we can restrict our discussion to $p = 1$.

Definition 3.1 (Uniform integrability) A collection of r.v.s $(X_\alpha)_{\alpha \in I}$ is uniformly integrable (u.i.), if

$$\lim_{n \rightarrow \infty} \sup_{\alpha \in I} \mathbb{E}|X_\alpha| \mathbb{1}_{\{|X_\alpha| \geq M\}} = 0. \quad (3.6)$$

Note that if X_α are u.i., then $\mathbb{E}|X_\alpha|$ are uniformly bounded, since

$$\sup_{\alpha} \mathbb{E}|X_\alpha| \leq M + \sup_{\alpha \in I} \mathbb{E}|X_\alpha| \mathbb{1}_{\{|X_\alpha| \geq M\}} < \infty.$$

Uniform integrability can be seen as a necessary and sufficient condition for (1.10) to hold. Therefore, it will be the last resort if conditions for Theorems 1.27 to 1.29 are not met.

Theorem 3.11 If $\mathbb{E}|X_n| < \infty$, $\mathbb{E}|X| < \infty$ and $X_n \rightarrow X$ in pr., then the following are equivalent:

1. $\{X_n\}_{n \geq 1}$ are u.i.;
2. $X_n \rightarrow X$ in L^1 ;
3. $\mathbb{E}|X_n| \rightarrow \mathbb{E}|X|$.

Proof: From 1 to 2. Let

$$\varphi_M(x) = (-M) \vee X \wedge M = \begin{cases} -M, & x \leq -M, \\ x, & x \in [-M, M], \\ M, & x \geq M. \end{cases}$$

(Here, “ $\vee$ ” and “ $\wedge$ ” are associative.) By definition we have $|X - \varphi_M(X)| \leq |X| \mathbb{1}_{\{|X| \geq M\}}$, and thus

$$\mathbb{E}|X_n - X| \leq \mathbb{E}|\varphi_M(X_n) - \varphi_M(X)| + \mathbb{E}|\varphi_M(X_n) - X_n| + \mathbb{E}|\varphi_M(X) - X|$$

Taking $n \rightarrow \infty$ and then $M \rightarrow \infty$, the first term goes to 0 by DCT, the second goes to zero since X_n are u.i., and the third goes to zero since $\mathbb{E}|X| < \infty$ which follows from Fatou's lemma and (3.6):

$$\mathbb{E}|X| \leq \liminf_{n \rightarrow \infty} \mathbb{E}|X_n| \leq \sup_n \mathbb{E}|X_n| < \infty.$$

From 2 to 3. It follows from $|\mathbb{E}X_n - X| \leq \mathbb{E}|X_n - X|$.

From 3 to 1. Let

$$\psi_M(x) = \begin{cases} x, & x \in [0, M-1], \\ 0, & x \geq M. \end{cases}$$

Let $\varepsilon > 0$. We have

$$\begin{aligned} \mathbb{E}|X_n| \mathbb{1}_{\{|X_n| \geq M\}} &\leq \mathbb{E}|X_n| - \mathbb{E}\psi_M(|X_n|) \\ &\leq (\mathbb{E}|X| + \varepsilon) - (\mathbb{E}\psi_M(|X|) - \varepsilon), \quad n \geq n_0, \end{aligned}$$

where such n_0 exists since $\mathbb{E}|X_n| \rightarrow \mathbb{E}X$ by the assumption and $\mathbb{E}\psi_M(|X_n|) \rightarrow \mathbb{E}\psi_M(|X|)$ by BCT. Since $\psi_M(t) \rightarrow t$ for every t and $\psi_M(|X|) \leq \mathbb{E}|X|$, by DCT there exists $M_0 > 0$ such that

$$\mathbb{E}|X| - \mathbb{E}\psi_M(|X|) \leq \varepsilon, \quad M \geq M_0,$$

Combining these we obtain that for every $\varepsilon > 0$, there exist n_0 and M_0 s.t.

$$\sup_{n \geq n_0} \mathbb{E}|X_n| \mathbb{1}_{\{|X_n| \geq M\}} \leq 3\varepsilon, \quad M \geq M_0.$$

It follows that $(X_n)_{n \geq 1}$ are u.i. □

3.4 Weak convergence

The limit of weak convergence is unique in the sense of distribution of the r.v.s.

Proposition 3.12 *If $\mathbb{E}f(X) = \mathbb{E}f(Y)$ for every bounded continuous function f , then $\mu_X = \mu_Y$ as probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.*

Proof: For every open interval (a, b) , there exist non-negative bounded continuous function f_n such that $f_n(x) \uparrow \mathbb{1}_{(a,b)}(x)$. Taking $n \rightarrow \infty$ in $\mathbb{E}f_n(X) = \mathbb{E}f_n(Y)$, by MCT, we have $\mathbb{E}\mathbb{1}_{(a,b)}(X) = \mathbb{E}\mathbb{1}_{(a,b)}(Y)$. Therefore, $\mu_X(I) = \mu_Y(I)$ for every open interval I . Since open intervals generate $\mathcal{B}(\mathbb{R})$, it follows that $\mu_X = \mu_Y$. □

As Proposition 3.12 suggests, the bounded continuous functions appearing in the definition of the weak convergence merely serve as test functions. In fact, we can characterize the weak convergence $X_n \Rightarrow_d X$ using solely the information of μ_{X_n} and μ_X , and that is why we also call it *convergence in distribution*. Second, when we consider the weak convergence of X_n to X , the r.v.s can live on totally different probability spaces, since what is under concern is their distributions μ_{X_n} and μ_X which are probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Third, it is not true that $\mu_{X_n}(A) \rightarrow \mu_X(A)$ for every $A \in \mathcal{B}(\mathbb{R})$ if $X_n \Rightarrow_d X$, even when A is an open interval. This is the reason why the convergence is *weak*. Functional analysis calls such convergence *weak-* convergence*, which we will sketch below.

Let $\mathcal{X}$ be the Banach space of all bounded continuous functions, and $\mathcal{X}^*$ be its *dual space*, consisting of all bounded linear functional from $\mathcal{X}$ to $\mathbb{R}$. By Riesz's representation theorem, $\mathcal{X}^*$ coincides with the space of all bounded *signed measures* on $\mathcal{B}(\mathbb{R})$, which contains all the probability measures. For a generic Banach space $\mathcal{X}$ and its dual $\mathcal{X}^*$, we say that $u_n \rightarrow u$ weakly in $\mathcal{X}$, if

$$\ell(u_n) \rightarrow \ell(u), \quad \forall \ell \in \mathcal{X}^*,$$

and we say that $\ell_n \rightarrow \ell$ weakly-* in $\mathcal{X}^*$, if

$$\ell_n(u) \rightarrow \ell(u), \quad \forall u \in \mathcal{X}.$$

Weak and weak-* convergence are equivalent if the space $\mathcal{X}$ is *reflective*, that is, $(\mathcal{X}^*)^* = \mathcal{X}$. While reflectivity holds for common L^p spaces, $1 \leq p < \infty$, it is not the case for $\mathcal{X}^*$ being the space of bounded continuous functions. So strictly speaking, $X_n \Rightarrow_d X$ means $\mu_{X_n} \rightarrow \mu_X$ weakly-*. It is in probability context that we drop the “*” and call it weak convergence. For weak convergence of probability measures, an good reference is [Bil99].

4 Law of large numbers

The goal of this section is to establish the following strong law of large numbers (SLLN).

Theorem 4.1 (Strong law of large number) *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}|X_i| < \infty$. Let $\mathbb{E}X_i = \mu$ and $S_n = X_1 + \dots + X_n$. Then $S_n/n \rightarrow \mu$ a.s. as $n \rightarrow \infty$.*

The above theorem is called “strong” because almost sure convergence is the best that one can hope. Similar statements where the convergence holds in a weaker sense, like in L^p or in probability are called “weak” law of large numbers.

In **Theorem 4.1**, the first moment condition $\mathbb{E}|X_i| < \infty$ will be optimal. But we will also introduce proofs under weaker assumptions, as an opportunity to introduce useful probabilistic techniques that may be useful for other problems.

4.1 L^2 -weak law of large numbers

Let X_n , $n \geq 1$, be i.i.d. r.v.s. For the discussion of law of large numbers, we assume, without loss of generality, that all X_n are *centered*, namely, $\mu := \mathbb{E}X_n = 0$. Otherwise, we can always center the r.v.s by setting $\tilde{X}_i = X_i - \mu$ and consider the centered case. For centered r.v.s, we have

$$\mathbb{E}X_i^2 = \text{Var}(X_i), \quad \mathbb{E}X_iX_j = \text{Cov}(X_i, X_j), \quad i \neq j.$$

We write $S_n = X_1 + \dots + X_n$.

The r.v.s $(X_i)_{i \in I}$ with $\mathbb{E}X_1^2 < \infty$ is *uncorrelated* if

$$\mathbb{E}(X_iX_j) = \mathbb{E}X_i\mathbb{E}X_j \quad \text{whenever } i \neq j. \quad (4.1)$$

We note that the second moment condition $\mathbb{E}X_i^2 < \infty$ ensures that expectations in (4.1) are all defined. When $\mu = 0$, (4.1) becomes

$$\mathbb{E}(X_iX_j) = 0, \quad \forall i \neq j. \quad (4.2)$$

Let a family of random variables $(X_n)_{n \geq 1}$ with $\mathbb{E}X_1^2 < \infty$ be uncorrelated. By linearity of expectation, we have

$$\mathbb{E}S_n = \mathbb{E}X_1 + \dots + \mathbb{E}X_n = n\mu = 0.$$

Using definition of the variance, we have

$$\text{Var}(S_n) = \mathbb{E}S_n^2 = \mathbb{E}\left(\sum_{i=1}^n X_i\right)\left(\sum_{j=1}^n X_j\right) = \sum_{i=1}^n \mathbb{E}X_i^2 = n\mathbb{E}X_1^2. \quad (4.3)$$

A key observation is that the variance grows linearly in n , although it is the expectation of the sum of n^2 terms. Assuming $\mathbb{E}X_1^4 < \infty$, we can further estimate the fourth moment of S_n :

$$\begin{aligned}\mathbb{E}S_n^4 &= \sum_{i_1, i_2, i_3, i_4} \mathbb{E}X_{i_1}X_{i_2}X_{i_3}X_{i_4} = \sum_{i=1}^n \mathbb{E}X_i^4 + 6 \sum_{i < j} \mathbb{E}X_i^2 X_j^2 \\ &\leq n\mathbb{E}X_1^4 + 3 \sum_{i < j} \mathbb{E}(X_i^4 + X_j^4) = (3n^2 - 2n)\mathbb{E}X_1^4 \leq Cn^2.\end{aligned}\tag{4.4}$$

Here, in the first line, if an index appears in i_1, i_2, i_3, i_4 once, then by (4.2), the expectation $\mathbb{E}X_{i_1}X_{i_2}X_{i_3}X_{i_4}$ will be zero and we can drop such terms from the sum; in the second line we use the elementary inequality $2ab \leq a^2 + b^2$. Again, we see that the growth rate of $\mathbb{E}S_n^4$ is $O(n^2)$ which is much less than the number of terms, n^4 . The discrepancy will get larger if we estimate higher moments of S_n . But the fourth moment is sufficient for us to use Borel–Cantelli to get the first strong law of large numbers.

Proposition 4.2 *Let $X_1, X_2, \dots$ be i.i.d. $\mathbb{E}X_1^4 < \infty$. Then $S_n/n \rightarrow 0$ a.s.*

Proof: Since $\mathbb{E}X_1^4 < \infty$, by (4.4) and Chebyshev’s inequality (Lemma 3.9), for some constant $C > 0$ we have

$$\mathbb{P}(|S_n| > n\varepsilon) = \mathbb{P}(|S_n|^4 > n^4\varepsilon^4) \leq \frac{Cn^2}{n^4\varepsilon^4} \leq \frac{C}{n^2\varepsilon^4}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$, by Borel–Cantelli lemma (Lemma 3.3), we have

$$\mathbb{P}(\{|S_n| > n\varepsilon \text{ i.o.}\}) = 0.$$

It follows from the ε - δ language formulation of limit

$$\left\{ \lim_{n \rightarrow \infty} \frac{S_n}{n} \neq 0 \right\} = \bigcup_{m=1}^{\infty} \left\{ \left| \frac{S_n}{n} \right| > \frac{1}{m} \text{ i.o.} \right\}.$$

Hence, by sub-additivity,

$$\mathbb{P}\left(\left\{ \lim_{n \rightarrow \infty} \frac{S_n}{n} \neq 0 \right\}\right) \leq \sum_{m=1}^{\infty} \mathbb{P}\left(\left\{ \left| \frac{S_n}{n} \right| > \frac{1}{m} \text{ i.o.} \right\}\right) = 0,$$

and this completes the proof. $\square$

Proposition 4.2 is enough for most practical examples, since most r.v.s in real life are bounded and thus have fourth moment. In fact, in (4.4) and **Proposition 4.2**, we did not rely on X_n having identical distribution; all we need is the independence of X_n and a uniform bound on their fourth moment. Similarly, assuming a uniform second moment condition, we can obtain the following *weak law of large numbers* without independence.

Theorem 4.3 (Weak law of large numbers) *Let $X_1, X_2, \dots$ be uncorrelated with $\mathbb{E}X_i^2 \leq C$ for some $C > 0$. Then as $n \rightarrow \infty$, $S_n/n \rightarrow 0$ in L^2 and in pr.*

Proof: Since X_i are uncorrelated, using (4.3) we have $\mathbb{E}S_n^2 \leq Cn$, and hence $\mathbb{E}S_n^2/n^2 \leq C/n$. It follows that $S_n/n \rightarrow 0$ in L^2 . By **Proposition 3.8**, this implies convergence in pr. $\square$

Using the second moment condition, it is also possible to obtain almost sure convergence.

Theorem 4.4 (SLLN with $\mathbb{E}X_1^2 < \infty$) *Let $X_n, n \geq 1$, be i.i.d. with $\mathbb{E}X_1^2 < \infty$. Then $\frac{S_n}{n} \rightarrow 0$, a.s.*

Proof: Let $M = \mathbb{E}X_1^2$. By (4.3) and Chebyshev's inequality, we have

$$\mathbb{P}(|S_{n^2}| > n^2\varepsilon) = \mathbb{P}(|S_{n^2}|^2 > n^4\varepsilon^2) \leq \frac{n^2M}{n^4\varepsilon^2} \leq \frac{M}{n^2\varepsilon^2},$$

which is summable. Hence, by Borel–Cantelli lemma, $\frac{S_{n^2}}{n^2} \rightarrow 0$ a.s. Let

$$D_n(\omega) = \max_{n^2 \leq k < (n+1)^2} |S_k - S_{n^2}| = \max_{1 \leq k \leq 2n} |X_{n^2+1} + \cdots + X_{n^2+k}|.$$

For every ω , we have

$$|D_n(\omega)|^2 \leq (|X_{n^2+1}| + \cdots + |X_{n^2+2n}|)^2 \leq 2n(X_{n^2+1}^2 + \cdots + X_{n^2+2n}^2)$$

and hence $\mathbb{E}D_n^2 \leq 4n^2M$. Then, by Chebyshev's inequality, we have for every $\varepsilon > 0$

$$\mathbb{P}(D_n \geq \varepsilon n^2) \leq \frac{\mathbb{E}D_n^2}{\varepsilon^2 n^4} \leq \frac{4M}{\varepsilon^2 n^2}.$$

It follows from Borel-Cantelli lemma $\mathbb{P}(\lim_{n \rightarrow \infty} D_n/n^2 = 0) = 1$.

To summarize, for almost every ω , we have

1. $\lim_{n \rightarrow \infty} \frac{S_{n^2}}{n^2} = 0$.
2. $\lim_{n \rightarrow \infty} \frac{D_n}{n^2} = 0$.

Since

$$\frac{S_{n^2} - D_n}{(n+1)^2} \leq \frac{S_k}{k} \leq \frac{S_{n^2} + D_n}{n^2}, \quad n^2 \leq k < (n+1)^2,$$

by the Squeeze Theorem, we have $S_k/k \rightarrow 0$. This completes the proof. $\square$

Remark 4.1 1. We only need $\mathbb{E}X_i X_j = 0$, $i \neq j$ (uncorrelated) and $\sup_n \mathbb{E}X_n^2 < \infty$.

2. The above condition can be further weakened to allow some finite-range correlation:

$$|\mathbb{E}X_i X_j| \leq M \cdot \mathbb{1}_{\{|i-j| \leq L\}}$$

for some $L > 0$ and $M > 0$.

Example 4.2 (Normal number) Every $\omega \in [0, 1)$ admits a decimal expansion

$$\omega = 0.x_1x_2x_3x_4 \cdots, \quad x_i = x_i(\omega) \in \{0, 1, \dots, 9\}.$$

Let

$$\nu_k^{(n)}(\omega) = |\{1 \leq i \leq n : x_i = k\}| = \sum_{i=1}^n \mathbb{1}_{\{x_i(\omega)=k\}}$$

be the number of occurrence of number k in the first n digits. It is clear that $x_i(\omega)$ are i.i.d., uniformly on $\{0, 1, \dots, 9\}$. Then $\xi_i = \mathbb{1}_{\{x_i(\omega)=k\}}$ are i.i.d. $\text{Ber}(1/10)$, and in particular $|\xi_i| \leq 1$. For every k , by SLLN, for almost every $\omega \in [0, 1)$,

$$\frac{\nu_k^{(n)}(\omega)}{n} = \frac{\sum_{i=1}^n \xi_i}{n} \rightarrow \mathbb{E}\xi_i = \frac{1}{10}, \quad k \in \{0, \dots, 9\}. \quad (4.5)$$

A number ω is called a *normal number* (Borel, 1909) if for its fractional part, the limit (4.5) holds. As a consequence of the SLLN, almost every number in $[0, 1)$ is normal. However, we do not know whether common transcendental numbers like π or e are normal.

We can also strengthen the definition slightly. A number $\omega \in [0, 1)$ is *completely normal*, if for every pattern $\vec{k} = (k_1, k_2, \dots, k_r) \in \{0, \dots, 9\}^r$,

$$\frac{\nu_{\vec{k}}^{(n)}(\omega)}{n} := \frac{|\{1 \leq i \leq n : (x_i, \dots, x_{i+r-1}) = \vec{k}\}|}{n} \rightarrow \frac{1}{10^r}, \quad n \rightarrow \infty.$$

Using the remark after **Theorem 4.4** with $L = r$ and $M = 1$, almost every $\omega \in [0, 1)$ is also completely normal.

As an illustration, if a monkey is typing randomly before a typewriter, then after waiting long enough it will produce all Shakespeare's works (more than once), since any pattern $\vec{k}$, even as long as all Shakespeare's works, has a small but positive probability of occurrence. This seems paradoxical, but note that the waiting time will be much longer than the age of the universe in this case, so it will not happen in real life.

Example 4.3 (Empirical distribution function) Let $X_1, X_2, \dots$ be i.i.d. samples with c.d.f. F and let

$$F_n(x) = n^{-1} \sum_{m=1}^n \mathbb{1}_{\{X_m \leq x\}}, \quad \forall x \in \mathbb{R}$$

be the *empirical distribution function* from n samples. For every x , the indicators $\xi_n(\omega) := \mathbb{1}_{X_n(\omega) \leq x}$ are i.i.d. r.v.s since they are Borel functions of X_n . By SLLN, we have

$$F_n(x) = \frac{\sum_{m=1}^n \xi_m}{n} \xrightarrow{\text{a.s.}} \mathbb{E} \xi_n = \mathbb{P}(\{X_n \leq x\}) = F(x).$$

Theorem 4.5 (Glivenko–Cantelli theorem) As $n \rightarrow \infty$, $\sup_x |F_n(x) - F(x)| \rightarrow 0$ a.s.

(To fill in the proof.)

Example 4.4 (Waiting time Paradox) This example is related to the renewal theory.

Let $X_1, X_2, \dots$ are i.i.d. Suppose that the n -th bus from the bus terminal at time S_n , where $S_n = X_1 + \dots + X_n$. For simplicity assume that $\mathbb{P}\{X_n = a\} = \mathbb{P}\{X_n = b\} = \frac{1}{2}$ for some $a < b$. We are trying to compute the “average waiting time” for a person randomly arriving at the terminal before departure.

We first compute how many buses departing in the time interval $[0, T]$. Let

$$N = N_T(\omega) = \text{the number of buses departing in } [0, T] = \max\{n : S_n(\omega) \leq T\}.$$

Since

$$\frac{X_1 + \dots + X_{N_T}}{N_T} < \frac{T}{N_T} < \frac{X_1 + \dots + X_{N_T+1}}{N_T + 1} \leq \frac{N_T + 1}{N_T}$$

it follows from the Squeeze Theorem and SLLN that

$$\frac{T}{N_T} \rightarrow \mathbb{E} X_1 = \frac{a+b}{2}, \quad \text{a.s.},$$

and hence

$$\frac{N_T}{T} \rightarrow \frac{1}{\mathbb{E} X_1} = \frac{2}{a+b}, \quad \text{a.s.} \quad (4.6)$$

We interpret the “average waiting time” as follows. Let a person arrive at the bus stop at time $\xi \sim U[0, 1]$, where ξ is independent of $(X_n)_{n \geq 1}$ (we can realize this by accommodate ξ and $(X_n)_{n \geq 1}$ on a bigger product probability space). The average waiting time Q is given by

$$Q = \frac{1}{T} \int_0^T (S_{n_\xi} - \xi) d\xi,$$

where $n_\xi = \min\{m : S_m > \xi\}$ is the departure time of the next bus after time ξ . Noting that $n_\xi = n$ if $\xi \in [S_{n-1}, S_n)$, we have

$$\begin{aligned} Q &= \frac{1}{T} \left[\sum_{n=1}^{N_T} \int_{S_{n-1}}^{S_n} (S_n - \xi) d\xi + \int_{S_{N_T}}^T (S_{N_T+1} - \xi) d\xi \right] \\ &= \frac{1}{T} \left[\sum_{n=1}^{N_T} \frac{(S_n - S_{n-1})^2}{2} + \int_{S_{N_T}}^T (S_{N_T+1} - \xi) d\xi \right] \\ &= \frac{1}{T} \sum_{n=1}^{N_T} \frac{X_n^2}{2} + \frac{1}{T} \int_{S_{N_T}}^T (S_{N_T+1} - \xi) d\xi. \end{aligned}$$

It follows from the SLLN for X_i^2 and (4.6), for the first term we have

$$\frac{1}{T} \sum_{n=1}^{N_T} \frac{X_n^2}{2} = \frac{X_1^2 + \dots + X_{N_T}^2}{X_T} \cdot \frac{N_T}{2T} \xrightarrow{\text{a.s.}} \mathbb{E} X_I^2 \cdot \frac{1}{a+b} = \frac{a^2 + b^2}{2(a+b)} = \frac{1}{2} \left(a \cdot \frac{a}{a+b} + b \cdot \frac{b}{a+b} \right), \quad (4.7)$$

while for the second term we have

$$\left| \frac{1}{T} \int_{S_{N_T}}^T (S_{N_T+1} - \xi) d\xi \right| \leq \frac{\max(a, b)^2}{T} \rightarrow 0, \quad T \rightarrow \infty.$$

Hence, we have

$$Q \rightarrow \frac{1}{2} \left(a \cdot \frac{a}{a+b} + b \cdot \frac{b}{a+b} \right), \quad (4.8)$$

How to understand (4.8)? If the time for the next departure is τ , then for a person arriving at a random time the average waiting time should be $\tau/2$. One would think naively that since τ takes the value a and b with probability $1/2$, then the average waiting time should be $(a+b)/2$. But this is WRONG. Indeed, the number of intervals with length a and b are around 50%, but since their lengths are different, the random arrival time hitting these two types of intervals are also different, or more precisely, proportional to their lengths. Therefore, the probability of the arrival time hitting $[S_{n-1}, S_n)$ with $X_{n-1} = a$ is asymptotically $\frac{a}{a+b}$, and $\frac{b}{a+b}$ otherwise. This explains the rightmost decomposition in (4.7).

4.2 Weak law for triangular arrays

A lot of classical limit theorems in probability concern arrays $X_{n,k}, 1 \leq k \leq n$, of random variables and investigate the limiting behavior of their row sums $S_n = X_{n,1} + \dots + X_{n,n}$.

Proposition 4.6 *Let $(X_{n,k})_{k=1}^n$ be independent and $\mu_n = \mathbb{E} S_n$, $\sigma_n^2 = \text{Var}(S_n)$. If $\sigma_n^2/b_n^2 \rightarrow 0$, then*

$$\frac{S_n - \mu_n}{b_n} \rightarrow 0, \quad \text{in probability.}$$

Proof: Chebyshev's inequality gives that for every $\varepsilon > 0$,

$$\mathbb{P} \left(\left| \frac{S_n - \mu_n}{b_n} \right| \geq \varepsilon \right) \leq \frac{\text{Var}(S_n)}{\varepsilon^2 b_n^2} = \frac{\sigma_n^2}{\varepsilon^2 b_n^2} \rightarrow 0.$$

□

Example 4.5 (Coupon collector) Let $\xi_1, \xi_2, \dots$ be i.i.d. uniform on $\{1, 2, \dots, n\}$. The numbers $1, \dots, n$ are thought of as “coupons” while ξ_m is the m -th coupon that one collects. Let

$$\tau_k^n = \min\{m : m \geq 0, |\{\xi_1, \dots, \xi_m\}| \geq k\}$$

be the first time that one collects k different coupons. For example, we always have $\tau_1^n = 1$. We Set $\tau_0^n = 0$ for consistency of notation.

For $1 \leq k \leq n$, let $X_{n,k} = \tau_k^n - \tau_{k-1}^n$ represent the time spent to collect the k -th coupon. We claim the following two facts without proof:

1. $X_{n,k}$ is independent of $X_{n,1}, \dots, X_{n,k-1}$;
2. $X_{n,k}$ has a geometric distribution with parameter $1 - (k-1)/n$.

Let $S_n = X_{n,1} + X_{n,2} + \dots + X_{n,n} = \tau_n^n$. We want to understand the asymptotic behavior of S_n , the time spent to collect all coupons.

To use the result from [Proposition 4.6](#), we need to compute $\mathbb{E}S_n$ and $\text{Var}(S_n)$. Note that if $Y \sim \text{Geo}(p)$, then $\mathbb{E}Y = 1/p$ and $\mathbb{E}Y^2 \leq 1/p^2$. We have

$$\mathbb{E}S_n = \sum_{k=1}^n X_{n,k} = \sum_{k=1}^n \left(1 - \frac{k-1}{n}\right)^{-1} = n \sum_{m=1}^n m^{-1} \sim n \log n, \quad (4.9)$$

and

$$\text{Var}(S_n) = \sum_{k=1}^n \text{Var}(X_{n,k}) \leq n^2 \sum_{m=1}^n \frac{1}{m^2} \leq Cn^2.$$

Hence, for $b_n = n \log n$, $\sigma_n/b_n \rightarrow 0$, and it follows from [Proposition 4.6](#)

$$\frac{S_n - \mathbb{E}S_n}{b_n} \rightarrow 0 \text{ in probability.}$$

Noting [\(4.9\)](#), we have $\frac{S_n}{b_n} \rightarrow 1$ in probability.

Let $\mathbb{E}|X| < \infty$ and $(X_{n,k})_{k=1}^n$, $1 \leq k \leq n$ be independent. Let $b_n > 0$ with $b_n \rightarrow \infty$. We introduce the “truncation” of $X_{n,k}$ as follows:

$$\bar{X}_{n,k} = X_{n,k} \mathbb{1}_{(|X_{n,k}| \leq b_n)} = \begin{cases} X_{n,k}, & \text{if } |X_{n,k}| \leq b_n \\ 0, & \text{if } |X_{n,k}| > b_n. \end{cases} \quad (4.10)$$

The truncation will help us to obtain the weak law to random variables without a finite second moment.

Theorem 4.7 (Weak LLN for triangular arrays) *Let $X_{n,k}$, $1 \leq k \leq n$, be independent. Let $b_n > 0$ with $b_n \rightarrow \infty$ and $X_{n,k}$ be defined in [\(4.10\)](#). Suppose that as $n \rightarrow \infty$,*

1. $\sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n) \rightarrow 0$, and
2. $b_n^{-2} \sum_{k=1}^n \mathbb{E}\bar{X}_{n,k}^2 \rightarrow 0$.

Then

$$(S_n - a_n)/b_n \rightarrow 0 \text{ in probability,}$$

where $S_n = X_{n,1} + \dots + X_{n,n}$ and $a_n = \sum_{k=1}^n \mathbb{E}\bar{X}_{n,k}$.

Proof: For every $\varepsilon > 0$, we have

$$\mathbb{P}\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \leq \mathbb{P}(S_n \neq \bar{S}_n) + \mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right)$$

To estimate the first term, we note that

$$\mathbb{P}(S_n \neq \bar{S}_n) \leq \mathbb{P}\left(\bigcup_{k=1}^n \{\bar{X}_{n,k} \neq X_{n,k}\}\right) \leq \sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n) \rightarrow 0$$

by the first condition. For the second term, we use Chebyshev's inequality to obtain

$$\begin{aligned} \mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right) &\leq \frac{1}{\varepsilon^2} \mathbb{E}\left|\frac{\bar{S}_n - a_n}{b_n}\right|^2 = \frac{\text{Var}(\bar{S}_n)}{\varepsilon^2 b_n^2} \\ &= \frac{\sum_{k=1}^n \text{Var}(\bar{X}_{n,k})}{\varepsilon^2 b_n^2} \leq \frac{\sum_{k=1}^n \mathbb{E}(\bar{X}_{n,k})^2}{\varepsilon^2 b_n^2} \rightarrow 0 \end{aligned}$$

by the second condition, and the proof is complete. $\square$

Theorem 4.8 Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}|X_i| < \infty$. Let $S_n = X_1 + \dots + X_n$ and let $\mu = \mathbb{E}X_1$. Then $S_n/n \rightarrow \mu$ in probability.

Proof: Let $X_{n,k} = X_k$ and $b_n = n$. We need to check the two conditions of [Theorem 4.7](#).

For the first condition, by DCT, we have

$$\sum_{k=1}^n \mathbb{P}(|X_k| > n) = n\mathbb{P}(|X_1| > n) \leq \mathbb{E}|X_1| \mathbb{1}_{\{|X_1| \geq n\}} \rightarrow 0, \quad (4.11)$$

since $\mathbb{1}_{\{|X_1| \geq n\}}|X_1| \xrightarrow{\text{a.s.}} 0$ and $\mathbb{E}|X_1| < \infty$.

For the second condition, we have

$$\frac{1}{n^2} \sum_{k=1}^n \mathbb{E}|X_k|^2 \mathbb{1}_{\{|X_k| \leq n\}} = \frac{1}{n} \mathbb{E}|X_1|^2 \mathbb{1}_{\{|X_1| \leq n\}}$$

and

$$\begin{aligned} \mathbb{E}|X_1|^2 \mathbb{1}_{\{|X_1| \leq n\}} &= \sum_{k=1}^n \mathbb{E}|X_1|^2 \mathbb{1}_{\{|X_1| \in [k-1, k]\}} \\ &\leq \sum_{k=1}^n k^2 \mathbb{P}(|X_1| \in [k-1, k]) \\ &= \mathbb{P}(|X_1| \in [0, 1]) + \sum_{k=1}^n ((k+1)^2 - k^2) \mathbb{P}(|X_1| \in [1, n]) \\ &\leq \mathbb{P}(|X_1| \in [0, 1]) + \sum_{k=1}^n 3k \mathbb{P}(|X_1| \geq k) \end{aligned}$$

By Stolz's theorem, we have $\frac{1}{n} \sum_{k=1}^n 3k \mathbb{P}(|X_1| \geq k) \rightarrow \lim_{n \rightarrow \infty} n \mathbb{P}(|X_1| \geq n) = 0$, again by [\(4.11\)](#).

Note that $a_n = n\mu_n$ where $\mu_n = \mathbb{E}X_1 \mathbb{1}_{\{|X_1| \leq n\}} \neq \mu$ due to the truncation. But by DCT,

$$\mu_n = \mathbb{E}X_1 \mathbb{1}_{\{|X_1| \leq n\}} \rightarrow \mathbb{E}X_1 = \mu.$$

□

Example 4.6 (St. Petersburg's game) Let $X_1, X_2, \dots$ be independent random variables with

$$\mathbb{P}(X_i = 2^j) = 2^{-j} \quad \text{for } j \geq 1. \quad (4.12)$$

Imagine you are playing a game continuously tossing a coin. You win 2^j dollars if it takes $j+1$ tosses to get a head, but if you can a head the first toss you leave without any reward. Now we want to determine what is the “fair” entry fee to play this game. Since $\mathbb{E}X_1 = \infty$, the LLN is useless, as it is not reasonable to ask ∞ dollars for the entry fee!

Now we will try to use [Theorem 4.7](#) to find out how much we should ask for the entry fee. The answer will depend on the total number of games to be played. Indeed, we are trying to find c_n where $S_n/nc_n \rightarrow 1$.

In the setting of [Theorem 4.7](#), let $X_{n,k} = X_k$. We need to determine $b_n = nc_n$. We observe that if m is an integer

$$\mathbb{P}(X_1 \geq 2^m) = \sum_{j=m}^{\infty} 2^{-j} = 2^{-m+1}$$

Let $m(n) = \log_2 n + K(n)$ where $K(n) \rightarrow \infty$ and is chosen so that $m(n)$ is an integer (and hence the displayed formula is valid). Letting $b_n = 2^{m(n)}$, we have

$$\mathbb{E} \bar{X}_{n,k}^2 = \sum_{j=1}^{m(n)} 2^{2j} \cdot 2^{-j} \leq 2^{m(n)} \sum_{k=0}^{\infty} 2^{-k} = 2b_n$$

The last two steps are to evaluate a_n and to apply the theorem.

$$\mathbb{E} \bar{X}_{n,k} = \sum_{j=1}^{m(n)} 2^j 2^{-j} = m(n)$$

so $a_n = nm(n)$. We have $m(n) = \log_2 n + K(n)$, so if we pick $K(n)/\log_2 n \rightarrow 0$ then $a_n/n \log_2 n \rightarrow 1$ as $n \rightarrow \infty$. Now we have

$$\frac{S_n - a_n}{n 2^{K(n)}} \rightarrow 0 \quad \text{in probability}$$

If we suppose that $K(N \leq \log_2 \log_2 n)$ for large n then the last conclusion holds with the denominator replaces by $n \log_2 n$, and it follows that $S_n/(n \log_2 n) \rightarrow 1$ in probability.

4.3 First proof of SLLN

4.3.1 Some preparation

We recall the (first) Borel-Cantelli Lemma: if $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$ then $\mathbb{P}(A_n \text{ i.o.}) = 0$. For the other direction, we have the following.

Theorem 4.9 (Second Borel–Cantelli lemma) *If the events A_n are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\{A_n, \text{ i.o.}\}) = 1$*

Proof: By definition of the i.o. sets, we have

$$\{A_n \text{ i.o.}\}^c = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m^c.$$

Using independence, it follows that

$$\mathbb{P}\left(\bigcap_{n=m}^{\infty} A_n^c\right) = \lim_{M \rightarrow \infty} \prod_{n=m}^M \mathbb{P}(A_n^c) = \lim_{M \rightarrow \infty} \prod_{n=m}^M (1 - \mathbb{P}(A_n)) = 0,$$

where the last limit is due to $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. □

The following proposition states that the $\mathbb{E}|X_1| < \infty$ is also necessary for the existence of $\lim_{n \rightarrow \infty} S_n/n$.

Proposition 4.10 *If $X_1, X_2 \dots$ are i.i.d. and $\mathbb{E}|X_i| = \infty$, then $\mathbb{P}(\lim S_n/n \text{ exists} \in (-\infty, \infty)) = 0$.*

Proof: Let $A_n = \{|X_n| \geq n\}$. We claim that on the event $\{A_n, \text{ i.o.}\}$, a *finite* limit $\lim_{n \rightarrow \infty} S_n/n$ cannot exist. Indeed, by Cauchy criterion, if such limit exists, for $\varepsilon_0 = \frac{1}{2}$, there exists $n_0 = n_0(\omega)$ such that $|X_n/n| = |\frac{S_n}{n} - \frac{S_{n-1}}{n}| < \frac{1}{2}$ for every $n > n_0$. This contradicts with $|X_n| \geq n$ for an infinite number of n 's.

By [Theorem 4.9](#), since

$$\infty = \mathbb{E}|X_1| \leq \sum_{n=0}^{\infty} \mathbb{P}(|X_1| > n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| \geq n)$$

and $X_1, X_2 \dots$ are i.i.d., it follows that $\mathbb{P}(\{A_n, \text{ i.o.}\}) = 1$. This completes the proof. □

Example 4.7 (St. Petersburg's game (continued)) Let X_n , $n \geq 1$, be i.i.d. with distribution given by (4.12). By Proposition 4.10, since $\mathbb{E}X_1 = \infty$, we know that S_n/n does not have a limit. But if we are more careful about the estimate, we have

$$\mathbb{P}(|X_n| \geq n \log_2 n) = \sum_{j \geq j_0 = \lceil \log_2(n \log_2 n) \rceil} 2^{-j} \sim 2^{-\log_2(n \log_2 n)} = \frac{1}{n \log_2 n}$$

which is not summable (one can compare this with $\int_1^\infty \frac{dx}{x \log_2 x}$). Hence, almost surely, for an infinite number of n 's, it happens that $(S_{n+1} - S_n)/n \log_2 n \geq 1$, and hence $S_n/n \log_2 n \not\rightarrow 1$.

With the finite first moment assumption, we need to truncate the r.v.s first. Let $Y_n = X_n \mathbb{1}_{\{|X_n| \leq n\}}$ and $T_n = Y_1 + \dots + Y_n$ be the partial sum of $(Y_n)_{n \geq 1}$. We still have the independence of Y_n , but they are no longer identically distributed. With the truncation we can estimate the second moment of Y_n . The following proposition show that the limits of T_n/n and S_n/n are the same,

Proposition 4.11 $T_n/n \rightarrow \mu$ a.s. if and only if $S_n/n \rightarrow \mu$ a.s.

Proof: We have

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) &= \sum_{n=1}^{\infty} \mathbb{P}(|X_n| \geq n) = \sum_{n=1}^{\infty} \int_0^{\infty} \mathbb{1}_{\{y \geq n\}} \mu_{|x|} dy \\ &= \int_0^{\infty} \sum_{n=1}^{\infty} \mathbb{1}_{\{y \geq n\}} \mu_{|x|} dy \\ &\leq \int_0^{\infty} [y] \mu_{|x|} dy = \mathbb{E}[|X_1|] \leq \mathbb{E}|X_1| + 1 < \infty. \end{aligned}$$

By Borel–Cantelli lemma, we have $\mathbb{P}(\{X_n \neq Y_n, \text{ i.o.}\}) = 0$. Therefore, almost surely, there exists $n_0 = n_0(\omega)$ such that $X_n = Y_n$ for all $n \geq n_0$, and when this happens, we have $\lim_{n \rightarrow \infty} \frac{T_n}{n} = \lim_{n \rightarrow \infty} \frac{S_n}{n}$, provided either of the two limits exists. This completes the proof. $\square$

For the proof of the SLLN we will need the following technical result.

Proposition 4.12 $\sum_{k=1}^{\infty} \frac{1}{n^2} \text{Var}(Y_k) \leq 4\mathbb{E}|X_1| < \infty$.

Proof: We start with

$$\text{Var}(Y_n) \leq \mathbb{E}|Y_n|^2 = \int_0^{\infty} 2y \mathbb{P}(|Y_n| > y) dy \leq \int_0^n 2y \mathbb{P}(|X_1| > y) dy.$$

Treating the sum as integration w.r.t. to the counting measure on $\mathbb{N}$ and using Fubini's theorem (since everything is non-negative), we have

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n^2} \mathbb{E}Y_n^2 &\leq \sum_{n=1}^{\infty} \frac{1}{n^2} \int_0^{\infty} \mathbb{1}_{\{y \leq n\}} 2y \mathbb{P}(|X_1| > y) dy \\ &= \int_0^{\infty} \left[\sum_{k=1}^{\infty} \frac{1}{n^2} \mathbb{1}_{\{y \leq n\}} \right] \cdot 2y \mathbb{P}(|X_1| > y) dy. \end{aligned}$$

Since $\mathbb{E}|X_1| = \int_0^{\infty} \mathbb{P}(|X_1| > y) dy$, it suffices to show

$$2y \sum_{n \geq y} \frac{1}{n^2} \leq 4, \quad \forall y > 0. \quad (4.13)$$

Noting that

$$\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n},$$

for $y \geq 2$, we have

$$2y \sum_{n \geq y} \frac{1}{n^2} \leq 2y \sum_{n \geq y} \left(\frac{1}{n-1} - \frac{1}{n} \right) \leq \frac{2y}{y-1} \leq 4,$$

and for $1 < y < 2$, we have

$$2y \sum_{n \geq y} \frac{1}{n^2} = 2y \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) \leq 2y \leq 4.$$

If $0 < y \leq 1$, then

$$2y \sum_{n \geq y} \frac{1}{n^2} \leq 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \leq 2 \cdot \frac{\pi^2}{6} \leq 4.$$

These establish (4.13) and complete the proof. $\square$

4.3.2 Etemadi's argument

The argument presented in this section was due to Etemadi (1981).

We have seen in the proof of Theorem 4.4, it is useful to first consider almost sure convergence along a subsequence (n_k) , then use other means to control what happens for $n \in (n_k, n_{k+1})$. Etemadi's idea is to use monotonicity of the partial sum when the summands are non-negative to control the intermediate terms.

Let us assume first that $X_n \geq 0$ (and hence $Y_n \geq 0$), and that for some subsequence (n_k) ,

$$T_{n_k}/n_k \rightarrow \mu, \quad \text{a.s.}, \quad (4.14)$$

and see how far we can get. Since Y_n are non-negative, for $n \in (n_k, n_{k+1})$ we have

$$\frac{T_{n_k}}{n_{k+1}} = \frac{T_{n_k}}{n_k} \cdot \frac{n_k}{n_{k+1}} \leq \frac{T_n}{n} \leq \frac{T_{n_{k+1}}}{n_k} = \frac{T_{n_{k+1}}}{n_{k+1}} \cdot \frac{n_{k+1}}{n_k}. \quad (4.15)$$

Taking the limit $k \rightarrow \infty$, we have

$$\mu \cdot \liminf_{k \rightarrow \infty} \frac{n_k}{n_{k+1}} \leq \liminf_{n \rightarrow \infty} \frac{T_n}{n} \leq \limsup_{n \rightarrow \infty} \frac{T_n}{n} \leq \mu \cdot \limsup_{k \rightarrow \infty} \frac{n_{k+1}}{n_k}. \quad (4.16)$$

Intuitively, the condition $n_{k+1}/n_k \approx 1$ will force $\liminf$ and $\limsup$ of T_n/n to be close, similar to the argument of the squeeze theorem. In fact, for any polynomial growth $n_k = k^p$, $p \geq 1$, the limit $\lim_{k \rightarrow \infty} n_{k+1}/n_k$ is 1, and (4.15) implies $\lim_{n \rightarrow \infty} T_n/n = \mu$. We have used such n_k in the proof of Theorem 4.4; as we will see, such growth cannot guarantee (4.15) when assuming merely finite first moment.

Let us go through our usual argument of combining Chebyshev's inequality and Borel–Cantelli lemma to see what is needed for the subsequence of (n_k) to guarantee (4.14). For every $\varepsilon > 0$, by Chebyshev's inequality, we have

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}(|T_{n_k} - \mathbb{E}T_{n_k}| > \varepsilon n_k) &\leq \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \text{Var}(T_{n_k})/n_k^2 = \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \frac{1}{n_k^2} \sum_{m=1}^{n_k} \text{Var}(Y_m) \\ &= \frac{1}{\varepsilon^2} \sum_{m=1}^{\infty} \text{Var}(Y_m) \sum_{n_k \geq m} \frac{1}{n_k^2} \end{aligned} \quad (4.17)$$

where we have used Fubini's theorem to interchange the two summations of nonnegative terms. Now, in light of [Proposition 4.12](#), we are hoping for

$$\sum_{n_k \geq m} \frac{1}{n_k^2} \leq \frac{C}{m^2}. \quad (4.18)$$

If n_k grows polynomially fast, that is, $n_k \sim k^p$ for some $p > 0$, then [\(4.18\)](#) cannot hold. For [\(4.18\)](#) to hold, we need to control the sum by the first (and largest) term, so n_k must grow *exponentially*. That means $\alpha = \liminf_{k \rightarrow \infty} n_{k+1}/n_k > 1$. Using [\(4.15\)](#), we can close the argument by taking α arbitrarily close to 1.

First proof of Theorem 4.1: Assume first $X_n \geq 0$.

Take $n_k = [\alpha^k]$, $k \geq 1$, where $\alpha > 1$ and $[\cdot]$ denotes the integer part. Let k_0 be the smallest k such that $n_k \geq m$. Then $k_0 \geq \log_\alpha m$. Since the sum of a geometric series is dominated by the largest term, we have

$$\sum_{n_k \geq m} \frac{1}{n_k^2} \leq \sum_{k=k_0}^{\infty} \frac{1}{[\alpha^k - 1]^2} \leq C_\alpha \frac{1}{\alpha^{2k_0}} \leq \frac{C_\alpha}{m^2}.$$

for some constant $C_\alpha > 0$. By [\(4.17\)](#) and Borel–Cantelli lemma, we have

$$\frac{T_{n_k} - \mathbb{E}T_{n_k}}{n_k} \rightarrow 0, \quad \text{a.s.}$$

By Stolz lemma and DCT,

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}T_n}{n} = \lim_{n \rightarrow \infty} \mathbb{E}Y_n = \lim_{n \rightarrow \infty} \mathbb{E}X_1 \mathbb{1}_{\{X_1 \leq 0\}} = \mathbb{E}X_1 = \mu.$$

Recalling $n_k = [\alpha^n]$, from [\(4.16\)](#) we have

$$\frac{\mu}{\alpha} \leq \liminf_{n \rightarrow \infty} \frac{T_n}{n} \leq \limsup_{n \rightarrow \infty} \frac{T_n}{n} \leq \alpha\mu. \quad (4.19)$$

Since [\(4.19\)](#) holds for arbitrary $\alpha > 1$, by letting $\alpha \downarrow 1$ we see that $\lim_{n \rightarrow \infty} T_n/n = \mu$ a.s., and by [Proposition 4.11](#) $\lim_{n \rightarrow \infty} S_n/n = \mu$ a.s. as desired.

For general X_n , let $X_n = X_n^+ - X_n^-$ be the decomposition into positive and negative parts, and let $S_n^\pm$ be the partial sums of $X_n^\pm$. Then

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \lim_{n \rightarrow \infty} \frac{S_n^+ - S_n^-}{n} = \mathbb{E}X_1^+ - \mathbb{E}X_1^- = \mathbb{E}X_1, \quad \text{a.s.}$$

The proof is complete. □

As a corollary, we can also treat the case when $\mathbb{E}X_1 = \pm\infty$.

Corollary 4.13 *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_i^+ = \infty$ and $\mathbb{E}X_i^- < \infty$. Then $S_n/n \rightarrow \infty$ a.s.*

Proof: Let $M > 0$ and $X_i^M = X_i \wedge M$. The X_i^M are i.i.d. with $\mathbb{E}|X_i^M| < \infty$. Let S_n^M be the partial sum of X_i^M . Using [Theorem 4.1](#) and $X_i \geq X_i^M$, we have

$$\liminf_{n \rightarrow \infty} S_n/n \geq \lim_{n \rightarrow \infty} S_n^M/n = \mathbb{E}X_1^M, \quad \text{a.s.}$$

The MCT implies $\mathbb{E}(X_1^M)^+ \uparrow \mathbb{E}X_1^+ = \infty$ as $M \uparrow \infty$, so $\mathbb{E}X_i^M = \mathbb{E}(X_i^M)^+ - \mathbb{E}(X_i^M)^- \uparrow \infty$. Hence $\liminf_{n \rightarrow \infty} S_n/n \geq \infty$ a.s., and the conclusion follows. □

4.4 Second proof of SLLN

In the section we follow Komolgorov's treatment of the SLLN.

4.4.1 Tail σ -algebras and zero-one law

In this section we will have a small detour to introduce the tail σ -algebras and Kolmogorov's zero-one law. These results are not directly used in the proof of SLLN, but they give useful intuition.

We first do some measure theory.

Let $(Y_n)_{n \in I}$ be r.v.s on $(\Omega, \mathcal{F}, \mathbb{P})$ where I is a countable index set. We introduce the "smallest σ -algebra" with respect to which all Y_n are measurable. It is defined by

$$\sigma(Y_n, n \in I) = \sigma(Y_n^{-1}(A), A \in \mathcal{B}(\mathbb{R}), n \in I). \quad (4.20)$$

The σ -algebra in (4.20) is generated by the semi-algebra

$$\mathcal{S} = \left\{ \bigcap_{n \in I_1} Y_n^{-1}(A_n) : A_n \in \mathcal{B}(\mathbb{R}), I_1 \subset I \text{ finite} \right\}. \quad (4.21)$$

We check that $\mathcal{S}$ is a semi-algebra using the following lemma.

Lemma 4.14 *Let I be a countable index set and $\mathcal{F}_n, n \in I$, be σ -algebras. Then*

$$\mathcal{S}_1 = \left\{ \bigcap_{n \in I_1} B_n, B_n \in \mathcal{F}_n, I_1 \subset I \text{ finite} \right\}$$

is a semi-algebra.

In particular, when $\mathcal{F}_n = \sigma(Y_n)$, $\mathcal{S}$ defined in (4.21) is a semi-algebra.

Proof: Let

$$A = \bigcap_{n \in I_1} A_n, \quad \tilde{A} = \bigcap_{m \in I_2} \tilde{A}_m, \quad (4.22)$$

be two sets in $\mathcal{S}_1$. We can replace I_1 and I_2 by their union $J = I_1 \cup I_2$ by adding Ω in both of the intersection in (4.22) if necessary. Note that J is the union of two finite sets I_1 and I_2 and thus is also finite. We have

$$A \cap \tilde{A} = \bigcap_{n \in J} (A_n \cap \tilde{A}_n) \in \mathcal{S}_1.$$

This shows that $\mathcal{S}_1$ is closed under intersections.

It remains to show that the complement of every set in $\mathcal{S}_1$ can be written as a finite disjoint union of sets in $\mathcal{S}_1$. This follows from

$$(A_{n_1} \cap \cdots \cap A_{n_k})^c = (A_{n_1}^c) \cup (A_{n_1} \cap A_{n_2}^c) \cup \cdots \cup (A_{n_1} \cap \cdots \cap A_{n_k}^c).$$

□

Proposition 4.15 *If $\mathcal{F}_n, n \in I$ and $\mathcal{G}_m, m \in J$ are independent where I, J are countable, then*

$$\sigma(\mathcal{F}_n, n \in I) \quad \text{and} \quad \sigma(\mathcal{G}_m, m \in J)$$

are independent.

In particular, if $X_n, n \in I$ and $Y_m, m \in J$ are all independent, then $\sigma(X_n, n \in I)$ and $\sigma(Y_m, m \in J)$ are independent.

Proof: Recall that two σ -algebras $\mathcal{F}$ and $\mathcal{G}$ are independent if $P(A \cap B) = P(A)P(B)$ for every $A \in \mathcal{F}$ and $B \in \mathcal{G}$. By [Lemma 4.14](#), $\mathcal{F}$ and $\mathcal{G}$ are generated by the semi-algebras

$$\mathcal{S}_1 = \left\{ \bigcap_{n \in I_1} A_n, A_n \in \mathcal{F}_n, I_1 \subset I \text{ finite} \right\}, \quad \mathcal{S}_2 = \left\{ \bigcap_{m \in J_1} B_m, B_m \in \mathcal{G}_m, J_1 \subset J \text{ finite} \right\}.$$

To show independent of $\mathcal{F}$ and $\mathcal{G}$, it suffices to show that

$$P(A \cap B) = P(A)P(B), \quad \forall A \in \mathcal{S}_1, B \in \mathcal{S}_2. \quad (4.23)$$

Extending [\(4.23\)](#) to arbitrary A and B can be done by the usual appropriate set arguments and continuity of probability measures.

For $A \in \mathcal{S}_1$ and $B \in \mathcal{S}_2$, the independence of $\mathcal{F}_n$ and $\mathcal{G}_m$ implies

$$\begin{aligned} P(A \cap B) &= P\left(\bigcap_{n \in I_1} A_n \cap \bigcap_{m \in J_1} B_m\right) = \prod_{n \in I_1} P(A_n) \prod_{m \in J_1} P(B_m) \\ &= P\left(\bigcap_{n \in I_1} A_n\right) P\left(\bigcap_{m \in J_1} B_m\right) = P(A)P(B). \end{aligned}$$

This establishes [\(4.23\)](#) and completes the proof. $\square$

Let $(X_n)_{n \geq 1}$ be independent on $(\Omega, \mathcal{F}, P)$. Let us introduce

$$\mathcal{F}_n = \sigma(X_1, \dots, X_n), \quad \mathcal{F}_{>n} = \sigma(X_{n+1}, X_{n+2}, \dots) = \sigma(X_m, m > n).$$

The σ -algebra $\mathcal{F}_n$, containing information before time n , should be regarded as the “past”, while $\mathcal{F}_{>n}$ should be regarded as the “future”. By [Proposition 4.15](#), $\mathcal{F}_n$ and $\mathcal{F}_{>n}$ are independent for all $n \geq 1$, which agrees with our intuition.

Definition 4.1 (Tail σ -algebra) *The tail σ -algebra is $\mathcal{T} = \bigcap_{n=0}^{\infty} \mathcal{F}_{>n}$.*

The tail σ -algebra should be regarded as the “remote future”, as it does not concern anything happening in finite time.

Example 4.8 (Examples of sets in $\mathcal{T}$) 1. $\{\lim_{n \rightarrow \infty} S_n \text{ exists}\} \in \mathcal{T}$.

2. $\{\limsup_{n \rightarrow \infty} \frac{S_n}{n} > x\} \in \mathcal{T}$ for any x .

We will verify for the first set. Since $\lim_{m \rightarrow \infty} S_m$ exists if and only if $\lim_{m \rightarrow \infty} (S_{n+m} - S_n)$ exists, for all $n \geq 0$, we have

$$\left\{ \lim_{m \rightarrow \infty} (S_{n+m} - S_n) \text{ exists} \right\} = \left\{ \lim_{m \rightarrow \infty} (X_{n+1} + X_{n+2} + \dots + X_{n+m}) \text{ exists} \right\} \in \mathcal{F}_{>n}.$$

Hence, $\{\lim_{m \rightarrow \infty} S_m\} \in \bigcap_{n \geq 0} \mathcal{F}_{>n} = \mathcal{T}$.

An important observation of Kolmogorov was that the tail σ -algebra is trivial, and thus it makes sense to study the almost sure convergence of random series.

Theorem 4.16 (Kolmogorov’s zero-one law) *If $X_1, X_2, \dots$ are independent and $A \in \mathcal{T}$ then $P(A) = 0$ or 1 .*

Proof: For all $m \geq n$, since $\mathcal{F}_n$ and $\mathcal{F}_{>m}$ are independent, so we have $\mathcal{F}_n \perp \bigcap_{m \geq n} \mathcal{F}_{>m}$, that is, for all $n \geq 1$, $\mathcal{F}_n \perp \mathcal{T}$. Let $\mathcal{F}_{\infty} = \sigma(X_1, X_2, \dots)$. For all $n \geq 1$, since $\mathcal{T}$ and $\sigma(X_n)$ are independent, so we have $\mathcal{T} \perp \sigma(X_1, X_2, \dots) = \mathcal{F}_{\infty}$, and thus $\mathcal{T} = \bigcap_{n=0}^{\infty} \mathcal{F}_{>n} \subset \mathcal{F}_{\infty}$. It follows that $\mathcal{T} \perp \mathcal{T}$. For all $A \in \mathcal{T}$, A is independent to itself, that is,

$$P(A \cap A) = P(A) = P(A)^2$$

it follows that $P(A) = 1$ or 0 . $\square$

4.4.2 Kolmogorov's proof

Proposition 4.17 (Kolmogorov's maximal inequality) *Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$ and $\text{Var}(X_i) < \infty$. Then*

$$\mathbb{P}(\max_{1 \leq k \leq n} |S_k| \geq x) \leq \frac{\text{Var}(S_n)}{x^2}. \quad (4.24)$$

Remark 4.9 1. Note that Chebyshev's inequality can only give (4.24) *without* the maximum inside the probability, so Proposition 4.17 is highly non-trivial.

2. In fact, $(S_n)_{n \geq 1}$ forms a martingale and Proposition 4.17 is a special case of the *Doob's maximal inequality* for martingales. In the proof, we will also use the idea of "stopping time", which is common in martingale analysis

Proof: Let $T(\omega) = \min\{k : k \geq 1, |S_k(\omega)| \geq x\}$ to be the first time that $|S_k|$ exceeds x . More precisely,

$$\{T(\omega) = k\} = \{|S_1|, |S_2|, \dots, |S_{k-1}| < x, |S_k| \geq x\},$$

for $k \in \{1, \dots, n\}$ and $T = \infty$ if the event in (4.24) does not happen. By definition we have $\{T = k\} \in \sigma(X_1, \dots, X_k) =: \mathcal{F}_k$ for $k \in \{1, \dots, n\}$.

We have

$$\mathbb{E}S_n^2 \geq \sum_{k=1}^n \mathbb{E}S_n^2 \mathbb{1}_{\{T=k\}} = \sum_{k=1}^n \mathbb{E}S_n^2 \mathbb{1}_{\{T=k\}}.$$

For $k \leq n$, we have

$$\begin{aligned} \mathbb{E}S_n^2 \mathbb{1}_{\{T=k\}} &= \mathbb{E}(S_k + (S_n - S_k))^2 \mathbb{1}_{\{T=k\}} \\ &\geq \mathbb{E}S_k^2 \mathbb{1}_{\{T=k\}} + 2\mathbb{E}S_k \mathbb{1}_{\{T=k\}} \cdot (S_n - S_k) + \mathbb{E}(S_n - S_k)^2 \mathbb{1}_{\{T=k\}} \\ &\geq \mathbb{E}S_k^2 \mathbb{1}_{\{T=k\}} + 2\mathbb{E}S_k \mathbb{1}_{\{T=k\}} \cdot \mathbb{E}(S_n - S_k) \\ &= \mathbb{E}S_k^2 \mathbb{1}_{\{T=k\}} \geq x^2 \mathbb{P}(T = k). \end{aligned}$$

Here, in the third line we use the independence of $S_k \mathbb{1}_{\{T=k\}} \in \sigma(X_1, \dots, X_k)$ and $S_n - S_k \in \sigma(X_{k+1}, \dots)$. Summing over $k \in \{1, \dots, n\}$, we have

$$\mathbb{E}S_n^2 \geq x^2 \mathbb{P}(T \leq n),$$

and this gives (4.24). $\square$

Proposition 4.18 (Kolmogorov's one-series theorem) *Let $X_1, X_2, \dots$ be independent with $\mathbb{E}X_n = 0$. If*

$$\sum_{n=1}^{\infty} \mathbb{E}X_n^2 < \infty, \quad (4.25)$$

then $\sum_{n=1}^{\infty} X_n(\omega)$ converges a.s.

Proof: For every $\varepsilon > 0$, by Proposition 4.17, for all $M, N > 0$, we have

$$\mathbb{P}(\max_{M \leq n \leq N} |S_n - S_M| \geq \varepsilon) \leq \frac{\text{Var}(S_n - S_M)}{\varepsilon^2}.$$

Letting $N \rightarrow \infty$, by MCT, we have

$$\mathbb{P}(u_M := \sup_{n \geq M} |S_n - S_M| \geq \varepsilon) \leq \frac{\sum_{n=M}^{\infty} \mathbb{E}X_n^2}{\varepsilon^2},$$

which goes to 0 by (4.25). Let $\tilde{u}_M = \sup_{m,n \geq M} |S_n - S_m|$. Then for every $\varepsilon > 0$,

$$\lim_{m \rightarrow \infty} \mathbb{P}(\tilde{u}_M \geq \varepsilon) \leq 2 \lim_{m \rightarrow \infty} \mathbb{P}(u_M \geq \frac{\varepsilon}{2}) = 0.$$

Hence, for every $\varepsilon > 0$, we have $\mathbb{P}(\lim_{n \rightarrow \infty} \tilde{u}_M \geq \varepsilon) = 0$ since $\tilde{u}_M$ is decreasing. Therefore, $\tilde{u}_M \downarrow 0$ as $M \rightarrow \infty$ almost surely, and $\lim_{n \rightarrow \infty} S_n$ exists by Cauchy criterion. $\square$

Example 4.10 It is well known that alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges conditionally. What if we put random $\pm$ signs before the harmonic series?

To model it, let $(\xi_n)_{n \geq 1}$ be i.i.d. with $\mathbb{P}\{\xi_n = \pm 1\} = \frac{1}{2}$. By Proposition 4.18, since $\sum_{n=1}^{\infty} \mathbb{E} \frac{|\xi_n|^2}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$, we have $\sum_{n=1}^{\infty} \frac{\xi_n}{n}$ converges a.s. In fact, the conclusion holds for $\sum_{n=1}^{\infty} \xi_n/n^p$ with $p \in (1/2, 1]$.

The next result is an immediate corollary of Proposition 4.18 and treat the case of non-centered r.v.s.

Proposition 4.19 (Kolmogorov's two-series theorem) *Let $X_1, X_2, \dots$ are independent with $\mathbb{E}|X_n| < \infty$. If*

$$\sum_{n=1}^{\infty} \mathbb{E}X_n \text{ exists, } \sum_{n=1}^{\infty} \mathbb{E}X_n^2 < \infty,$$

then $\sum_{n=1}^{\infty} X_n(\omega)$ converges a.s.

For the almost sure convergence of random series, the final theorem provides necessary and sufficient conditions.

Theorem 4.20 (Kolmogorov's three-series theorem) *Let $A > 0$. Let $X_1, X_2, \dots$ be independent and $Y_i = X_i \mathbb{1}_{(|X_i| \leq A)}$. Then, $\sum_{n=1}^{\infty} X_n$ converges a.s. if and only if all of the following conditions hold:*

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| \geq A) < \infty, \tag{4.26a}$$

$$\sum_{n=1}^{\infty} \mathbb{E}Y_n \text{ exists,} \tag{4.26b}$$

$$\sum_{n=1}^{\infty} \text{Var}(Y_n) < \infty. \tag{4.26c}$$

Proof: **The “if” part.** By Borel-Cantelli lemma, (4.26a) implies that $\mathbb{P}(\{|X_n| > A, \text{ i.o.}\}) = 0$. On the event $\{|X_n| > A, \text{ i.o.}\}^c$, there exists $n_0 = n_0(\omega)$ such that $Y_n = X_n$ for every $n > n_0$, and hence $\sum_{n=1}^{\infty} X_n$ converges if and only if $\sum_{n=1}^{\infty} Y_n$ does; on the other hand, the latter random series converges a.s. by Proposition 4.19, (4.26b) and (4.26c).

The “only if” part. Assume now that $\sum_{n=1}^{\infty} X_n$ converges a.s. If (4.26a) fails, by the second Borel-Cantelli lemma (Theorem 4.9), we have $\mathbb{P}(\{|X_n| \geq A, \text{ i.o.}\}) = 1$. But on $\{|X_n| \geq A, \text{ i.o.}\}^c$, the series $\sum_{n=1}^{\infty} X_n(\omega)$ cannot converge as the Cauchy criterion is violated. Hence, (4.26a) must hold. Then, as has been proven, (4.26a) implies that $\sum_{n=1}^{\infty} Y_n$ also converges a.s.

Now we have $|Y_n - \mathbb{E}Y_n| \leq 2A$. By Lemma 4.21 proven below, we have

$$\mathbb{P}\left(\max_{M \leq n \leq N} \left| \sum_{k=M}^n Y_k \right| \leq 1\right) \leq \frac{(2A+1)^2}{\sum_{n=M}^{N-1} \text{Var}(Y_n)}. \tag{4.27}$$

If (4.26c) fails and $\sum_{n=1}^{\infty} \text{Var}(Y_n) = \infty$, then (4.27) implies

$$\mathbb{P}\left(\sup_{n \geq M} \left| \sum_{k=M}^n Y_k \right| \leq 1\right) = 0, \quad \forall M \geq 1,$$

which implies that $\sum_{n=1}^{\infty} Y_n$ diverges a.s. and leads to a contradiction. Thence, (4.26c) also holds.

It remains to show (4.26c). By Proposition 4.18, (4.26c) implies that $\sum_{n=1}^{\infty} (Y_n - \mathbb{E}Y_n)$ converges a.s., and hence

$$\sum_{n=1}^{\infty} \mathbb{E}Y_n = \sum_{n=1}^{\infty} Y_n - \sum_{n=1}^{\infty} (Y_n - \mathbb{E}Y_n)$$

also converges. This completes the proof of the “only if” part. $\square$

For (4.27) we use the following results, which says if a random walk have large variance in each step, then it is unlikely that it will stay in a small region; this is the opposite direction of Proposition 4.17.

Lemma 4.21 *Let $Z_1, \dots, Z_n$ be independent with $\mathbb{E}Z_i = 0$ and $|Z_i| \leq L$ for some $L > 0$. Let $S_n = Z_1 + \dots + Z_n$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \leq \varepsilon\right) \leq \frac{(L + \varepsilon)^2}{\text{Var}(S_n)}.$$

Proof: Let

$$T = \min\{k : |S_k| > \varepsilon\} \in \{1, 2, \dots, n, \infty\},$$

with the convention $T = \infty$ if $\max_{1 \leq k \leq n} |S_k| \leq \varepsilon$. We note that $\{T = k\}, \{T \geq k+1\} \in \sigma(X_1, \dots, X_k)$ for every $1 \leq k \leq n$.

Let $a_k = \mathbb{E}S_k^2 \mathbb{1}_{\{T \geq k+1\}}$, $0 \leq k \leq n$. Since $\{T \geq k+2\} = \{T \geq k+1\} \setminus \{T = k+1\}$, we have

$$a_{k+1} = \mathbb{E}S_{k+1}^2 \mathbb{1}_{\{T \geq k+1\}} - \mathbb{E}S_{k+1}^2 \mathbb{1}_{\{T = k+1\}}. \quad (4.28)$$

Using independence of X_{k+1} and $S_k, \mathbb{1}_{\{T \geq k+1\}}$ (both measurable w.r.t. $\sigma(X_1, \dots, X_k)$), the first term in (4.28) is bounded below by

$$\begin{aligned} \mathbb{E}S_{k+1}^2 \mathbb{1}_{\{T \geq k+1\}} &= \mathbb{E}S_k^2 \mathbb{1}_{\{T \geq k+1\}} + 2\mathbb{E}X_{k+1} \cdot \mathbb{E}S_k \mathbb{1}_{\{T \geq k+1\}} + \mathbb{E}X_{k+1}^2 \cdot \mathbb{P}(T \geq k+1) \\ &\geq a_k + \mathbb{E}X_{k+1}^2 \cdot \mathbb{P}(T \geq k+1). \end{aligned} \quad (4.29)$$

For the second term in (4.28), since

$$\{T = k+1\} = \{|S_1| \leq \varepsilon, \dots, |S_k| \leq \varepsilon, |S_{k+1}| > \varepsilon\},$$

when $T = k+1$, we have $|S_{k+1}| \leq |S_k| + |X_{k+1}| \leq L + \varepsilon$, and hence

$$\mathbb{E}S_{k+1}^2 \mathbb{1}_{\{T = k+1\}} \leq (L + \varepsilon)^2 \mathbb{P}(T = k+1). \quad (4.30)$$

Combining (4.29) and (4.30), we have

$$(a_{k+1} - a_k) + (L + \varepsilon)^2 \mathbb{P}(T = k+1) \geq \mathbb{E}X_{k+1}^2 \cdot \mathbb{P}(T \geq k+1), \quad k = 0, \dots, n-1.$$

Summing the above inequality over k , and using that $\mathbb{P}(T \geq k+1) \geq \mathbb{P}(T = \infty)$, we have

$$\text{Var}(S_n) \cdot \mathbb{P}(T = \infty) \leq (L + \varepsilon)^2 \mathbb{P}(T \leq n) + \mathbb{E}S_n^2 \mathbb{1}_{\{T \geq n+1\}}.$$

Noting that when $T \geq n+1$, $S_n^2 \leq \varepsilon^2$, the last term in the last display is bounded by $\varepsilon^2 \cdot \mathbb{P}(T \geq n+1)$, so we get

$$\text{Var}(S_n) \cdot \mathbb{P}(T = \infty) \leq (L + \varepsilon)^2.$$

The desired inequality follows. $\square$

Proposition 4.22 (Kronecker's lemma) *If $a_n \uparrow \infty$ and $\sum_{n=1}^{\infty} x_n/a_n$ converges, then*

$$a_n^{-1} \sum_{m=1}^n x_m \rightarrow 0$$

Proof: Let $S_n := \sum_{m=1}^n x_m/a_m$ and $S := \lim_{n \rightarrow \infty} S_n$. Using Summation by parts (a.k.a. Abel's transformation) we have

$$\frac{1}{a_n} \sum_{m=1}^n a_m(S_m - S_{m-1}) = S_n - \sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} S_{m-1}$$

By using Generalized Stolz's Lemma (Lemma 4.23) below with $\rho_{n,k} = (a_k - a_{k-1})/a_n$, we have

$$\lim_{n \rightarrow \infty} \frac{1}{a_n} \sum_{m=1}^n a_m(S_m - S_{m-1}) = \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} \sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} S_{m-1} = S - S = 0.$$

□

Lemma 4.23 (Generalized Stolz) *Let $\rho_{n,k} \geq 0$, $1 \leq k \leq n$, be such that*

$$\lim_{n \rightarrow \infty} \sum_{k=K}^n \rho_{n,k} = 1, \quad \lim_{n \rightarrow \infty} \sum_{k=1}^{K-1} \rho_{n,k} = 0,$$

for every $K > 0$. Then

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \rho_{n,k} x_k = \lim_{n \rightarrow \infty} x_n$$

provided that second limit exists.

In particular, when $\rho_{n,k} = 1/n$, this is the Stolz's Lemma.

Proof: Let $L = \lim_{n \rightarrow \infty} x_n$. For simplicity we assume $|L| < \infty$, but the lemma also holds for $L = \infty$ or $-\infty$.

For every $\varepsilon > 0$, there exists $K > 0$ such that $x_k \geq L - \varepsilon$ for $k \geq K$. Hence, we have

$$\sum_{k=1}^n \rho_{n,k} x_k \geq -(\sup_m |x_m|) \sum_{k=1}^{K-1} \rho_{n,k} + (L - \varepsilon) \sum_{k=K}^n \rho_{n,k}.$$

Taking $n \rightarrow \infty$ and using the assumption on $\rho_{n,k}$ we get $\liminf_{n \rightarrow \infty} \sum_{k=1}^n \rho_{n,k} x_k \geq L - \varepsilon$, and since $\varepsilon > 0$ is arbitrary we have $\liminf_{n \rightarrow \infty} \sum_{k=1}^n \rho_{n,k} x_k \geq L$. Similarly, we can show $\limsup_{n \rightarrow \infty} \sum_{k=1}^n \rho_{n,k} x_k \leq L$. The conclusion follows. □

We can now give the proof of Theorem 4.1.

Proof of Theorem 4.1: By Proposition 4.10, it suffices to show that $(T_n - \mu_n)/n \rightarrow 0$, a.s., where $T_n = \sum_{k=1}^n Y_n$ and $Y_n = X_n \mathbb{1}_{\{|X_n| \leq n\}}$. By Proposition 4.22, it suffices to show that $\sum_{n=1}^{\infty} \frac{Y_n}{n}$ converges a.s., and this follows from Proposition 4.18 and Proposition 4.12. □

The above proof also gives us a way to estimate the rate of convergence, as the next result shows.

Proposition 4.24 *Let $X_1, X_2, \dots$ are i.i.d. r.v.s with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = \sigma^2 < \infty$. Let $S_n = X_1 + \dots + X_n$. Then, for every $\varepsilon > 0$,*

$$\frac{S_n}{\sqrt{n}(\log n)^{1/2+\varepsilon}} \rightarrow 0 \quad \text{a.s.}$$

Proof: Let $a_n = n^{\frac{1}{2}}(\log n)^{\frac{1}{2}+\varepsilon}$, $n \geq 2$. We have

$$\sum_{n=2}^{\infty} \text{Var}\left(\frac{x_n}{a_n}\right) = \sigma^2 \sum_{n=2}^{\infty} \frac{1}{a_n^2} = \sigma^2 \sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1+2\varepsilon}} < \infty.$$

By [Proposition 4.18](#), the series $\sum_{n=1}^{\infty} \frac{x_n}{a_n}$ converges a.s., and it follows from [Proposition 4.22](#) that $\frac{1}{a_n} \sum_{k=1}^n x_k \rightarrow 0$ a.s. $\square$

To conclude our discussion of the SLLN, we cite a result given by Feller (see also [\[Dur19, Thm 2.5.13\]](#)), which says when the r.v.s are not integrable, SLLN cannot hold.

Proposition 4.25 *Let $X_1, X_2, \dots$ are i.i.d. r.v.s with $E|X_1| = \infty$. Let a_n be a sequence of positive numbers with a_n/n increasing. Then $\limsup_{n \rightarrow \infty} |S_n|/a_n = 0$ or ∞ according as $\sum_{n=1}^{\infty} P(|X_1| \geq a_n) < \infty$ or $= \infty$.*

5 Weak convergence and central limit theorem

Let $\mathcal{P}(\mathbb{R})$ be the set of all probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. To goal of this section is to study the weak convergence of probability measure. To start, we mention the *total variation distance*, a natural metric on $\mathcal{P}(\mathbb{R})$.

Definition 5.1 *Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$. The total variation distance between μ and ν is given by*

$$\|\mu - \nu\|_{TV} := 2 \cdot \sup_{A \in \mathcal{B}(\mathbb{R})} |\mu(A) - \nu(A)| \in [0, 2]. \quad (5.1)$$

It is not hard to check that [\(5.1\)](#) defines a metric: (a) it is positive definite: $\|\mu - \nu\|_{TV} \geq 0$, with $\|\mu - \nu\|_{TV} = 0$ if and only if $\mu(A) = \nu(A)$ for all A , which means $\mu = \nu$; (b) it is symmetric: $\|\mu - \nu\|_{TV} = \|\nu - \mu\|_{TV}$; (c) it satisfies the triangle inequality by taking supremum of $|\mu(A) - \nu(A)| \leq |\mu(A) - \lambda(A)| + |\lambda(A) - \nu(A)|$.

Example 5.1 Let μ and ν are mutually singular (see [Definition 1.13](#)) and let $A \in \mathcal{B}(\mathbb{R})$ be such that $\mu(A) = 0$ and $\nu(A) = 1$. Then $|\mu(A) - \nu(A)| = 1$, and hence $\|\mu - \nu\|_{TV} = 2$ since 2 is the maximum for total variation distance.

Example 5.2 Let μ and ν are absolutely continuous and $f, g \in L^1(\mathbb{R})$ be their densities. Then

$$\|\mu - \nu\|_{TV} = \|f - g\|_{L^1(\mathbb{R})},$$

where the supremum in [\(5.1\)](#) is achieved by $A = \{x : f(x) \geq g(x)\}$ in [\(5.1\)](#). More generally, let F and G be the c.d.f.s of μ and ν ; then

$$\|\mu - \nu\|_{TV} = \sup_{t_1 < \dots < t_n} \sum_{i=1}^{n-1} \left| (F - G)(t_{i+1}) - (F - G)(t_i) \right|,$$

which is the *total variation* of $F - G$ over $\mathbb{R}$.

By [\(5.1\)](#), convergence in total variational distance implies convergence of the set function.

Proposition 5.1 *If $\|\mu_n - \mu\|_{TV} \rightarrow 0$, then*

$$\lim_{n \rightarrow \infty} \mu_n(A) = \mu(A), \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

In particular, by taking $A = (-\infty, x]$,

$$\lim_{n \rightarrow \infty} F_{\mu_n}(x) = F_{\mu}(x), \quad \forall x. \quad (5.2)$$

However, as can be seen from the next two examples, the convergence in total variation distance is too restrictive.

Example 5.3 Let $\mu_n = \delta_{1/n}$ and $\mu = \delta_0$. We expect $\mu_n \rightarrow \mu$ since $1/n \rightarrow 0$. On the other hand, $\|\mu_n - \mu\|_{TV} = 2$ so no convergence in total variation distance.

Example 5.4 Let $\mu_n = \text{Unif}\{0, \frac{1}{n}, \dots, \frac{n-1}{n}\}$. We expect $\mu_n \rightarrow \mu = \text{Unif}[0, 1]$, which will justify the standard procedure to generate $\text{Unif}[0, 1]$ r.v.s on computers mentioned in [Section 1.4](#). On the other hand, we have $\mu_n(\mathbb{Q}) = 1$ while $\mu(\mathbb{Q}) = 0$, so μ_n and μ are mutually singular and thus $\|\mu_n - \mu\|_{TV} \equiv 2$.

Weak convergence is one way to give a more relaxed mode of convergence for measures.

5.1 Definition of weak convergence

Throughout this section, μ_n, μ will be probability measures and F_n, F be their c.d.f.s.

Definition 5.2 (Weak convergence) We say that μ_n converge to μ weakly, written $\mu_n \Rightarrow \mu$, if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x), \quad \text{almost every } x. \quad (5.3)$$

With abuse of notation we also write $F_n \Rightarrow F$ for (5.3).

Let X_n, X be r.v.s. We say that X_n converge to X in distribution/law, or weakly, written $X_n \Rightarrow X$, if $\mu_{X_n} \Rightarrow \mu_X$.

By directly computing the c.d.f's, one can verify that

$$\delta_{1/n} \Rightarrow \delta_0, \quad \text{Unif}\left\{0, \frac{1}{n}, \dots, \frac{n-1}{n}\right\} \Rightarrow \text{Unif}[0, 1].$$

(5.3) is *weaker* than (5.2) since it allows an exceptional set of measure zero. The first question we ask is the uniqueness of such limit.

Proposition 5.2 If $F_n \Rightarrow F$ and $F_n \Rightarrow G$, then $F = G$.

Proof: Implicitly here, we require that both F and G are c.d.f.s, that is, right continuous and increasing functions. We know that such functions are determined by its value on a dense set. By the assumption, there exist zero measure sets N_1 and N_2 such that $F_n(x) \rightarrow F(x)$ for $x \notin N_1$ and $F_n(x) \rightarrow G(x)$ for $x \notin N_2$, and hence $F(x) = G(x)$ for $x \notin N_1 \cup N_2$. The measure of $N_1 \cup N_2$ is 0, and the complement of any zero measure set is dense, so $F = G$ as desired. $\square$

In fact, we know precisely what is the exceptional set in (5.3).

Proposition 5.3 (Also an alternative definition for $F_n \Rightarrow F$) The condition (5.3) is equivalent to

$$\lim_{n \rightarrow \infty} F_n(x) = F(x), \quad \forall \text{ continuous point } x \text{ of } F. \quad (5.4)$$

Proof: (5.3) follows from (5.4) since discontinuous point of c.d.f. is at most countable, and a countable set has measure 0.

For the other direction, let x_0 be a continuous point of F . For every $\varepsilon > 0$, there exists $\delta > 0$ such that $|F(x) - F(x_0)| < \varepsilon$ for $|x - x_0| < \delta$. Then, we can find $y_1 \in (x_0 - \delta, x_0)$ and $y_2 \in (x_0, x_0 + \delta)$ such that y_1 and y_2 are not in the exceptional set in (5.3). We also have, by the monotonicity of F_n ,

$$\begin{aligned} F(x_0) - \varepsilon < F(y_1) &= \lim_{n \rightarrow \infty} F_n(y_1) \leq \liminf_{n \rightarrow \infty} F_n(x_0) \\ &\leq \limsup_{n \rightarrow \infty} F_n(x_0) \leq \lim_{n \rightarrow \infty} F_n(y_2) = F(y_2) < F(x_0) + \varepsilon. \end{aligned} \quad (5.5)$$

By letting $\varepsilon \downarrow 0$, we obtain $\lim_{n \rightarrow \infty} F_n(x_0) = F(x_0)$. $\square$

The real power of weak convergence is the extraction of convergence subsequence (so-called *sequential pre-compactness*) under minimum assumption.

Definition 5.3 We say that $(\mu_n)_{n \in I}$ is *tight*, if for every $\varepsilon > 0$, there exists a compact set $K = K_\varepsilon$ such that

$$\mu_n(K^c) \leq \varepsilon, \quad \forall n \in I. \quad (5.6)$$

Compact sets in $\mathbb{R}$ are bounded closed sets, and the above condition can be reformulated as the existence of $L > 0$ such that

$$\mu_n[-L, L]^c < \varepsilon, \quad \forall n \in I. \quad (5.7)$$

Note that using (5.6), the notion of tightness can be generalized to arbitrary metric spaces. [We will take up this again in XXX.]

Theorem 5.4 (Helly selection theorem) If $(\mu_n)_{n \geq 1}$ is tight, then there exists a subsequence $(\mu_{n_k})_{k \geq 1}$ and a probability measure μ_∞ such that $\mu_{n_k} \Rightarrow \mu_\infty$.

Proof: For every $q \in \mathbb{Q}$, the sequence $(F_n(q))_{n \geq 1} \subset [0, 1]$ has a convergent subsequence. Such subsequence depends on $q \in \mathbb{Q}$, but since $\mathbb{Q}$ is countable, by a standard diagonal sequence argument, there exists a common subsequence $(F_{n_k})_{k \geq 1}$ such that

$$\lim_{k \rightarrow \infty} F_{n_k}(q) =: \bar{F}(q), \quad \forall q \in \mathbb{Q}.$$

The limiting function $\bar{F}$ is increasing on $\mathbb{Q}$, so we can use it to define a right continuous, increasing function by

$$F(x) := \inf\{\bar{F}(q) : q \geq x\}.$$

We first show that $\lim_{k \rightarrow \infty} F_{n_k}(x) = F(x)$ for every continuous point of F . Let x_0 be a continuous point of F . Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|\bar{F}(q) - F(x_0)| < \varepsilon$ for all $|q - x_0| < \delta$. Pick any $q_1 \in (x_0 - \delta, x_0) \cap \mathbb{Q}$ and $q_2 \in (x_0, x_0 + \delta)$. Similar to (5.5), we have

$$F(x_0) - \varepsilon < \bar{F}(q_1) \leq \liminf_{k \rightarrow \infty} F_{n_k}(x_0) \leq \limsup_{k \rightarrow \infty} F_{n_k}(x_0) \leq \bar{F}(q_2) < F(x_0) + \varepsilon.$$

By sending $\varepsilon \rightarrow 0$ we prove the desired limit.

Second, we need to show that F is a c.d.f. It suffices to verify $\bar{F}(-\infty) = 0$ and $\bar{F}(\infty) = 1$, for which we will use tightness. Indeed, since $(\mu_n)_{n \geq 1}$ is tight, for every ε , there exists $L \in \mathbb{Q}$ such that $F_n(L) - F_n(-L) \geq 1 - \varepsilon$ (see (5.7)). Letting $n = n_k \rightarrow \infty$, we have

$$\bar{F}(L) - \bar{F}(-L) \geq 1 - \varepsilon.$$

Therefore,

$$\lim_{q \rightarrow -\infty} \bar{F}(q) = -\infty, \quad \lim_{q \rightarrow \infty} \bar{F}(q) = \infty,$$

and this show that F is a c.d.f. $\square$

Remark 5.5 Tightness is necessary to prevent the “escape of mass to infinity”. Consider $\mu_n = \text{Unif}[n, n+1]$, then $F_n(x) \rightarrow 0$ for every x , but the limiting function is 0, and cannot be a distribution function.

Proposition 5.5 (necessity of tightness) If $\mu_n \Rightarrow \mu$, then $(\mu_n)_{n \geq 1}$ is tight.

Proof: This will be an easy consequence of [Theorem 5.7](#), but we give another proof here by analyzing the c.d.f.s.

The first observation is that a single probability measure is tight. Therefore, for every $\varepsilon > 0$, there exists $L > 0$ such that $\pm L$ are both continuous points of F and $F(-L) \leq \varepsilon/4$, $F(L) \geq 1 - \varepsilon/4$. Since $\lim_{n \rightarrow \infty} F_n(\pm L) = F(\pm L)$, there exists n_0 such that $F_n(-L) \leq \varepsilon/2$ and $F_n(L) \geq 1 - \varepsilon/2$. Also, for each $1 \leq k \leq n_0$, there exist L_k such that $F_k(-L_k) \leq \varepsilon/2$, $F_k(L_k) \geq 1 - \varepsilon/2$. Let $\bar{L} = \max\{L, L_1, \dots, L_{n_0}\}$. Then $\mu_n[-\bar{L}, \bar{L}]^c \leq \varepsilon$ for all $n \geq 1$, and this proves the tightness. $\square$

Without the tightness, the convergence in the first part in the proof of [Theorem 5.4](#) is sometimes called *vague convergence*.

Definition 5.4 We say that $F_n \rightarrow F$ vaguely if $\lim_{n \rightarrow \infty} F_n(x) = F(x)$ for almost every x .

It may happen that F is a sub-probability measure, that is,

$$\lim_{n \rightarrow +\infty} F(x) - \lim_{n \rightarrow -\infty} F(x) < 1.$$

We can reformulate the previous results.

Proposition 5.6 Every sequence of probability measures $(\mu_n)_{n \geq 1}$ has a vaguely convergent subsequence. The limit is a probability measure if and only if the subsequence is tight.

5.2 Other characterizations of weak convergence

The concept of weak convergence can be generalized to arbitrary metric spaces. An excellent reference in this account is [\[Bil99\]](#). We only present a topological way to define weak convergence, which is applicable to more general settings.

Theorem 5.7 Let μ_n, μ be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The following statements are equivalent.

1. $\mu_n \Rightarrow \mu$, that is, [\(5.4\)](#) holds.
2. For every bounded continuous function g ,

$$\lim_{n \rightarrow \infty} \int g d\mu_n = \int g d\mu. \quad (5.8)$$

3. For every open set G ,

$$\liminf_{n \rightarrow \infty} \mu_n(G) \geq \mu(G). \quad (5.9)$$

4. For every closed set K ,

$$\limsup_{n \rightarrow \infty} \mu_n(K) \leq \mu(K). \quad (5.10)$$

5. For every A with $\mu(\partial A) = 0$, $\lim_{n \rightarrow \infty} \mu_n(A) = \mu(A)$.

Proof: From [Item 1](#) to [Item 2](#). We will start from simplest forms of g .

First, consider

$$g(x) = \begin{cases} 0, & x < a \\ (b - a), & x > b \\ \text{linear interpolation,} & x \in [a, b]. \end{cases} \quad (5.11)$$

Then $g(x) = \int_{-\infty}^x \mathbb{1}_{(a,b)}(y) dy$. Using Fubini's theorem we have

$$\begin{aligned} \int g d\mu_n &= \int \left[\int_{-\infty}^{\infty} \mathbb{1}_{(a,b)}(y) \mathbb{1}_{\{y < x\}} dy \right] d\mu_n(x) \\ &= \int_{-\infty}^{\infty} \mathbb{1}_{(a,b)}(y) dy \cdot \int_{-\infty}^{\infty} \mathbb{1}_{\{y < x\}} d\mu_n(x) \\ &= \int_a^b [1 - F_n(y)] dy \\ &\rightarrow \int_a^b [1 - F(y)] dy = \int g d\mu, \end{aligned}$$

where the last line is due to $F_n(y) \rightarrow F(y)$ for a.e. y and BCT on the finite interval $[a, b]$.

Second, we consider g being a piecewise linear function with compact support. Then such g can be written as a linear combination of functions in the form (5.11), so (5.8) holds.

Third, let K be a compact set and consider

$$g \in \mathcal{C}_K = \{g : \text{continuous, } \text{supp } g \subset K\}.$$

Then there exist $g_m \in \mathcal{C}_K$ piecewise linear with compact support such that $g_m \rightarrow g$ uniformly on K , by uniform continuity of g . To estimate the difference of the terms in (5.8), we replace g by g_m , with error controlled by the triangle inequality. We have for every m ,

$$\limsup_{n \rightarrow \infty} \left| \int g d\mu_n - \int g d\mu \right| \leq \limsup_{n \rightarrow \infty} \left| \int g_m d\mu_n - \int g_m d\mu \right| + \int |g_m - g| (d\mu_m + d\mu) \leq 2 \cdot \sup |g_m - g|. \quad (5.12)$$

Letting $g_m \rightarrow g$ we see that the LHS must be zero.

Finally, let g be bounded and continuous. For every compact set K , there exists $g_K \in \mathcal{C}_K$ such that g_K has compact support, $g_K \equiv g$ on K and $\sup |g_K| \leq \sup |g|$. By tightness, for every $\varepsilon > 0$, there exists a compact set K such that $\mu_n(K^c), \mu(K^c) < \varepsilon$. Similar to (5.12), and using that $g_K = g$ on K , we have

$$\limsup_{n \rightarrow \infty} \left| \int g d\mu_n - \int g d\mu \right| \leq \sup |g_K - g| \cdot \limsup_{n \rightarrow \infty} (\mu(K^c) + \mu_n(K^c)) \leq 4 \sup |g| \varepsilon.$$

Letting $\varepsilon \downarrow 0$, the limit on the LHS is 0. This proves (5.8).

From Item 2 to Item 3. For every G open, there exists $g_m \geq 0$, bounded and continuous such that $g_n \uparrow \mathbb{1}_G$. For every g_m , by (5.8) we have

$$\liminf_{n \rightarrow \infty} \mu_n(G) \geq \liminf_{n \rightarrow \infty} \int g_m d\mu_n = \int g_m d\mu.$$

The right hand side increases to $\int \mathbb{1}_G d\mu = \mu(G)$ by MCT, so (5.9) holds.

Note that Item 3 and Item 4 are equivalent since K is closed if and only if $G = K^c$ is open, and $\mu(K) = 1 - \mu(K^c)$.

From Items 3 and 4 to Item 5. For any set A , let $\text{int } A$ be the *interior* of A , defined by

$$\text{int } A = \{x \in A : \exists r > 0 \text{ s.t. } B_r(x) \subset A\},$$

and $\bar{A}$ the *closure* of A . Then $\text{int } A \subset A \subset \bar{A}$, and $\partial A = \bar{A} \setminus \text{int } A$. By (5.9) and (5.10),

$$\mu(\text{int } A) \leq \liminf_{n \rightarrow \infty} \mu_n(A) \leq \limsup_{n \rightarrow \infty} \mu_n(A) \leq \mu(\bar{A}).$$

But $\mu(\partial A) = \mu(\bar{A}) - \mu(\text{int } A) = 0$, so $\mu(\text{int } A) = \mu(\bar{A}) = \mu(A)$, and the conclusion follows.

From Item 5 to Item 1. If x_0 is a continuous point of F , then $\mu\{x_0\} = 0 = \mu(\partial(-\infty, x_0))$. Hence,

$$\lim_{n \rightarrow \infty} F_n(x_0) = \lim_{n \rightarrow \infty} \mu_n(-\infty, x_0] = \mu(-\infty, x_0] = F(x_0).$$

□

Weak convergence can also be characterized using metrics on $\mathcal{P}(\mathbb{R})$.

From the proof of [Theorem 5.7](#), (5.8) holds if and only if it holds for all compactly supported continuous functions, denoted by $\mathcal{C}_c(\mathbb{R})$. There is a countable dense subset $(g_m)_{m \geq 1} \subset \mathcal{C}_c(\mathbb{R})$, in the sense that for every $\varepsilon > 0$ and every $g \in \mathcal{C}_c(\mathbb{R})$, there exists g_{m_0} such that $\sup|g_{m_0} - g| < \varepsilon$. One such subset is all the piecewise linear continuous functions, whose graphs are obtained by connecting points $(x_i, g(x_i)) \in \mathbb{Q}^2$. Fixing such a dense subset $(g_m)_{m \geq 1}$, we have $\mu_n \Rightarrow \mu$ if and only if

$$\int g_m d\mu_n \rightarrow \int g_m d\mu, \quad \forall g_m.$$

This means that weak convergence is characterized by the following metric:

$$d(\mu, \nu) = \sum_{m=1}^{\infty} 2^{-m} \left(1 \wedge \left| \int g_m d\mu - \int g_m d\nu \right| \right),$$

that is, $\mu_n \Rightarrow \mu$ if and only if $d(\mu_n, \mu) \rightarrow 0$.

Another useful metric is called the *Lévy distance* between monotone function. To motivate it, let us consider the graph of any increasing function F , defined by

$$\Gamma_F = \{(x, y) : x \in \mathbb{R}, F(x-) \leq y \leq F(x+)\} \subset \mathbb{R}^2.$$

The distinction of continuous and discontinuous points in (5.4) is caused by the potential discontinuous point in F ; but from the point of view of the graphs, (5.4) just means that the graphs of F_n converge to that of F . To make this idea precise, we can use the *Hausdorff distance* to measure closeness between any $A, B \subset \mathbb{R}^2$ ($\mathbb{R}^2$ could be any metric space):

$$\begin{aligned} d_H(A, B) &= \inf\{\varepsilon > 0 : A \subset B_\varepsilon, B \subset A_\varepsilon\}, \quad D_\varepsilon = \bigcup_{x \in D} \{y : |y - x| \leq \varepsilon\}, \\ &= \inf\{\varepsilon > 0 : d(a, B) \leq \varepsilon, \forall a \in A, d(b, A) \leq \varepsilon, \forall b \in B\}, \end{aligned} \tag{5.13}$$

where D_ε is also known as the ε -neighborhood of D . The Lévy distance can be defined by

$$d_L(F, G) = d_H(\Gamma_F, \Gamma_G).$$

The more well-known form of Lévy distance is

$$d_L(F, G) = \inf\{\varepsilon > 0 : F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon\}.$$

The two definitions are exactly the same if we use ℓ^∞ distance in $\mathbb{R}^2$ in (5.13).

We conclude this section by two simple properties of weak convergence.

Proposition 5.8 *If $X_n \rightarrow X$ in probability, then $X_n \Rightarrow X$.*

Proof: It suffices to show that $\text{E}g(X_n) \rightarrow \text{E}g(X)$ for any bounded continuous function g , which follows from DCT. $\square$

The converse is not true, unless the limit is a δ -measure.

Proposition 5.9 *If $X_n \Rightarrow X$ where $\text{P}(X = c) = 1$ for some $c \in \mathbb{R}$, then $X_n \rightarrow c$ in probability.*

Proof: Since $\mu_X\{c - \varepsilon, c + \varepsilon\} = 0$, by Item 5 in Theorem 5.7, we have

$$\lim_{n \rightarrow \infty} \text{P}(|X_n - c| \geq \varepsilon) = \lim_{n \rightarrow \infty} \mu_{X_n}(c - \varepsilon, c + \varepsilon)^c = \mu_X(c - \varepsilon, c + \varepsilon)^c = 0.$$

$\square$

5.3 Characteristic functions

In this section we introduce the *characteristic function (ch.f.)* of a r.v. X , defined by

$$\varphi_X(\xi) = \text{E}e^{i\xi X} = \text{E}\cos(\xi X) + i\text{E}\sin(\xi X).$$

The integration of the complex-valued r.v. $e^{i\xi X}$ can be done by integrating the real and imaginary parts separately, that is, $\text{E}X := \text{E}X_1 + i\text{E}X_2$ if X_1 and X_2 are real and complex parts of X . We also recall the norm of a complex number $z = a + bi$ is $|z| = \sqrt{a^2 + b^2}$. Like absolute values we have the following inequality for complex norms.

Lemma 5.10 *Let $X \in \mathbb{C}$ be a complex-valued r.v. Then $|\text{E}X| \leq \text{E}|X|$.*

Proof: Write $X = X_1 + iX_2$. Since $\phi(a, b) = \sqrt{a^2 + b^2}$ is convex, by Jensen's inequality we have

$$|\text{E}X| = \sqrt{(\text{E}X_1)^2 + (\text{E}X_2)^2} = \phi(\text{E}X_1, \text{E}X_2) \leq \text{E}\phi(X_1, X_2) \leq \text{E}\sqrt{X_1^2 + X_2^2} = \text{E}|X|.$$

$\square$

The ch.f. of a r.v. X is just the *Fourier transform* (up to some constants and signs) of this distribution μ_X . Unsurprisingly, a probability measure is uniquely determined by its Fourier transform, and we will assume this fact without proof in this section.

We have some basic properties of the ch.f.s.

Proposition 5.11 *Let $\varphi(\xi) = \text{E}e^{i\xi X}$. Then*

1. $\varphi(-\xi) = \overline{\varphi(\xi)}$, where $\bar{\cdot}$ denotes complex conjugate.
2. $\text{E}e^{i\xi(aX+b)} = e^{ib\xi}\varphi(a\xi)$.
3. $\varphi(0) = 1$ and $|\varphi(\xi)| \leq 1$.
4. $\xi \mapsto \varphi(\xi)$ is uniformly continuous.

Proof: We will only prove the last one; the others are straightforward.

For any ξ_1 and ξ_2 , we have

$$|\varphi(\xi_1) - \varphi(\xi_2)| \leq \text{E}|e^{i\xi_1 X} - e^{i\xi_2 X}| = \text{E}|e^{i(\xi_1 - \xi_2)X} - 1|.$$

Since $|e^{iy} - 1| \leq 2$ for any $y \in \mathbb{R}$, by BCT, we have

$$\lim_{h \rightarrow 0} \text{E}|e^{ihX} - 1| = \text{E} \lim_{h \rightarrow 0} |e^{ihX} - 1| = \text{E}|e^0 - 1| = 0.$$

Since the upper bound only depends on $\xi_1 - \xi_2$, the conclusion follows. $\square$

Characteristic functions are useful in the study of the sum of independent r.v.s, since the Fourier transform turns convolutions into products.

Proposition 5.12 *Let X, Y be independent. Then $\varphi_{X+Y} = \varphi_X \cdot \varphi_Y$.*

Proof: For every fixed ξ , the function $h(x) = e^{i\xi x}$ is bounded and continuous. **Proposition 2.2** also holds for complex-valued functions, and hence

$$\varphi_{X+Y}(\xi) = \mathbb{E}h(X)h(Y) = \mathbb{E}h(X) \cdot \mathbb{E}h(Y) = \varphi_X(\xi)\varphi_Y(\xi).$$

□

Since $h(x) = e^{i\xi x}$ is a bounded continuous function for every ξ , we know $\varphi_{X_n} \rightarrow \varphi_X$ pointwise if $X_n \Rightarrow X$. The converse is true if $(\mu_{X_n})_{n \geq 1}$ is tight. This is the next result.

Theorem 5.13 (continuity theorem) *If $\varphi_{X_n}(\xi) \rightarrow \varphi(\xi)$ for every ξ , and φ is continuous at $\xi = 0$, then there exists a r.v. X such that $X_n \Rightarrow X$ and $\varphi_X = \varphi$.*

Proof: We first show that the continuity of φ at 0 implies the tightness of $(\mu_{X_n})_{n \geq 1}$. We will use **Lemma 5.14** proven below, which gives

$$\mu_{X_n}\{x : |x| \geq 2/u\} \leq \frac{1}{u} \int_{-u}^u (1 - \varphi_{X_n}(\xi)) d\xi. \quad (5.14)$$

Note that $\varphi(0) = \lim_{n \rightarrow \infty} \varphi_{X_n}(0) = 1$. By continuity, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|1 - \varphi(\xi)| \leq \varepsilon$ when $|\xi| \leq \delta$. Taking $u = \delta$ in (5.14), since $|1 - \varphi_{X_n}| \leq 2$, by BCT we have

$$\begin{aligned} \limsup_{n \rightarrow \infty} \mu_{X_n}\{x : |x| \geq 2/\delta\} &\leq \lim_{n \rightarrow \infty} \frac{1}{\delta} \int_{-\delta}^{\delta} |1 - \varphi_{X_n}(\xi)| d\xi \\ &= \frac{1}{\delta} \int_{-\delta}^{\delta} \lim_{n \rightarrow \infty} |1 - \varphi_{X_n}(\xi)| d\xi \\ &= \frac{1}{\delta} \int_{-\delta}^{\delta} |1 - \varphi(\xi)| d\xi \leq 2\varepsilon. \end{aligned}$$

This implies the tightness of $(\mu_{X_n})_{n \geq 1}$.

Since $(\mu_{X_n})_{n \geq 1}$ is tight, by **Theorem 5.4** there exists a subsequence (X_{n_k}) such that $X_{n_k} \Rightarrow X$ for some r.v. X . Then $\mathbb{E}e^{i\xi X_{n_k}} \rightarrow \mathbb{E}e^{i\xi X}$ and hence $\varphi_X \equiv \varphi$. Next we will show that $X_n \Rightarrow X$ along the full sequence. If not, then there exists f bounded, continuous and $\varepsilon_0 > 0$ such that

$$|\mathbb{E}f(X_{m_k}) - f(X)| \geq \varepsilon_0, \quad \text{along some subsequence } (X_{m_k}). \quad (5.15)$$

Since $\mu_{X_{m_k}}$ are also tight, there exists a further subsequence $\mu_{X_{m'_k}}$ such that $X_{m'_k} \Rightarrow Y$ for some Y . But then we have $\varphi_Y = \varphi = \varphi_X$, this contradicts with (5.15). □

Lemma 5.14 *Let ν be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and φ be its ch.f. Then*

$$\nu\{x : |x| \geq 2/u\} \leq \frac{1}{u} \int_{-u}^u (1 - \varphi(\xi)) d\xi.$$

(Since $\varphi(-\xi) = \overline{\varphi(\xi)}$ and the domain is symmetric, the integral on the right side is real.)

Proof: Using Fubini's Theorem, we have

$$\begin{aligned}
\int_{-u}^u (1 - \varphi(\xi)) d\xi &= \int \nu(dx) \int_{-u}^u (1 - e^{i\xi x}) d\xi \\
&= \int \nu(dx) \int_{-u}^u (1 - \cos \xi x) d\xi \\
&= \int \left(2u - \frac{2 \sin ux}{x}\right) \nu(dx) \\
&= 2u \int \left(1 - \frac{\sin ux}{ux}\right) \nu(dx) \\
&\geq u \int_{\{x: |ux| \geq 2\}} \frac{1}{2} \nu(dx) = u \cdot \nu\{x : |ux| \geq 2\}
\end{aligned}$$

Here, we use $1 - \frac{\sin ux}{ux} \geq 1 - |ux|^{-1} \geq 1/2$ if $|ux| \geq 2$. □

5.4 *Notes on Fourier transform

This section will give a brief introduction to the Fourier transform. The goal is to help the readers to understand characteristic functions in a more general context.

Fourier transform is first defined for functions. The *Fourier transform* of a function $g \in L^1(\mathbb{R})$ is defined by

$$(\mathbb{F}g)(\xi) := \int e^{i\xi x} g(x) dx. \quad (5.16)$$

The integrability condition $g \in L^1(\mathbb{R})$ is to ensure the integral in (5.16) to be defined.

Remark 5.6 In general, one needs to decide where to put constants and plus/minus signs in defining the Fourier transform; for example, more common definitions in harmonic analysis are

$$(\mathbb{F}g)(\xi) = \frac{1}{\sqrt{2\pi}} \int e^{-i\xi x} g(x) dx, \quad \text{or} \quad (\mathbb{F}g)(\xi) = \int e^{-2\pi i \xi x} g(x) dx.$$

But (5.16) agrees with the form of characteristic functions used in the probability theory so we will stick to it.

One can also define the *inverse Fourier transform* by

$$(\mathbb{F}^{-1}h)(x) := \frac{1}{2\pi} \int e^{-i\xi x} h(\xi) d\xi. \quad (5.17)$$

Note that like $\mathbb{F}$, the natural domain for $\mathbb{F}^{-1}$ are functions in $L^1(\mathbb{R})$. However, if $g \in L^1(\mathbb{R})$, then in general we merely have $\mathbb{F}g \in L^\infty(\mathbb{R})$, so $\mathbb{F}^{-1}$ is not a true “inverse” (but it will be after a proper adjustment). When it happens that $\mathbb{F}g \in L^1(\mathbb{R})$, the map $\mathbb{F}^{-1}$ indeed takes $\mathbb{F}g$ back to g . Here, the form of $\mathbb{F}^{-1}$ in (5.17) depends on the choice we made in (5.16) to define $\mathbb{F}$.

Proposition 5.15 *If $g \in L^1(\mathbb{R})$ and $\mathbb{F}g \in L^1(\mathbb{R})$, then $(\mathbb{F}^{-1} \circ \mathbb{F})g = g$.*

The proof usually involves some integration tricks, and can be found in most analysis/PDE textbooks that present the Fourier transform. We skip the proof here since the most important thing for us is to know that the Fourier transform does have an inverse, at least in some sense.

The next question is that we need to define the Fourier transform for objects other than L^1 functions, like the probability measures. One can say that probability measures are like L^1 functions, but

we will see below that the Fourier transform can even be defined for unbounded functions/measures. The key are the “Schwartz space” and its dual space, the “tempered distributions”.

The *Schwartz space* contains smooth functions that decays fast at ∞ ; more precisely,

$$\mathcal{S} = \{g \in \mathcal{C}^\infty(\mathbb{R}) : \lim_{|x| \rightarrow \infty} |x|^k |g^{(m)}(x)| = 0, \forall k, m \geq 0\}.$$

We call functions in $\mathcal{S}$ *Schwartz functions*. We can talk about convergence in $\mathcal{S}$: $g_n \rightarrow g$ in $\mathcal{S}$ if for every $k, m \geq 0$, $\sup_x |x|^k |g_n^{(m)}(x) - g^{(m)}(x)| \rightarrow 0$. The convergence can also be characterized by the metric

$$d(f, g) = \sum_{k, m=0}^{\infty} \frac{|f - g|_{k, m} \wedge 1}{2^{m+k}}, \quad |h|_{k, m} := \sup_x |x|^k |h^{(m)}(x)|.$$

A nice thing about the Fourier transform is that it turns differentiation ∂_x^k into multiplication $(-i\xi)^k$ and vice versa.

Proposition 5.16 *Let $g \in \mathcal{S}$. Then for $k \geq 1$,*

$$(\mathbb{F}g^{(k)})(\xi) = (-i\xi)^k (\mathbb{F}g)(\xi), \quad \mathbb{F}((-ix)^k g) = \mathbb{F}g^{(k)}.$$

Hence, the Schwartz space $\mathcal{S}$ is invariant under $\mathbb{F}$. In light of [Proposition 5.15](#), it is a bijection on $\mathcal{S}$.

Proposition 5.17 *The Fourier transform $\mathbb{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a bijection.*

Another obvious fact is that $\mathbb{F}$ is linear: $\mathbb{F}(f + g) = \mathbb{F}f + \mathbb{F}g$. It is natural to consider the action of $\mathbb{F}$ on the dual of $\mathcal{S}$, called the *tempered distribution*, defined by

$$\begin{aligned} \mathcal{S}' &:= \{\text{continuous, linear functional on } \mathcal{S}\} \\ &= \{\ell \text{ linear} : \mathcal{S} \rightarrow \mathbb{R}, |\ell(g)| \leq C_{m,k} |g|_{k,m}, \forall k, m \geq 0\}. \end{aligned}$$

The space $\mathcal{S}'$ contains all probability measures μ , identified with the linear functional

$$\ell_\mu(g) := \int g(x) d\mu(x).$$

It also contains $\mathcal{S}$ itself, identified with the linear functionals defined by taking L^2 inner product:

$$\ell_h(g) := \int g(x) h(x) dx, \quad h \in \mathcal{S}.$$

The Fourier transform can be defined on $\mathcal{S}'$ by duality:

$$(\mathbb{F}\ell)(g) := \ell(\mathbb{F}g).$$

For example, if μ is a probability measure on $\mathbb{R}$, then by Fubini's Theorem,

$$(\mathbb{F}\mu)(g) = \mu(\mathbb{F}g) = \int \left[\int e^{i\xi x} dx \right] d\mu(\xi) = \int \left[\int e^{i\xi x} d\mu(\xi) \right] g(x) dx = \int \varphi_\mu(x) g(x) dx, \quad \forall g \in \mathcal{S},$$

where φ_μ is the ch.f. of μ . Hence, the ch.f. φ_μ is $\mathbb{F}(\mu)$, when μ is treated as an element in $\mathcal{S}'$. Since $\mathbb{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a bijection, it is also a bijection on $\mathcal{S}'$. Therefore, a probability measure is *uniquely* determined by its ch.f.

If one needs more information, the inversion from ch.f.s to probability measures can also be done via the *inversion formula*, which is more or less equivalent to [Proposition 5.15](#).

Theorem 5.18 (Inversion formula) *Let μ be a probability measure on $\mathbb{R}$ and φ its ch.f. Then for every $a < b$,*

$$\lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-itb} - e^{-ita}}{it} \varphi(t) dt = \mu(a, b) + \frac{1}{2} \mu\{a, b\}.$$

5.5 Central limit Theorem

5.5.1 CLT for i.i.d. random variables

We will use ch.f.s to prove central limit theorems. An important fact is that the derivatives of the ch.f. is related to the moments of the r.v.; there is a more general result for the Fourier transform, see [Proposition 5.16](#).

Proposition 5.19 *If $E|X|^n < \infty$, then $\varphi^{(n)}(\xi) = E(iX)^n e^{i\xi X}$.*

Remark 5.7 Let $g(x, \xi) = e^{i\xi x}$. Then $\frac{\partial^n}{\partial \xi^n} g = (ix)^n e^{i\xi x}$, so [Proposition 5.19](#) gives conditions to guarantee the “exchange of differentiation and integral (expectation)”

$$\frac{d^n}{d\xi^n} E g(X, \xi) = E \frac{\partial^n}{\partial \xi^n} g(X, \xi).$$

Proof: We will only prove the case $n = 1$. For $n \geq 2$, the proof uses induction and a similar argument.

Since $E|X| < \infty$, we can define $h_1(\xi) = E(iX)e^{i\xi X}$ as $|(iX)e^{i\xi X}| \leq |X|$. Also, $|h_1(\xi)| \leq E|X|$, and similar to [Proposition 5.11](#), one can show that $\xi \mapsto h_1(\xi)$ is (uniformly) continuous.

By Fubini's Theorem, for every $a < b$, we have

$$\int_a^b h_1(\xi) d\xi = \int_a^b E(iX)e^{i\xi X} d\xi = E \int_a^b (iX)e^{i\xi X} d\xi = E(e^{ibX} - e^{iaX}) = \varphi(b) - \varphi(a).$$

Since h_1 is continuous, φ is the anti-derivative of h_1 by the fundamental theorem of calculus, that is, $\varphi' = h_1$. $\square$

To deal with complex logarithm we often use the following lemma.

Lemma 5.20 *If $|z| \leq 1/2$, then $|\log(1+z) - z| \leq |z|^2$.*

Proof: The function $\log(1+z)$ is analytic for $|z| < 1$, and hence we have the Taylor expansion

$$\log(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} z^k.$$

Therefore,

$$|\log(1+z) - z| \leq \sum_{k=2}^{\infty} \frac{|z|^k}{k} \leq |z|^2 \sum_{k=2}^{\infty} \frac{1}{2^{k-2}k} \leq |z|^2.$$

$\square$

We say that X has normal distribution $\mathcal{N}(\mu, \sigma^2)$ if X is a continuous r.v. with density

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Let us compute the ch.f. for normal distributions.

Proposition 5.21 *Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then*

$$\varphi_X(\xi) = e^{i\mu\xi - \frac{1}{2}\sigma^2\xi^2}.$$

Proof: By Proposition 5.11, without loss of generality we can assume $\mu = 0$ and $\sigma = 1$. We need to show

$$\int_{\mathbb{R}} e^{i\xi x} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = e^{-\frac{\xi^2}{2}}.$$

Completing the square, the left hand side is

$$\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-i\xi)^2}{2}} \cdot e^{-\frac{\xi^2}{2}} dx,$$

so it suffices to show

$$1 = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-i\xi)^2}{2}} dx. \quad (5.18)$$

If $i\xi$ is a real number, then (5.18) follows from a change of variables, but it is not. We need some contour integral trick from complex variables.

Assume $\xi \geq 0$. Let $h(z) = \frac{1}{2\pi} e^{-z^2/2}$, $z \in \mathbb{C}$. Then $h(z)$ is an entire function (since the exponential function is nice), and by Cauchy integral theorem, $\int_{\Gamma} h(z) dz = 0$ for any closed contour Γ . Consider the contour $\Gamma = \bigcup_{k=1}^4 \Gamma_k$, where

$$\Gamma_1 = [-L, L], \quad \Gamma_2 = \{L + iy : y \in [0, \xi]\}, \quad \Gamma_3 = \{x + i\xi : x \in [-L, L]\}, \quad \Gamma_4 = \{-L + iy : y \in [0, \xi]\}$$

with proper orientation (counter-clockwise). Then

$$\left| \int_{\Gamma_2} h(z) dz \right| \leq \int_0^\xi |h(L + iy)| dy \leq \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(L^2 - \xi^2)} \cdot \xi \rightarrow 0, \quad L \rightarrow \infty,$$

and similar limit holds for Γ_4 . Hence,

$$\lim_{L \rightarrow \infty} \int_{\Gamma_1} h(z) dz = - \lim_{L \rightarrow \infty} \int_{\Gamma_3} h(z) dz. \quad (5.19)$$

The right hand side of (5.19) is right hand side of (5.18), while the left hand side of (5.19) is the integration of the density of $\mathcal{N}(0, 1)$, which is 1. This completes the proof. $\square$

Theorem 5.22 Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_1 = \mu$ and $\text{Var}(X_1) = \sigma^2$. Then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0, 1).$$

Proof: By Theorem 5.13 and Proposition 5.21, it suffices to show that for every $\xi \in \mathbb{R}$,

$$\mathbb{E} e^{i\xi \frac{S_n - n\mu}{\sigma\sqrt{n}}} \rightarrow e^{-\frac{1}{2}\xi^2}, \quad n \rightarrow \infty.$$

Rearranging, the LHS becomes

$$\mathbb{E} e^{i \frac{\xi}{\sqrt{n}} \sum_{m=1}^n \frac{X_m - \mu}{\sigma}} = \mathbb{E} e^{i \frac{\xi}{\sqrt{n}} \sum_{m=1}^n Y_m},$$

where $Y_m = \frac{X_m - \mu}{\sigma}$ is the normalized r.v.s, with $\mathbb{E}Y_m = 0$, $\mathbb{E}Y_m^2 = 1$. The r.v.s Y_m are i.i.d. Let φ be the ch.f. of Y_1 . Then by independence and Proposition 5.12, we need to show

$$\left[\varphi(\xi/\sqrt{n}) \right]^n \rightarrow e^{-\frac{1}{2}\xi^2},$$

or equivalently, since the limit is positive and exponential function is continuous,

$$n \log \varphi(\xi/\sqrt{n}) \rightarrow -\frac{1}{2}\xi^2.$$

By [Proposition 5.19](#), since Y_1 has second moment, its ch.f. φ is twice differentiable, and $\varphi(0) = 1$, $\varphi'(0) = 1$, $\varphi''(0) = -1$. In particular, we have Taylor expansion for φ at 0 with Peano remainder:

$$\varphi(\eta) = 1 - \frac{1}{2}\eta^2 + \eta^2\alpha(\eta), \quad \lim_{\eta \rightarrow 0} |\alpha(\eta)| = 0. \quad (5.20)$$

Note that the $o(1)$ term $\alpha(\eta)$ is complex.

For n large enough, $|\varphi(\xi/\sqrt{n}) - 1| \leq 1/2$, and hence by [Lemma 5.20](#) and [\(5.20\)](#), we have

$$\begin{aligned} \left| n \log \varphi(\xi/\sqrt{n}) + \frac{1}{2}\xi^2 \right| &\leq \left| n \log \varphi(\xi/\sqrt{n}) - n(\varphi(\xi/\sqrt{n}) - 1) \right| + \left| n(\varphi(\xi/\sqrt{n}) - 1) + \frac{1}{2}\xi^2 \right| \\ &\leq n|\varphi(\xi/\sqrt{n}) - 1|^2 + \xi^2|\alpha(\xi/\sqrt{n})|. \end{aligned}$$

The second term converges to 0; the first term is

$$\left| -\frac{1}{2}\xi^2 + \xi^2\alpha(\xi/\sqrt{n}) \right| \cdot |\varphi(\xi/\sqrt{n}) - 1| \leq C|\varphi(\xi/\sqrt{n}) - 1|$$

and also converges to 0. This completes the proof. $\square$

5.5.2 CLT for triangular arrays

The motivation to study CLT for triangular arrays is that normal distributions in real life, such as height, weights and so on, are results of many independent, yet *not identically distributed* small factors. When the r.v.s are not i.i.d., we need more delicate control of the ch.f.s.

The following result is useful.

Proposition 5.23 *Let $h(z) = e^{iz}$ and $P_k(z)$ be the k -th order Taylor polynomials of $h(z)$ at $z = 0$. Then*

$$|\varphi(\xi) - \mathbb{E}P_k(\xi X)| \leq \mathbb{E}\left(\frac{|\xi X|^{k+1}}{(k+1)!} \wedge \frac{2|\xi X|^k}{k!}\right).$$

In particular, when $k = 2$, we have $P_k(z) = 1 + iz - \frac{z^2}{2}$ and

$$|\varphi(\xi) - (1 + i\xi\mathbb{E}X - \frac{\xi^2}{2}\mathbb{E}X^2)| \leq \xi^2\mathbb{E}\left(\frac{|\xi X^3|}{6} \wedge |X|^2\right).$$

Proof: Since $|h^{(k+1)}| \leq 1$, we have

$$|h(\xi X) - P_k(\xi X)| = \left| \int_0^\xi \frac{(iX)^{k+1}\theta^k h^{(k+1)}(\theta X)}{k!} d\theta \right| \leq \frac{|\xi X|^{k+1}}{(k+1)!}. \quad (5.21)$$

The bound is bad if $|\xi X|$ is large. Using [\(5.21\)](#), we also have

$$|h(\xi X) - P_k(\xi X)| \leq |h(\xi X) - P_{k-1}(\xi X)| + \left| \frac{(i\xi X)^k}{k!} \right| \leq 2\frac{|\xi X|^k}{k!}. \quad (5.22)$$

The conclusion follows from combining [\(5.21\)](#) and [\(5.22\)](#), and then taking expectation. $\square$

Theorem 5.24 (Linderburg-Feller) *Let $(X_{n,m})_{m=1}^n$ be independent with $\mathbb{E}X_{n,m} = 0$. Assume that*

$$\sum_{m=1}^n \mathbb{E}X_{n,m}^2 \rightarrow \sigma^2, \quad n \rightarrow \infty, \quad (5.23)$$

and the so-called “Linderburg’s condition”:

$$\forall \varepsilon > 0, \quad M_n := \sum_{m=1}^n \mathbb{E}X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}} \rightarrow 0, \quad n \rightarrow \infty. \quad (5.24)$$

Then $S_n = X_{n,1} + \dots + X_{n,n} \Rightarrow \mathcal{N}(0, \sigma^2)$.

Proof: By independence,

$$\mathbb{E}e^{i\xi S_n} = \prod_{m=1}^n \mathbb{E}e^{i\xi X_{n,m}} =: \prod_{m=1}^n \varphi_{n,m}(\xi).$$

By [Theorem 5.13](#) and [Proposition 5.21](#), it suffices to show that for every $\xi \in \mathbb{R}$,

$$\sum_{m=1}^n \log \varphi_{n,m}(\xi) \rightarrow -\frac{1}{2}\sigma^2\xi^2. \quad (5.25)$$

The idea is to use the approximations $\log \varphi_{n,m}(\xi) \approx \varphi_{n,m}(\xi) - 1 \approx -\mathbb{E}X_{n,m}^2$ and sum over m . To control the accumulated error after all these approximations, we need to use the Linderburg’s condition [\(5.24\)](#) and [Lemma 5.20](#).

By [Proposition 5.23](#), we have

$$|\varphi_{n,m}(\xi) - 1 + \frac{\xi^2}{2}\mathbb{E}X_{n,m}^2| \leq \xi^2 \mathbb{E}(|\xi X_{n,m}^3|) \wedge |X_{n,m}|^2 \leq \xi^2(\varepsilon \mathbb{E}|X_{n,m}^2| + \mathbb{E}X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}})$$

We also have

$$\mathbb{E}X_{n,m}^2 \leq \varepsilon^2 + \mathbb{E}X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}}.$$

Therefore, for some constant $C = C(\xi)$,

$$|\varphi_{n,m}(\xi) - 1|, |\varphi_{n,m}(\xi) - 1 + \frac{\xi^2}{2}\mathbb{E}X_{n,m}^2| \leq C(\varepsilon + \mathbb{E}X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}}) \leq C(\varepsilon + M_n).$$

By first choosing ε small enough and then n large enough, we can ensure that $|\varphi_{n,m}(\xi) - 1| \leq 1/2$ for all m .

Using [Lemma 5.20](#), we have

$$\begin{aligned} \sum_{m=1}^n \left| \log \varphi_{n,m}(\xi) + \frac{\xi^2}{2}\mathbb{E}X_{n,m}^2 \right| &\leq \sum_{m=1}^n \left| \log \varphi_{n,m}(\xi) - \varphi_{n,m}(\xi) + 1 \right| + \sum_{m=1}^n \left| \varphi_{n,m}(\xi) - 1 + \frac{\xi^2}{2}\mathbb{E}X_{n,m}^2 \right| \\ &\leq \sum_{m=1}^n |\varphi_{n,m}(\xi) - 1|^2 + C_1 \sum_{m=1}^n (\varepsilon \mathbb{E}X_{n,m}^2 + \mathbb{E}X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}}). \end{aligned}$$

By [\(5.23\)](#) and [\(5.24\)](#), the lim sup of second term is bounded by $C_2\varepsilon$ as $n \rightarrow \infty$. For the first term, we have

$$\sum_{m=1}^n |\varphi_{n,m}(\xi) - 1|^2 \leq \max_{1 \leq m \leq n} |\varphi_{n,m}(\xi) - 1| \cdot \sum_{m=1}^n |\varphi_{n,m}(\xi) - 1| \leq C(\varepsilon + M_n) \cdot \sum_{m=1}^n |\varphi_{n,m}(\xi) - 1|,$$

which is bounded by $C_3\varepsilon$ since the summation is bounded by

$$\sum_{m=1}^n \left[\frac{\xi^2}{2} \mathbb{E} X_{n,m}^2 + \left| \varphi_{n,m}(\xi) - 1 + \frac{\xi^2}{2} \mathbb{E} X_{n,m}^2 \right| \right]$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n \left| \log \varphi_{n,m}(\xi) + \frac{\xi^2}{2} \mathbb{E} X_{n,m}^2 \right| = 0.$$

Then (5.25) follows from this and (5.23). $\square$

Example 5.8 We can recover Theorem 5.22 from Theorem 5.24.

Let Y_n be i.i.d. with $\mathbb{E} Y_n = 0$ and $\mathbb{E} Y_n^2 = 1$. Let $X_{n,m} = \frac{Y_m}{\sqrt{n}}$. Then (5.23) is satisfied. For (5.24), we have

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n \mathbb{E} X_{n,m}^2 \mathbb{1}_{\{|X_{n,m}| \geq \varepsilon\}} = \lim_{n \rightarrow \infty} \sum_{m=1}^n \mathbb{E} \frac{Y_m^2}{n} \mathbb{1}_{\{|Y_m| \geq \sqrt{n}\varepsilon\}} = \lim_{n \rightarrow \infty} \mathbb{E} Y_1^2 \mathbb{1}_{\{|Y_1| \geq \sqrt{n}\varepsilon\}} = \mathbb{E} \lim_{n \rightarrow \infty} \mathbb{E} Y_1^2 \mathbb{1}_{\{|Y_1| \geq \sqrt{n}\varepsilon\}} = 0,$$

where the exchange of limit and expectation is due to $\mathbb{E} Y_1^2 < \infty$ and DCT.

Example 5.9 Theorem 5.24 can treat the case where the r.v.s are not identically distributed. Note that $\mathbb{E} Y_n^2 \leq C$ along cannot guarantee that $X_{n,m} = \frac{Y_m}{\sqrt{n}}$ satisfies the Linderburg's condition (5.24). A sufficient condition is that Y_n has uniform $(2 + \delta)$ -moment for any $\delta > 0$, that is, $\mathbb{E} |Y_n|^{2+\delta} \leq C$ for some $C > 0$ and $\delta > 0$.

Indeed, for such Y_n , we have

$$\sum_{m=1}^n \mathbb{E} \frac{Y_m^2}{n} \mathbb{1}_{\{|Y_m| \geq \sqrt{n}\varepsilon\}} \leq \sum_{m=1}^n \mathbb{E} \frac{|Y_m|^{2+\delta}}{n \cdot (\sqrt{n}\varepsilon)^\delta} \mathbb{1}_{\{|Y_m| \geq \sqrt{n}\varepsilon\}} \leq n \cdot \frac{C}{n \cdot (\sqrt{n}\varepsilon)^\delta} = \frac{C}{(\sqrt{n}\varepsilon)^\delta} \rightarrow 0, \quad n \rightarrow \infty.$$

Next we use Theorem 5.24 to derive a CLT-type limit theorem for i.i.d. r.v.s with infinite variance. We should mention a result due to Lévy.

Theorem 5.25 Let X_n be i.i.d. and S_n be its partial sum. Then there exist a_n, b_n such that $\frac{S_n - a_n}{b_n} \Rightarrow \mathcal{N}(0, 1)$ if and only if

$$\frac{y^2 \mathbb{P}(|X_1| \geq y)}{\mathbb{E} |X_1|^2 \mathbb{1}_{\{|X_1| \leq y\}}} \rightarrow 0. \quad (5.26)$$

The idea behind (5.26) is that to have normal distribution as the limit, each term in the partial sum cannot be too large, while here the “largeness” is measured by $y^2 \mathbb{P}(|X_1| \geq y)$, compared to the truncated second moment of X_1 .

Let us consider i.i.d. r.v.s X_n where $\mathbb{P}(X_1 < -x) = \mathbb{P}(X_1 > x) = x^{-2}$, $x \geq 1$. We will show that

$$\frac{X_1 + \cdots + X_n}{\sqrt{n \log n}} \Rightarrow \mathcal{N}(0, 1).$$

To apply Theorem 5.25, we need to compute variance. Since $\mathbb{E} X_1^2 = \infty$, we need to apply truncation first. Let $Y_{n,m} = X_m \mathbb{1}_{\{|X_m| \leq c_n\}}$ and $\tilde{S}_n = Y_{n,1} + \cdots + Y_{n,n}$. We first prove a simple result, saying that if the truncation does not affect S_n much, then it will not affect the weak convergence.

Proposition 5.26 If $\mathbb{P}(S_n \neq \tilde{S}_n) \rightarrow 0$ and $\frac{\tilde{S}_n}{b_n} \Rightarrow \mathcal{N}(0, 1)$, then $\frac{S_n}{b_n} \Rightarrow \mathcal{N}(0, 1)$.

Proof: Let g be a bounded continuous function and $N \sim \mathcal{N}(0, 1)$. We have

$$\begin{aligned} \left| g\left(\frac{S_n}{b_n}\right) - g(N) \right| &\leq \left| g(S_n/b_n) - g(\tilde{S}_n/b_n) \right| + \left| g(\tilde{S}_n/b_n) - g(N) \right| \\ &\leq 2 \sup |g| \cdot \mathbb{P}(S_n \neq \tilde{S}_n) + \left| g(\tilde{S}_n/b_n) - g(N) \right| \rightarrow 0. \end{aligned}$$

□

To have $\mathbb{P}(\tilde{S}_n \neq S_n) \rightarrow 0$, a sufficient condition is

$$\sum_{m=1}^n \mathbb{P}(Y_{n,m} \neq X_m) = n \mathbb{P}(|X_1| \geq c_n) = \frac{n}{c_n^2} \rightarrow 0.$$

We will choose $c_n = n^{1/2} \log \log n$. The reason for double logarithm will be clear in a moment.

Now let us verify the two conditions (5.23) and (5.24).

For (5.23), we have

$$\begin{aligned} \sum_{m=1}^n \mathbb{E} Y_{n,m}^2 &= n \int_1^{c_n} 2y \mathbb{P}(|X_1| \geq y) dy = n \int_1^{c_n} \frac{2dy}{y} \\ &= 2n \log(n^{1/2} \log \log n) = n \log n + 2n \log \log \log n. \end{aligned}$$

So

$$\sum_{m=1}^n \mathbb{E} \left| \frac{Y_{n,m}}{\sqrt{n \log n}} \right|^2 \rightarrow 1.$$

For (5.24), we have

$$\sum_{m=1}^n \mathbb{E} \left| \frac{Y_{n,m}}{\sqrt{n \log n}} \right|^2 \mathbb{1}_{\{|Y_{n,m}| \geq \varepsilon \sqrt{n \log n}\}} = 0$$

for large n , since $|Y_m| \leq c_n = n^{1/2} \log \log n \ll \sqrt{n \log n}$.

5.5.3 Multidimensional CLT

In this section we discuss how to generalize the CLT to $\mathbb{R}^d$. First we need to introduce the weak convergence and the ch.f. in $\mathbb{R}^d$.

We write $X_n = (X_{n,1}, \dots, X_{n,d}) \in \mathbb{R}^d$ for i.i.d. random vectors in $\mathbb{R}^d$. We say that $X_n \Rightarrow X$ if $\mathbb{E}g(X_n) \rightarrow \mathbb{E}g(X)$ for every bounded continuous $g : \mathbb{R}^d \rightarrow \mathbb{R}$. A family of probability measures on $\mathbb{R}^d$, $(\mu_n)_{n \geq 1}$, are *tight*, if for every $\varepsilon > 0$, there exists a compact set K so that $\mu_n(K^c) \leq \varepsilon$ for all $n \geq 1$. Since compact sets in $\mathbb{R}^d$ are bounded and closed, we can take $K = [-L, L]^d$ in this case.

Let $X \in \mathbb{R}^d$ be a random vector. Its characteristic function φ_X is

$$\varphi_X(\xi) = \mathbb{E} e^{i\xi \cdot X}, \quad \xi \in \mathbb{R}^d.$$

Here, $\cdot$ denotes the inner/dot product in $\mathbb{R}^d$: $\xi \cdot x := \xi_1 x_1 + \dots + \xi_d x_d$. We have a version of [Theorem 5.13](#) in $\mathbb{R}^d$.

Theorem 5.27 *Let X_n, X_∞ be random vectors in $\mathbb{R}^d$. Then $X_n \Rightarrow X_\infty$ if and only if $\varphi_{X_n}(\xi) \rightarrow \varphi_{X_\infty}(\xi)$ for every $\xi \in \mathbb{R}^d$.*

Proof: The “only if” part follows from the definition of weak convergence and that $x \mapsto e^{i\xi \cdot x}$ is bounded continuous.

For the “if” part, similar to the proof of [Theorem 5.13](#), it suffices to show that $(\mu_n = \mu_{X_n})$ is tight, and we can use a subsequence argument to finish the proof.

Let e_k be the unit vector in the k -th direction. Then $(e_k \cdot X_n)_{n \geq 1}$ is a family of r.v.s, with ch.f.s

$$\varphi_n^{(k)}(\eta) = \mathbb{E} e^{i\eta e_k \cdot X_n} = \varphi_{X_n}(\eta e_k) \rightarrow \varphi_{X_\infty}(\eta e_k),$$

where the limit is the ch.f. of $e_k \cdot X_\infty$ and hence continuous at $\eta = 0$. By [Theorem 5.13](#), the distribution of $e_k \cdot X_n$ is tight, namely, for $\varepsilon/d > 0$, there exists $L_k > 0$ such that

$$\mu_n\{e_k \cdot X_n \notin [-L_k, L_k]\} \leq \frac{\varepsilon}{d}, \quad \forall n \geq 1.$$

Do this for every $k \in \{1, \dots, d\}$, and let $L = \max\{L_1, \dots, L_d\}$, we have

$$\mu_n\{X_n \notin [-L, L]^d\} \leq \sum_{k=1}^d \mu_n\{e_k \cdot X_n \notin [-L_k, L_k]\} \leq d \cdot \frac{\varepsilon}{d} = \varepsilon,$$

and hence $(\mu_n)_{n \geq 1}$ is tight. $\square$

In fact in the proof we have characterize weak convergence in $\mathbb{R}^d$ via weak convergence in $\mathbb{R}$.

Proposition 5.28 *The weak convergence $X_n \Rightarrow X$ holds for random vectors in $\mathbb{R}^d$ if and only if $\theta \cdot X_n \Rightarrow \theta \cdot X$ as r.v.s for every $\theta \in \mathbb{R}^d$.*

We are ready to give a multidimensional version of CLT.

Theorem 5.29 (CLT in $\mathbb{R}^d$) *Let X_n be i.i.d. random vectors in $\mathbb{R}^d$, with $\mathbb{E}X_1 = \mu \in \mathbb{R}^d$, and covariance matrix $\Gamma = \mathbb{E}(X_1 - \mu) \cdot (X_1^T - \mu)$, that is,*

$$\Gamma_{jk} = \text{Cov}(X_{1,j}, X_{1,k}), \quad 1 \leq j, k \leq d.$$

Then

$$\mathbb{E} e^{i \frac{S_n - n\mu}{\sqrt{n}} \cdot \xi} \rightarrow e^{-\frac{1}{2} \xi^T \Gamma \xi}, \quad \xi \in \mathbb{R}^d,$$

where $e^{-\frac{1}{2} \xi^T \Gamma \xi}$ is the ch.f. of the multi-variate normal distribution $\mathcal{N}(0, \Gamma)$.

Proof: Assume $\mu = 0$. Let $N \sim \mathcal{N}(0, \Gamma)$. Then $\theta \cdot N \sim \mathcal{N}(0, \theta^T \Gamma \theta)$. We have

$$e^{i \frac{S_n}{\sqrt{n}} \cdot \theta} = e^{i \frac{\eta}{\sqrt{n}} \sum_{m=1}^n (X_m \cdot \theta)} \rightarrow e^{-\frac{\eta^2}{2} \cdot \theta^T \Gamma \theta}$$

by [Theorem 5.22](#) and

$$\mathbb{E}(\theta \cdot X_1)^2 = \mathbb{E} \theta^T X_1 X_1^T \theta = \theta^T \Gamma \theta.$$

This and [Proposition 5.28](#) prove the theorem. $\square$

To allow possible degeneracy, the most convenient way to define the multi-variate normal $\mathcal{N}(\mu, \Gamma)$ is to use the ch.f.:

$$\varphi_{\mathcal{N}(\mu, \Gamma)}(\xi) = e^{i\xi \cdot \mu - \frac{1}{2} \xi^T \Gamma \xi}.$$

Note that the covariance matrix Γ is always semi-positive definite symmetric. If all eigenvalues of Γ are positive, then $\mathcal{N}(0, \Gamma)$ has a density given by

$$\frac{1}{(\sqrt{2\pi})^d \sqrt{\det(\Gamma)}} e^{-\frac{(x-\mu)^T \Gamma^{-1} (x-\mu)}{2}}.$$

In general, one can diagonalize Γ to get $\Gamma = \sum_{k=1}^d \lambda_k v_k^T v_k$ where $\vec{v}_k$ form an orthonormal basis in $\mathbb{R}^d$ and $\lambda_k \geq 0$. Then $\mathcal{N}(0, \Gamma)$ can be realized as

$$N = \sum_{k=1}^d \lambda_k \varepsilon_k \vec{v}_k,$$

where ε_k are i.i.d. $\mathcal{N}(0, 1)$ r.v.s. Another way to define multi-variate normal is by projection: we say that $N \sim \mathcal{N}(\mu, \Gamma)$ if $\theta \cdot N \sim \mathcal{N}(\theta \cdot \mu, \theta^T \Gamma \theta)$ for every $\theta \in \mathbb{R}^d$.

6 Weak convergence on general spaces and functional CLT

6.1 Preliminaries for probability measures on metric spaces

Let (M, d) be a generic metric space, where $d : M \times M \rightarrow [0, \infty)$ is the metric. Examples include:

- $M = \mathbb{R}^d$, the d -dimensional Euclidean space, with $d(x, y) = |x - y|_p$, $p \in [1, \infty]$.
Here, all ℓ_p norms are *equivalent*, that is, they generate the same open sets.
- $M = \mathcal{C}[0, 1]$, the space of continuous function on $[0, 1]$, with $d(x, y) = \sup_{t \in [0, 1]} |x(t) - y(t)|$.
- $M = L^p(\Omega, \mathcal{F}, \mu)$, the space of L^p -functions, with $d_{L^p}(f - g) = \int |f(x) - g(x)|^p \mu(dx)$.

We first recall some basic concepts for metric spaces.

- **Open sets.** A set $G \subset M$ is *open* if G contains an ε -ball around every $x \in G$, that is, $\forall x \in G$, $\exists \varepsilon > 0$ s.t. $y \in G$ whenever $d(y, x) < \varepsilon$.
- **Closed set.** A set $F \subset M$ is *closed* if F^c is open.
- **Completeness.** The space M is said to be *complete* if every Cauchy sequence $(x_n) \subset M$ has a limit point in M , that is, if $\lim_{N \rightarrow \infty} \sup_{n, m \geq N} d(x_n, x_m) = 0$, then there exists $x_\infty \in M$ such that $\lim_{n \rightarrow \infty} x_n = x_\infty$.

Most metric spaces are complete, like $\mathbb{R}^d$, L^p and $\mathcal{C}[a, b]$. If starting from a non-complete metric space, one can always *complete* it using Cauchy sequences, as one did in completing $\mathbb{Q}$ to get $\mathbb{R}$.

- **Compact sets.** A set $K \subset M$ is *compact*, if every open cover $\bigcup_{i \in I} G_i \supset K$ contains a finite subcover $G_{i_1} \cup \dots \cup G_{i_n} \supset K$.
- **Separable.** The space M is called separable if there exists a countable dense subset D . We say that D is dense in M , if for every $\varepsilon > 0$ and $x \in M$, there exists $y \in D$ such that $d(x, y) < \varepsilon$.

The Euclidean space $\mathbb{R}^d$ is separable by taking $D = \mathbb{Q}^d$.

The space $\mathcal{C}[0, 1]$ is separable by taking $D = \bigcup_{n=1}^\infty D_n$, where

$$D_n = \left\{ x \in \mathcal{C}[0, 1] : x(t) \in \frac{1}{n}\mathbb{Z}, t \in \frac{1}{n}\mathbb{Z}, \text{ and linear on } [i/n, (i+1)/n] \right\}.$$

The space $L^p(\mathbb{R}^d)$ is separable when $p \neq \infty$.

Another class of non-separable spaces are linear operators on Banach spaces. For example, all the bounded linear map from $L^p(\mathbb{R})$ ($p \in [1, \infty)$) into itself, equipped with the operator norm, is non-separable, even if $L^p(\mathbb{R})$ is separable. To see this, the translation operators $(\tau_s f)(x) = f(s + x)$

are bounded and linear on $L^p(\mathbb{R})$, while $\|\tau_s - \tau_{s'}\| = 1$ whenever $s \neq s'$, so $(\tau_s)_{s \in \mathbb{R}}$ cannot be close to a countable set.

One way to use compactness and separability is as follows: if M is a complete, separable, compact metric space, then $\mathcal{C}(M)$ is compact.

- **Sequentially compact.** A set K is *sequentially compact*, if every sequence $(x_n) \subset K$ has a subsequence (x_{n_k}) such that $x_{n_k} \rightarrow x_*$ for some $x_* \in K$.

On separable metric spaces, compactness is equivalent to sequentially compactness, so we do not distinguish between them hereafter.

On $\mathbb{R}^d$, compact sets are bounded, closed sets.

On $\mathcal{C}[0, 1]$, by Arzelà–Ascoli, $(x_n)_{n \in I}$ are sequentially compact if and only if they are *uniformly bounded*,

$$\exists C > 0, \text{ s.t. } \sup_{n \in I} \sup_{t \in [0, 1]} |x_n(t)| \leq C, \quad (6.1)$$

and *equi-continuous*,

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } |x_n(t_1) - x_n(t_2)| \leq \varepsilon, \quad \forall n \in I, \forall |t_1 - t_2| < \delta. \quad (6.2)$$

Assuming (6.2), (6.1) can also be replaced by

$$\exists C > 0, \text{ s.t. } \sup_{n \in I} |x_n(0)| \leq C, \quad (6.3)$$

Let μ_n, μ_∞ be probability measures on (M, d) . We say that μ_n converges to μ_∞ weakly, denoted by $\mu_n \Rightarrow \mu_\infty$, if for every bounded continuous function $g : M \rightarrow \mathbb{R}$,

$$\int_M g d\mu_n \rightarrow \int_M g d\mu.$$

We say that $(\mu_n)_{n \in I}$ is *tight*, if for every $\varepsilon > 0$, there exists a compact set $K \subset M$ such that

$$\mu_n(K^c) \leq \varepsilon, \quad n \in I.$$

We have seen these definitions for $M = \mathbb{R}$. For general metric spaces we have the following result.

Theorem 6.1 (Prohorov Theorem) *Let (M, d) be a separable and complete metric space (a.k.a. Polish space). Let $(\mu_n)_{n \geq 1}$ be probability measures on (M, d) . If (μ_n) is tight, then there exists a subsequence (μ_{n_k}) and μ_∞ such that $\mu_{n_k} \Rightarrow \mu_\infty$.*

6.2 Donsker invariance principle

Let X_n be i.i.d. r.v.s with $\mathbb{E}X_1 = 0$ and $\mathbb{E}X_1^2 = 1$. We can extend the partial sum $S_n = X_1 + \dots + X_n$ to a continuous function on $[0, 1]$ by defining

$$\tilde{S}_n(t) = \begin{cases} S_m, & t = \frac{m}{n}, m = 0, 1, \dots, n, \\ \text{linear}, & \frac{m}{n} < t < \frac{m+1}{n}. \end{cases}$$

Then $\frac{\tilde{S}_n}{\sqrt{n}}$ is a random element in $\mathcal{C}[0, 1]$.

Theorem 6.2 (Donsker's Invariance Principle/functional CLT) *The law of $\frac{\tilde{S}_n}{\sqrt{n}}$, as a probability distribution on $\mathcal{C}[0, 1]$, converges weakly to the Wiener measure, the law of the Brownian motion.*

In this section we will not rigorously define what is the Brownian motion, or the Wiener measure, as it is a large subject studied in details in stochastic analysis. We will be content with establishing the tightness of the law of $\frac{\tilde{S}_n}{\sqrt{n}}$, and have a better understanding of the central limit theorem.

First we want to reformulate the equi-continuity condition (6.2). For $x \in \mathcal{C}[0, 1]$ and $\delta > 0$, we define the *modules of continuity* of x at δ by

$$\omega(x; \delta) = \sup_{|t_1 - t_2| < \delta} |x(t_1) - x(t_2)|.$$

Then (6.2) is equivalent to

$$\lim_{\delta \rightarrow 0} \sup_n \omega(x_n, \delta) = 0. \quad (6.4)$$

Proposition 6.3 *Let μ_n be the law of random elements $x_n \in \mathcal{C}[0, 1]$. The $(\mu_n)_{n \geq 1}$ is tight if and only if for every $\varepsilon > 0$, there exists $C > 0$ such that*

$$\mu_n\{|x_n(0)| > C\} < \varepsilon, \quad \forall n \geq 1, \quad (6.5)$$

and for every $\eta > 0$, there exists $\delta > 0$ such that

$$\mu_n\{\omega(x_n, \delta) > \eta\} < \varepsilon, \quad \forall n \geq 1. \quad (6.6)$$

Proof: The “only if” part is simple. We will prove the “if” part.

Let $\varepsilon > 0$. By (6.5), there exists $C > 0$ such that

$$\mu_n(A_0) \geq 1 - \varepsilon/2, \quad n \geq 1, \quad A_0 = \{x : |x(0)| < C\}.$$

By (6.6), there exist δ_k for all $k \geq 1$ such that

$$\mu_n(A_k) \geq 1 - \varepsilon/2^{k+1}, \quad n \geq 1, \quad A_k = \{x : \omega(x, \delta_k) < \frac{1}{k}\}.$$

Now let $A = \bigcap_{k=0}^{\infty} A_k$ and let $\bar{A}$ be its closure. By subadditivity, for all $n \geq 1$ we have

$$\mu_n(\bar{A}) \geq \mu_n(A) \geq 1 - \sum_{k=0}^{\infty} \mu_n(A_k^c) \geq 1 - \varepsilon.$$

We will establish the tightness, by showing that $\bar{A}$ is a compact set in $\mathcal{C}[0, 1]$.

It suffices to check the two conditions (6.3) and (6.4) on A . Let $x \in A$. We have $|x(0)| \leq C$ since $x \in A_0$. Since $x \in A_k$, $\omega(x, \delta_k) \leq \frac{1}{k}$ for all k . Since $\omega(x, \delta)$ is decreasing in δ , we have

$$\lim_{\delta \rightarrow 0} \sup_{x \in A} \omega(x, \delta) \leq \lim_{k \rightarrow \infty} \sup_{x \in A} \frac{1}{k} = 0.$$

□

Now we will use Proposition 6.3 to show that $\mu_n = \mathcal{L}(\frac{\tilde{S}_n(\cdot)}{\sqrt{n}})$ is tight.

(6.5) holds since $\tilde{S}_n(0) \equiv 0$.

For (6.6), let $\eta, \delta > 0$, and let m be such that $\frac{m}{2n} < \delta \leq \frac{m}{n}$. We claim that

$$\mu_n\{\omega(x, \delta) \geq \eta\} \leq \sum_{k=0}^{\lfloor n/m \rfloor} \mu_n\left(\max_{km \leq \ell \leq (k+1)m} \left| \frac{S_\ell - S_{km}}{\sqrt{n}} \right| \geq \eta/3\right). \quad (6.7)$$

Indeed, let us decompose $[0, 1]$ into intervals $[0, m/n], [m/n, 2m/n], \dots, [(n-1)/n, 1]$. Then for $|t_1 - t_2| < \delta \leq m/n$, the points t_1 and t_2 either fall into the same interval, or into two adjacent intervals, or otherwise their distance will be larger than m/n which is impossible. On the union of the events at the RHS of (6.7), if $t_1, t_2 \in [km, (k+1)/n]$, then

$$|\tilde{S}_n(t_1) - \tilde{S}_n(t_2)| \leq |\tilde{S}_n(t_1) - S_k| + |\tilde{S}_n(t_2) - S_k| \leq \frac{2\sqrt{n}\eta}{3},$$

or if $km \leq t_1 < (k+1)m \leq t_2 \leq (k+2)m$,

$$|\tilde{S}_n(t_1) - \tilde{S}_n(t_2)| \leq |\tilde{S}_n(t_1) - S_k| + |S_{k+1} - S_k| + |\tilde{S}_n(t_2) - S_{k+1}| \leq \sqrt{n}\eta.$$

Now let us continue (6.7). Since X_i are i.i.d., all the summands in the sum are the same and equal to the first one. We have

$$\begin{aligned} \mu_n\{\omega(x, \delta) \geq \eta\} &\leq \frac{2}{\delta} \mu_n\left(\max_{\ell \leq m} \frac{|S_\ell|}{\sqrt{n}} \geq \eta/3\right) \\ &\leq \frac{2}{\delta} \mu_n\left(\max_{\ell \leq m} \frac{|S_\ell|}{\sqrt{m}} \geq \eta/3\sqrt{\delta}\right) \\ &= \frac{C}{\lambda^2} \mathbf{P}\left(\max_{\ell \leq m} |S_\ell| \geq \lambda\sqrt{m}\right), \end{aligned}$$

where $\lambda = \eta/3\sqrt{\delta}$. It remains to show for every $\varepsilon > 0$, there exists $\lambda > 0$ such that

$$\limsup_{m \rightarrow \infty} \lambda^2 \mathbf{P}\left(\max_{\ell \leq m} |S_\ell| \geq \lambda\sqrt{m}\right) \leq \varepsilon. \quad (6.8)$$

Note that by Kolmogorov's maximal inequality [Proposition 4.17](#), at best we have

$$\lambda^2 \mathbf{P}\left(\max_{\ell \leq m} |S_\ell| \geq \lambda\sqrt{m}\right) \leq \text{Var}(S_m)/m = \mathbf{E}X_1^2,$$

which cannot be made arbitrarily small.

We will use the following improvement of [Proposition 4.17](#).

Lemma 6.4 *Let X_m be independent with $\mathbf{E}X_m = 0$ and $D_n^2 = \text{Var}(S_n)$. Then*

$$\mathbf{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda D_n\right) \leq 2\mathbf{P}\left(|S_n| \geq (\lambda - \sqrt{2})D_n\right). \quad (6.9)$$

Let us postpone the proof of [Lemma 6.4](#) and see first why it is helpful. If applying Chebyshev's inequality on the RHS of (6.9), it is not better than Kolmogorov's inequality. However, if X_m are i.i.d., then by the central limit theorem, $S_n/\sqrt{n} \Rightarrow \mathcal{N}(0, 1)$, and hence

$$\limsup_{n \rightarrow \infty} \frac{1}{\lambda^2} \mathbf{P}\left(|S_n| \geq (\lambda - \sqrt{2})\sqrt{n}\right) = \frac{2}{\lambda^2} \int_{(\lambda - \sqrt{2})}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \leq C \frac{1}{\lambda^2} e^{-\frac{(\lambda - \sqrt{2})^2}{2}}.$$

And (6.8) follows.

Proof of Lemma 6.4: Let $T = \inf\{T : |S_k| \geq \lambda\sqrt{n}D_n\}$. Then we have

$$\begin{aligned} P(T \leq n) &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \sum_{k=1}^{n-1} P(T = k, S_n < (\lambda - \sqrt{2})D_n) \\ &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \sum_{k=1}^n P(T = k, |S_n - S_k| > \sqrt{2}D_n). \end{aligned}$$

Since $\{T = k\} \in \sigma(X_1, \dots, X_k)$ and $|S_n - S_k| \in \sigma(X_{k+1}, \dots, X_n)$, they are independent, so we can continue to get

$$\begin{aligned} P(T \leq n) &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \sum_{k=1}^n P(T = k)P(|S_n - S_k| > \sqrt{2}D_n) \\ &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \sum_{k=1}^n P(T = k) \frac{1}{2D_n^2} \text{Var}(S_n - S_k) \\ &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \sum_{k=1}^n P(T = k) \frac{1}{2} \\ &\leq P(S_n > (\lambda - \sqrt{2})D_n) + \frac{1}{2}P(T \leq n). \end{aligned}$$

Rearranging the terms, we obtain (6.9). □

Finally, let us mention that any sequential limit of μ_n has the same finite dimensional distribution, and hence the limit point is unique. Indeed, consider the bounded, continuous function

$$F(x) := \exp\left(i(\xi_1 x(t_1) + \xi_2(x(t_2) - x(t_1)) + \dots + \xi_m(x(t_m) - x(t_{m-1})))\right),$$

where $\xi_k \in \mathbb{R}$ and $0 \leq t_1 < \dots < t_m \leq 1$. Then by the functional CLT, $\int F(x) d\mu_n$ converge. On the other hand, by CLT, $\int F(x) d\mu_n$ as the ch.f. of the random vector $(x_n(t_1), x_n(t_2) - x_n(t_1), \dots, x_n(t_m) - x_n(t_{m-1}))$ will converge to the ch.f. of $\mathcal{N}(0, \text{diag}\{t_1, t_2 - t_1, \dots, t_m - t_{m-1}\})$. The characterizes the f.d.d. of the Brownian motion.

6.3 *Tightness and weak-* convergence

(to be completed)

7 Poisson limit theorem and stable laws

After studying the CLT, one may wonder why is the normal distribution is so special, and what will happen if the i.i.d. r.v.s do not have second moment. In this section, we will investigate this problem.

As a prototype, we consider X_n i.i.d. with $P(|X_1| > x) \sim x^{-\alpha}$ for some $\alpha < 2$. Note that $\alpha < 2$ implies that $E|X_1|^2 = \infty$ since

$$E|X_1|^2 = \int_0^\infty 2yP(|X_1| \geq y) dy.$$

We want to study the weak limit of $\frac{S_n - b_n}{a_n}$, where a_n, b_n are properly chosen so that the limiting distribution is non-degenerate. Motivated by the functional CLT, we can be more ambitious by asking what is the function limit

$$L(t) = \lim_{n \rightarrow \infty} \frac{S_{[nt]} - b_n}{a_n}$$

as a random function of t .

The answer is that when $\mathbb{E}|X_1|^2 = \infty$, the function $L(t)$ is no longer continuous. If we record the locations, t , and the sizes, ℓ , of all the jumps, then we get a random point process in the (t, ℓ) -plane. This point process is a so-called *Poisson Point Process*, which enjoys the maximal degree of independence of all point processes.

7.1 Poisson limit theorem and Poisson point processes

Recall that $X \sim \text{Poi}(\lambda)$ if $\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$, $k \geq 0$. Here, the probabilities sum up to one thanks to the Taylor expansion $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$. The ch.f. of X is then given by

$$\mathbb{E} e^{i\xi X} = \sum_{k=0}^{\infty} (e^{i\xi})^k \frac{\lambda^k}{k!} e^{-\lambda} = e^{\lambda(e^{i\xi} - 1)}. \quad (7.1)$$

The Poisson distribution models the cumulative effect of many rare events, as the following result shows.

Theorem 7.1 (Poisson Limit Theorem) *Let $X_{n,m} \sim \text{Ber}(p_{n,m})$ be independent. Assume that*

$$\sum_{m=1}^n p_{n,m} \rightarrow \lambda, \quad n \rightarrow \infty,$$

and

$$\max_{1 \leq m \leq n} |p_{n,m}| \rightarrow 0, \quad n \rightarrow \infty. \quad (7.2)$$

Then $S_n := X_{n,1} + \dots + X_{n,n} \Rightarrow \text{Poi}(\lambda)$.

Proof: We will prove this by computing the ch.f.

By independence, we have

$$\mathbb{E} e^{i\xi S_n} = \prod_{m=1}^n [(1 - p_{n,m}) + p_{n,m} e^{i\xi}] = \prod_{m=1}^n [1 + p_{n,m}(e^{i\xi} - 1)].$$

By (7.2), $|p_{n,m}(e^{i\xi} - 1)| \leq 1/2$ for large n . Using Lemma 5.20, we have

$$\left| \log \mathbb{E} e^{i\xi S_n} - \sum_{m=1}^n p_{n,m}(e^{i\xi} - 1) \right| \leq \sum_{m=1}^n |p_{n,m}(e^{i\xi} - 1)|^2 \leq \max |p_{n,m}| \cdot \sum_{m=1}^n p_{n,m} \rightarrow 0.$$

This completes the proof. $\square$

Next we define the Poisson point process on the measurable space $(H, \mathcal{H}) = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. In the sequel the cases $d = 1$ or 2 are most relevant to us.

A *Poisson point process (PPP)* on $(H, \mathcal{H})$ is a random counting measure ν on H , that is, for every $C \in \mathcal{H}$, $\nu(C)$ is a r.v. taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$. The quantity $\nu(C)$ counts how many points fall into the set C , since the location of points are random, $\nu(C)$ is also random. In addition, the distribution of all $\nu(C)$ is characterized by the following.

1. For every $C \in \mathcal{H}$, either $\mathbb{E}\nu(C) = \infty$, or $\nu(C) \sim \text{Poi}(\mathbb{E}\nu(C))$. We define $\lambda(C) := \mathbb{E}\nu(C)$. Then $\lambda(C)$ is a deterministic measure. We call λ the *intensity* of the PPP ν .
2. For disjoint $C_1, \dots, C_n \in \mathcal{H}$, the r.v.s $\nu(C_1), \dots, \nu(C_n)$ are independent.

Example 7.1 (Poisson process) Let $H = \mathbb{R}$ and λ be the Lebesgue measure on $\mathbb{R}$. The process $N_t = \nu(0, t]$ contains all the information of the PPP ν , and we call N_t a *Poisson process*. The point process models the arrival times of customers, where the waiting time for the next customer are i.i.d. $\text{Exp}(1)$ r.v.s.

Example 7.2 (Compound Poisson) Let Y_n be i.i.d. and N_t be the Poisson process, independent of all Y_n . The r.v.

$$Z_t = \sum_{m=1}^{N_t} Y_m$$

is called a *compound Poisson*. Note that Z_t can be represented as an integral against a PPP on $\mathbb{R}^2$:

$$Z_t = \int_{[0,t] \times \mathbb{R}} \ell \nu(dtd\ell),$$

where ν is a PPP on $\mathbb{R}^2$ with intensity $\lambda = \text{Leb} \otimes \mu_Y$.

Example 7.3 We can further generalize the compound Poisson as follows. Let ν be a PPP with intensity λ , and let f be a measurable function such that $\int_H |f(z)| \lambda(dz) < \infty$. Then we can study the r.v.

$$Z = \int f(z) \nu(dz). \quad (7.3)$$

7.2 stable law limit theorem

Let $X_n \geq 0$ be i.i.d. with $\mathbb{P}(X_1 > x) \sim x^{-\alpha}$ where $\alpha < 2$.

Proposition 7.2 Let $0 < a < b$. Then

$$\#\{1 \leq m \leq n : X_m/n^{1/\alpha} \in (a, b)\} \Rightarrow \text{Poi}(a^{-\alpha} - b^{-\alpha}). \quad (7.4)$$

Proof: The LHS of (7.4) can be written as sum of i.i.d. Bernoulli random variables, $\sum_{m=1}^n \xi_{n,m}$, where

$$\xi_{n,m} = \mathbb{1}_{\{X_m/n^{1/\alpha} \in (a,b)\}} \sim \text{Ber}(p_n),$$

and

$$p_n = \mathbb{P}(X_1/n^{1/\alpha} \in (a, b)) \sim (a^{-\alpha} - b^{-\alpha})n^{-1}.$$

The conclusion then follows from **Theorem 7.1**. □

In fact, a much stronger statement holds. Consider the rescaled point process given by

$$\nu_n = \sum_{m=1}^n \delta_{(\frac{m}{n}, \frac{X_m}{n^{1/\alpha}})}. \quad (7.5)$$

Proposition 7.2 says that for every rectangle $R = (a, b) \times [0, 1]$, $\nu_n(R) \Rightarrow \text{Poi}(\lambda(R))$, where

$$\lambda(dtd\ell) = \text{Leb} \otimes (\mathbb{1}_{\ell > 0} \alpha \ell^{-\alpha+1} d\ell). \quad (7.6)$$

One can show that the $\nu_n \Rightarrow$ a PPP with intensity λ given in (7.6).

With the representation (7.5), we can express $\frac{S_{[nt]}}{n^{1/\alpha}}$ as

$$\frac{S_{[nt]}}{n^{1/\alpha}} = \int_{[0,t] \times \mathbb{R}} \ell \nu_n(dtd\ell).$$

In particular, the limit of $\frac{S_n}{\sqrt{n^{1/\alpha}}}$ should be related to

$$\int_{[0,1] \times \mathbb{R}} \ell \nu(dtd\ell), \quad (7.7)$$

where ν is the PPP with intensity (7.6). This is a r.v. taking the form (7.3).

If we consider a more general tail condition

$$\mathbb{P}(X_1 > x) \sim \theta x^{-\alpha}, \quad \mathbb{P}(X_1 < -x) \sim (1 - \theta)x^{-\alpha}, \quad (7.8)$$

then the intensity of the corresponding PPP will be $\lambda = \text{Leb} \otimes \lambda_{\alpha,\theta}$ where

$$\lambda_{\alpha,\theta} := \left(\mathbb{1}_{\{\ell < 0\}} \alpha (1 - \theta) \ell^{-\alpha-1} + \mathbb{1}_{\{\ell > 0\}} \alpha \theta \ell^{-\alpha-1} \right) d\ell. \quad (7.9)$$

Theorem 7.3 (Stable law limit theorem) *Let X_n be i.i.d. that satisfy the tail condition (7.8). Let*

$$a_n = \inf\{x : \mathbb{P}(|X_1| > x) \leq n^{-1}\} \sim n^{1/\alpha}, \quad b_n = n \mathbb{E}X_1 \mathbb{1}_{\{|X_1| < a_n\}}.$$

Then $\frac{S_n - b_n}{a_n} \Rightarrow Y$, where Y is a non-degenerate r.v. with ch.f.

$$\mathbb{E}e^{i\xi Y} = \exp\left(i\xi c + \int_{-\infty}^{\infty} (e^{i\xi x} - 1 - \frac{i\xi x}{1+x^2}) \lambda_{\alpha,\theta}(dx)\right) \quad (7.10)$$

where $c \in \mathbb{R}$ and $\lambda_{\alpha,\theta}$ is given by (7.9).

The proof of Theorem 7.3 is quite technical and we will try to understand the statement rather than prove it. The connection between (7.10) and (7.7) can be seen through the following computation.

Proposition 7.4 *Let ν be a PPP with intensity λ , and let f be a measurable function such that $\int |f| d\lambda < \infty$. Then*

$$\mathbb{E}e^{i\xi \int f d\nu} = \exp\left(\int (e^{i\xi f(x)} - 1) \lambda(dx)\right). \quad (7.11)$$

Proof: We will prove (7.11) for f an indicator function, a simple function, and then a general function in $L^1(d\lambda)$.

First, let $f = \mathbb{1}_A$. Then $\int f d\nu = \nu(A) \sim \text{Poi}(\lambda(A))$ by the definition of PPP. Hence, by (7.1),

$$\mathbb{E}e^{i\xi \int f d\nu} = \exp(\lambda(A)(e^{i\xi} - 1)) = \exp\left(\int (e^{i\xi \mathbb{1}_A(z)} - 1) \lambda(dz)\right),$$

since $e^{i\xi \mathbb{1}_A(z)} - 1 = 0 \Leftrightarrow \mathbb{1}_A(z) = 0$.

Next, let $f(x) = \sum_{k=1}^n c_k \mathbb{1}_{A_k}(x)$ to be a simple function. We can assume that A_k are disjoint. By definition of PPP, $\nu(A_k)$ are independent $\text{Poi}(\lambda(A_k))$ r.v.s, so we have

$$\mathbb{E}e^{i\xi \sum_{k=1}^n c_k \nu(A_k)} = \prod_{k=1}^n e^{\lambda(A_k)(e^{i\xi c_k} - 1)}.$$

It is easy to check that

$$\sum_{k=1}^n \lambda(A_k)(e^{i\xi c_k} - 1) = \int_H (e^{i\xi f(z)} - 1) \lambda(dz).$$

Finally, for a general function f , we can approximate it by f_n simple, with $|f_n| \leq |f|$ and $f_n(z) \rightarrow f(z)$ for every z . It suffices to show that we can pass the limit on both sides of (7.11).

Note that $\int |f| d\lambda < \infty$ implies that

$$\mathbb{E} \int |f| d\nu = \int |f| d\lambda < \infty,$$

so $\int |f| d\nu < \infty$ for almost every ν , and hence by DCT on such ν ,

$$\lim_{n \rightarrow \infty} \int f_n(z) \nu(dz) = \int f(z) \nu(dz).$$

Then by BCT,

$$\lim_{n \rightarrow \infty} \mathbb{E} e^{i\xi \int f_n(z) \nu(dz)} = \mathbb{E} e^{i\xi \int f(z) \nu(dz)}.$$

For the right hand side, since

$$|e^{i\xi f_n(z)} - 1| \leq |\xi| \cdot |f_n(z)| \leq |\xi| \cdot |f_n(z)|,$$

it follows from the DCT that

$$\lim_{n \rightarrow \infty} \int (e^{i\xi f_n(z)} - 1) \lambda(dz) = \int (e^{i\xi f(z)} - 1) \lambda(dz).$$

□

From Proposition 7.2, in (7.10) the term

$$\int_{-\infty}^{\infty} (e^{i\xi x} - 1) \lambda_{\alpha, \theta}(dx)$$

will correspond to

$$\int_{[0,1] \times \mathbb{R}} \ell \nu(dt d\ell) \approx S_n / n^{1/\alpha},$$

where ν is PPP with intensity $\lambda_{\alpha, \theta}$. The extra term $\frac{i\xi x}{1+x^2}$ is to compensate the asymmetry of the tail of X_1 ; in the symmetric case $\theta = 1/2$, the term can be dropped, since it is odd and $\lambda_{\alpha, \theta}$ is even.

How do we understand the index α ? In the proof of Theorem 7.3, one needs the contribution from small jumps and from big jumps. In (7.10) only contribution from big jumps matters; they appear in the limiting PPP. We can compare the contribution from small jumps and big jumps from the form of $\lambda_{\alpha, \theta}$. For simplicity, let us say the law of X_1 is symmetric, and thus $\theta = 1/2$. When $\alpha < 1$, the sum of small jumps are negligible compared to large jumps, since

$$\mathbb{E} \left| \int_{-\varepsilon}^{\varepsilon} \ell \nu(d\ell) \right| \leq \int_0^{\varepsilon} \ell \cdot \frac{\alpha}{\ell^{-\alpha-1}} d\ell < \infty, \quad (7.12)$$

while the large jumps contribute much more since

$$\int_{\varepsilon}^{\infty} \ell^{-\alpha} d\ell = \infty.$$

When $\alpha \in (1, 2]$, the RHS of (7.12) is also ∞ , but if we consider the cancellation of positive and negative jumps, as the Kolmogorov's one-series theorem suggest, we should integrate ℓ^2 rather than ℓ . Since

$$\int_0^\varepsilon \ell^2 \ell^{-\alpha-1} d\ell < \infty, \quad (7.13)$$

the sum of small jumps still converges, while the sum of large jumps diverges since

$$\int_\varepsilon^\infty \ell^2 \ell^{-\alpha-1} d\ell = \infty. \quad (7.14)$$

But when $\alpha > 2$, the integral (7.13) will diverge, while the integral (7.14) is finite, so the main contribution to the sum S_n is from small jumps, and since all jumps are small, we see a continuous distribution as the limit; this intuition agrees with the Linderburg's condition (5.24) and the equicontinuity estimate (6.2) when we derive the functional CLT.

7.3 Stable laws and infinite divisible laws

Random variables with ch.f.s of the form (7.10) are called *stable laws*. Besides the ch.f., an “official” definition for the stable law is the following.

Definition 7.1 A r.v. Y has stable law if for every k , there exist a_k, b_k such that

$$\frac{Y_1 + \cdots + Y_k - b_k}{a_k} \stackrel{d}{=} Y,$$

where Y_j are i.i.d. and $Y_j \stackrel{d}{=} Y$. Here, $X_1 \stackrel{d}{=} X_2$ means that X_1 and X_2 have the same distribution.

The next theorem explains the word “stable”.

Theorem 7.5 A r.v. Y has stable law if and only if there exist i.i.d. r.v. X_n , constants a_n and b_n such that

$$\frac{X_1 + \cdots + X_n - b_n}{a_n} \Rightarrow Y.$$

Proof: The “only if” part follows from the definition with $X_n \stackrel{d}{=} Y$.

For the “if” part, we only explain the intuition.

If X_n, a_n and b_n exist, then for each k ,

$$\begin{aligned} Y &\stackrel{d}{=} \lim_{n \rightarrow \infty} \frac{X_1 + \cdots + X_{kn} - b_{kn}}{a_{kn}} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{X_1 + \cdots + X_n - b_n}{a_n} + \cdots + \frac{X_{(k-1)n+1} + \cdots + X_{kn} - b_n}{a_n} + \frac{kb_n - b_{kn}}{a_{kn}}}{a_{kn}/a_n} \\ &\stackrel{d}{=} \frac{Y_1 + \cdots + Y_k - \tilde{a}_k}{\tilde{b}_k}, \end{aligned}$$

where

$$\tilde{a}_k = \lim_{n \rightarrow \infty} a_{kn}/a_n, \quad \tilde{b}_k = \lim_{n \rightarrow \infty} \frac{kb_n - b_{kn}}{a_{kn}}. \quad (7.15)$$

So after we show that the two limits in (7.15) indeed exist, we know that Y has stable law. $\square$

A closely related concept is *infinitely divisible law*. A r.v. Y has infinitely divisible law if for every $n \geq 1$, there exists $X_{n,1}, \dots, X_{n,n}$ i.i.d. such that $Y \stackrel{d}{=} X_{n,1} + \cdots + X_{n,n}$. Clearly, a stable law is

infinitely divisible, by taking $X_{n,k} = \frac{Y_k - b_n/n}{a_n}$. Also, Y is infinitely divisible if and only if for every $n \geq 1$, the n -th root of its ch.f. is also a ch.f. for some r.v. $(X_{n,1})$. There is a characterization of infinitely divisible laws similar to [Theorem 7.5](#).

Theorem 7.6 *A r.v. Y has infinitely divisible law if and only if there exist i.i.d. $X_{n,k}$ such that $X_{n,1} + \dots + X_{n,n} \Rightarrow Y$.*

The celebrated Levy–Khinchin Theorem completely characterized the ch.f.s for infinitely divisible law.

Theorem 7.7 [*Levy–Khinchin*] *A r.v. Y has infinitely divisible law if and only if its ch.f. takes the form*

$$\log \varphi(\xi) = ic\xi - \frac{\sigma^2}{2}\xi^2 + \int (e^{i\xi x} - 1 - \frac{i\xi x}{1+x^2}) \mu(dx),$$

where $c, \sigma \in \mathbb{R}$ and μ is a measure with $\mu\{0\} = 0$ and $\int \frac{x^2}{1+x^2} \mu(dx) < \infty$.

We conclude by some examples of infinite divisible laws.

Example 7.4 1. Normal distribution. The ch.f. is $e^{i\mu\xi - \frac{1}{2}\sigma^2\xi^2}$.

2. Stable laws. The ch.f. is given in [\(7.10\)](#).

3. Poisson distribution. The ch.f. is $e^{\lambda(e^{i\xi} - 1)}$.

4. Compound Poisson or integral against a PPP. The ch.f. is given in [Proposition 7.4](#).

8 Martingales

8.1 Conditional expectation

8.1.1 Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra. Let X be a r.v. with $\mathbb{E}|X| < \infty$. The *conditional expectation* $Y = \mathbb{E}[X | \mathcal{G}]$ is a r.v. that satisfies the following two properties:

Y is $\mathcal{G}$ -measurable,

$$\mathbb{E}Y\mathbb{1}_A = \mathbb{E}X\mathbb{1}_A, \quad \forall A \in \mathcal{G}. \quad (8.1a)$$

Such r.v.s Y are not unique. We call each of them a *version* of $\mathbb{E}[X | \mathcal{G}]$.

Let us look at a simple σ -algebra $\mathcal{G} = \{\emptyset, B, B^c, \Omega\}$ and take $X = \mathbb{1}_A$. It is not hard to show that any $\mathcal{G}$ -measurable map is a linear combination of $\mathbb{1}_B$ and $\mathbb{1}_{B^c}$. Therefore,

$$\mathbb{E}[\mathbb{1}_A | \mathcal{G}](\omega) = c_1\mathbb{1}_B(\omega) + c_2\mathbb{1}_{B^c}(\omega),$$

And we need to determine c_1 and c_2 . Since B and B^c are in $\mathcal{G}$, by [\(8.1a\)](#) we have

$$\mathbb{E}\mathbb{1}_A\mathbb{1}_B = \mathbb{E}(c_1\mathbb{1}_B + c_2\mathbb{1}_{B^c})\mathbb{1}_B = c_1\mathbb{P}(B), \quad \mathbb{E}\mathbb{1}_A\mathbb{1}_{B^c} = \mathbb{E}(c_1\mathbb{1}_B + c_2\mathbb{1}_{B^c})\mathbb{1}_{B^c} = c_2\mathbb{P}(B^c).$$

So

$$c_1\mathbb{P}(B) = \mathbb{P}(AB), \quad c_2\mathbb{P}(B^c) = \mathbb{P}(AB^c).$$

There are three cases.

1. If $P(B), P(B^c) \neq 0$, then $c_1 = \frac{P(AB)}{P(B)} = P[A|B]$, $c_2 = \frac{P(AB^c)}{P(B^c)} = P[A|B^c]$ are the classical conditional probabilities.
2. If $P(B) = 0$ (and hence $P(B^c) = 1$), then c_1 can be arbitrary and $c_2 = P(A)$, but the conditional expectation is only undetermined on a zero measure set B .
3. If $P(B^c) = 0$ (and hence $P(B) = 1$), then c_2 can be arbitrary and $c_1 = P(A)$, and this is similar to the previous case.

Note that from elementary probability, we also avoid $P[A|B]$ if $P(B) = 0$.

As a generalization of the previous example, suppose we can partition the sample space Ω into disjoint union of at most countably many sets $\Omega = \bigcup_{n=1}^N \Omega_n$, where $P(\Omega_n) > 0$ and $1 \leq N \leq \infty$. Let $\mathcal{G} = \sigma(\Omega_n, n \geq 1)$. Then we have

$$E[X|\mathcal{G}](\omega) = \frac{EX\mathbb{1}_{\Omega_n}}{P(\Omega_n)}, \quad \omega \in \Omega_n.$$

In particular, when $X = \mathbb{1}_A$, we define the *conditional probability* of A w.r.t. $\mathcal{G}$ to be

$$P[A|\mathcal{G}](\omega) := P[A|\Omega_n] = \frac{P(A\Omega_n)}{P(\Omega_n)}, \quad \omega \in \Omega_n.$$

8.1.2 Uniqueness and Existence

Since r.v.s are indistinguishable up to modification on zero measure sets, our best hope is that conditional expectation is unique in the almost sure sense. We start with a simple lemma.

Lemma 8.1 *Let Y be a version of $E[X|\mathcal{G}]$. Then $E|Y| \leq E|X|$.*

Proof: Since $Y \in \mathcal{G}$, we have $A = \{Y \geq 0\} \in \mathcal{G}$. By (8.1a), we have

$$EY^+ = EY\mathbb{1}_A = EX\mathbb{1}_A \leq E|X|\mathbb{1}_A.$$

Similarly, $A^c \in \mathcal{G}$ and we have

$$EY^- = -EY\mathbb{1}_{A^c} = -EX\mathbb{1}_{A^c} \leq E|X|\mathbb{1}_{A^c}.$$

□

Proof of a.s. uniqueness of conditional expectation: Let Y and Y' be two versions of $E[X|\mathcal{G}]$. For every $\varepsilon > 0$, let $A_\varepsilon = \{Y - Y' \geq \varepsilon\} \in \mathcal{G}$. By (8.1a), we have

$$P(A_\varepsilon) \leq E(Y - Y')\mathbb{1}_{A_\varepsilon} = EX\mathbb{1}_{A_\varepsilon} - EX\mathbb{1}_{A_\varepsilon} = 0.$$

Therefore,

$$P(Y - Y' > 0) \leq \sum_{n=1}^{\infty} P(A_{1/n}) = 0.$$

By symmetry we also have $P(Y' - Y > 0) = 0$. Hence, $P(Y' = Y) = 1$, as desired. □

For the existence of conditional expectation, we can use the Radon–Nikodym Theorem, **Theorem 1.18**. The setup is as follows.

$$\nu_\pm(A) = EX^\pm \mathbb{1}_A, \quad A \in \mathcal{G}.$$

For any disjoint $A_n \in \mathcal{G}$, since $\mathbb{E}|X| < \infty$ and $|X| \geq \sum_{n=1}^N X^\pm \mathbb{1}_{A_n}$, by DCT, we have

$$\nu_\pm(\bigcup_{n=1}^{\infty} A_n) = \mathbb{E} \lim_{N \rightarrow \infty} \sum_{n=1}^N X^\pm \mathbb{1}_{A_n} = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{E} X^\pm \mathbb{1}_{A_n} = \sum_{n=1}^{\infty} \nu_\pm(A_n).$$

Also, $\nu_\pm(\Omega) \leq \mathbb{E}|X| < \infty$. So $\nu_\pm$ are finite measures. Clearly, $\nu_\pm \ll \mathbb{P}$. By [Theorem 1.18](#), there exist r.v.s $Y^\pm \in \mathcal{G}$ such that $\nu_\pm(A) = \mathbb{E} Y^\pm \mathbb{1}_A$. Let $Y = Y^+ - Y^-$. Then Y is a version of $\mathbb{E}[X | \mathcal{G}]$.

Alternatively, we present a proof using projection in the Hilbert space $H = L^2(\Omega, \mathcal{F}, \mathbb{P})$, which in fact leads to a proof for the Radon–Nikodym theorem as we..

Lemma 8.2 *Consider the Hilbert space $H = L^2(\Omega, \mathcal{F}, \mathbb{P})$. For any $\mathcal{G} \subset \mathcal{F}$, the space $H_0 = L^2(\Omega, \mathcal{G}, \mathbb{P})$ is a linear, closed subspace of H_1 , and for every $X \in H$, there is a unique $Y = \pi X \in H_0$ such that*

$$\mathbb{E}(X - Y) \cdot Z = 0, \quad \forall Z \in H_0. \quad (8.2)$$

The r.v. Y is the projection of X onto H_0 .

Proof: It is easy to show that H_0 is a closed linear subspace of H .

Let

$$d = \inf_{Y \in H_0} \mathbb{E}|X - Y|^2. \quad (8.3)$$

We will show that the infimum in (8.3) is achieved by some Y , which satisfies (8.2).

Let $Y_n \in H_0$ be a minimizing sequence of (8.3). We have

$$2\mathbb{E}|X - Y_n|^2 + 2\mathbb{E}|X - Y_m|^2 = \mathbb{E}|Y_n - Y_m|^2 + \mathbb{E}[(X - Y_n) + (X - Y_m)]^2 \geq \mathbb{E}|Y_n - Y_m|^2 + 4d,$$

and hence

$$\limsup_{n, m \rightarrow \infty} \mathbb{E}|Y_n - Y_m|^2 \leq 2 \lim_{n, m \rightarrow \infty} \mathbb{E}|X - Y_n|^2 + \mathbb{E}|X - Y_m|^2 - 4d = 0.$$

Therefore, (Y_n) forms a Cauchy sequence in $L^2(\Omega, \mathcal{F}, \mathbb{P})$. By taking a subsequence we may assume $Y_n \rightarrow Y$ almost surely as well. Let Y be its limit in $L^2(\Omega, \mathcal{F}, \mathbb{P})$ and in the almost sure sense. Then by closedness of H_0 , we have $Y \in H_0$. Moreover, by Fatou's lemma,

$$\mathbb{E}|X - Y|^2 \leq \liminf_{n \rightarrow \infty} \mathbb{E}|X - Y_n|^2 = d.$$

Hence, the infimum in (8.3) is achieved.

Now for every $Z \in H_0$, since $Y + \varepsilon Z \in H_0$ and Y achieves the minimum in (8.3), we have

$$\mathbb{E}|X - Y + \varepsilon Z|^2 \geq \mathbb{E}|X - Y|^2, \quad \forall \varepsilon \in \mathbb{R}.$$

As this holds for all ε , we have $\mathbb{E}(X - Y) \cdot Z = 0$. □

Proposition 8.3 *The projection Y constructed above is $\mathbb{E}[X | \mathcal{G}]$.*

Proof: It suffices to verify (8.1a). Indeed, since $A \in \mathcal{G}$, $Z = \mathbb{1}_A \in L^2(\Omega, \mathcal{G}, \mathbb{P})$ and hence by (8.2),

$$\mathbb{E}(X - Y)\mathbb{1}_A = 0.$$

□

Let $X \in L^1$ and Y be an arbitrary r.v. We write $\mathbb{E}[X | \sigma(Y)]$ as $\mathbb{E}[X | Y]$, since any $\sigma(Y)$ -measurable map has the form $h(Y)$ where h is a Borel measurable function. As another example, let (X, Y) be 2d random vector with density $f(x, y)$, that is, for any $B \in \mathcal{B}(\mathbb{R}^2)$,

$$\mathbb{P}((X, Y) \in B) = \int_B f(x, y) dx dy.$$

Proposition 8.4 Let g be bounded measurable. Then $\mathbb{E}[g(X) | Y] = h(Y)$, where

$$h(y) = \begin{cases} \frac{\int g(x)f(x,y)dx}{\int f(x',y)dx'}, & \int f(x,y)dx \neq 0, \\ c, & \int f(x,y)dx = 0, \end{cases} \quad (8.4)$$

where $c \in \mathbb{R}$ is arbitrary.

This means that the conditional law " $\mathcal{L}[\cdot | Y]$ " has density

$$\rho_{X|Y}(x|y) = \frac{f(x,y)}{\int f(x',y)dx'}.$$

Proof: Since $\sigma(Y) = \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^2)\}$, for any $A \in \sigma(Y)$, there exists some B_0 such that $A = Y^{-1}(B_0)$, and $\mathbb{1}_A(\omega) = \mathbb{1}_{B_0}(Y(\omega))$. We have

$$\begin{aligned} \int h(Y(\omega))\mathbb{1}_A(\omega) P(d\omega) &= \int h(Y(\omega))\mathbb{1}_{B_0}(Y(\omega)) P(d\omega) \\ &= \int h(y)\mathbb{1}_{B_0}(y)f(x,y) dx dy \\ &= \int \mathbb{1}_{B_0}(y) dy \left[\int h(y)f(x,y) dx \right] \\ &= \int \mathbb{1}_{B_0}(y) dy \int g(x)f(x,y) dx \\ &= \int g(x)\mathbb{1}_{B_0}(y)f(x,y) dx dy = \int g(X(\omega))\mathbb{1}_A(Y(\omega)) P(d\omega). \end{aligned}$$

That is, $\mathbb{E}h(Y)\mathbb{1}_A = \mathbb{E}g(X)\mathbb{1}_A$. In the fourth line, we have used

$$\int h(y)f(x,y) dx = \int g(x)f(x,y) dx, \quad \forall y,$$

which follows from (8.4). Therefore, $h(Y) = \mathbb{E}[X | Y]$. This completes the proof. $\square$

8.1.3 Properties of conditional expectation

Proposition 8.5 Let $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$.

1. (linearity) For all $a, b \in \mathbb{R}$

$$\mathbb{E}[aX + bY | \mathcal{G}] = a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}].$$

2. (order preserving) If $X \leq Y$, then

$$\mathbb{E}[X | \mathcal{G}] \leq \mathbb{E}[Y | \mathcal{G}], \quad a.s.$$

3. (conditional MCT) Let $X_n \geq 0$, $X_n \uparrow X$ and $\mathbb{E}X < \infty$. Then

$$\mathbb{E}[X_n | \mathcal{G}] \uparrow \mathbb{E}[X | \mathcal{G}], \quad a.s.$$

Proof: For **Item 1**, we have for every $A \in \mathcal{G}$,

$$\begin{aligned} \mathbb{E}(\mathbb{1}_A \cdot \mathbb{E}[aX + bY | \mathcal{G}]) &= \mathbb{E}\mathbb{1}_A(aX + bY) = a\mathbb{E}\mathbb{1}_AX + b\mathbb{E}\mathbb{1}_AY \\ &= a\mathbb{E}(\mathbb{1}_A \cdot \mathbb{E}[X | \mathcal{G}]) + b\mathbb{E}(\mathbb{1}_A \cdot \mathbb{E}[Y | \mathcal{G}]) = \mathbb{E}(\mathbb{1}_A \cdot (a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}])). \end{aligned}$$

For **Item 2**, consider $A_\varepsilon = \{\mathbb{E}[X | \mathcal{G}] - \mathbb{E}[Y | \mathcal{G}] > \varepsilon\}$ and proceed as in the proof of uniqueness.

For **Item 3**, let $Y_n = \mathbb{E}[X_n | \mathcal{G}]$. Then by **Item 2**, $Y_n \uparrow$ almost surely. Let Y be the a.s. limit of Y_n .

For any $A \in \mathcal{G}$, since $Y_n \mathbb{1}_A \uparrow Y \mathbb{1}_A$ a.s., by MCT we have

$$\lim_{n \rightarrow \infty} \mathbb{E}Y_n \mathbb{1}_A = \mathbb{E}Y \mathbb{1}_A.$$

Since $X_n \mathbb{1}_A \uparrow X \mathbb{1}_A$, by MCT we have

$$\lim_{n \rightarrow \infty} \mathbb{E}X_n \mathbb{1}_A = \mathbb{E}X \mathbb{1}_A.$$

But $\mathbb{E}X_n \mathbb{1}_A = \mathbb{E}Y_n \mathbb{1}_A$ by definition. Therefore, $\mathbb{E}X \mathbb{1}_A = \mathbb{E}Y \mathbb{1}_A$ for every $A \in \mathcal{G}$, so $Y = \mathbb{E}[X | \mathcal{G}]$. $\square$

Using **Item 3**, it is not hard to establish Fatou's lemma and DCT for conditional expectations.

Proposition 8.6 (Jensen inequality for conditional expectation) *Let φ be convex. Suppose that $\mathbb{E}|X| < \infty$ and $\mathbb{E}\varphi(X) < \infty$. Then*

$$\mathbb{E}[\varphi(X) | \mathcal{G}] \geq \varphi(\mathbb{E}[X | \mathcal{G}]), \quad \text{a.s.} \quad (8.5)$$

Remark 8.1 Recall that in proving the unconditional version **Proposition 1.31**, we took expectation of the inequality $\varphi(x) \geq ax + b$, where we chose a and b so that $ax + b$ is a tangent line at $x = \mathbb{E}X$. For the conditional expectation, such strategy has a measurability problem. To take the conditional expectation of the inequality $\varphi(x) \geq ax + b$, the numbers a and b will vary with the choice of $\mathbb{E}[X | \mathcal{G}]$, which is not a fixed number and can take values in an uncountable set such as $\mathbb{R}$. The resulting inequality $\mathbb{E}[\varphi(X) | \mathcal{G}] \geq a\mathbb{E}[X | \mathcal{G}] + b$ only holds outside a zero measure set depending on a and b , and since the uncountable union of negligible sets can fail to be negligible (or even measurable), we cannot argue (8.5) holds a.s.

The idea of actual proof is deal with this issue.

Proof: All the straight lines below a convex function φ fully characterize it. We can do better by using a countable number of them, that is,

$$\varphi(x) = \sup\{ax + b : a, b \in \mathbb{Q}, \varphi(t) \geq at + b, \forall t\}. \quad (8.6)$$

For every (a, b) in (8.6), by **Proposition 8.5** there is a $N_{a,b}$ with $\mathbb{P}(N_{a,b}) = 0$ such that

$$\mathbb{E}[\varphi(X) | \mathcal{G}](\omega) \geq \mathbb{E}[aX + b | \mathcal{G}] = a\mathbb{E}[X | \mathcal{G}](\omega) + b, \quad \forall \omega \in N_{a,b}^c. \quad (8.7)$$

Hence, when $x \notin N = \bigcup_{a,b} N_{a,b}$, by (8.6) and (8.7) we have

$$\mathbb{E}[\varphi(X) | \mathcal{G}](\omega) \geq \varphi(\mathbb{E}[X | \mathcal{G}](\omega)). \quad (8.8)$$

On the other hand, by σ -subadditivity, $\mathbb{P}(N) \leq \sum_{a,b} \mathbb{P}(N_{a,b}) = 0$, and hence (8.8) holds a.s. $\square$

Below we prove some other useful properties for conditional expectation.

Proposition 8.7 $\mathbb{E}(\mathbb{E}[X | \mathcal{G}]) = \mathbb{E}X$.

Proof: It follows from (8.1a) by taking $A = \Omega$. $\square$

Proposition 8.8 Let $p \geq 1$. Then $E|X|^p \geq E|E[X | \mathcal{G}]|^p$.

When $p = 1$, this is contained in the proof of [Lemma 8.1](#).

Proof: Since $x \mapsto |x|^p$ is convex for $p \geq 1$, by [Proposition 8.6](#) we have

$$E[|X|^p | \mathcal{G}] \geq |E[X | \mathcal{G}]|^p.$$

Taking expectation of both sides and using [Proposition 8.7](#), we obtain the statement. $\square$

Proposition 8.9 If $Y \in \mathcal{G}$, then

$$E[XY | \mathcal{G}] = YE[X | \mathcal{G}]. \quad (8.9)$$

Proof: Let $\mathcal{H}$ be the collection of Y 's such that (8.9) holds. Since we have linearity and MCT for conditional expectation ([Proposition 8.5](#)), it suffices to show that $\mathcal{H}$ contains all indicator functions.

Let $Y = \mathbb{1}_B$ where $B \in \mathcal{G}$. Let $A \in \mathcal{G}$ and we need to check (8.1a). Indeed,

$$E(\mathbb{1}_A \cdot E[XY | \mathcal{G}]) = EXY\mathbb{1}_A = EX\mathbb{1}_{A \cap B} = E(\mathbb{1}_{A \cap B} E[X | \mathcal{G}]) = E(\mathbb{1}_A \cdot (YE[X | \mathcal{G}])).$$

This completes the proof. $\square$

Proposition 8.10 If X is independent of $\mathcal{G}$, then $E[X | \mathcal{G}] = EX$, and if $X \in \mathcal{G}$, then $E[X | \mathcal{G}] = X$.

Proof: Let $A \in \mathcal{G}$. If X and $\mathcal{G}$ are independent, then

$$E(\mathbb{1}_A \cdot E[X | \mathcal{G}]) = EX\mathbb{1}_A = EX \cdot E\mathbb{1}_A = E(\mathbb{1}_A \cdot EX).$$

This proves the first statement. The second statme is obvious. $\square$

Proposition 8.11 Let $\mathcal{G}_1 \subset \mathcal{G}_2$. Then

$$E[E[X | \mathcal{G}_1] | \mathcal{G}_2] = E[E[X | \mathcal{G}_2] | \mathcal{G}_1] = E[X | \mathcal{G}_1].$$

Proof: Since $E[X | \mathcal{G}_1] \in \mathcal{G}_1 \subset \mathcal{G}_2$, we have $E[E[X | \mathcal{G}_1] | \mathcal{G}_2] = E[X | \mathcal{G}_1]$.

For the second one, let $A \in \mathcal{G}_1$, and we have

$$E(\mathbb{1}_A \cdot E[X | \mathcal{G}_2]) = EE[\mathbb{1}_A X | \mathcal{G}_2] = E\mathbb{1}_A X = E(\mathbb{1}_A \cdot E[X | \mathcal{G}_1]),$$

and hence $E[E[X | \mathcal{G}_2] | \mathcal{G}_1] = E[X | \mathcal{G}_1]$. $\square$

The conditional expectation can also be understood as a projection in a Hilbert space. This also leads to a proof of [Theorem 1.18](#) using Hilbert space theory. See LAX.

Proposition 8.12 If $EX^2 < \infty$, then

$$E(X - E[X | \mathcal{G}])^2 = \min_{Y \in \mathcal{G}} E(X - Y)^2. \quad (8.10)$$

Proof: The space

$$H = \{Z : EZ^2 < \infty\}$$

is a Hilbert space, with the inner product $EX \cdot Y$. The space $H_1 = \{Z \in H : Z \in \mathcal{G}\}$ is a linear subspace of H . By standard Hilbert space theory, the orthogonal projection $Y = \pi_{H_1}(X)$ exists and achieves the minimum in (8.10). In addition, the projection is characterized by

$$E(X - Y)Z = 0, \quad \forall Z \in H_1.$$

In particular, taking $Z = \mathbb{1}_A$, $A \in \mathcal{G}$, we see that $Y = E[X | \mathcal{G}]$. $\square$

8.1.4 Regular conditional expectation

Let $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ be a measurable map. One can take $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, but we allow more generality here. Let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra.

We note that for every set $A \in \mathcal{F}$, the conditional probability $P[X \in A | \mathcal{G}]$ exists a.s. We also know that for any disjoint A_n ,

$$P\left[X \in \bigcup_{n=1}^{\infty} A_n \mid \mathcal{G}\right] = \sum_{n=1}^{\infty} P[X \in A_n | \mathcal{G}], \quad \text{a.s.}, \quad (8.11)$$

where the zero measure exceptional set in (8.11) will depend on $A_1, A_2, \dots$.

It is tempting to say that $P[X \in \cdot | \mathcal{G}](\omega)$ defines a probability measure on $\mathcal{F}$. However, the σ -additivity may not hold, since there are uncountably many ways of choosing $A_1, A_2, \dots$, so the union of all exceptional sets in (8.11), may not be negligible. If one succeeds in finding a common negligible set, it is called the *regular conditional probability*.

Definition 8.1 (regular conditional probability) A map $\mu : \Omega \times \mathcal{S} \rightarrow [0, 1]$ is called a regular conditional probability of $P(X \in \cdot)$ with respect to $\mathcal{G}$, if

1. for a.e. ω , $\mu(\omega, \cdot)$ is a probability measure on $(S, \mathcal{S})$;
2. for every $A \in \mathcal{S}$, $\mu(\omega, A)$ is a version of $P[X \in A | \mathcal{G}]$.

Example 8.2 Let (X, Y) have density $f(x, y)$ and $\mathcal{G} = \sigma(Y)$. Then

$$\mu(\omega, A) = \begin{cases} \frac{\int_A f(x, Y(\omega)) dx}{\int_{\mathbb{R}} f(x, Y(\omega)) dx}, & \int_{\mathbb{R}} f(x, Y(\omega)) dx \neq 0, \\ 0, & \text{else,} \end{cases}$$

is a regular conditional probability of μ_X w.r.t. $\sigma(Y)$.

For existence of the regular conditional probability, the key is to find a common negligible set. This is possible when, say, a measure can be determined by its value on countably many sets A , by more generally, by countable many test functions. For example, a measure μ on $\mathbb{R}$ is uniqueness determined by $\mu(-\infty, q]$ where $q \in \mathbb{Q}$; A measure μ on a complete separable metric space M is uniquely determined by $\int f d\mu$ where $f \in \mathcal{C}_c(M)$, the space of continuous functions on M which have compact support. Note that $\mathcal{C}_c(M)$ is also separable, so this imposes countably many conditions.

The technical requirement for existence of regular conditional probability is for the space $(S, \mathcal{S})$ to be *Borel*, namely, there exists a map $\varphi : (S, \mathcal{S}) \rightarrow ([0, 1], \mathcal{B}[0, 1])$ so that φ is 1-1 and both φ, φ^{-1} are measurable. Complete, separable metric spaces, like $\mathbb{R}$ and $\mathcal{C}[a, b]$, are Borel spaces.

8.2 Basic martingale theory

A *filtration* $(\mathcal{F}_n)_{n \geq 0}$ on $(\Omega, \mathcal{F}, P)$ is an increasing sub- σ -algebra of $\mathcal{F}$.

Definition 8.2 A family of random variables $(M_n)_{n \geq 1}$ is a $(\mathcal{F}_n)$ -martingale if $M_n \in \mathcal{F}_n$ and $E[M_n] < \infty$ for all n , and

$$E(M_{n+1} | \mathcal{F}_n) = M_n, \quad n \geq 1. \quad (8.12)$$

If “=” in (8.12) is replaced by “ $\leq$ ” or “ $\geq$ ”, then (M_n) is called a *super-martingale* or a *sub-martingale*.

If the filtration is not specified, we take the *natural filtration* $\mathcal{F}_n = \mathcal{F}_n^M := \sigma(M_1, \dots, M_n)$. If $X_n \in \mathcal{F}_n$ for all $n \geq 1$, we say that (X_n) is *adapted to* the filtration $(\mathcal{F}_n)$. We also note that (8.12) can be replaced by

$$\mathbb{E}[M_{n+m} | \mathcal{F}_n] = M_n, \quad n \geq 1, \quad m \geq 1,$$

since by Proposition 8.11 and the increasing property of $(\mathcal{F}_n)$,

$$\mathbb{E}[M_{n+n} | \mathcal{F}_n] = \mathbb{E}[\mathbb{E}[M_{n+m} | \mathcal{F}_{n+m-1}] | \mathcal{F}_n] = \mathbb{E}[M_{n+m-1} | \mathcal{F}_n] = \dots = \mathbb{E}[M_{n+1} | \mathcal{F}_n].$$

Example 8.3 Let X_i be independent random variables with $\mathbb{E}X_i = 0$. Then the partial sum $S_n = X_1 + \dots + X_n$ forms a martingale with respect to $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, since by independence,

$$\mathbb{E}[S_{n+m} | \mathcal{F}_n] = X_1 + \dots + X_n + \mathbb{E}(X_{n+1} + \dots + X_m) = S_n.$$

We say that such process S_n has “mean zero independent increment”.

Proposition 8.13 *Let $(X_n)_{n \geq 0}$ has mean zero independent increments. Then*

1. $(X_n)_{n \geq 0}$ is a martingale.
2. If $X_n \in L^2$ for all $n \geq 0$, then $(X_n^2 - \mathbb{E}X_n^2)_{n \geq 0}$ is a martingale.
3. If for some $\lambda \in \mathbb{R}$, $\mathbb{E}e^{\lambda X_n} < \infty$ for all $n \geq 0$, then $(\frac{e^{\lambda X_n}}{\mathbb{E}e^{\lambda X - t}})_{n \geq 0}$ is a martingale.

Proof:

1. This is obvious.
2. We have for all $n \geq 1$,

$$\begin{aligned} \mathbb{E}[X_{n+1}^2 - X_n^2 | \mathcal{F}_n] &= \mathbb{E}[(X_{n+1} - X_n + X_n)^2 - X_n^2 | \mathcal{F}_n] \\ &= \mathbb{E}[(X_{n+1} - X_n)^2 | \mathcal{F}_n] + 2X_n \mathbb{E}[X_{n+1} - X_n | \mathcal{F}_n] \\ &= \mathbb{E}(X_{n+1} - X_n)^2 = \mathbb{E}(X_{n+1} - X_n)(X_{n+1} + X_n) - 2\mathbb{E}X_n(X_{n+1} - X_n) \\ &= \mathbb{E}X_{n+1}^2 - \mathbb{E}X_n^2. \end{aligned}$$

3. We have for $n \geq 1$,

$$\mathbb{E}[e^{\lambda X_{n+1}} | \mathcal{F}_n] = e^{\lambda X_n} \mathbb{E}[e^{\lambda(X_{n+1} - X_n)} | \mathcal{F}_n] = e^{\lambda X_n} \mathbb{E}e^{\lambda(X_{n+1} - X_n)} = e^{\lambda X_n} \frac{\mathbb{E}e^{\lambda X_{n+1}}}{\mathbb{E}e^{\lambda X_n}}.$$

□

We can use convex/concave function to generate new super- or sup-martingales.

Proposition 8.14 *If $(M_n)_{n \geq 1}$ is a martingale, and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function, then $(\varphi(M_n))_{n \geq 1}$ is a sub-martingale.*

Proof: By Proposition 8.6, we have for all $n \geq 1$ and $m \geq 1$,

$$\mathbb{E}[\varphi(M_{n+m}) | \mathcal{F}_n] \geq \varphi(\mathbb{E}[M_{n+m} | \mathcal{F}_n]) = \varphi(M_n). \quad (8.13)$$

□

The function $|x|^p$ ($p \geq 1$) is convex. So if (M_n) is a martingale, then $|M_n|^p$ is a sub-martingale.

Proposition 8.15 *If $(M_n)_{n \geq 1}$ is a sub-martingale and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and increasing, then $(\varphi(M_n))_{n \geq 1}$ is also a sub-martingale.*

Proof: Since φ is increasing and $(M_n)_{n \geq 1}$ is a sub-martingale, the last equality in (8.13) will become

$$\varphi(\mathbb{E}[X_{n+m} | \mathcal{F}_n]) \geq \varphi(X_n),$$

and this completes the proof. $\square$

The functions $x \vee a$ ($a \in \mathbb{R}$) and in particular $x^+ = x \vee 0$ are convex and increasing. So if (M_n) is a sub-martingale, then $M_n \vee a$ and M_n^+ are also sub-martingales.

Another way to create new smartingales is to use stopping times.

Definition 8.3 (stopping time) *We say that a r.v. $T \in \{0, 1, \dots\} \cup \{\infty\}$ is a stopping time w.r.t. $(\mathcal{F}_n)$ if $\{T \leq n\} \in \mathcal{F}_n$ for every $n \geq 1$.*

Proposition 8.16 *If $(M_n)_{n \geq 1}$ is a sub-martingale and T is a stopping time which is almost surely finite, then $(M_{n \wedge T})_{n \geq 1}$ is also sub-martingale.*

In particular, if $(M_n)_{n \geq 1}$ is a martingale, then $(M_{n \wedge T})_{n \geq 1}$ is also a martingale.

Proof: By definition we have

$$M_{n \wedge T} = \sum_{k=0}^n \mathbb{1}_{\{T=k\}} M_k + \mathbb{1}_{\{T \geq n+1\}} M_n.$$

Therefore,

$$\begin{aligned} \mathbb{E}[M_{(n+1) \wedge T} - M_{n \wedge T} | \mathcal{F}_n] &= \mathbb{E}[\mathbb{1}_{\{T \geq n+2\}} M_{n+1} + \mathbb{1}_{\{T=n+1\}} M_{n+1} - \mathbb{1}_{\{T \geq n+1\}} M_n | \mathcal{F}_n] \\ &= \mathbb{E}[\mathbb{1}_{\{T \geq n+1\}} M_{n+1} - \mathbb{1}_{\{T \geq n+1\}} M_n | \mathcal{F}_n]. \end{aligned} \quad (8.14)$$

But $\{T \geq n+1\} = \{T \leq n\}^c \in \mathcal{F}_n$, so the last line of (8.14) is

$$\mathbb{1}_{\{T \geq n+1\}} \mathbb{E}[M_{n+1} - M_n | \mathcal{F}_n] \geq \mathbb{1}_{\{T \geq n+1\}} \cdot 0 = 0.$$

For the last statement, note that if (M_n) is a martingale if and only if (M_n) and $(-M_n)$ are both sub-martingales.

$\square$

A more general way to generate new sub-martingales is to use a (discrete) martingale integration.

We say that a process (H_n) is *predictable* if $H_n \in \mathcal{F}_{n-1}$ for all n . We define

$$(H \cdot X)_n = \sum_{m=1}^n H_m (X_m - X_{m-1}), \quad (8.15)$$

which can be thought of as the discrete Riemann–Stieljes integration $\int_0^t H_s dX_s$. Clearly, $(H \cdot X)_n \in \mathcal{F}_n$ for all n .

Proposition 8.17 *If (H_n) is predictable and non-negative, and (X_n) is a sub-martingale, then $(H \cdot X)_n$ is a sub-martingale.*

If (H_n) is predictable and (X_n) is a martingale, then $(H \cdot X)_n$ is a martingale.

Proof: By (8.15), we have

$$\mathbb{E}[(H \cdot X)_{n+1} - (H \cdot X)_n \mid \mathcal{F}_n] = \mathbb{E}[H_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n] = H_n \mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] \geq 0, \quad (8.16)$$

if $H_n \geq 0$ and (X_n) is a sub-martingale. If instead (X_n) is martingale, then the RHS of (8.16) is 0 regardless of the sign of H_n , so $(H \cdot X)_n$ is a martingale. $\square$

Example 8.4 If T is a stopping time, then Let $H_n = \mathbb{1}_{\{T \geq n\}}$ where T is a stopping time. Since $\{T \geq n\} = \{T \leq n-1\}^c \in \mathcal{F}_{n-1}$, the process (H_n) is predictable. Now

$$(H \cdot X)_n = \sum_{m=1}^n \mathbb{1}_{\{T \geq m\}}(X_m - X_{m-1}) = X_{n \wedge T} - X_0,$$

so by Proposition 8.17, the process $(H \cdot X_n)$ is a sub-martingale if (X_n) is a sub-martingale, and we recover Proposition 8.16.

8.3 Convergence of martingales

In this section we discuss the almost sure and L^1 -limits of martingales. The main tools are Doob's Up-crossing Theorem and uniform integrability.

Let (X_n) be $(\mathcal{F}_n)$ -adapted and $a < b$. Consider the following stopping times: $T_b^{(0)} = -\infty$,

$$T_a^{(\ell)} = \inf\{t \geq T_b^{(\ell-1)} : X_n \leq a\}, \quad T_b^{(\ell)} = \inf\{t \geq T_a^{(\ell)} : X_n \geq b\}, \quad \ell \geq 1. \quad (8.17)$$

In every interval $[T_a^{(\ell)}, T_b^{(\ell)}]$, the process (X_n) completes an up-crossing of $[a, b]$. The total number of up-crossing in a given time interval $[0, n]$ is defined by

$$U_{ab}^X[0, n] = \max\{k : T_b^{(k)} \leq n\}.$$

Theorem 8.18 (Doob's up-crossing inequality) Let $(X_n)_{n \geq 1}$ be a sub-martingale, then

$$\mathbb{E}U_{ab}^X[0, n] \leq \frac{1}{b-a} \left(\mathbb{E}(X_n - a)_+ - \mathbb{E}(X_0 - a)_+ \right).$$

Proof: Let $Y_n = (X_n - a)_+$. If (X_n) is a sub-martingale, then (Y_n) is also a sub-martingale since $x \mapsto (x-a)_+$ is convex and increasing. In addition, X_n and (Y_n) have the same number of up-crossing, that is

$$U_{ab}^X[0, n] = U_{ab}^Y[0, n], \quad \forall n \geq 1, \forall a < b. \quad (8.18)$$

Let us define $T_a^{(\ell)}$ and $T_b^{(\ell)}$ using (Y_n) in (8.17), and estimate $U_{ab}^Y[0, n]$.

Let

$$H_m = \begin{cases} 1, & T_a^{(\ell)} < m \leq T_b^{(\ell)}, \\ 0, & \text{else.} \end{cases} \quad (8.19)$$

Since all $T_a^{(\ell)}$ and $T_b^{(\ell)}$ are stopping times, we have

$$\{H_m = 1\} = \bigcup_{\ell=1}^{\infty} \{T_a^{(\ell)} < m\} \cap \{m \leq T_b^{(\ell)}\} = \bigcup_{\ell=1}^{\infty} \{T_a^{(\ell)} \leq m-1\} \cap \{m-1 \geq T_b^{(\ell)}\}^c \in \mathcal{F}_{m-1}.$$

Hence (H_m) is predictable.

Let $K_m = 1 - H_m$. By [Proposition 8.17](#), both $(H \cdot Y)_n$ and $(K \cdot Y)_n$ are sub-martingales, so $E(K \cdot Y)_n \geq E(K \cdot Y)_0 = 0$. On the other hand, $H_n + K_n \equiv 1$. Combining these, we have

$$E(H \cdot Y)_n \leq E(H \cdot Y)_n + E(K \cdot Y)_n = \sum_{m=1}^n E(Y_m - Y_{m-1}) = E(X_n - a)_+ - E(X_0 - a)_+. \quad (8.20)$$

Note that by [\(8.19\)](#),

$$E(H \cdot Y)_n \geq U_{ab}^Y[0, n] \cdot (b - a). \quad (8.21)$$

The conclusion follows from [\(8.18\)](#), [\(8.20\)](#) and [\(8.21\)](#). $\square$

An important observation is that there must be infinitely many up-crossing for a divergent sequence.

Proposition 8.19 *If (X_n) is a sub-martingale, and $\sup_n EX_n^+ < \infty$. Then there exists X such that $X_n \rightarrow X$ a.s.*

Proof: The up-crossing number is increasing in n , and hence by assumption and [Theorem 8.18](#),

$$EU_{ab}^X[0, \infty) = \lim_{n \rightarrow \infty} EU_{ab}^X[0, n] \leq \frac{\sup_n EX_n^+ + |a|}{b - a} < \infty.$$

This implies that $U_{ab}^X[0, \infty)$ is a.s. finite r.v., with probability one, any interval $[a, b]$ is being up-crossed by at most finitely many times. As a consequence, for any fixed a and b , there exists $N_{a,b}$ with $P(N_{a,b}) = 0$ such that

$$\liminf_{n \rightarrow \infty} X_n(\omega) < a < b < \limsup_{n \rightarrow \infty} X_n(\omega), \quad (8.22)$$

cannot happen on $N_{a,b}^c$.

Then, on N^c where $N = \bigcup_{a,b \in \mathbb{Q}} N_{a,b}$, [\(8.22\)](#) does not happen for all $a, b \in \mathbb{Q}$, and hence

$$\limsup_{n \rightarrow \infty} X_n(\omega) = \liminf_{n \rightarrow \infty} X_n(\omega), \quad \forall \omega \in N^c.$$

Note that $P(N) \leq \sum_{a,b \in \mathbb{Q}} P(N_{a,b}) = 0$. This shows that $\lim_{n \rightarrow \infty} X_n$ exists a.s. $\square$

Example 8.5 If a martingale $(X_n)_{n \geq 0}$ is non-negative, then $EX_n^+ = EX_n = EX_0$, and hence $\lim_{n \rightarrow \infty} X_n$ exists by [Proposition 8.19](#).

Next we will discuss the L^1 -convergence of smartingales. Recall the definition of uniform integrability [Definition 3.1](#), which gives a sufficient and necessary condition of L^1 -convergence for a.s. convergence r.v.s ([Theorem 3.11](#)).

Proposition 8.20 *Let $Z \in L^1(\Omega, \mathcal{F}, P)$. Then the collection of r.v.s*

$$E[Z | \mathcal{G}], \quad \mathcal{G} \text{ is a sub-}\sigma\text{-field of } \mathcal{F},$$

is uniformly integrable.

Proof: Since $Z \in L^1(\Omega, \mathcal{F}, P)$, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $E|Z|1_A < \varepsilon$ whenever $P(A) < \delta$.

We write $Y = E[Z | \mathcal{G}]$. By Jensen inequality, we have $|Y| \leq E[|Z| | \mathcal{G}]$, and hence for every $A \in \mathcal{G}$,

$$E1_A|Y| \leq E(1_A \cdot E[|Z| | \mathcal{G}]) = E|Z|1_A.$$

When $A = \Omega$, the above inequality gives $E|Y| \leq E|Z|$ (or this is [Proposition 8.8](#)). Then by Chebyshev's inequality,

$$P(|Y| \geq M) \leq \frac{E|Y|}{M} \leq \frac{E|Z|}{M},$$

uniformly for all sub- σ -field $\mathcal{G}$. Combining all these together we prove the statement. $\square$

Proposition 8.21 *A martingale (X_n) is uniformly integrable, if and only if there exists $X_\infty \in L^1$ such that $X_n = E[X_\infty | \mathcal{F}_n]$.*

Proof: The “only if” direction. Uniform integrability implies that $\sup_n E|X_n| < \infty$, hence [Proposition 8.19](#) implies that there exists X_∞ such that $X_n \rightarrow X_\infty$ a.s. But (X_n) is also uniformly integrable, so the limit is also in L^1 by [Theorem 3.11](#).

For any $A \in \mathcal{F}_n$, since $E[X_\infty | \mathcal{F}_n] \in \mathcal{F}_n$ and $X_{n+m}\mathbb{1}_A \rightarrow X_\infty\mathbb{1}_A$ in L^1 , we have

$$E(E[X_\infty | \mathcal{F}_n]\mathbb{1}_A) = EX_\infty\mathbb{1}_A = \lim_{m \rightarrow \infty} EX_{n+m}\mathbb{1}_A = \lim_{m \rightarrow \infty} E(E[X_{n+m} | \mathcal{F}_n]\mathbb{1}_A) = EX_n\mathbb{1}_A.$$

Since $X_n \in \mathcal{F}_n$, by the definition of the conditional expectation, we have

$$E[X_\infty | \mathcal{F}_n] = X_n, \quad \text{a.s.}$$

The “if” direction. It follows from [Proposition 8.20](#). $\square$

8.4 Optional Sampling Theorem

In this section, we assume all stopping times are a.s. finite.

Recall from [Proposition 8.16](#) that if (X_n) is a martingale and T is a stopping time, then $(X_{n \wedge T})_{n \geq 1}$ is also a martingale. In particular,

$$EX_{n \wedge T} = EX_{0 \wedge T} = EX_0. \quad (8.23)$$

Assume that X_n is bounded, then as $n \rightarrow \infty$, by BCT the LHS of (8.23) converges to EX_T . Thus we obtain the simplest form of the *optional sampling theorem*

Theorem 8.22 (optional sampling theorem) *Let X_n be a $(\mathcal{F}_n)$ -martingale and T an $(\mathcal{F}_n)$ -stopping time. Assume that X_n is bounded,*

$$EX_T = EX_0.$$

The optional sampling theorem says that no strategy can guarantee profit in a fair game.

To prove a more general form of optional sampling theorem, let us introduce the *stopping σ -algebra*.

Definition 8.4 *Let T be a stopping time. The stopping σ -algebra is*

$$\mathcal{F}_T = \{A \in \mathcal{F} : A \cap \{T \leq n\} \in \mathcal{F}_n, \forall n\}.$$

Intuitively, $\mathcal{F}_T$ contains the information before a stopping time T .

Example 8.6 Let $m \geq 0$ and consider $T = m$ (a constant time). Then T is a stopping time and $\mathcal{F}_T = \mathcal{F}_m$.

We can compare the stopping σ -algebras for different stopping time, or extract information from the stopping σ -algebra.

Proposition 8.23 *If $S \leq T$ are two stopping times, then $\mathcal{F}_S \subset \mathcal{F}_T$.*

Remark 8.7 Since $S \leq T$, “information before S ” is less than “information before T ”.

Proof: If $A \subset \mathcal{F}_S$, then for every n ,

$$A \cap \{T \leq n\} = (A \cap \{S \leq n\}) \cap \{T \leq n\} \in \mathcal{F}_n.$$

So $A \subset \mathcal{F}_T$. This completes the proof. $\square$

Proposition 8.24 If T is a stopping time and $S \geq T$ is random time such that S is $\mathcal{F}_T$ -measurable, then S is also a stopping time.

Proof: For each $n \geq 0$, since $\{S \leq n\} \in \mathcal{F}_T$,

$$\{S \leq n\} = \{S \leq n\} \cap \{T \leq n\} \in \mathcal{F}_n.$$

This completes the proof. $\square$

Remark 8.8 The stopping time S will take the form $S = f(T)$ for some function $f : \mathbb{N} \rightarrow \mathbb{N}$ with $f(m) \geq m$.

Theorem 8.25 Let $(X_n)_{n \geq 0}$ be a martingale, and $S \leq T$ be two stopping times. Suppose that either

1. S, T are bounded, that is, there is a constant $N > 0$ such that $S, T \leq N$, or
2. $(X_n)_{n \geq 1}$ is uniformly integrable.

Then

$$X_S = \mathbb{E}[X_T | \mathcal{F}_S].$$

In particular, $\mathbb{E}X_S = \mathbb{E}X_T = \mathbb{E}X_0$.

Remark 8.9 The first condition implies that $X_n = \mathbb{E}[X_N | \mathcal{F}_n]$, and the second condition by [Proposition 8.21](#) implies that

$$X_n = \begin{cases} \mathbb{E}[X_\infty | \mathcal{F}_n], & n < \infty \\ X_\infty, & n = \infty. \end{cases} \quad (8.24)$$

So both conditions implies that there is a r.v. $Z \in L^1$ such that $X_n = \mathbb{E}[Z | \mathcal{F}_n]$ for all n that we care about.

Proof: Let $Z = X_N$ if the first condition holds and $Z = X_\infty$ if the second condition holds. Then [\(8.24\)](#) holds with $X_\infty = Z$. It suffices to show

$$X_T = \mathbb{E}[Z | \mathcal{F}_T]. \quad (8.25)$$

Indeed, if [\(8.25\)](#) holds, since $\mathcal{F}_S \subset \mathcal{F}_T$, we have

$$\mathbb{E}[X_T | \mathcal{F}_S] = \mathbb{E}[\mathbb{E}[Z | \mathcal{F}_T] | \mathcal{F}_S] = \mathbb{E}[Z | \mathcal{F}_S] = X_S.$$

Now let us prove [\(8.25\)](#). For all $A \in \mathcal{F}_S$, we have

$$\begin{aligned} \mathbb{E}(\mathbb{E}[Z | \mathcal{F}_S] \mathbb{1}_A) &= \mathbb{E}Z \mathbb{1}_A = \sum_{n=1}^{\infty} \mathbb{E}Z \mathbb{1}_{A \cap \{T=n\}} + \mathbb{E}Z \mathbb{1}_{A \cap \{T=\infty\}} \\ &= \sum_{n=1}^{\infty} \mathbb{E}(\mathbb{1}_{A \cap \{T=n\}} \cdot \mathbb{E}[Z | \mathcal{F}_n]) + \mathbb{E}Z \mathbb{1}_{A \cap \{T=\infty\}} \\ &= \sum_{n=1}^{\infty} \mathbb{E} \mathbb{1}_{A \cap \{T=n\}} X_n + \mathbb{E}Z \mathbb{1}_{A \cap \{T=\infty\}} \\ &= \mathbb{E}X_T \mathbb{1}_A, \end{aligned}$$

where in the second line we use that $A \cap \{T = n\} \in \mathcal{F}_n$ since T is a stopping time. $\square$

Example 8.10 We can recover **Proposition 8.16**. If T is a stopping time, $(M_n)_{n \geq 1}$ is a martingale, then $(M_{n \wedge T})_{n \geq 1}$ is also a martingale, since

$$\mathbb{E}[M_{n \wedge T} | \mathcal{F}_{m \wedge T}] = M_{m \wedge T}, \quad \forall n > m,$$

by **Theorem 8.25** and the boundedness of the stopping time $n \wedge T$, $m \wedge T$.

Note: convert this example to simple random walk. We will also mention the Optional Sampling Theorem for sub-/super-martingales.

Definition 8.5 A smartingale $(X_n)_{n \geq 1}$ has a last element/is closed by X_∞ , if there exists $X_\infty \in L^1$ such that $(X_n)_{0 \leq t \leq \infty}$ forms a smartingale.

Example 8.11 If $(M_n)_{n \geq 1}$ is a martingale, then by **Proposition 8.21**, it has a last element if and only if it is uniformly integrable. In addition, M_∞ is the a.s. and L^1 limit of M_n .

Example 8.12 If $(X_n)_{n \geq 1}$ is a non-negative super-martingale, then it always has a last element $X_\infty = 0$, since it is trivially true that

$$X_n \geq 0 = \mathbb{E}[X_\infty | \mathcal{F}_n], \quad \forall n \geq 1.$$

But having a last element is weaker than uniform integrability. Consider $X_n = 1 + S_{n \wedge T-1}$ which is a martingale and hence super-martingale. It is non-negative. It is easy to see that

$$X_\infty = \lim_{t \rightarrow \infty} X_n = 1 + S_{T-1} = 0,$$

but $1 = \lim_{t \rightarrow \infty} \mathbb{E}X_n \neq \mathbb{E}X_\infty = 0$, so it cannot be uniformly integrable.

Theorem 8.26 Let $(X_n)_{n \geq 1}$ is a sub-martingale and $S \leq T$ be two stopping times. If either

1. S, T are bounded, or
2. $(X_n)_{n \geq 1}$ has a last element $X_\infty \in L^1$,

then

$$\mathbb{E}[X_T | \mathcal{F}_S] \geq X_S. \tag{8.26}$$

A similar statement also holds for super-martingale.

Remark 8.13 The argument in **Theorem 8.25** no longer works since the conclusion of the theorem cannot be derived from $\mathbb{E}[X_\infty | \mathcal{F}_T] \geq X_T$.

Proof: Let $A \in \mathcal{F}_S$ and

$$H_n = \mathbb{1}_A \cdot \mathbb{1}_{\{S < n \leq T\}}.$$

Then (H_n) is predictable since

$$\{H_n = 1\} = (A \cap \{S \leq n-1\}) \cap \{T \geq n-1\}^c \in \mathcal{F}_{n-1}, \quad \forall n \geq 1.$$

By **Proposition 8.17**,

$$(H \cdot X)_0 = 0, \quad (H \cdot X)_n = \sum_{k=1}^n H_k (X_k - X_{k-1}), \quad n \geq 1,$$

is a sub-martingale. In particular, for all $n \geq 1$

$$0 = \mathbb{E}(H \cdot X)_0 \leq \mathbb{E}(H \cdot X)_n = \mathbb{E}(X_{T \wedge n} - X_{S \wedge n}) \mathbb{1}_{A \cap \{S < T\}}.$$

But $X_{T \wedge n} = X_{S \wedge n}$ on $\{S = T\}$, so we obtain

$$\mathbb{E}(X_{T \wedge n} - X_{S \wedge n})\mathbb{1}_A \geq 0, \quad \forall n \geq 1, A \in \mathcal{F}_S. \quad (8.27)$$

If T is bounded, then there exists N such that $T \leq N$ a.s., and taking $n = N + 1$ in (8.27) gives

$$\mathbb{E}(X_T - X_S)\mathbb{1}_A \geq 0.$$

and this proves (8.26).

If (X_n) has a last element, without loss of generality we can assume $X_\infty = 0$, otherwise we can consider $X'_n = X_n - \mathbb{E}[X_\infty | \mathcal{F}_n]$, and (8.26) is equivalent to

$$\mathbb{E}[X'_T | \mathcal{F}_S] \geq X'_S,$$

by Theorem 8.25, and (X'_n) has a last element 0.

If T is unbounded but $S \leq N$ is bounded, then since $-X_n \geq 0$, by Fatou we have

$$-\mathbb{E}X_S\mathbb{1}_A = -\mathbb{E}X_{S \wedge (N+1)}\mathbb{1}_A \geq \liminf_{n \rightarrow \infty} \mathbb{E}(-X_{T \wedge n})\mathbb{1}_A \geq \mathbb{E}(-X_T)\mathbb{1}_{A \cap \{T < \infty\}}.$$

By adding $0 = \mathbb{E} - X_\infty\mathbb{1}_{A \cap \{T = \infty\}}$ to both sides, we obtain

$$-\mathbb{E}X_S\mathbb{1}_A \geq -\mathbb{E}X_T\mathbb{1}_A. \quad (8.28)$$

This proves (8.26).

Now we need to treat the case where S is unbounded. For every $m \geq 1$, $S \wedge m$ is a bounded stopping time. In addition, $\{S = m\} \cap A \in \mathcal{F}_{S \wedge m}$ since

$$(\{S = m\} \cap A) \cap \{S \wedge m \leq n\} \begin{cases} = \{S = m\} \cap A \in \mathcal{F}_m \subset \mathcal{F}_n, & m \leq n, \\ = \emptyset \in \mathcal{F}_n, & m \geq n + 1. \end{cases}$$

From what we have proven, we have

$$-\mathbb{E}X_S\mathbb{1}_{\{S=m\} \cap A} = -\mathbb{E}X_{S \wedge m}\mathbb{1}_{\{S=m\} \cap A} \geq -\mathbb{E}X_T\mathbb{1}_{\{S=m\} \cap A}. \quad (8.29)$$

Summing (8.29) over $m \in \{0, 1, \dots\}$, we have

$$-\mathbb{E}X_S\mathbb{1}_{\{S < \infty\} \cap A} \geq -\mathbb{E}X_T\mathbb{1}_{\{S < \infty\} \cap A}.$$

Noting that $X_\infty = 0$, and $S = \infty$ implies $T = \infty$, we can remove $\{S < \infty\}$ on both sides to obtain (8.28). This completes the proof. $\square$

8.5 Doob's Maximal inequality

We will state the maximal inequality for sub-martingales. Similar statements also hold for super-martingales.

Theorem 8.27 *Let $(X_n)_{n \geq 1}$ be a sub-martingale and $\lambda > 0$. Then*

$$\lambda \mathbb{P}\left(\max_{0 \leq m \leq n} X_m > \lambda\right) \leq \mathbb{E}X_n^+, \quad (8.30)$$

$$\lambda \mathbb{P}\left(\inf_{0 \leq m \leq n} X_m < -\lambda\right) \leq \mathbb{E}X_n^+ - \mathbb{E}X_0. \quad (8.31)$$

Proof: Write the event in (8.30) as A and let $T = \min\{m : X_m \geq \lambda\}$. Then $A = \{T \leq n\}$. Since X is a sub-martingale, X^+ is also a sub-martingale. By Theorem 8.26 we have

$$\mathbb{E}X_n^+ \geq \mathbb{E}X_{n \wedge T}^+ \geq \mathbb{E}X_{n \wedge T}^+ \mathbb{1}_{\{T \leq n\}} = \lambda \mathbb{P}(A).$$

This proves (8.30).

Write the event in (8.31) by B and let $S = \min\{m : X_m \leq \lambda\}$. Then $B = \{S \leq n\}$. Again by Theorem 8.26, we have

$$\begin{aligned} \mathbb{E}X_0 &\leq \mathbb{E}X_{n \wedge S} = \mathbb{E}X_n \mathbb{1}_{\{T > n\}} + \mathbb{E}X_T \mathbb{1}_{\{T \leq n\}} \\ &\leq \mathbb{E}X_n \mathbb{1}_{\{T > n\}} - \lambda \mathbb{P}(B) \leq \mathbb{E}X_n^+ - \lambda \mathbb{P}(B), \end{aligned}$$

and (8.31) follows. $\square$

Proposition 8.28 *Let $(M_n)_{n \geq 1}$ be a continuous martingale. Then for every $\lambda > 0$,*

$$\lambda \mathbb{P}\left(\max_{0 \leq m \leq n} |M_m| \geq \lambda\right) \leq \mathbb{E}|X_n|.$$

Proof: We apply (8.30) in Theorem 8.27 to the sub-martingale $(|M_n|)_{n \geq 1}$. $\square$

For martingales, we also have the control on the maximum of L^p norm.

Theorem 8.29 *Let $(M_n)_{n \geq 1}$ be a martingale. Then for every $p > 1$,*

$$\mathbb{E} \max_{0 \leq m \leq n} |M_m|^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|X_n|^p.$$

Proof: Let $Y = \max_{0 \leq m \leq n} |M_m|$. Since $(|M_n|)_{n \geq 1}$ is a sub-martingale, we have

$$\lambda \mathbb{P}(Y \geq \lambda) + \mathbb{E}|M_n| \mathbb{1}_{\{Y < \lambda\}} \leq \mathbb{E}|M_n|,$$

and hence

$$\mathbb{P}(Y \geq \lambda) \leq \frac{1}{\lambda} \mathbb{E}|M_n| \mathbb{1}_{\{Y \geq \lambda\}}.$$

Now

$$\begin{aligned} \mathbb{E}Y^p &= p \int_0^\infty \lambda^{p-1} \mathbb{P}(Y \geq \lambda) d\lambda \\ &\leq p \int_0^\infty \lambda^{p-2} \mathbb{E}(|M_n| \mathbb{1}_{\{Y \geq \lambda\}}) d\lambda \\ &= \mathbb{E}(|M_n| \int_0^Y p \lambda^{p-2} d\lambda) \\ &= \frac{p}{p-1} \cdot \mathbb{E}(|M_n| \cdot Y^{p-1}) \\ &\leq \frac{p}{p-1} (\mathbb{E}|M_n|^p)^{1/p} (\mathbb{E}Y^p)^{p/(p-1)}. \end{aligned}$$

The last line is Hölder's inequality. Hence, if $\mathbb{E}Y^p < \infty$, then we can divide both sides by $(\mathbb{E}Y^p)^{p/(p-1)}$ and then take the p -th power to get $\mathbb{E}Y^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|M_n|^p$. To treat the general case where $\mathbb{E}Y^p < \infty$ is not known, we use truncation, that is, we first get the estimate

$$\mathbb{E}(Y \wedge m)^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|M_n|^p$$

for the bounded r.v. $(Y \wedge m)$ with any $m > 0$. Then we let $m \rightarrow \infty$ and get the desired conclusion. $\square$

9 Examples of martingales

9.1 Radon–Nikodym derivatives

Proposition 9.1 *Let $N \in \mathbb{N} \cup \{\infty\}$.*

1. *Let $\mathbb{P}$ and $\tilde{\mathbb{P}}$ be two probability measures on $(\Omega, \mathcal{F})$ and $(\mathcal{F}_n)_{0 \leq n \leq N}$ a filtration. Let $\mathbb{P}_n$ and $\tilde{\mathbb{P}}_n$ be the restriction of $\mathbb{P}$ and $\tilde{\mathbb{P}}$ on $\mathcal{F}_n$. Suppose that $\tilde{\mathbb{P}} \ll \mathbb{P}$. Then $\tilde{\mathbb{P}}_n \ll \mathbb{P}_n$, and the Radon–Nikodym derivatives are*

$$Z_n = \frac{d\tilde{\mathbb{P}}_n}{d\mathbb{P}_n} = \mathbb{E}\left[\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \middle| \mathcal{F}_n\right], \quad 0 \leq n \leq N, \quad (9.1)$$

which form a martingale.

2. *Let $(Z_n)_{0 \leq n < N}$ be a $(\mathcal{F}_n)$ -martingale on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathcal{F} = \sigma(\mathcal{F}_n, n \geq 0)$. Then*

$$\tilde{\mathbb{P}}(A) = \mathbb{E}1_A(\omega)Z_N(\omega), \quad \forall A \in \mathcal{F}_n, \quad 0 \leq n < N, \quad (9.2)$$

defines a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F})$.

In addition, if $(Z_n)_{0 \leq n < N}$ is u.i. and thus $Z_N = \lim_{n \rightarrow N} Z_n$ exists in L^1 and a.s., then $\tilde{\mathbb{P}} \ll \mathbb{P}$ and $Z_N = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$.

Proof: **Part 1.** Let $A \in \mathcal{F}_n$. We have

$$\mathbb{P}_n(A) = 0 \quad \Rightarrow \quad \mathbb{P}(A) = 0 \quad \Rightarrow \quad \tilde{\mathbb{P}}(A) = 0 \quad \Rightarrow \quad \tilde{\mathbb{P}}_n(A) = 0.$$

Hence, $\tilde{\mathbb{P}} \ll \mathbb{P}$ implies that $\tilde{\mathbb{P}}_n \ll \mathbb{P}_n$.

To show that $(Z_n)_{n \geq 0}$ is a martingale, it suffices to show the second equality in (9.1). Let $A \in \mathcal{F}_n$. Then by the definition of Radon–Nikodym derivatives,

$$\tilde{\mathbb{P}}_n(A) = \mathbb{E}1_A \frac{d\tilde{\mathbb{P}}_n}{d\mathbb{P}_n}, \quad \tilde{\mathbb{P}}(A) = \mathbb{E}1_A \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}.$$

Hence, for all $A \in \mathcal{F}_n$,

$$\mathbb{E}1_A \frac{d\tilde{\mathbb{P}}_n}{d\mathbb{P}_n} = \mathbb{E}1_A \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$$

Therefore, the second equality in (9.1) follows from the definition of conditional expectation.

Part 2. First, we need to check that $\tilde{\mathbb{P}}$ is well-defined: for $m < n$, (9.2) gives two definitions for $\tilde{\mathbb{P}}(A)$ if $A \in \mathcal{F}_m \subset \mathcal{F}_n$,

$$\tilde{\mathbb{P}}(A) = \mathbb{E}1_A Z_n, \quad \tilde{\mathbb{P}}(A) = \mathbb{E}1_A Z_m;$$

they are equal since Z_n is $(\mathcal{F}_n)$ -martingale.

Suppose now that Z_N exists. For any $A \in \mathcal{F}_n$, $\{1_A Z_m, m \geq n\}$ is u.i. since Z_m are u.i. Then,

$$\tilde{\mathbb{P}}(A) = \lim_{m \rightarrow N} \mathbb{E}1_A Z_m = \mathbb{E}1_A Z_N.$$

Since $\tilde{\mathbb{P}}(A) = \mathbb{E}1_A Z_N$ holds for any $A \in \mathcal{F}_n, n \geq 0$, it holds for any $A \in \mathcal{F} = \sigma(\mathcal{F}_n, n \geq 0)$. Therefore, we have $\tilde{\mathbb{P}} \ll \mathbb{P}$ and Z_N is the Radon–Nikodym derivative. $\square$

Specialized to the case of product measures $(\Omega, \mathcal{F}) = (\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R}^{\mathbb{N}}))$, we have the following *Kakutani's dichotomy*.

Let us consider two product measures on $(\Omega, \mathcal{F}) = (\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R}^{\mathbb{N}}))$:

$$\mathbf{P} = G_1 \otimes G_2 \otimes G_3 \otimes \cdots, \quad \tilde{\mathbf{P}} = F_1 \otimes F_2 \otimes F_3 \otimes \cdots.$$

Assume that $F_n \ll G_n$, and $q_n = \frac{dF_n}{dG_n} > 0$, G_n -a.s. Then,

$$X_n = \frac{d\tilde{\mathbf{P}}_n}{d\mathbf{P}_n} = q_1 q_2 \cdots q_n$$

is a $(\mathcal{F}_n)$ -martingale. Since $\mathbf{P}$ and $\tilde{\mathbf{P}}$, the r.v.s q_n are independent. By [Proposition 9.1](#), (X_n) is a martingale. Since $X_n \geq 0$, by [Proposition 8.19](#) there exists an almost sure limit X of X_n . What's more,

$$\{\lim_{n \rightarrow \infty} X_n > 0\} = \left\{ \sum_{n=1}^{\infty} \log q_n > -\infty \right\}$$

is a tail event and has probability 0 or 1 by Kolmogorov's zero-one law [Theorem 4.16](#).

Theorem 9.2 $\tilde{\mathbf{P}} \ll \mathbf{P}$ or $\tilde{\mathbf{P}} \perp \mathbf{P}$, according as $\prod_{m=1}^{\infty} \int \sqrt{q_m} dG_m > 0$ or $= 0$.

See [[Dur19](#), Theorem 4.3.8] for a proof.

9.2 Simple random walks on $\mathbb{R}^d$

Let ξ_n be i.i.d. with $\mathbf{P}(\xi_n = \pm 1) = 1/2$. For $x \in \mathbb{Z}$, we call

$$X_0 = x, \quad X_n := x + \xi_1 + \cdots + \xi_n, \quad n \geq 1, \tag{9.3}$$

a *simple random walk* on $\mathbb{Z}$ starting from x .

For $y \in \mathbb{Z}$, the *hitting time* of y is

$$T_y = \inf\{n \geq 0 : X_n = y\} \in \mathbb{N} \cup \{\infty\}.$$

The hitting times T_y are stopping times. We ask two questions about the stopping times T_y : what is the hitting probability $\mathbf{P}(T_y < \infty)$, and what is the distribution of T_y if it is finite a.s.?

We can also define the simple random walk on $\mathbb{Z}^d$. Let e_i , $1 \leq i \leq d$, be the unit vectors in $\mathbb{Z}^d$. For a simple random walk X_n in $\mathbb{Z}^d$, in [\(9.3\)](#) ξ_k will be i.i.d. random vectors with distribution

$$\mathbf{P}(\xi_k = \pm e_i) = \frac{1}{2d}, \quad 1 \leq i \leq d.$$

We say that X_n is *recurrent* if

$$\mathbf{P}(T_a < \infty) = 1, \quad \forall a \in \mathbb{Z}^d,$$

and *transient* if

$$\mathbf{P}(T_a = \infty) > 0, \quad \forall a \in \mathbb{Z}^d \setminus \{x\}.$$

We will show that the simple random walks is recurrent if $d \leq 2$, and transient if $d \geq 3$.

We first investigate the case of dimension one. Let $x = 0$ and $a < 0 < b$. Since X_n is a martingale, by [Theorem 8.25](#) applied to the bounded stopping time $T_a \wedge T_b \wedge n$, we have

$$\mathbf{E}X_{T_a \wedge T_b \wedge n} = \mathbf{E}X_0 = 0. \tag{9.4}$$

To take the limit $n \rightarrow \infty$ in [\(9.4\)](#), we need to show first $\mathbf{P}(T_a \wedge T_b < \infty) = 1$.

Proposition 9.3 *There exists $\rho < 1$ such that*

$$\mathbb{P}(T_a \wedge T_b \geq k(b-a)) \leq \rho^k.$$

In particular, $\mathbb{P}(T_a \wedge T_b < \infty) = 1$.

Proof: Let $M = b - a$. For every $x \in [a, b]$, we have

$$\mathbb{P}[X_{n+M} \notin (a, b) \mid X_n = x] \geq \frac{1}{2^M}.$$

Writing $T = T_a \wedge T_b$, we have

$$\mathbb{P}[T \geq (k+1)M \mid T \geq kM] \leq \max_{x \in [a, b]} \mathbb{P}[X_{(k+1)M} \in (a, b) \mid X_{kM} = x] \leq (1 - 2^{-M}) =: \rho < 1.$$

Therefore,

$$\mathbb{P}(T \geq kM) = \mathbb{P}[T \geq kM \mid T \geq (k-1)M] \cdot \mathbb{P}[T \geq (k-1)M \mid T \geq (k-2)M] \cdots \mathbb{P}(T \geq M) \leq \rho^k.$$

□

Assume that the random walk X_n starts from $x = 0$. Applying [Theorem 8.25](#) to the martingale [\(9.3\)](#) and the bounded stopping time $T_a \wedge T_b \wedge n$, we obtain

$$\mathbb{E}X_{T_a \wedge T_b \wedge n} = \mathbb{E}X_0 = 0. \quad (9.5)$$

Since $T_a \wedge T_b$ is finite a.s. by [Proposition 9.3](#), we have

$$\lim_{n \rightarrow \infty} X_{T_a \wedge T_b \wedge n} = X_{T_a \wedge T_b}, \quad \text{a.s.}$$

Using the trivial bound $|X_{T_a \wedge T_b \wedge n}| \leq |a| \vee |b|$, by BCT we can take the limit $n \rightarrow \infty$ in [\(9.5\)](#) to obtain

$$0 = \mathbb{E} \lim_{n \rightarrow \infty} X_{T_a \wedge T_b \wedge n} = \mathbb{E}X_{T_a \wedge T_b} = a\mathbb{P}(T_a < T_b) + b\mathbb{P}(T_a > T_b). \quad (9.6)$$

The event $\{T_a = T_b\}$ can be possible, if $T_a = T_b = \infty$, but it has zero probability due to [Proposition 9.3](#) again. Since the two probabilities on the RHS of [\(9.6\)](#) sum up to one, we have from solving a linear equation

$$\mathbb{P}(T_a < T_b) = \frac{b}{b-a}, \quad \mathbb{P}(T_a > T_b) = \frac{-a}{b-a}. \quad (9.7)$$

In [\(9.7\)](#) taking $b \uparrow \infty$, since $T_b \geq b$, we have $T_b \uparrow \infty$ and hence

$$\mathbb{P}(T_a < \infty) = \lim_{b \uparrow \infty} \mathbb{P}(T_a < T_b) = \lim_{b \uparrow \infty} \frac{b}{b-a} = 1.$$

Similarly $\mathbb{P}(T_b < \infty) = 1$.

We can use the martingale $(X_n^2 - n)$ (see [Proposition 8.13](#)) to compute expectation of hitting times. As before we start with an identity obtained from the optional sampling theorem:

$$\mathbb{E}X_{T_a \wedge T_b \wedge n}^2 - (T_a \wedge T_b \wedge n) = 0. \quad (9.8)$$

Since $T_a \wedge T_b \wedge n$ is bounded we can rewrite [\(9.8\)](#) to obtain

$$\mathbb{E}X_{T_a \wedge T_b \wedge n}^2 = \mathbb{E}(T_a \wedge T_b \wedge n).$$

We can pass the limit $n \rightarrow \infty$ into the expectation, using MCT on the LHS, and BCT on the right hand side. Combining the resulting equation with (9.7) we get

$$\mathbb{E}(T_a \wedge T_b) = \mathbb{E}X_{T_a \wedge T_b}^2 = \frac{ba^2 - ab^2}{b - a} = -ab.$$

Taking $a \downarrow -\infty$ or $b \uparrow \infty$, by MCT we have $\mathbb{E}T_a = \mathbb{E}T_b = \infty$.

Next we investigate the recurrence of simple random walks in dimensions $d \geq 2$. The goal is to use optional sampling theorem to show that the random walk is recurrent in $d = 2$ and transient in $d \geq 3$.

To mimic the case of dimension one, we need to define certain hitting times and find a proper function f so that $f(X_n)$ is a martingale to apply the optional sampling theorem. Let $|\mathbf{x}|_\infty = \max_{1 \leq i \leq d} |x_i|$ be ℓ^∞ -norm of $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{Z}^d$. Let

$$B_R = \{\mathbf{x} \in \mathbb{Z}^d : |\mathbf{x}|_\infty \leq R\}$$

be the ℓ^∞ -ball of radius R , and

$$\partial B_R = \{\mathbf{x} \in \mathbb{Z}^d : |\mathbf{x}|_\infty = R\}$$

be its boundary. For $0 < a < |X_0| < b$, we define

$$T_a = \inf\{n \geq 0 : X_n \in B_a\} = \inf\{n \geq 0 : X_n \in \partial B_a\}, T_b = \inf\{n \geq 0 : X_n \notin B_b\} = \inf\{n \geq 0 : X_n \in \partial B_b\}.$$

Here, we can choose other ℓ^p -norm, but then the boundary of balls and the hitting times will have a less cleaner form.

What makes $f(X_n)$ a martingale? A direct computation gives

$$\mathbb{E}[f(X_{n+1}) - f(X_n) | \mathcal{F}_n] = \mathbb{E}[f(X_{n+1}) - f(X_n) | X_n] = \frac{1}{2d} \left[\sum_{i=1}^d f(X_n + e_i) + f(X_n - e_i) \right] - f(X_n) =: (\bar{\Delta}f)(X_n),$$

where e_i is the unit vector in the i -th direction. We call $\bar{\Delta}$ the *discrete Laplacian*. The name comes from the following computation. Let $f \in \mathcal{C}^3$. Then by Taylor expansion with the Lagrange remainder, for $h > 0$,

$$\frac{1}{2d} \sum_{i=1}^d [f(x + he_i) + f(x - he_i) - 2f(x)] = \frac{1}{2d} (\Delta f)(x) h^2 + \frac{1}{2d \cdot 6} \sum_{i=1}^d [\partial_{x_i}^3 f(\xi_i^+) + \partial_{x_i}^3 f(\xi_i^-)] h^3. \quad (9.9)$$

This means that the difference between $\frac{1}{2d} \Delta f$ and $\bar{\Delta}f$ is of higher order, hence the name of $\bar{\Delta}$.

The computation (9.9) suggests that Δf should be close to 0. In fact, functions with $\Delta f = 0$ is *harmonic functions*, and radially symmetric harmonic functions in $\mathbb{R}^d$ is the so-called *fundamental solutions* for the Laplace equation whose explicit forms are known:

$$\Phi(x) = \begin{cases} |x|, & d = 1, \\ c \log|x|, & d = 2, \\ c|x|^{2-d}, & d \geq 3. \end{cases}$$

Here $|x|$ is the Euclidean norm (that is, the ℓ^2 -norm).

Assume for a moment that $\Phi(X_n)$ is a martingale, and that $\Phi(\partial B_a) = \Phi(a)$ and $\Phi(\partial B_b) = \Phi(b)$. Similar to (9.6), we obtain

$$\mathbb{E}\Phi(x) = \mathbb{E}\Phi(X_{T_a \wedge T_b}) = \Phi(a)\mathbb{P}(T_a < T_b) + \Phi(b)\mathbb{P}(T_b < T_a),$$

where $x = X_0 \in B_b \setminus B_a$. Again, from [Proposition 9.3](#) we have $T_b < \infty$ almost surely. Then

$$\mathbb{P}(T_a < T_b) = \frac{\Phi(x) - \Phi(b)}{\Phi(a) - \Phi(b)}, \quad \mathbb{P}(T_a > T_b) = \frac{\Phi(a) - \Phi(x)}{\Phi(a) - \Phi(b)}.$$

When taking the limit $b \rightarrow \infty$, depending on the form of Φ , there are two cases. When $d = 2$, since $\Phi(b) \uparrow \infty$, we have

$$\mathbb{P}(T_a < \infty) = \lim_{b \rightarrow \infty} \frac{\Phi(x) - \Phi(b)}{\Phi(a) - \Phi(b)} = 1. \quad (9.10)$$

This means that the random walk will return to a ball almost surely, which implies recurrence. When $d \geq 3$, since $\Phi(b) \downarrow 0$, we have

$$\mathbb{P}(T_a < \infty) = \frac{\Phi(x)}{\Phi(a)} < 1. \quad (9.11)$$

This means that there is positive probability that the random walk escape to ∞ , which is transience. In conclusion, the key difference between $d = 2$ and $d \geq 3$, is that the fundamental solutions have different asymptotic behaviour at ∞ .

Now let us give a more rigorous argument, in which we need to control the difference between f and Φ . We want to find $f = f_{a,b}$ defined on $B_b \setminus B_a$ such that

$$\bar{\Delta}f = 0, \quad B_b \setminus B_a, \quad f = \Phi, \quad \partial B_b \cup \partial B_a. \quad (9.12)$$

Writing $f(x) = \Phi(x) + v(x)$, we need to find $v(x)$ that satisfies the discrete Laplace equation

$$\begin{cases} \bar{\Delta}v(x) = -\bar{\Delta}\Phi(x) =: g(x), & x \in B_b \setminus B_a, \\ v(x) = 0, & x \in \partial B_b \cup \partial B_a. \end{cases}$$

A direct computation gives

$$|\bar{\Delta}\Phi(x)| \leq C \max |\partial_{x_i}^3 \Phi| \leq \frac{C_1}{|x|^{d+1}}$$

for some constant $C_1 > 0$.

Next, we perform the argument of comparison principle/maximum principle, which is common in studying elliptic PDEs. Let $\tilde{v}(x) = \frac{M}{|x|^{d-1}} \pm v(x)$ with M to be chosen. A direct computation gives

$$\bar{\Delta} \frac{1}{|x|^{d-1}} \geq \frac{C_2}{|x|^{d+1}}$$

for some constant $C_2 > 0$. Then, by choosing M sufficiently large, we have $\bar{\Delta}\tilde{v} \geq 0$ in $B_b \setminus B_a$. For such $\tilde{v}$, the maximum of the function must be achieved on the boundary, since for an interior point x_0 ,

$$\tilde{v}(x_0) \leq \max\{\tilde{v}(x_0 \pm e_i)\}.$$

Hence,

$$\max_{B_b \setminus B_a} \tilde{v}(x_0) \leq \max_{\partial B_b \cup \partial B_a} \tilde{v}(x_0) \leq \frac{M}{a^{d-1}},$$

and we have

$$|\Phi(x) - f(x)| \leq Ca^{-d+1}. \quad (9.13)$$

If f solves [\(9.12\)](#), then $f(X_n)$ is a martingale. By optional sampling theorem we have

$$f(x) = \mathbb{E}\Phi(X_{T_a})\mathbb{1}_{\{T_a < T_b\}} + \Phi(X_{T_b})\mathbb{1}_{\{T_a > T_b\}}. \quad (9.14)$$

Concerning the discrepancy between the ℓ^2 -norm and the ℓ^∞ -norm, we define

$$\Phi^+(R) = \max_{x \in \partial B_R} \Phi(x), \quad \Phi^-(R) = \min_{x \in \partial B_R} \Phi(x).$$

From (9.14) we obtain

$$\Phi^-(a)\mathbb{P}(T_a < T_b) + \Phi^-(b)\mathbb{P}(T_a > T_b) \leq f(x) \leq \Phi^+(a)\mathbb{P}(T_a < T_b) + \Phi^+(b)\mathbb{P}(T_a > T_b).$$

Using $\mathbb{P}(T_a < T_b) + \mathbb{P}(T_a > T_b) = 1$, we have

$$\frac{\Phi^-(b) - f(x)}{\Phi^-(b) - \Phi^-(a)} \leq \mathbb{P}(T_a < T_b) \leq \frac{\Phi^+(b) - f(x)}{\Phi^+(b) - \Phi^+(a)} \quad (9.15)$$

when $d = 2$ (and thus $\Phi^\pm(b) \geq \Phi^\pm(a)$), and

$$\frac{\Phi^+(b) - f(x)}{\Phi^+(b) - \Phi^+(a)} \leq \mathbb{P}(T_a < T_b) \leq \frac{\Phi^-(b) - f(x)}{\Phi^-(b) - \Phi^-(a)} \quad (9.16)$$

when $d \geq 3$. We obtain (9.10) and (9.11) from (9.13), (9.15) and (9.16).

9.3 Random walks on $\mathbb{R}$

In this section we consider random walks in $\mathbb{R}$. As our state space is simpler, we can consider random walks that take more general jumps. To model it, let ξ_n be i.i.d. r.v.s, and we define

$$X_n = \xi_1 + \xi_2 + \cdots + \xi_n. \quad (9.17)$$

Our goal is to prove the following.

Theorem 9.4 *Consider X_n in (9.17). One of the four scenarios will happen.*

1. *Almost surely, $X_n = 0$ for all $n \geq 0$.*
2. *Almost surely, $X_n \rightarrow \infty$ as $n \rightarrow \infty$.*
3. *Almost surely, $X_n \rightarrow -\infty$ as $n \rightarrow \infty$.*
4. *Almost surely,*

$$\limsup_{n \rightarrow \infty} X_n = \infty, \quad \liminf_{n \rightarrow -\infty} X_n = -\infty. \quad (9.18)$$

Recall that the SLLN says, if $\mathbb{E}\xi_1 \in [-\infty, \infty]$ makes sense, then

$$\frac{X_n}{n} = \frac{\xi_1 + \cdots + \xi_n}{n} \rightarrow \mathbb{E}\xi_1 \quad (9.19)$$

almost surely. If $\mathbb{E}\xi_1 > 0$ or $\mathbb{E}\xi_1 < 0$, then (9.19) means that Item 2 or Item 3 holds; if $\mathbb{P}(\xi_1 = 0) = 1$, then Item 1 holds. The most complicate case Item 4 happens when $\mathbb{E}\xi_1 = 0$ or $\mathbb{E}\xi_1$ undefined.

We will give two proofs of (9.18). The first proof is based on the Hewitt–Savage zero-one law, and the second on analysis of a bounded increment martingale.

Let $\mathcal{F} = \sigma(\xi_1, \xi_2, \dots)$. We say that an event $A \in \mathcal{F}$ is *permutable* if it is not changed under any finite permutation π of $\xi_1, \xi_2, \dots$, that is,

$$A = f(\xi_1, \xi_2, \dots) = f(\xi_{\pi(1)}, \xi_{\pi(2)}, \dots) =: A \circ \pi$$

where $\pi : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection so that $\pi(i) = i$, $i \geq i_0$ for some $i_0 \geq 1$. We define the *exchangeable σ -algebra* to be

$$\mathcal{E} = \sigma(A; A \text{ permutable}).$$

Example 9.1 1. The tail σ -algebra is contained in $\mathcal{E}$.

To see this, let $A \in \mathcal{T}$ and π be a finite permutation. Since $A \in \mathcal{T}$, we know $A \in \sigma(\xi_{i_0}, \xi_{i_0+1}, \dots)$. Since π does not exchange ξ_i after i_0 , we have $A = A \circ \pi$. Since this is true for all π , we conclude that $A \in \mathcal{E}$.

2. The inclusion $\mathcal{T} \subset \mathcal{E}$ is strict. In fact, let $c(\omega) = \limsup_{n \rightarrow \infty} X_n(\omega)$, then

$$\{c(\omega) > a\} \in \mathcal{E} \setminus \mathcal{T}.$$

The Hewitt–Savage zero-one law guarantees that $\mathcal{E}$ is trivial if ξ_n are i.i.d.; see [Dur19, Theorem 2.5.4] Hence, there exists a constant c_* such that

$$\limsup_{n \rightarrow \infty} X_n(\omega) = c_*,$$

since any $\mathcal{E}$ -measurable r.v.s are constant. To see that $c_* \in \{-\infty, \infty\}$ almost surely, let us consider

$$\tilde{X}_n(\omega) = \xi_2 + \xi_3 + \dots = X_n(\omega) - \xi_1. \quad (9.20)$$

By Hewitt–Savage, there exists a constant $\tilde{c}_*$ such that $\limsup_{n \rightarrow \infty} \tilde{X}_n(\omega) = \tilde{c}_*$ almost surely. Since ξ_n are i.i.d., $(\xi_1, \xi_2, \dots)$ and $(\xi_2, \xi_3, \dots)$ have the same law, so c_* and $\tilde{c}_*$ have the same law as r.v.s., and hence $c_* = \tilde{c}_*$. Using (9.20), we obtain $c_* = \xi_1 + c_*$ almost surely. Since ξ_1 is not identically zero, this can happen only if $c_* \in \{-\infty, \infty\}$.

For the second proof, we will impose a stronger condition that $|\xi_n| \leq M$ for some $M > 0$. Then X_n is a martingale with $|X_{n+1} - X_n| \leq M$. We say that X_n is a martingale with bounded increments.

Proposition 9.5 *If (X_n) is a martingale with bounded increment, then almost surely, either*

$$\lim_{n \rightarrow \infty} X_n \in (-\infty, \infty) \text{ exists,} \quad (9.21)$$

or

$$\limsup_{n \rightarrow \infty} X_n = \infty, \quad \liminf_{n \rightarrow \infty} X_n = -\infty. \quad (9.22)$$

As an application, if $X_n = \xi_1 + \xi_2 + \dots + \xi_n$ where ξ_i are i.i.d. and not identical, then (9.21) cannot happen since for some $\varepsilon > 0$,

$$\mathbb{P}(|X_n| \geq \varepsilon, \text{ i.o.}) = 1,$$

by the second Borel–Cantelli lemma. Hence (9.22) will happen which is (9.18).

Proof: Fixing $K > 0$, let

$$T_K = \inf\{n : X_n \leq -K\} \in \{0, 1, \dots\} \cup \{\infty\}.$$

Then $X_{T \wedge n}$ is a martingale. In addition, when $T < \infty$,

$$X_T \geq X_{T-1} - K - M,$$

and hence $X_{T \wedge n} \geq -K - M$. Since $X_{T \wedge n}$ is bounded from below, it has an almost sure limit.

We write $T = T_K$ for the dependence on K . On $\{T_K = \infty\}$, $X_n = X_{T \wedge n}$, so

$$\lim_{n \rightarrow \infty} X_n = \lim_{n \rightarrow \infty} X_{T \wedge n} \in (-K, \infty) \text{ exists.}$$

Hence,

$$\{\liminf_{n \rightarrow \infty} X_n > -\infty\} \subset \bigcup_{K=1}^{\infty} \{T_K = \infty\} \subset \{\lim_{n \rightarrow \infty} X_n \text{ exists and is finite}\}. \quad (9.23)$$

Similarly,

$$\{\limsup_{n \rightarrow \infty} X_n < \infty\} \subset \{\lim_{n \rightarrow \infty} X_n \text{ exists and is finite}\}. \quad (9.24)$$

Combining (9.23) and (9.24) we complete the proof. $\square$

Here is another application of [Proposition 9.5](#).

Theorem 9.6 (Borel–Cantelli) *Let $\mathcal{F}_0 = \emptyset \subset \mathcal{F}_1 \subset \dots$ be a filtration, and let $B_n \in \mathcal{F}_n$ be a sequence of events. Then*

$$\{B_n, \text{ i.o.}\} = \left\{ \sum_{n=1}^{\infty} \mathbb{P}[B_n | \mathcal{F}_{n-1}] = \infty \right\}. \quad (9.25)$$

As consequences of [Theorem 9.6](#), when B_n are independent and $\mathcal{F}_n = \sigma(B_k, k \leq n)$, the condition $\sum_{n=1}^{\infty} \mathbb{P}(B_n) = \infty$ implies that $\mathbb{P}(B_n, \text{ i.o.}) = 1$ and we recover the second Borel–Cantelli lemma; when $\sum_{n=1}^{\infty} \mathbb{P}(B_n) < \infty$, by Jensen’s inequality,

$$\sum_{n=1}^{\infty} \mathbb{P}[B_n | \mathcal{F}_{n-1}] \leq \sum_{n=1}^{\infty} \mathbb{P}(B_n) < \infty,$$

so $\mathbb{P}(B_n, \text{ i.o.}) = 0$ and we recover the first Borel–Cantelli lemma.

Proof: Let

$$M_n = \sum_{k=1}^n \mathbb{1}_{B_k} - \mathbb{E}[\mathbb{1}_{B_k} | \mathcal{F}_{k-1}]. \quad (9.26)$$

Then (M_n) is a martingale and $|M_n - M_{n-1}| \leq 1$. By [Proposition 9.5](#), almost surely one of (9.21) and (9.22) will happen.

We note that

$$\sum_{k=1}^{\infty} \mathbb{1}_{B_k} = \infty \Leftrightarrow B_k \text{ happens infinitely often.}$$

When $\lim_{n \rightarrow \infty} M_n$ exists, then by (9.26)

$$\sum_{k=1}^{\infty} \mathbb{1}_{B_k} = \infty \Leftrightarrow \sum_{k=1}^{\infty} \mathbb{E}[\mathbb{1}_{B_k} | \mathcal{F}_{k-1}] = \infty,$$

on the other hand,

$$\limsup_{n \rightarrow \infty} M_n = \infty \Rightarrow \sum_{k=1}^{\infty} \mathbb{1}_{B_k} = \infty, \quad \liminf_{n \rightarrow \infty} M_n = -\infty \Rightarrow \sum_{k=1}^{\infty} \mathbb{E}[\mathbb{1}_{B_k} | \mathcal{F}_{k-1}] = \infty.$$

So in both cases, the conditions inside the events of (9.25) are equivalent and this completes the proof. $\square$

9.4 Branching process

The branching process Z_n models the number of individuals at the n -th generation of a family tree. We define Z_n recursively by $Z_0 = 1$,

$$Z_{n+1} = \begin{cases} \xi_1^{n+1} + \dots + \xi_{Z_n}^{n+1}, & Z_n > 0, \\ 0, & Z_n = 0. \end{cases}$$

The r.v.s ξ_k^{n+1} models the number of offsprings from the k -th individual at the n -th generation. We assume that ξ_k^n , $n \geq 1, k \geq 1$ are i.i.d. with finite expectation $\mu = E\xi \in (0, \infty)$. We also introduce the filtration $\mathcal{F}_n = \sigma(\xi_i^m, i \geq 1, 1 \leq m \leq n)$.

To produce a martingale, we have the following observation.

Proposition 9.7 *The process Z_n/μ^n is a $(\mathcal{F}_n)$ -martingale.*

Proof: We have

$$\begin{aligned} E[Z_{n+1} | \mathcal{F}_n] &= E[\xi_1^{n+1} + \dots + \xi_{Z_n}^{n+1} | \mathcal{F}_n] \\ &= E\left[\sum_{k=0}^{\infty} \mathbb{1}_{\{Z_n=k\}}(\xi_1^{n+1} + \dots + \xi_k^{n+1}) \mid \mathcal{F}_n\right] \\ &= \sum_{k=0}^{\infty} E[\mathbb{1}_{\{Z_n=k\}}(\xi_1^{n+1} + \dots + \xi_k^{n+1}) | \mathcal{F}_n] \\ &= \sum_{k=0}^{\infty} \mathbb{1}_{\{Z_n=k\}} k \cdot E\xi = \mu Z_n. \end{aligned}$$

□

Since Z_n/μ^n is a non-negative martingale, there exists a r.v. $a(\omega)$ such that $Z_n/\mu^n \rightarrow a$ almost surely.

An important question concerning the branching process is the probability of distinction, the event where $Z_n = 0$ for all $n \geq n_0$. If $\mu < 1$, the almost surely,

$$\limsup_{n \rightarrow \infty} Z_n \leq \lim_{n \rightarrow \infty} \mu^n \cdot a = 0 \cdot a = 0.$$

Since $Z_n \in \mathbb{Z}$, this implies that the distinction probability is 1.

Theorem 9.8 *If $\mu = 1$ and $P(\xi = 1) < 1$, then the distinction probability is 1.*

Proof: Since $Z_n \in \mathbb{Z}$ and $Z_n \rightarrow a$, there exists $n_0 = n_0(\omega)$ such that $Z_n = a$ for all $n \geq n_0$. The goal is to show a has to be 0.

Indeed, the conditions $E\xi = 1$ and $P(\xi = 1) < 1$ implies that $P(\xi = 0) > 0$, and hence

$$P[Z_{m+1} = k | Z_m = k] \leq 1 - P[Z_{m+1} = 0 | Z_m = k] \leq 1 - P(\xi = 0)^k.$$

Hence, for all $k \neq 0, n_0 \geq 0$,

$$P(Z_n = k, \forall n \geq n_0) \leq P(Z_{n_0} = k)P[Z_{n_0+1} = k | Z_{n_0} = k] \cdots P[Z_{n_0+m} = k | Z_{n_0+m-1} = k] \leq P(Z_{n_0} = k) \cdot \rho^m, \quad (9.27)$$

for some $\rho < 1$. Since $m \geq 1$ is arbitrary, the LHS of (9.27) is 0. Therefore,

$$P(a \neq 0) \leq \sum_{k \neq 0, n_0 \geq 0} P(Z_n = k, \forall n \geq n_0) = 0,$$

and this completes the proof. □

For the case $\mu > 1$, we set $\rho := P(Z_n = 0 \text{ for some } n)$ to be the distinction probability.

Theorem 9.9 *The number ρ is less than 1, and is the solution of $\varphi(\rho) = \rho$ where*

$$\varphi(s) = \sum_{k=0}^{\infty} p_k s^k, \quad p_k = P(\xi = k).$$

Proof: By differentiation under summation, we have

$$\varphi'(s) = \sum_{k=1}^{\infty} p_k k s^{k-1}, \quad \varphi''(s) = \sum_{k=2}^{\infty} p_k k(k-1) s^{k-2}.$$

Since $E\mu = \sum_{k=0}^{\infty} p_k k < \infty$, the power series $\varphi'(s)$ is uniformly convergent on $[0, 1]$; also $\varphi''(s)$ is convergent on $[0, 1)$. Since $\varphi'' \geq 0$, the function φ is convex.

The point $s = 1$ is one fixed point of φ . Since $\varphi'(1) > 1$ and $\varphi(0) \geq 0$, by convexity there exists another fixed point in $[0, 1)$, and we call it ρ . We will show that it is the distinction probability.

Let $\theta_m = P(Z_m = 0)$. Noting that $Z_m = 0$ implies that $Z_{m+1} = 0$, we have

$$\theta_m \uparrow P(Z_n = 0 \text{ for some } n).$$

We have

$$\theta_{m+1} = P(Z_{m+1} = 0) = \sum_{k=0}^{\infty} P[Z_{m+1} = 0 | Z_1 = k] P(Z_1 = k) = \sum_{k=0}^{\infty} p_k (\theta_m)^k = \varphi(\theta_m). \quad (9.28)$$

We have $\theta_0 = 0 \leq \rho$. By induction we can show that $\theta_m \leq \rho$ for all ρ . Hence, $\lim_{m \rightarrow \infty} \theta_m \leq \rho$. By (9.28) the limit of θ_m must be a fixed point of φ . This completes the proof. $\square$

10 Markov process

10.1 Markov property

Let X_n be r.v.s on $(\Omega, \mathcal{F}, P)$ and let $(\mathcal{F}_n)$ be a filtration. We say that (X_n) is a *Markov chain*, if

1. $X_n \in \mathcal{F}_n$,
2. the *Markov property* holds, that is,

$$P[X_{n+1} \in \cdot | \mathcal{F}_n] = P[X_{n+1} \in \cdot | X_n]. \quad (10.1)$$

For simplicity, we assume that X_n takes value in a countable set S called the *state space*. We write $\mathcal{S}$ for the Borel σ -algebra of S ; if S is countable, then $\mathcal{S}$ is the collection of all subsets of S , so the measurable condition is simple.

For the existence of Markov chains, we introduce the notion of *Markov kernel*. We say that $p(\cdot, \cdot) : S \times S$ is a Markov kernel if

1. for every $x \in S$, the set function $p(x, \cdot)$ is a probability measure on $(S, \mathcal{S})$;
2. for every $A \in \mathcal{S}$, the map $x \mapsto p(x, A)$ is measurable.

We say that (X_n) is a Markov chain with transition kernel p , if

$$P[X_{n+1} \in A | \mathcal{F}_n](\omega) = p(X_n(\omega), A). \quad (10.2)$$

(10.2) implies (10.1), since $X_n \in \mathcal{F}_n$ implies

$$E[p(X_n, A) | \mathcal{F}_n] = p(X_n, A).$$

The other direction of implication is also true, where we construct p using the regular conditional probability. We point out the kernel p does not depend on n , and we call such Markov chain *homogeneous*.

When S is countable, the transition kernel is determined by the *transitional probability*

$$p(i, j) := p(i, \{j\}), \quad i, j \in S.$$

The transitional probability is any non-negative function $p : S \times S$ satisfying $\sum_{j \in S} p(i, j) = 1$ for all $i \in S$. If S is finite, then p is a non-negative matrix with constant row sum 1.

Example 10.1 SRW on $\mathbb{Z}$ is a Markov chain, with

$$p(i, j) = \begin{cases} 1/2, & |i - j| = 1, \\ 0, & \text{else.} \end{cases}$$

The branching process is a Markov chain, with

$$p(i, j) = \mathbf{P}\left(\sum_{k=1}^i \xi_k = j\right) = (\mu_\xi)^{*i}(j),$$

where μ_ξ is the offspring distribution and $*i$ denotes the i -th fold convolution.

Let μ be a probability measure on $(S, \mathcal{S})$. Let $\mathbf{P}^\mu$ be a probability measure on $(S^\infty, \mathcal{S}^\infty)$, with finite-dimensional distribution

$$\mathbf{P}^\mu(X_j \in B_j, 0 \leq j \leq n) = \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \int_{B_2} p(x_1, dx_2) \cdots \int_{B_{n-1}} p(x_{n-2}, dx_{n-1}) p(x_{n-1}, B_n). \quad (10.3)$$

One can check that the finite-dimensional distributions (10.3) satisfy the consistency condition. Hence by the Kolmogorov's Extension Theorem, the probability measure $\mathbf{P}^\mu$ exists.

Proposition 10.1 (Existence of Markov chain) Under $\mathbf{P}^\mu$ given in (10.3), the process $X_n(\omega) = \omega_n$ is a Markov chain with transitional kernel p , with initial condition μ .

When $\mu = \delta_x$ is a Dirac measure, we write $\mathbf{P}^x$ instead of $\mathbf{P}^{\delta_x}$.

The Markov property (10.1) states that the status at time $n + 1$, conditioned on $\mathcal{F}_n$, the past, is the same as conditioned on X_n , the present. The information of X_{n+1} is part of the future. We can generalize this notion to include all the times after n , not only $n + 1$. To this end, we introduce the *shift operator* θ^n on S^∞ , defined by

$$\theta^n(\omega_0, \omega_1, \dots) = (\omega_n, \omega_{n+1}, \dots). \quad (10.4)$$

In other words, $\theta^n \omega$ is the *future after time n* . We have the a general version of the Markov property.

Theorem 10.2 (Markov property) Let $F : (S^\infty, \mathcal{S}^\infty) \rightarrow \mathbb{R}$ be bounded measurable. Then

$$\mathbf{E}^\mu[F(\theta^n \omega) | \mathcal{F}_n] = \varphi(X_n), \quad \varphi(x) = \mathbf{E}^x F(\theta).$$

The non-random function $x \mapsto \varphi(x)$ is measurable. We also write $\varphi(X_n)$ as

$$\varphi(X_n) = \mathbf{E}^{X_n} F(\theta) = [\mathbf{E}^x F(\theta)]_{x=X_n}.$$

Example 10.2 When $F(\omega) = \mathbb{1}_A(\omega_1)$, we have

$$\mathbb{E}^\mu[F(\theta^n \omega) | \mathcal{F}_n] = \mathbb{E}[F(\omega_{n+1}) | \mathcal{F}_n] = \mathbb{P}^\mu[X_{n+1} \in A | \mathcal{F}_n] = p(X_n, A) = \mathbb{E}^{X_n} \mathbb{1}_A(\omega_1). \quad (10.5)$$

This is (10.1).

Sketch of the proof: We can establish (10.5) for more and more general forms of F .

1. $F(\omega) = \mathbb{1}_{A_1}(\omega_1) \mathbb{1}_{A_2}(\omega_2) \cdots \mathbb{1}_{A_m}(\omega_m)$. For this step, the proof is similar to the argument to verify that (10.3) is consistent.
2. $F(\omega) = \mathbb{1}_A(\omega_1, \omega_2, \dots, \omega_m)$ where $A \in \mathcal{S}^m$.
3. $F(\omega) = \mathbb{1}_A(\omega)$ where $A \in \mathcal{S}^\infty$.
4. $F(\omega)$ is a simple function.
5. $F(\omega)$ is a bounded function.

□

10.2 Strong Markov property

The Markov property **Theorem 10.2** says that conditioned on the present (X_n) , the future $(\theta^n \omega)$ is independent of the past $(\mathcal{F}_n)$. Similar to the optional sampling theorem, we can use a stopping time to define the “past” and “future”. This leads to the *strong Markov property*.

Theorem 10.3 Let T be an almost finite stopping time and $F : (S, \mathcal{S}) \rightarrow \mathbb{R}$ be bounded measurable. Then

$$\mathbb{E}^\mu[F(\theta^T \omega) | \mathcal{F}_T] = [\mathbb{E}^x F(\theta)]_{x=X_T}.$$

Proof: Let $A \in \mathcal{F}_T$. Then

$$\begin{aligned} \mathbb{E}^\mu F(\theta^T \omega) \mathbb{1}_A &= \mathbb{E}^\mu \sum_{k=0}^{\infty} \mathbb{1}_{A \cap \{T=k\}} F(\theta^T \omega) \\ &= \mathbb{E}^\mu \sum_{k=0}^{\infty} \mathbb{1}_{A \cap \{T=k\}} F(\theta^k \omega) \\ &= \sum_{k=0}^{\infty} \mathbb{E}^\mu \mathbb{1}_{A \cap \{T=k\}} F(\theta^k \omega) \\ &= \sum_{k=0}^{\infty} \mathbb{E}^\mu \mathbb{1}_{A \cap \{T=k\}} \mathbb{E}^\mu[F(\theta^k \omega) | \mathcal{F}_k] \\ &= \sum_{k=0}^{\infty} \mathbb{E}^\mu \mathbb{1}_{A \cap \{T=k\}} \mathbb{E}^{X_k} F(\omega) \\ &= \mathbb{E}^\mu (\mathbb{1}_A \mathbb{E}^{X_T} F(\omega)). \end{aligned}$$

□

Example 10.3 Let X_n be the SRW on $\mathbb{Z}$. For $a \in \mathbb{Z}$,

$$T_a(\omega) = \min\{n \geq 0 : X_n = a\}.$$

The random time T_a is almost surely finite from our discussion in [Section 9.2](#).

Let

$$F(\omega) = \mathbb{1}_{\{T_2(\omega) - T_1(\omega) \leq m\}}.$$

Applying [Theorem 10.3](#) to the stopping time T_1 and $\mu = \delta_x$ where $x \leq 0$, we obtain

$$\mathbb{E}^x[F(\theta^{T_1}\omega) | \mathcal{F}_{T_1}] = \mathbb{E}^1 F(\omega).$$

Since $x \leq 0$, the RW is below 1 before T_1 , and hence

$$F(\theta^{T_1}\omega) = F(\omega). \quad (10.6)$$

When the RW starting from 1, that is, $\omega_0 = 1$ we have $T_1(\omega) = 0$, and hence

$$\mathbb{E}^1 F(\omega) = \mathbb{E}^1 \mathbb{1}_{\{T_2(\omega) \leq m\}} = \mathbb{E}^0 \mathbb{1}_{\{T_1(\omega) \leq m\}}, \quad (10.7)$$

where the last equality is due to the translation invariance of the RW. Combining [\(10.6\)](#) and [\(10.7\)](#), we obtain

$$\mathbb{P}^x[T_2 - T_1 \leq m | \mathcal{F}_{T_1}] = \mathbb{P}^0(T_1 \leq m)$$

Since the RHS is independent of $\mathcal{F}_{T_1}$, we conclude that $T_2 - T_1$ is independent of $\mathcal{F}_{T_1}$; moreover it has the same distribution of T_1 when the RW starting from 0.

The proof of [Theorem 10.3](#) is similar to that of the optional sampling theorem for martingales. In fact, we can characterize the Markov property using martingales, which can be used to give another proof of [Theorem 10.3](#).

Proposition 10.4 *Let f be bounded measurable. Then*

$$M_n^f = \sum_{m=1}^n f(X_m) - f(X_{m-1}) - \mathbb{E}[f(X_m) - f(X_{m-1}) | \mathcal{F}_{m-1}]. \quad (10.8)$$

is a martingale.

Introducing

$$(Lf)(x) := \int p(x, dy)(f(y) - f(x)),$$

we can rewrite [\(10.8\)](#) as

$$M_n^f = f(X_n) - f(X_0) - \sum_{m=0}^{n-1} (Lf)(X_m). \quad (10.9)$$

We point out that [\(10.9\)](#) depends only on the Markov kernel p .

Theorem 10.5 *Let $\tilde{\mathbb{P}}$ be a probability measure on $(S^\infty, \mathcal{S}^\infty)$. Let $X_n(\omega) = \omega_n$. Assume that $\tilde{\mathbb{P}}(X_0 \in \cdot) = \mu$ and that M_n^f given in [\(10.9\)](#) is a martingale under $\tilde{\mathbb{P}}$ for every bounded measurable f . Then $\tilde{\mathbb{P}} = \mathbb{P}^\mu$.*

Proof Idea: From $\tilde{\mathbb{E}}[M_1^f | \mathcal{F}_0] = M_0^f = 0$, we obtain

$$\tilde{\mathbb{E}}[f(X_1) | \mathcal{F}_0] - f(X_0) = (Lf)(X_0) = \int p(X_0, dy)(f(y) - f(X_0)),$$

and hence

$$\tilde{\mathbb{E}}[f(X_1) | \mathcal{F}_0] = \int p(X_0, dy) f(y).$$

In particular, taking $f = \mathbb{1}_A$, we obtain

$$\tilde{\mathbb{P}}[X_1 \in A | \mathcal{F}_0] = p(X_0, A).$$

□

Now we use [Theorem 10.5](#) to give another proof of [Theorem 10.3](#). For simplicity we assume that T is bounded.

Second proof of Theorem 10.3: For every bounded measurable f , by optional sampling theorem,

$$\mathbb{E}^\mu[M_{n+T} - M_T | \mathcal{F}_T] = 0.$$

We have

$$M_{n+T}^f - M_T^f = \sum_{m=T}^{n+T-1} f(X_{m+1}) - f(X_m) - (Lf)(X_m) = M_n^f(\theta^T \omega)$$

Hence, under the measure

$$\tilde{\mathbb{P}}(\omega \in \cdot) = \mathbb{P}^\mu[\theta^T \omega \in \cdot | \mathcal{F}_T],$$

the process M_n^f is a martingale for every bounded measurable f . By [Theorem 10.5](#), we have $\tilde{\mathbb{P}} = \mathbb{P}^\nu$ where $\nu(\cdot) = \mathbb{P}^\mu(X_T \in \cdot)$ and this completes the proof. □

10.3 Invariant measure

Let μ be a measure on $(S, \mathcal{S})$. For $n \geq 1$, we define

$$\mu \mathbb{P}^n(A) = \int \mu(dx) \mathbb{P}^x(X_n \in A). \quad (10.10)$$

If μ is a probability measure, then $\mu \mathbb{P}^n$ is $\mathbb{P}^\mu(X_n \in \cdot)$. The point of [\(10.10\)](#) is to allow the more general case where the initial condition is not a finite measure.

We say that μ is an *invariant distribution* if it is an invariant measure with total mass 1. Note that if μ is an invariant measure, then $c\mu$ is also an invariant measure for all $c > 0$; hence when μ is a finite measure, we can normalize it to get an invariant distribution.

As another example, let $S = \{1, 2, \dots, N\}$ be finite, and write the Markov kernel as a matrix $\mathbb{P} = (p_{ij})_{i,j=1}^N$. Then any measure μ on $(S, \mathcal{S})$ should be a row vector, and

$$(\mu \mathbb{P}^n)_i = \sum_{j_0, j_1, \dots, j_n \in \{1, \dots, N\}} \mu_{j_0} p_{j_0 j_1} \cdots p_{j_{n-1} j_n}, \quad j_n = i. \quad (10.11)$$

We can also view [\(10.11\)](#) as a matrix multiplication.

Next, let us compute the invariant measures for some Markov chains explicitly.

SRW on $\mathbb{Z}$. The invariant measure $\mu = (\mu_i)_{i \in \mathbb{Z}}$ satisfies

$$\mu_i = (\mu \mathbb{P})_i = \frac{1}{2}(\mu_{i-1} + \mu_{i+1}),$$

Then μ_i is an arithmetic sequence so $\mu_i = a + bi$ for some $a, b \in \mathbb{R}$. Since $\mu_i \geq 0$ for all i , the common difference b must be 0. Hence, all invariant measures are multiple of a constant, and there is no invariant distribution.

Asymmetric RW on $\mathbb{Z}$. Let $p > 1/2$ and consider a RW X_n given by

$$\mathbb{P}[X_{n+1} = X_n + 1 | X_n] = p, \quad \mathbb{P}[X_{n+1} = X_n - 1 | X_n] = 1 - p.$$

Then invariant measures will satisfy

$$\mu_i = (\mu P)_i = \mu_{i-1}p + (1-p)\mu_{i+1}. \quad (10.12)$$

This is a second order difference equation, whose characteristic equation $(1-p)\lambda^2 - \lambda + p = 0$ gives two distinct roots $\lambda = 1$ and $\lambda = p/(1-p)$. Hence,

$$\mu_i = a + b\left(\frac{p}{1-p}\right)^i, \quad i \in \mathbb{Z}$$

for some $a, b \in \mathbb{R}$. There will be two linearly independent invariant measures; one is a constant, the other is $\mu_i = (p/(1-p))^i$. None of them and their linear combinations are summable, so again, there is no invariant distribution.

RW on the half line $\mathbb{Z}_{\geq 0}$. Let $p \in (0, 1)$ and consider a RW X_n on $\mathbb{Z}_{\geq 0}$ with sticky reflecting boundary at 0, that is,

$$\begin{aligned} \mathbb{P}[X_{n+1} = X_n + 1 | X_n] &= p, & \mathbb{P}[X_{n+1} = X_n - 1 | X_n] &= 1 - p, & X_n \geq 1, \\ \mathbb{P}[X_{n+1} = 1 | X_n = 0] &= p, & \mathbb{P}[X_{n+1} = 0 | X_n = 0] &= 1 - p. \end{aligned}$$

Then invariant measure will satisfy

$$p\mu_{i-1} + (1-p)\mu_{i+1} = \mu_i, \quad i \geq 1, \quad \mu_0 = (1-p)\mu_0 + p\mu_1.$$

Similarly to (10.12), we obtain

$$\mu_i = a + b\left(\frac{p}{1-p}\right)^i, \quad i \geq 1, \quad \mu_0 = \mu_1.$$

Then μ_i is summable only if $a = 0$ and $p < 1/2$. Intuitively, the condition $p < 1/2$ means that the RW will be pushed toward the origin, which is balanced by the reflecting boundary condition at 0, and thus an invariant distribution exists.

An important question is to understand the invariant measures/distribution of a Markov chain. In the rest of the section we consider a finite state space. An invariant distribution μ satisfies $\mu P = \mu$ and thus is a left eigenvector of the transitional matrix P with eigenvalue 1.

Let $\nu = (\nu_1, \dots, \nu_N)^T$ be a row vector; its ℓ^1 -norm is

$$|\nu|_1 = \sum_{k=1}^N |\nu_k|.$$

If $\mu \geq 0$ and $|\mu|_1 = 1$, then μ can be viewed as a probability measure on $\{1, 2, \dots, N\}$. Moreover, if μ, ν are two probability measures on $\{1, 2, \dots, N\}$, then

$$\|\mu - \nu\|_{TV} = |\mu - \nu|_1.$$

The next theorem characterizes the invariant measure of a Markov chain with a finite state space.

Theorem 10.6 (Perron–Frobenius) Let $P = (p_{ij})_{i,j=1}^N$ be the transitional matrix.

1. $\lambda = 1$ is an eigenvalue of P .
2. Let $\lambda \in \mathbb{C}$ be an eigenvalue of P . Then $|\lambda| \leq 1$.

Assume additionally that

$$\delta := \min_{i,j} p_{ij} > 0. \quad (10.13)$$

3. There exists non-zero eigenvector $\mu \geq 0$ of P with eigenvalue 1, and the eigenspace has dimension 1.
4. There exists $\rho < 1$ such that for any probability measure ν on $\{1, 2, \dots, N\}$,

$$|\nu P^n - \mu|_1 \leq 2\rho^n. \quad (10.14)$$

Proof: Since P has row sum 1, the column vector $1 = (1, \dots, 1)$ satisfies $P1 = 1$. Hence $\lambda = 1$ is an eigenvalue of P .

Let $v = (v_1, \dots, v_N) \neq 0$ and $Pv = \lambda v$. Let

$$|v_j| = \max\{|v_1|, \dots, |v_N|\}.$$

By multiplying an imaginary rotation $e^{i\theta}$ we can assume $v_j \in \mathbb{R}$ and $v_j > 0$. Using that the row sum of P is 1 and the triangle inequality for complex norms, we have

$$|\lambda v_j| = \left| \sum_{k=1}^N p_{jk} v_k \right| \leq \sum_{k=1}^N p_{jk} |v_k| \leq v_j.$$

Hence $|\lambda| \leq 1$.

By **Item 1**, there exists a non-trivial left eigenvector μ . The question is whether all entries of μ have the same sign.

We write $\mu = \mu_+ - \mu_-$ to decompose μ into positive and negative parts. We first establish a contraction in ℓ^1 -norm: for any μ ,

$$|\mu P|_1 \leq |\mu|. \quad (10.15)$$

Indeed, since $\mu_{\pm}$ are non-negative vectors,

$$|\mu P|_1 \leq |\mu_+ P|_1 + |\mu_- P|_1 = \mu_+ P1 + \mu_- P1 = \mu_+ 1 + \mu_- 1 = |\mu_+|_1 + |\mu_-|_1 = |\mu|. \quad (10.16)$$

If $\mu = \mu P$, then the inequality in (10.15) must be equality. In particular, for all i , one of the two numbers $(\mu_{\pm} P)_i$ must be zero. On the other hand, under our assumption (10.13),

$$(\mu_{\pm} P)_i \geq \delta |\mu_{\pm}|_1.$$

Hence one of $\mu_{\pm}$ must be zero.

For the uniqueness of μ and the last part, we will prove a strong contraction in ℓ^1 -norm. Let ν_1, ν_2 be two probability measures on $\{1, \dots, N\}$. Then for some $\rho < 1$.

$$|\nu_1 P - \nu_2 P|_1 \leq \rho |\nu_1 - \nu_2|_1. \quad (10.17)$$

Indeed, by (10.13) we can write $P = \delta E + (1 - N\delta)\tilde{P}$, where E is the $N \times N$ matrix with all entries 1, and $\tilde{P}$ is Markov matrix (non-negative and has row sums 1). Since $\nu_i E = 1^T$, we have

$$|\nu_1 P - \nu_2 P|_1 = |\delta(\nu_1 - \nu_2)E + (1 - N\delta)(\nu_1 - \nu_2)\tilde{P}|_1 = (1 - N\delta)|(\nu_1 - \nu_2)\tilde{P}|_1 \leq (1 - N\delta)|\nu_1 - \nu_2|_1,$$

where we apply the weak ℓ^1 -contraction (10.16) to $\tilde{P}$. If μ_1 and μ_2 are two left eigenvectors with $|\mu_i|_1 = 1$, then (10.17) implies

$$|\mu_1 - \mu_2|_1 = |\mu_1 P - \mu_2 P|_1 \leq \rho |\mu_1 - \mu_2|_1,$$

so $\mu_1 = \mu_2$ and this is the uniqueness. For (10.14), we have

$$|\mu - \nu P^n|_1 = |\mu P^n - \nu P^n|_1 \leq \rho^n |\mu - \nu|_1 \leq 2\rho^n.$$

□

Remark 10.4 Item 2 is a more general fact about matrices: the spectral norm of a matrix (the largest eigenvalue in norm) is less than any of its ℓ^p -norm, while matrices with row sum 1 has ℓ^∞ -norm 1.

10.4 Recurrence

In this section we consider a countable state space $S = \{1, 2, \dots\}$. For every $i \in S$, let

$$T_i^{(1)} = T_i = \inf\{n \geq 1 : X_n = i\}, \quad T_i^{(m+1)} = \inf\{n > T_i^{(m)} : X_n = i\}.$$

The time $T_i^{(m)}$ is the m -th visit of the Markov chain of the state i .

We classify any state $i \in S$ as follows: i is

- *transient* if $\mathbf{P}^i(T_i = \infty) > 0$,
- *null recurrent* if $\mathbf{P}^i(T_i < \infty) = 1$ but $\mathbf{E}^i T_i = \infty$,
- *positive recurrent* if $\mathbf{E}^i T_i < \infty$.

We write $\rho_{ij} = \mathbf{P}^i(T_j < \infty)$. We say that j is *accessible* from i , denoted by $i \rightarrow j$, if there exists $j_0 = i, j_1, j_2, \dots, j_m = j$ such that $p(j_k, j_{k+1}) > 0$. Note that $i \rightarrow j$ if and only if $\rho_{ij} > 0$. We say that i and j *commutes*, denoted by $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$. The “ $\leftrightarrow$ ” relation is an equivalence relation.

The main result of this section is the following.

Theorem 10.7 *If $i \leftrightarrow j$, then i and j have the same type (transient, null recurrent, positive recurrent).*

We start from a lemma following from the strong Markov property.

Lemma 10.8 *For all $i, j \in S$,*

$$\mathbf{P}^j(T_i^{(m)} < \infty) = \rho_{ij} \rho_{ii}^{m-1}.$$

In particular,

$$\mathbf{P}^i(T_i^{(m)} < \infty) = \rho_{ii}^m.$$

Proof: We will prove by induction. The case $m = 1$ is definition.

By definition, the condition $T_i^{(m+1)} < \infty$ implies $T_i^{(m)} < \infty$. Hence, by strong Markov property,

$$\mathbf{P}^j(T_i^{(m+1)} < \infty) = \mathbf{P}^j[T_i^{(m+1)} < \infty | T_i^{(m)} < \infty] \mathbf{P}(T_i^{(m)} < \infty) = \mathbf{P}^i(T_i^{(1)} < \infty) \mathbf{P}^j(T_i^{(m)} < \infty) = \rho_{ij} \rho_{ii}^m.$$

This is the induction step and we complete the proof. □

Let $N(i)$ be the total number of visits to the state i , that is,

$$N(i) = \sum_{n=1}^{\infty} \mathbb{1}_{\{X_n = i\}}.$$

Proposition 10.9 *A state i is recurrent if and only if $\mathbb{E}^i N(i) = \infty$.*

Proof: By Lemma 10.8, we have

$$\mathbb{P}^i(N(i) \geq m) = \mathbb{P}^i(T_i^{(m)} < \infty) = \rho_{ii}^m.$$

So $N(i)$ is a geometric r.v. under $\mathbb{P}^i$, and we have

$$\mathbb{E}^i N(i) = \frac{1}{1 - \rho_{ii}} \begin{cases} = \infty, & \rho_{ii} = 1, \\ < \infty, & \rho_{ii} < 1. \end{cases}$$

□

By Fubini, we have

$$\mathbb{E}^i N(i) = \sum_{n=1}^{\infty} \mathbb{P}^i(X_n = i) = \sum_{n=1}^{\infty} p^n(i, i). \quad (10.18)$$

We can use (10.18) to determine recurrence type of the SRW on $\mathbb{Z}^d$. Indeed, since by CLT,

$$\frac{X_n - i}{\sqrt{n}} = \frac{\xi_1 + \xi_2 + \cdots + \xi_n}{\sqrt{n}} \rightarrow \mathcal{N}(0, \sigma^2),$$

we have

$$\mathbb{P}^i(X_n = i) \approx f_\sigma(0)(1/\sqrt{n})^d = cn^{-d/2}, \quad (10.19)$$

where f_σ is the density of $\mathcal{N}(0, \sigma^2)$, and $(1/\sqrt{n})^d$ is the volume of the d -dimensional box with length $1/\sqrt{n}$. Then (10.19) is summable if and only if $d \leq 2$, and we recover our result on the recurrence of the SRW.

Proposition 10.10 *If i is recurrent and $\rho_{ij} > 0$, then j is recurrent and $\rho_{ji} = 1$.*

Proof: Since $\rho_{ij} > 0$, we have $\mathbb{P}^i(T_j < \infty, T_j < T_i) > 0$. By strong Markov property, we have

$$0 = \mathbb{P}^i(T_i = \infty) \geq \mathbb{P}^i(T_j < \infty, T_j < T_i) \mathbb{P}^j(T_i = \infty).$$

Hence $\mathbb{P}^j(T_i = \infty) = 0$ and $\rho_{ji} = 1$.

Since $\rho_{ij}, \rho_{ji} > 0$, there exists $L, K > 0$ such that

$$p^L(i, j) > 0, \quad p^K(j, i) > 0. \quad (10.20)$$

Hence,

$$p^{L+n+K}(j, j) \geq p^K(j, i) p^n(i, i) p^L(i, j), \quad \forall n \geq 1.$$

Using (10.18) and (10.20), we have

$$\mathbb{E}^j N(j) \geq \sum_{n=1}^{\infty} p^{L+n+K}(j, j) \geq p^K(j, i) \mathbb{E}^i N(i) p^L(i, j) = \infty.$$

This completes the proof. □

Using Proposition 10.10 we can show that i and j must be simultaneously transient or recurrent if $i \leftrightarrow j$. Next we will distinguish between positive and null recurrence.

Proposition 10.11 *If $\mathbb{E}^i T_i < \infty$, then there exists an invariant distribution μ such that $\mu(i) > 0$.*

Proof: Let

$$\pi_j = \mathbb{E}^i \sum_{n=0}^{T_i-1} \mathbb{1}_{\{X_n=j\}}. \quad (10.21)$$

Then $\pi_i = 1$ and

$$\sum_{j \in S} \pi_j = \mathbb{E}^i \sum_{n=0}^{T_i-1} \sum_{j \in S} \mathbb{1}_{\{X_n=j\}} = \mathbb{E}^i T_i < \infty.$$

Then $\mu(j) = \pi_j / \mathbb{E}^i T_i$ defines a probability measure and $\mu(i) > 0$.

We claim that π (and hence μ) is invariant. Using Fibini, we can rewrite (10.21) as

$$\pi_j = \sum_{n=0}^{\infty} \mathbb{E}^i \mathbb{1}_{\{X_n=j, n \leq T_i-1\}} = \sum_{n=0}^{\infty} \mathbb{P}^i(X_n = j, n \leq T_i - 1).$$

Using Markov property, we have

$$\begin{aligned} \sum_{j \in S} \pi_j p(j, k) &= \sum_{n=0}^{\infty} \sum_{j \in S} \mathbb{P}^i(X_n = j, n \leq T_i - 1) p(j, k) \\ &= \sum_{n=0}^{\infty} \mathbb{P}^i(X_{n+1} = k, n+1 \leq T_i) \\ &= \sum_{n=1}^{\infty} \mathbb{P}^i(X_n = k, n \leq T_i) \\ &= \mathbb{E}^i \sum_{n=1}^{T_i} \mathbb{1}_{\{X_n=k\}}. \end{aligned}$$

To conclude the RHS is π_k , we use that

$$\mathbb{1}_{\{X_{T_i}=k\}} = \mathbb{1}_{\{i=k\}} = \mathbb{1}_{\{X_0=k\}}.$$

□

Proposition 10.12 *If $\mathbb{P}^\mu(T_i < \infty) = 1$ and $\mathbb{P}^i(T_i < \infty)$. Then*

$$\frac{1}{N} \sum_{n=1}^N \mathbb{1}_{\{X_n=i\}} \rightarrow \frac{1}{\mathbb{E}^i T_i}, \quad \mathbb{P}^\mu\text{-a.s.} \quad (10.22)$$

Proof: By strong Markov property, the r.v.s

$$T_i^{(1)}, \quad T_i^{(2)} - T_i^{(1)}, \quad T_i^{(3)} - T_i^{(2)}, \dots \quad (10.23)$$

are independent; moreover, for $k \geq 1$,

$$\mathbb{P}^\mu(T_i^{(k+1)} - T_i^{(k)} \in \cdot) = \mathbb{P}^i(T_i \in \cdot), \quad (10.24)$$

so the r.v.s in (10.23) except for the first one all have the same distribution.

Let k be the integer such that

$$T_i^{(k)} \leq N < T_i^{(k+1)}.$$

Since $T_i^{(k+1)} - T_i^{(k)}$ are a.s. finite by (10.24) and the assumption, as $N \rightarrow \infty$, we have $k \rightarrow \infty$. We can estimate the LHS of (10.22) using

$$\frac{k}{T_i^{(k+1)}} < \frac{1}{N} \sum_{n=1}^N \mathbb{1}_{\{X_n=i\}} \leq \frac{k}{T_i^{(k)}}. \quad (10.25)$$

On the other hand, by SLLN, $\mathbf{P}^\mu$ almost surely,

$$\frac{T_i^{(k)}}{k} = \frac{T_i^{(1)} + (T_i^{(2)} - T_i^{(1)}) + \cdots + (T_i^{(k)} - T_i^{(k-1)})}{k} \rightarrow \mathbf{E} T_i^{(2)} - T_i^{(1)} = \mathbf{E}^i T_i, \quad (10.26)$$

even when $\mathbf{E}^i T_i = \infty$. Then (10.22) follows from (10.25) and (10.26), and we complete the proof. $\square$

We can now show that positive recurrence can be propagated. To have a clean setup, for a positive recurrent state i , let

$$S_0 = \{j \in S : i \rightarrow j\} = \{j \in S : i \leftrightarrow j\}.$$

The two definitions of S_0 are equivalent by Proposition 10.10. Sometimes S_0 is called the *commuting class* containing A . Without loss of generality we can assume $S_0 = S$, since we do not care about states inaccessible from i at this moment.

Proposition 10.13 *Let i be positive recurrent. Suppose that $i \leftrightarrow j$ for all $j \in S$. Then all $j \in S$ are positive recurrent.*

Proof: Since i is positive recurrent, Proposition 10.11 gives an invariant measure μ . Examining the construction (10.21), we see that $\mu(j) > 0$ if j is accessible from i ; under our setting, we have $\mu(j) > 0$.

Let us fix a $j \in S$. By the assumption, we have $\rho_{kj} = 1$ for all $k \in S$. Hence,

$$\mathbf{P}^\mu(T_j < \infty) = \sum_{k \in S} \mu(k) \rho_{kj} = 1.$$

By Proposition 10.12, we have $\mathbf{P}^\mu$ almost surely,

$$\frac{1}{N} \sum_{n=1}^N \mathbb{1}_{\{X_n=j\}} \rightarrow \frac{1}{\mathbf{E}^j T_j}. \quad (10.27)$$

The LHS of (10.27) is bounded by 1, and thus by BCT, we have

$$\lim_{N \rightarrow \infty} \mathbf{E}^\mu \frac{1}{N} \sum_{n=1}^N \mathbb{1}_{\{X_n=j\}} = \frac{1}{\mathbf{E}^j T_j}.$$

Using the invariance of μ , we have

$$\mathbf{E}^\mu \mathbb{1}_{\{X_n=j\}} = \mathbf{P}^\mu(X_n = j) = \mathbf{P}^\mu(X_0 = j) = \mu(j).$$

Hence $\mu(j) = 1/\mathbf{E}^j T_j > 0$, and j is positive recurrent. $\square$

As a corollary, invariant distribution on a commuting state space is unique if exists. Note that at this point we have established Theorem 10.7, so we can talk about the “type” of the entire commuting class, not just a single state.

Theorem 10.14 *Suppose $i \leftrightarrow j$ for all $i, j \in S$ (that is, S is irreducible). Then the Markov chain is positive recurrent, if and only if there exists an invariant distribution. The invariant distribution μ is unique and given by*

$$\mu(i) = \frac{1}{\mathbf{E}^i T_i}.$$

10.5 Ergodic theorem

We start with a corollary of [Proposition 10.13](#).

Proposition 10.15 *Let S be irreducible and μ be an invariant distribution for the Markov chain. Let $f : S \rightarrow \mathbb{R}$ be bounded, measurable. For all initial condition ν , the limit*

$$\frac{1}{N} \sum_{n=1}^N f(X_n) \rightarrow \int f d\mu \quad (10.28)$$

holds P^ν -a.s.

Proof: (10.28) is true when $f(x) = \mathbb{1}_{\{x=i\}}$ by [Proposition 10.12](#). Hence it is true for all $f(x) = \sum_i f(i) \mathbb{1}_{\{x=i\}}$ which is a (countable) linear combination of indicator functions. $\square$

[Proposition 10.15](#) is an example of *ergodic theorem*, which says that the *time average* of certain statistic (f) of a dynamics (X_n) , converges to the *ensemble average*. “Ensemble average” is a term in statistical mechanics, which means average over the probability space in our context; the sample space will be the “ensemble”.

The LHS of (10.28) is similar to what appears in the SLLN. In fact, ergodic theorem answers the question under what general condition, the almost sure limit

$$\lim_{N \rightarrow \infty} \frac{X_1 + X_2 + \cdots + X_N}{N}$$

exists. We already know some sufficient conditions, for example, X_n being i.i.d., or X_n from an irreducible positive recurrent Markov chain. The ergodic theorem generalizes the condition to include *stationary sequences*.

We need to introduce some definitions first.

We say that $A \in \mathcal{S}$ is *invariant* if

$$p(i, A) = 1, \quad i \in A, \quad p(i, A) = 0, \quad i \notin A. \quad (10.29)$$

The sets S and $\emptyset$ are always invariant, and if the Markov chain is irreducible, then these are the only invariant sets. The condition (10.29) can also be written as

$$(P\mathbb{1}_A)(i) = \mathbb{1}_A(i), \quad \forall i.$$

It is convenient to allow exceptional points where (10.29) fails. We measure “exceptional” point using measure: we say that A is (μ) -almost invariant, if $P\mathbb{1}_A = \mathbb{1}_A$ holds μ -a.s.

Definition 10.1 *An invariant distribution μ is ergodic, if $\mu(A) \in \{0, 1\}$ for every (almost) invariant set A .*

To motivate the definition of stationary sequence, let us consider the measure P^μ on $(S^\infty, \mathcal{S}^\infty)$ where μ is an invariant distribution. Then P^μ is *stationary*, in the sense that for every $n \geq 0$ and $k \geq 0$,

$$P^\mu(X_0 \in A_0, X_1 \in A_1, \dots, X_k \in A_k) = P^\mu(X_n \in A_0, X_{n+1} \in A_1, \dots, X_{n+k} \in A_k). \quad (10.30)$$

Recall that the shift operator θ^n in (10.4) (with the convention $\theta = \theta^1$). (10.30) can be rephrased as $P^\mu \circ \theta^{-n} = P^\mu$ for all $n \geq 1$. The definition of stationarity does not depend on the Markov chain, and it can be stated for any probability measure P on $(S^\infty, \mathcal{S}^\infty)$. We will consider general stationary sequence and use P instead of P^μ when the Markov chain structure is irrelevant.

We also introduce a notion of invariant set on $\mathcal{S}^\infty$. We say that $B \in \mathcal{S}^\infty$ is *invariant* if $\theta^{-1}B = B$, and B is *almost invariant* if $\theta^{-1}B = B$ P-a.s., that is $P(B \Delta \theta^{-1}B) = 0$, or $\mathbb{1}_B = \mathbb{1}_{\theta^{-1}B}$ almost surely. Let

$$\mathcal{I} = \{B \in \mathcal{S}^\infty : B \text{ invariant}\}, \quad \mathcal{I}_* = \{B \in \mathcal{S}^\infty : B \text{ almost invariant}\}.$$

Both $\mathcal{I}$ and $\mathcal{I}_*$ are σ -algebras. The next result characterizes the relation between $\mathcal{I}$ and $\mathcal{I}_*$.

Proposition 10.16 *If $B \in \mathcal{I}_*$, then there exists $C \in \mathcal{I}$ such that $P(B \Delta C) = 0$.*

Proof: Let

$$C = \liminf_{n \rightarrow \infty} \theta^{-n}B = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \theta^{-k}B.$$

We have

$$\theta^{-1}C = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \theta^{-k-1}B = \bigcup_{n=2}^{\infty} \bigcap_{k=n}^{\infty} \theta^{-k}B = C,$$

so $C \in \mathcal{I}$.

Since the map θ preserves P , we have

$$P(\theta^{-(n+1)}B \Delta \theta^{-n}B) = P \circ \theta^{-n}(\theta^{-1}B \Delta B) = P(\theta^{-1}B \Delta B) = 0,$$

and hence

$$P(\theta^{-n}B \Delta B) \leq \sum_{k=1}^{n-1} P(\theta^{-k-1}B \Delta \theta^{-k}B) = 0. \quad (10.31)$$

We have

$$B \setminus C = B \cap \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (\theta^{-k}B)^c \right) \subset \bigcup_{k=1}^{\infty} B \cap (\theta^{-k}B)^c = \bigcup_{k=1}^{\infty} (B \setminus \theta^{-k}B), \quad (10.32)$$

and

$$C \setminus B = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} (\theta^{-k}B \setminus B) \subset \bigcup_{n=1}^{\infty} (\theta^{-n}B \setminus B). \quad (10.33)$$

By (10.31) to (10.33), we have $P(B \Delta C) = 0$ and this completes the proof. $\square$

Theorem 10.17 (Birkhoff ergodic theorem) *Let P be a probability measure on $(S^\infty, \mathcal{S}^\infty)$ such that $P \circ \theta^{-1} = P$. Let $f : S^\infty \rightarrow \mathbb{R}$ be $L^1(P)$. Then P-a.s.,*

$$\frac{1}{N} \sum_{n=1}^N f(\theta^n \omega) \rightarrow E[f | \mathcal{I}] = E[f | \mathcal{I}_*]. \quad (10.34)$$

We postpone the proof of **Theorem 10.17** to the end of this section. First we point out the relation of **Theorem 10.17** to **Proposition 10.15**.

Proposition 10.18 *If μ is ergodic, then $\mathcal{I}$ (and $\mathcal{I}_*$) is trivial under P^μ .*

Proof: Let $B \in \mathcal{I}$. We consider

$$A = \{\omega_0 \in S : (\omega_0, \omega_1, \omega_2, \dots) \in B\}.$$

Then

$$\theta^{-1}B \stackrel{\text{a.s.}}{=} \{(\omega_0, \eta_0, \eta_1, \eta_2, \dots) : (\eta_0, \eta_1, \dots) \in B, p(\omega_0, \eta_0) > 0\}. \quad (10.35)$$

Noting that $\eta_0 \in A$ on the RHS of (10.35), since $B = \theta^{-1}B$, we have ω_0 on the RHS almost surely belongs to A , that is, we have for $\mathbf{P}^\mu$ -a.s. ω_0 ,

$$p(\omega_0, A) = 1, \quad \omega_0 \in A, \quad p(\omega_0, A) = 0, \quad \omega_0 \notin A.$$

So A is almost invariant. By ergodicity of μ , we have $\mu(A) \in \{0, 1\}$. If $\mu(A) = 0$, then

$$\mathbf{P}^\mu(B) \leq \mathbf{P}^\mu(A \times S \times S \times \cdots) = 0,$$

and if $\mu(A) = 1$,

$$\mathbf{P}^\mu(B) \geq \mathbf{P}^\mu(A \times A \times A \times \cdots) = 1.$$

Therefore, we show that $\mathbf{P}^\mu(B) \in \{0, 1\}$ if $B \in \mathcal{I}$, and this completes the proof. $\square$

Let us look at some examples.

Markov chain. Consider an irreducible, positive recurrent Markov chain. Then the only invariant sets are $\emptyset$ and S by irreducibility, so $\mathcal{I}$ is trivial. Hence, any invariant distribution is ergodic. We know that invariant distribution exists since the Markov chain is positive recurrent. As $\mathcal{I}$ is trivial, the conditional expectation in (10.34) becomes expectation, and we recover Proposition 10.15 for ν being the invariant distribution. Proposition 10.15 is stronger as we can start from any initial condition; this is the *mixing property*.

SLLN. Consider the infinite product measure $\mathbf{P} = \mu^{\otimes \infty}$ on (S^∞, S^∞) . If $B \in \mathcal{I}$, then $B = \theta^{-n}B \in \mathcal{F}_{\geq n}$ for every n , and hence B belongs to the tail σ -algebra. By Kolmogorov's zero-one law, the tail σ -algebra is trivial, so $\mathcal{I}$ is trivial. Then Theorem 10.17 recovers the SLLN.

Irrational rotation on the circle. We look at a “deterministic” example. Let S be the circle, parametrized by $S = \{e^{2\pi i \omega}, \omega \in \mathbb{R}\}$. The Markov kernel is deterministic, given by

$$p(x, \cdot) = \delta_{x+\rho},$$

that is, $X_n = X_0 + n\rho$. The Lebesgue measure is invariant, since θ is a rotation of the circle by angle ρ . We will show that this measure is ergodic.

Indeed, let A be an invariant set. Since $\mathbb{1}_A \in L^2([0, 1])$, it has a Fourier series expansion

$$\mathbb{1}_A(\omega) \stackrel{\text{a.s.}}{=} \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \omega}.$$

The invariance of A under rotation implies that $\mathbb{1}_A(\omega) = \mathbb{1}_A(\omega + \rho)$, and hence

$$\sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \omega} \stackrel{\text{a.s.}}{=} \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k (\omega + \rho)} = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \rho} \cdot e^{2\pi i k \omega}.$$

Since Fourier series expansion is unique, we have

$$c_k = c_k e^{2\pi i k \rho}, \quad \forall k \in \mathbb{Z}.$$

Recall that ρ is irrational, so $e^{2\pi i k \rho} \neq 1$ unless $k = 0$. Hence $c_k = 0$ for all $k \neq 0$, and $\mathbb{1}_A(\omega) \stackrel{\text{a.s.}}{=} c_0$. So $\mathbb{1}_A(\omega) \stackrel{\text{a.s.}}{=} 1$ or 0, and thus $\text{Leb}(A) \in \{0, 1\}$ (the total length of circle is 1 by our parametrization).

As a consequence of Theorem 10.17, if $f(\omega) = \mathbb{1}_{[a, b]}(\omega)$, then for Lebesgue almost every $x \in S$, we have

$$\frac{\#\{1 \leq n \leq N : x_n = x + n\rho \in [a, b]\}}{N} \rightarrow b - a.$$

This says that irrational rotation tries to evenly distribute points in the limit.

There are more than one proofs of [Theorem 10.17](#), but none of them is easy, as one may expect from the simplicity of the theorem. We include the following proof for the completeness of this note. **Proof of Theorem 10.17:** Without loss of generality we can assume $f \geq 0$. Let us write $(A_N f)(\omega) = \frac{1}{N} \sum_{n=1}^N f(\theta^n \omega)$ and define

$$\bar{f}(\omega) = \limsup_{n \rightarrow \infty} (A_n f)(\omega), \quad \underline{f}(\omega) = \liminf_{n \rightarrow \infty} (A_n f)(\omega).$$

First, both $\bar{f}$ and $\underline{f}$ are measurable with respect to $\mathcal{I}_*$. Indeed, from the structure of Cesaro sums, we have

$$|(A_n f)(\omega) - (A_n f)(\theta \omega)| \leq \frac{1}{n} f(\omega) + \frac{1}{n} f(\theta^n \omega). \quad (10.36)$$

For fixed ω , the first term goes to 0. Since $E f < \infty$, we have

$$\sum_{n=1}^{\infty} P(f(\theta^n \omega) \geq \varepsilon n) = \sum_{n=1}^{\infty} P(f(\omega) \geq \varepsilon n) < \infty,$$

and hence by Borel–Cantelli, the second term in [\(10.36\)](#) also goes to zero.

In the rest of the proof, we will show that

$$E \bar{f} \mathbb{1}_B \leq E f \mathbb{1}_B \leq E \underline{f} \mathbb{1}_B, \quad \forall B \in \mathcal{I}. \quad (10.37)$$

If [\(10.37\)](#) holds, then $P(\underline{f} = \bar{f}) = 1$, and $\bar{f} = E[f | \mathcal{I}]$; this proves [\(10.34\)](#).

[To be filled in]

□

10.6 Ergodic measures for Markov chains

In this section we will use the general result on ergodic theorem from last section to have a better understanding of invariant measures of the Markov chain.

Theorem 10.19 *let μ and $\tilde{\mu}$ be different ergodic measures. Then μ and $\tilde{\mu}$ are mutually singular.*

Proof: Since $\mu \neq \tilde{\mu}$, there exists a bounded measurable function f such that $\int f d\mu \neq \int f d\tilde{\mu}$. Let

$$B = \left\{ \frac{1}{N} \sum_{n=1}^N f(X_n) \rightarrow \int f d\mu \right\}.$$

By ergodicity of μ and $\tilde{\mu}$, the invariant σ -algebra $\mathcal{I}$ is trivial, and hence [Theorem 10.17](#) implies $P^\mu(B) = 1$ and $P^{\tilde{\mu}}(B) = 0$. Let $A = \pi_0^{-1} B$ be the projection of B onto the zero-th coordinate. Then $\mu(A) = 1$ and $\tilde{\mu}(A) = 0$. This completes the proof. □

If μ_1 and μ_2 are invariant distributions, then their convex combinations $\alpha \mu_1 + (1 - \alpha) \mu_2$, $\alpha \in (0, 1)$ are also invariant distribution, simply due to the linearity of the condition $\mu P = \mu$ for invariance. In other words, the space of invariant distribution, denoted by $\mathcal{P}_I$, is a convex set. We say that $\mu \in \mathcal{P}_I$ is *extremal*, if there is no non-trivial convex combination to represent μ , that is, if $\alpha \in (0, 1)$ and $\nu_1, \nu_2 \in \mathcal{P}_I$ are such that

$$\mu = \alpha \nu_1 + (1 - \alpha) \nu_2,$$

then $\mu = \nu_1 = \nu_2$.

The next result gives geometric meaning to ergodic measures.

Theorem 10.20 A probability measure $\mu \in \mathcal{P}_I$ is ergodic, if and only if μ is an extremal point of $\mathcal{P}_I$.

Proof: The “if” part. Suppose that μ is not ergodic, then there exists an invariant set A such that $\mu(A) \in (0, 1)$. Let

$$\nu_1(C) = \mu[C | A] = \frac{\mu(C \cap A)}{\mu(A)}, \quad \nu_2(C) = \mu[C | A^c] = \frac{\mu(C \cap A^c)}{\mu(A^c)}.$$

Then $\mu = \alpha\nu_1 + (1 - \alpha)\nu_2$ where $\alpha = \mu(A) \in (0, 1)$. We will show that $\nu_i \in \mathcal{P}_I$ and thus arrive at a contradiction that μ is extremal.

For $C \in \mathcal{S}$, we have

$$(\nu_1 P)(C) = \int \nu_1(dx) p(x, C) = \int \frac{\mu(dx) \mathbb{1}_A(x)}{\mu(A)} p(x, C). \quad (10.38)$$

We claim that

$$\mathbb{1}_A(x) p(x, C) = p(x, A \cap C). \quad (10.39)$$

Indeed, if $x \in A$, then by invariance of A , $p(x, A) = 1$, so $p(x, C) = p(x, C \cap A)$; if $x \notin A$, then $p(x, A) = 0 = p(x, A \cap C)$, so both sides of (10.39) is 0. Using (10.38), we can continue (10.38) to obtain

$$(\nu_1 P)(C) = \frac{1}{\mu(A)} \int \mu(dx) p(x, A \cap C) = \frac{1}{\mu(A)} \mu(A \cap C) = \nu_1(C).$$

Similarly, we can show that ν_2 is invariant. This proves the “if” direction.

The “only if” part. Let μ be ergodic and $\mu = \alpha\nu_1 + (1 - \alpha)\nu_2$ where $\nu_i \in \mathcal{P}_I$ and $\alpha \in (0, 1)$. Then $\nu_i \ll \mu$, and hence $P^\nu \ll P^\mu$. By Theorem 10.17, for every bounded measurable f , we have

$$\frac{1}{N} \sum_{n=1}^N f(X_n) \rightarrow \int f d\mu, \quad P^\mu\text{-a.s.}$$

The limit is a constant since μ is ergodic. Applying Theorem 10.17 to ν_i , we obtain

$$\frac{1}{N} \sum_{n=1}^N f(X_n) \rightarrow E^{\nu_i}[f | \mathcal{I}] \quad P^{\nu_i}\text{-a.s.}$$

Since P^μ -a.s. limit implies P^{ν_i} -a.s. limit by absolute continuity, we have

$$E^{\nu_i}[f | \mathcal{I}] = \int f d\mu, \quad P^{\nu_i}\text{-a.s.}$$

Taking another expectation, we have

$$\int f d\nu_i = \int f d\mu.$$

As this holds for arbitrary f , we see that $\mu = \nu_1 = \nu_2$. So μ is extremal. $\square$

Now let us consider an irreducible, positive recurrent Markov chain. From a fixed state i , by Proposition 10.11 we obtain an invariant distribution μ_* , with $\mu_*(i) > 0$. Note that

$$\mathcal{P}_I(i) = \{\mu \in \mathcal{P}_I : \mu(i) > 0\}$$

is also a convex set, and its extremal points, denoted by $\mathcal{P}_E(i)$, by a similar argument to Theorem 10.20, are ergodic. But ergodic measures are mutually singular due to Theorem 10.19, so there can be at most one ergodic measure μ such that $\mu(i) > 0$. Since $\mu_* \in \mathcal{P}_I(i)$, we know that $\mathcal{P}_I(i)$ is non-empty; a non-empty convex set has one extremal point, if and only if the convex set is single-point set. Therefore, $\mathcal{P}_I(i) = \{\mu_*\} = \mathcal{P}_E(i)$, and we arrive at the conclusion that the Markov chain has a unique invariant measures, which is also ergodic.

Proposition 1.16

11 Notation

11.1 Abbreviations

i.i.d.	independent, identically distributed
r.v.	random variable
p.m.	probability measure
c.d.f.	cumulative distribution function
f.d.d.	finite-dimensional distribution
ch.f.	characteristic function
u.i.	uniformly integrable

11.2 Relations

$\Rightarrow_d$ or $\Rightarrow$	convergence in distribution/law
$\stackrel{d}{=}$	equal in law

11.3 Functional spaces

$\mathcal{C}[a, b]$	continuous function defined on the interval $[a, b]$
$\mathcal{C}^\alpha[a, b]$	α -Hölder continuous function defined on the interval $[a, b]$
$\mathcal{M}(E)$	probability measures on a metric space E

11.4 Miscellaneous

$\mathcal{L}(X)$	distribution/law of a random variable/element X
$\mathcal{N}(\mu, \sigma^2)$	normal distribution
$\text{Exp}(\lambda)$	exponential distribution
$\text{Poi}(\lambda)$	Poisson distribution

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